GeoTaichi User API, Keyword, and Theory Manual

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1 Scope and Reading Guide

This manual documents the user-facing functions, configuration keywords, and numerical theory of the current GeoTaichi repository. It is organized for lookup: first the theory and derivations, then the API reference. Theory sections link forward to the API entries that activate each method; API entries link back to the derivations that explain the numerical meaning of their options.

The printed table of contents and PDF outline use multiple heading levels: parts separate theory and API, sections separate modules, subsections separate method families or API groups, subsubsections identify individual formulas, schemas, functions, models, and keywords, and paragraph-level headings are available for local derivation steps when a topic needs another layer.

The word "keyword" means a key passed by users through a Python dictionary or keyword argument. The word "function" means a public method normally called from example scripts. Low-level Taichi kernels are listed only when a kernel defines a user-visible method, option, compatibility rule, or formula.

Part I

Theory and Numerical Methods

2 Theory-to-API Locator

This locator is the first page a user should scan when starting from formulas and looking for the corresponding Python calls or keywords. The theory part is organized by physical modules, geometry modules, coupled solvers, and cross-cutting numerical modules.

Theory topic	Theory section	Related API entries
Core MPM weak form	MPM	mpm-set-configuration, mpm-add-element, mpm -add-body, mpm-run
Implicit solid and fluid MPM	Implicit solid MPM, MPM IPC CCD/ACCD, Incompressible fluid MPM, double-layer semi-implicit two-phase MPM	mpm-set-implicit-solver-parameters, key-fluid-level -set, key-density-projection
Adaptive MPM and virtual stress	Adaptive MPM, Virtual stress boundary	key-refinethreshold, key-hangingconstraintmode, mpm-add-virtual-stress-field
SDF and level-set geometry	SDF and Level Set Geometry, Redistance and advection	geometry API addendum, key-fluid-level-set, key-solid -sdf-cut-cell

Theory topic	Theory section	Related API entries
DEM rigid-body dynamics	DEM, Affine Body	dem-add-body, dem-set-affine-body-parameters, dem-run
LSMPM soft bodies	Level-set MPM soft bodies, MPM soft-particle backend, contact search implementation	key-lsmpm, mpm-soft-particle-api, mpdem-soft-affine-ipc-workflow
Finite elements, cloth, and FEM contact	Finite Element Method, FEM collision culling and CCD	fem-create-mesh, fem-set-solver, fem-add-contact, fem-run
DEM/LSDEM/AffineBody–FEM surface coupling	FEM–DEM coupling	fedem-user-api
MPM point–FEM surface coupling	FEM–MPM coupling	fempm-user-api
IGA basis, NURBS utilities, and weak form	IGA, NURBS utility library	iga-add-primitives, iga-run, nurbs-api
MPM-DEM and CFDEM coupling	MPM-DEM	dempm-cfdem-choose-contact-model, key-dragforcemodel, key-coupling-scheme
IGA-MPM coupling	IGA-MPM, IGA-MPM CCD/ACCD	igampm-build, igampm-run, iga-mpm-ipc-coupling-monolithic-barrier-system
Contact search	Contact Search, ACCD	dem-choose-neighbor, key-bvh, key-bvh-rebuild-interval
Contact models	Contact Model, soft-affine IPC, strict contact derivatives	dem-choose-contact-model, key-kappa, key-dhat, key-friction, key-epsv
Constitutive equations	Constitutive Equations, Random-field materials	mpm-add-material, model-mohrcoulomb, model-druckerprager, model-modifiedcamclay, random-field schema
Return mapping and substepping	Return mapping and explicit/substepping stress integration	key-stress-integration, key-softtype, key-ratedependent
Linear solvers and sparse assembly	Linear Solver	key-assemble-type, key-matrix-symmetric, key-multilevel
Optimization methods	Optimization Methods	key-max-iteration-number, key-line-search-work-tol, iga-mpm-ipc-coupling-monolithic-barrier-system

3 MPM

Related API. MPM.add_element, MPM.add_body, MPM.run.

3.1 Backend and Formulation Compatibility

Native MPM, Direct MPM, and LSMPM soft-particle mechanics are distinct runtimes rather than interchangeable names for one solver. Their supported formulations are summarized before the common balance laws.

Route	Integration and formulation	Materials and plasticity	Contact and principal limits
Native solid ULMPM	Explicit or implicit; two or three dimensions	Shared elastic and incremental elastoplastic models	Conventional grid contact; no Native solid MPM–AffineBody coupling
Native solid TLMPM	Explicit only	Shared solid material models	Reference-support dynamics; no implicit TL engine
Native incompressible fluid	FDM pressure projection in two or three dimensions	Fluid response; plasticity is not applicable	The Native/-common AffineBody route is a three-dimensional fixed-step volume-fraction immersed-boundary method, without Barrier IPC or DEM subcycling
Direct explicit MPM	Explicit ULMPM or TLMPM	Dedicated elastic, elastoplastic, and viscous models	Point-set workflow; no axisymmetry and no Native region/template lifecycle
Direct implicit ULMPM	Implicit finite strain in two or three dimensions	Neo-Hookean, finite-strain Drucker–Prager, von Mises, and Modified Cam–Clay	Ground and cross-solver IPC; MPM–AffineBody is three-dimensional with lagged friction
Direct implicit TLMPM	Implicit total Lagrangian	Neo-Hookean elasticity	Finite-strain plasticity and Direct MPM–AffineBody require ULMPM

Route	Integration and formulation	Materials and plasticity	Contact and principal limits
LSMPM soft particle	Explicit level-set contact or monolithic soft–AffineBody IPC	Neo-Hookean, Hencky, Mooney–Rivlin, Gent, or Hydrogel hyperelasticity; no plasticity	Owned by the DEM/MPDEM `scheme="LSMPM"` runtime; multiple soft materials in the IPC route must all be Neo-Hookean

3.2 Strong Form and Weak Form

For a continuum body occupying Ω_t, the momentum equation is

$$\rho \frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = \nabla \cdot \boldsymbol{\sigma} + \rho \boldsymbol{b}, \qquad \boldsymbol{x} \in \Omega_t, \tag{1}$$

with traction $\boldsymbol{\sigma}\boldsymbol{n} = \bar{\boldsymbol{t}}$ on Γ_t and prescribed velocity or displacement on Γ_u. Multiplying by a test function $\boldsymbol{w}$ and integrating gives

$$\int_{\Omega_t} \rho \boldsymbol{w} \cdot \frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} \,\mathrm{d}\Omega = -\int_{\Omega_t} \nabla \boldsymbol{w} : \boldsymbol{\sigma} \,\mathrm{d}\Omega + \int_{\Omega_t} \rho \boldsymbol{w} \cdot \boldsymbol{b} \,\mathrm{d}\Omega + \int_{\Gamma_t} \boldsymbol{w} \cdot \bar{\boldsymbol{t}} \,\mathrm{d}\Gamma. \tag{2}$$

MPM approximates mass and material history by particles and uses a background grid for solving. With nodal basis $N_i(\boldsymbol{x})$ and particle volume V_p, mass m_p, stress $\boldsymbol{\sigma}_p$, and position $\boldsymbol{x}_p$, the standard lumped nodal quantities are

$$m_i = \sum_p m_p N_i(\boldsymbol{x}_p), \tag{3}$$

$$\boldsymbol{p}_i = \sum_p m_p \boldsymbol{v}_p N_i(\boldsymbol{x}_p), \qquad \boldsymbol{v}_i = \frac{\boldsymbol{p}_i}{m_i}, \tag{4}$$

$$\boldsymbol{f}_i^{\mathrm{int}} = -\sum_p V_p \boldsymbol{\sigma}_p \nabla N_i(\boldsymbol{x}_p), \tag{5}$$

$$\boldsymbol{f}_i^{\mathrm{ext}} = \sum_p m_p \boldsymbol{b}_p N_i(\boldsymbol{x}_p) + \int_{\Gamma_t} N_i \bar{\boldsymbol{t}} \,\mathrm{d}\Gamma. \tag{6}$$

The discrete nodal equation is therefore

$$m_i \boldsymbol{a}_i = \boldsymbol{f}_i^{\mathrm{int}} + \boldsymbol{f}_i^{\mathrm{ext}}. \tag{7}$$

3.3 Particle-Grid Transfers

The particle-to-grid stage computes m_i, $\boldsymbol{p}_i$, forces, and active-grid state. The grid-to-particle stage updates particle kinematics:

$$\boldsymbol{v}_p^{n+1,\mathrm{PIC}} = \sum_i N_i(\boldsymbol{x}_p^n)\boldsymbol{v}_i^{n+1}, \tag{8}$$

$$\boldsymbol{v}_p^{n+1,\mathrm{FLIP}} = \boldsymbol{v}_p^n + \sum_i N_i(\boldsymbol{x}_p^n)\left(\boldsymbol{v}_i^{n+1} - \boldsymbol{v}_i^n\right), \tag{9}$$

$$\boldsymbol{v}_p^{n+1} = \alpha\boldsymbol{v}_p^{n+1,\mathrm{PIC}} + (1-\alpha)\boldsymbol{v}_p^{n+1,\mathrm{FLIP}}, \tag{10}$$

$$\boldsymbol{x}_p^{n+1} = \boldsymbol{x}_p^n + \Delta t \sum_i N_i(\boldsymbol{x}_p^n)\boldsymbol{v}_i^{n+1}. \tag{11}$$

The keyword `velocity_projection` selects the projection form; `alphaPIC` is the blending value for `PIC/FLIP`. `Affine` and `Taylor` activate APIC/TPIC-style enriched particle velocity information.

The particle velocity gradient is

$$\boldsymbol{L}_p = \sum_i \boldsymbol{v}_i \otimes \nabla N_i(\boldsymbol{x}_p), \tag{12}$$

which gives the small-strain increment $\Delta\boldsymbol{\varepsilon}_p = \Delta t\,\mathrm{sym}(\boldsymbol{L}_p)$ or the finite-strain deformation update

$$\dot{\boldsymbol{F}}_p = \boldsymbol{L}_p\boldsymbol{F}_p, \qquad \boldsymbol{F}_p^{n+1} \approx (\boldsymbol{I} + \Delta t\boldsymbol{L}_p)\boldsymbol{F}_p^n. \tag{13}$$

3.4 Shape Functions

`Linear` uses the standard multilinear hat basis. In one dimension,

$$N_i(x_p) = \max\left(0, 1 - \frac{|x_p - x_i|}{h}\right). \tag{14}$$

`GIMP` replaces the point value by the cell average over the particle domain Ω_p:

$$\bar{N}_i(x_p) = \frac{1}{|\Omega_p|}\int_{\Omega_p} N_i(\boldsymbol{x})\,\mathrm{d}\Omega, \tag{15}$$

which reduces cell-crossing noise. `QuadBSpline` and `CubicBSpline` increase support width and smoothness. Larger support improves smoothness but increases assembly stencil size, especially for 3D implicit MPM and adaptive shape-function hanging constraints.

Direct `QuadBSpline` uses a three-node tensor-product stencil per axis. On a finite grid, the first two and last two nodes use the existing boundary-modified polynomials, and the stencil is shifted inside the grid. There must be at least four nodes on each axis. For particles inside the grid, the corrected basis satisfies

$$\sum_i N_i = 1, \qquad \sum_i N_i\boldsymbol{x}_i = \boldsymbol{x}_p, \qquad \sum_i \nabla N_i = \boldsymbol{0}, \qquad \sum_i \boldsymbol{x}_i \otimes \nabla N_i = \boldsymbol{I}. \tag{16}$$

The IPC position Hessian differentiates this same boundary polynomial; truncating an interior stencil or mixing its derivatives with a corrected shape value violates these identities.

Endpoint gradients use the one-sided derivative from inside the grid. Particles outside the grid remain an error. In axisymmetry, $u_r = ar$ and $u_z = bz$ must reproduce $\boldsymbol{F} = \mathrm{diag}(1+a, 1+b, 1+a)$, including near the axis. The reference volume and pressure-area measure include $2\pi R$, while the interpolation basis does not. See `QuadBSpline`.

The implicit IGA–MPM flexible-barrier example uses this basis in its default 2D plane-strain section, with a solid IGA plate and unit out-of-plane depth. Its two coordinates represent physical (x, z); reported mass and forces are per unit depth. Constant nonassociated DP retains its physical Jacobian and the independent lagged-friction iteration. The absolute linear tolerance must be smaller than the force-residual target of the inner Newton solve.

3.5 Explicit Integration

The explicit nodal update is

$$\boldsymbol{v}_i^{n+1} = \boldsymbol{v}_i^n + \Delta t \frac{\boldsymbol{f}_i^{\mathrm{int}} + \boldsymbol{f}_i^{\mathrm{ext}}}{m_i}. \tag{17}$$

The stable timestep is limited by wave speed and element length:

$$\Delta t \le C_{\mathrm{CFL}} \min_e \frac{h_e}{c_e + |\boldsymbol{v}|}, \qquad c_e \approx \sqrt{\frac{K + 4G/3}{\rho}}, \tag{18}$$

with keyword `CFL` providing the safety factor.

3.6 Stabilization

The `B-BarMethod` replaces the volumetric strain part with an element average:

$$\nabla N_i \to \nabla N_i^{\mathrm{dev}} + \overline{\nabla N_i}^{\mathrm{vol}}, \tag{19}$$

which reduces volumetric locking for nearly incompressible materials. The `F-BarMethod` modifies the deformation gradient through a volume-average Jacobian:

$$\bar{\boldsymbol{F}}_p = \left(\frac{\bar{J}}{J_p}\right)^{1/d} \boldsymbol{F}_p, \tag{20}$$

where d is the spatial dimension. `fbar_fraction` and `jacobian_clamp` control the amount and bounds of this correction.

For the current two-dimensional plane-strain B-Bar kernels, let $\boldsymbol{g}_i = \nabla N_i$, $\bar{\boldsymbol{g}}_i$ be the cell-averaged gradient, and

$$\boldsymbol{t}_i = \frac{1}{2}(\bar{\boldsymbol{g}}_i - \boldsymbol{g}_i). \tag{21}$$

The projected velocity-gradient contribution of node i is

$$\boldsymbol{L}_i = \boldsymbol{v}_i \otimes \boldsymbol{g}_i + (\boldsymbol{t}_i \cdot \boldsymbol{v}_i)\boldsymbol{I}_2. \tag{22}$$

For in-plane stress vector $\boldsymbol{\sigma} = [\sigma_{xx}, \sigma_{yy}, \sigma_{zz}, \sigma_{xy}]^T$, the matching internal-force contribution is

$$\boldsymbol{f}_i^{\mathrm{int}} = -V_p \begin{bmatrix} (g_{ix} + t_{ix})\sigma_{xx} + t_{ix}\sigma_{yy} + g_{iy}\sigma_{xy} \\ t_{iy}\sigma_{xx} + (g_{iy} + t_{iy})\sigma_{yy} + g_{ix}\sigma_{xy} \end{bmatrix}. \tag{23}$$

The out-of-plane plane-strain stress σ_{zz} therefore does not enter two-dimensional virtual work. Explicit velocity-gradient and implicit displacement-gradient kernels use this same projected operator, which keeps force and kinematics work-conjugate. Particle volume uses the finite incremental area ratio $\det(\boldsymbol{I}_2 + \Delta t \boldsymbol{L})$, rather than adding an artificial three-dimensional trace term.

3.7 Implicit Solid MPM

Related API. MPM.set_implicit_solver_parameters, assemble_type, calculate_reaction_force.

3.7.1 Newmark Discretization

Implicit solid MPM solves for the displacement increment $\Delta \boldsymbol{u}$. With Newmark parameters γ and β,

$$\boldsymbol{a}_{n+1}(\Delta \boldsymbol{u}) = \frac{1}{\beta \Delta t^2} \Delta \boldsymbol{u} - \frac{1}{\beta \Delta t} \boldsymbol{v}_n - \left(\frac{1}{2\beta} - 1 \right) \boldsymbol{a}_n, \tag{24}$$

$$\boldsymbol{v}_{n+1}(\Delta \boldsymbol{u}) = \boldsymbol{v}_n + \Delta t \left((1-\gamma) \boldsymbol{a}_n + \gamma \boldsymbol{a}_{n+1} \right). \tag{25}$$

The nonlinear residual is

$$\boldsymbol{R}(\Delta \boldsymbol{u}) = \boldsymbol{M} \boldsymbol{a}_{n+1} + \boldsymbol{C} \boldsymbol{v}_{n+1} + \boldsymbol{f}_{\text{int}}(\boldsymbol{u}_{n+1}) - \boldsymbol{f}_{\text{ext},n+1}. \tag{26}$$

Newton iteration solves

$$\boldsymbol{K}_{\text{eff}} \delta \boldsymbol{u} = -\boldsymbol{R}, \tag{27}$$

where

$$\boldsymbol{K}_{\text{eff}} = \frac{1}{\beta \Delta t^2} \boldsymbol{M} + \frac{\gamma}{\beta \Delta t} \boldsymbol{C} + \boldsymbol{K}_{\text{int}}. \tag{28}$$

3.7.2 Material, Geometric, and Jaumann Tangents

For hyperelastic energy density $\Psi(\boldsymbol{F})$,

$$\boldsymbol{P} = \frac{\partial \Psi}{\partial \boldsymbol{F}}, \qquad \mathbb{A} = \frac{\partial \boldsymbol{P}}{\partial \boldsymbol{F}} = \frac{\partial^2 \Psi}{\partial \boldsymbol{F}^2}. \tag{29}$$

For small-strain materials,

$$\boldsymbol{\sigma} = \mathbb{C} : \boldsymbol{\varepsilon}^e, \qquad \Delta \boldsymbol{f}_i^{\text{int}} = -\sum_p V_p \left(\mathbb{C} : \Delta \boldsymbol{\varepsilon}_p \right) \nabla N_i. \tag{30}$$

The element block for nodes i, j follows

$$\boldsymbol{K}_{ij}^{\text{cons}} = \sum_p V_p \, \boldsymbol{B}_{ip}^T \mathbb{C}_p^{\text{alg}} \boldsymbol{B}_{jp}, \tag{31}$$

where $\mathbb{C}^{\text{alg}}$ is the elastic or return-mapping consistent tangent. This constitutive block is only one part of the updated Lagrangian small-strain tangent. Linearizing the internal force with the current Cauchy stress also gives the initial-stress or geometric stiffness

$$K^{\text{geo}}_{i\alpha,j\beta} = \sum_p V_p \, \delta_{\alpha\beta} \nabla N^T_{ip} \boldsymbol{\sigma}_p \nabla N_{jp}. \tag{32}$$

This term is present even if the constitutive model is linear elastic; it comes from differentiating the spatial gradients, volume mapping, or current configuration stress term in the updated Lagrangian residual.

Small-strain stress updates in the source also add an objective Jaumann rotation increment

$$\Delta\boldsymbol{\sigma}^J = \Delta t(\boldsymbol{\sigma}\boldsymbol{W} - \boldsymbol{W}\boldsymbol{\sigma}), \qquad \boldsymbol{W} = \frac{1}{2}(\boldsymbol{L} - \boldsymbol{L}^T). \tag{33}$$

Because $\boldsymbol{W}$ depends on the trial displacement gradient, this contributes another tangent block:

$$K^J_{i\alpha,j\beta} = \sum_p V_p \, B_{ip,\mu} \frac{\partial(\Delta\sigma^J_{\mu\alpha})}{\partial u_{j\beta}}. \tag{34}$$

In Voigt notation with $\boldsymbol{\sigma} = [\sigma_{11}, \sigma_{22}, \sigma_{33}, \sigma_{12}, \sigma_{23}, \sigma_{13}]^T$ and $\Delta\boldsymbol{\omega} = [\omega_{12}, \omega_{23}, \omega_{31}]^T$, the `Sigrot` helper evaluates

$$\Delta\sigma^J_{11} = 2(-\omega_{31}\sigma_{13} + \omega_{12}\sigma_{12}), \tag{35}$$
$$\Delta\sigma^J_{22} = 2(-\omega_{12}\sigma_{12} + \omega_{23}\sigma_{23}), \tag{36}$$
$$\Delta\sigma^J_{33} = 2(-\omega_{23}\sigma_{23} + \omega_{31}\sigma_{13}), \tag{37}$$
$$\Delta\sigma^J_{12} = -\omega_{31}\sigma_{23} + \omega_{23}\sigma_{13} + \omega_{12}(\sigma_{22} - \sigma_{11}), \tag{38}$$
$$\Delta\sigma^J_{23} = -\omega_{12}\sigma_{13} + \omega_{31}\sigma_{12} + \omega_{23}(\sigma_{33} - \sigma_{22}), \tag{39}$$
$$\Delta\sigma^J_{13} = -\omega_{23}\sigma_{12} + \omega_{12}\sigma_{23} + \omega_{31}(\sigma_{11} - \sigma_{33}). \tag{40}$$

The internal-force tangent is therefore

$$\boldsymbol{K}_{\text{int}} = \boldsymbol{K}^{\text{cons}} + \boldsymbol{K}^{\text{geo}} + \boldsymbol{K}^J, \tag{41}$$

when the selected assembly kernel includes the small-strain stress and spin linearization. In the current MPM kernels, `assemble_elastic_stiffness_matrix`, its 2D variant, and the 2D full `assemble_stiffness_matrix_2D` include explicit current-stress and Jaumann terms next to the constitutive contraction. The 3D full `assemble_stiffness_matrix` contracts the supplied full $\mathbb{C}^{alg}$ without appending a separate stress-rate block, so comparisons with continuum formulas must use the same kernel path. For associated or small-strain elastic updates the constitutive tangent can be symmetric, but geometric stress terms, Jaumann terms, non-associated plastic flow, and frictional contact can make the assembled tangent nonsymmetric; the `matrix_symmetric` HashTriplet option must match the assembled matrix, not only the material tensor.

3.7.3 Assembly Types

assemble_type	Meaning
MatrixFree	Does not store a global matrix. The solver applies $\boldsymbol{K}_{\text{eff}}\boldsymbol{x}$ through particle/grid loops. For solid MPM, Dirichlet conditions currently use the existing penalty path.
COO	Builds an assembled triplet/coordinate matrix, then applies strong Dirichlet constraints by modifying rows/columns and right-hand side.
HashTriplet	Accumulates matrix blocks through a hash-based triplet reduction. It reduces duplicate triplets and memory pressure for large implicit systems. The symmetric-storage flag should follow the material tangent symmetry.
MatrixFreeMGPCG	Legacy semi-implicit pressure alias; internally maps to MatrixFree plus pressure_solver="MGPCG". It is rejected for implicit solid MPM.
CSR or csr	Alias to COO in current option normalization.

3.7.4 Dirichlet Constraints and Reaction Force

For assembled systems with constrained dofs c, strong constraints enforce $u_c = \bar{u}_c$. A common row/column modification is

$$K_{cc} \leftarrow 1, \qquad K_{cj} \leftarrow 0, \qquad K_{ic} \leftarrow 0, \tag{42}$$

$$b_c \leftarrow \bar{u}_c. \tag{43}$$

The physical reaction force should be computed from the unconstrained operator:

$$\boldsymbol{r}_c = \left(\boldsymbol{K}_{\text{raw}}\boldsymbol{u} - \boldsymbol{b}_{\text{raw}}\right)_c. \tag{44}$$

For MatrixFree penalty constraints, the constrained equation is approximated as

$$\left(\boldsymbol{K} + \alpha \boldsymbol{E}_c^T \boldsymbol{E}_c\right)\boldsymbol{u} = \boldsymbol{b} + \alpha \boldsymbol{E}_c^T \bar{\boldsymbol{u}}, \tag{45}$$

so the reaction is approximately

$$\boldsymbol{r}_c \approx \alpha(\boldsymbol{u}_c - \bar{\boldsymbol{u}}_c) \tag{46}$$

plus any raw residual contribution. This is why assembled COO/HashTriplet strong constraints and MatrixFree penalty constraints can be close but not bit-identical.

3.8 Implicit MPM IPC Contact and CCD/ACCD

Related API. dhat, kappa, line_search_work_tol, MPM.set_implicit_solver_parameters.

The direct/soft-particle implicit MPM IPC branch treats contact as part of the nonlinear incremental potential. The active contact set is built from surface particles: point-ground pairs for rigid walls and point-point pairs for surface particles from different bodies. For a Newton increment $\Delta\boldsymbol{u}$, the temporary surface-particle position is

$$\boldsymbol{x}_p(\alpha) = \boldsymbol{x}_p^n + \alpha \sum_{i \in S(p)} N_i(\boldsymbol{x}_p^n)\Delta\boldsymbol{u}_i, \tag{47}$$

where $S(p)$ is the MPM stencil of particle p. Contact CCD therefore operates on particle positions induced by grid dofs, not on independent particle unknowns.

For a plane ground with signed distance $d_p^n > 0$, ground normal $\boldsymbol{n}_g$, and relative displacement

$$\Delta \boldsymbol{x}_{rel,p} = \sum_i N_i(\boldsymbol{x}_p)\Delta \boldsymbol{u}_i - \boldsymbol{v}_g \Delta t, \tag{48}$$

the code computes a closing speed $s_p = \boldsymbol{n}_g \cdot \Delta \boldsymbol{x}_{rel,p}$. If $s_p < 0$, the admissible fraction is

$$\alpha_{ground,p} = \eta_g \frac{d_p^n}{-s_p}, \qquad \eta_g < 1, \tag{49}$$

and the kernel takes the minimum over active surface-particle/wall pairs. This matches `ground_ccd` and its full-scan fallback.

For two MPM surface particles a, b, define

$$\boldsymbol{r}^n = \boldsymbol{x}_a^n - \boldsymbol{x}_b^n, \qquad \Delta \boldsymbol{r} = \sum_i N_i(\boldsymbol{x}_a)\Delta \boldsymbol{u}_i - \sum_j N_j(\boldsymbol{x}_b)\Delta \boldsymbol{u}_j. \tag{50}$$

The point-point CCD solves the quadratic

$$\|\boldsymbol{r}^n + \alpha \Delta \boldsymbol{r}\|^2 = \eta_p \|\boldsymbol{r}^n\|^2, \qquad \eta_p < 1, \tag{51}$$

and uses the smallest positive real root. In the current source this is `particle_ccd`; if the candidate set appears too optimistic relative to the CFL cap, `particle_full_ccd` scans all surface-particle pairs.

The final contact-safe step for MPM IPC is

$$\alpha_0 = \min(\alpha_{ground}, \alpha_{particle}, \alpha_{material}, \alpha_{CFL}), \tag{52}$$

where $\alpha_{material}$ prevents element inversion and $\alpha_{CFL} = 0.5\hat{d}/\max_p \|\Delta \boldsymbol{x}_p\|$ keeps the step small relative to the barrier thickness. Energy backtracking is then applied only inside this interval.

ACCD is the conservative generalization used when an exact root is unavailable or too expensive. For a distance $d(\alpha)$ and safe distance d_{safe}, let an upper bound for speed be

$$L(\alpha) \geq \left| \frac{\mathrm{d}d}{\mathrm{d}\alpha} \right| \leq \sum_{m \in \mathcal{P}} \left\| \frac{\partial d}{\partial \boldsymbol{x}_m} \right\| \|\Delta \boldsymbol{x}_m\|, \tag{53}$$

where $\mathcal{P}$ is the primitive vertex set. Additive conservative advancement repeats

$$\alpha_{k+1} = \alpha_k + \eta_{accd} \frac{d(\alpha_k) - d_{safe}}{L(\alpha_k) + \epsilon}, \qquad 0 < \eta_{accd} < 1, \tag{54}$$

until $\alpha \geq 1$, $d - d_{safe}$ is below tolerance, or the maximum ACCD iteration is reached. This is the correct extension for curved MPM contact surfaces, particle-level set contact, and any future non-planar wall contact. The existing MPM point-ground and point-point kernels are faster closed-form special cases of this ACCD pattern.

3.9 Incompressible Fluid MPM

Related API. fluid_level_set, density_projection, multilevel.

3.9.1 Projection Equation

The incompressible fluid branch is selected by `solver_type="Implicit"`, `material_type="Fluid"`, and `discretization="FDM"`. The velocity is stored on a staggered/MAC grid. Starting from the incompressible Navier-Stokes equations,

$$\rho\left(\frac{\partial \boldsymbol{u}}{\partial t}+\boldsymbol{u}\cdot\nabla\boldsymbol{u}\right)=-\nabla p+\mu\nabla^2\boldsymbol{u}+\rho\boldsymbol{b}+\boldsymbol{f}_c, \tag{55}$$

$$\nabla\cdot\boldsymbol{u}=0, \tag{56}$$

the solver first computes an intermediate velocity $\boldsymbol{u}^*$ without the new pressure. Then

$$\boldsymbol{u}^{n+1}=\boldsymbol{u}^*-\Delta t\frac{1}{\rho}\nabla p^{n+1}. \tag{57}$$

Taking divergence and enforcing $\nabla\cdot\boldsymbol{u}^{n+1}=0$ gives the pressure Poisson equation

$$\nabla\cdot\left(\frac{1}{\rho}\nabla p^{n+1}\right)=\frac{1}{\Delta t}\nabla\cdot\boldsymbol{u}^*. \tag{58}$$

On a MAC grid, each pressure cell couples to neighboring pressure cells through face-centered velocity differences. Free surfaces and solid cut cells modify the face coefficient by the open fluid fraction.

3.9.2 Particle-to-MAC-Grid Reduction

The incompressible FDM path transfers particle mass and momentum to staggered faces before the pressure projection. Direct particle scatter is mathematically

$$m_f=\sum_p N_{fp}m_p,\qquad (m\boldsymbol{u})_f=\sum_p N_{fp}m_p\boldsymbol{u}_p, \tag{59}$$

but many particles in one pressure cell contend on the same face atomics. The current implementation can first bin particles by Cartesian cell, reduce the contributions of a cell tile, and issue one reduced update per tile/face. With `GEOTAICHI_INCOMPRESSIBLE_P2G=auto`, CUDA selects the tile-local shared-memory kernel and other backends retain direct atomic scatter. The diagnostic values `atomic`, `shared`, `shared_memory`, and `cell_reduced` select an explicit implementation: `shared` aliases use CUDA shared memory on CUDA and a cell-reduced kernel on other backends. These routes implement the same transfer; the environment variable is an A/B and performance control, not a different physical model.

Cell binning allocates integer counts, cursors, a particle permutation, and prefix-scan workspace proportional to the number of pressure cells and the configured `max_particle_number`. This extra workspace is the cost of reducing atomic contention at high particles-per-cell ratios.

3.9.3 Level Set and Free Surface

The signed distance function ϕ satisfies

$$\phi(\boldsymbol{x})=\begin{cases}-\operatorname{dist}(\boldsymbol{x},\Gamma_f), & \boldsymbol{x}\in\Omega_f,\\ \operatorname{dist}(\boldsymbol{x},\Gamma_f), & \boldsymbol{x}\notin\Omega_f.\end{cases} \tag{60}$$

The pressure projection uses ϕ to detect fluid, air, and cut faces. If a cell-centered pressure unknown p_i has a neighbor p_j across a face, the fluid fraction can be approximated by

$$\theta_{ij} = \text{clamp}\left(\frac{\phi_i}{\phi_i - \phi_j}, \theta_{\min}, 1\right) \tag{61}$$

when the interface crosses the segment. The discrete pressure gradient across the face is then scaled by $1/\theta_{ij}$ or by an equivalent cut-cell coefficient.

The current optimized level-set path avoids a global particle SDF rebuild when the solver only needs cell type information. Surface tension and non-interior density projection still require particle/interface geometry, so particle SDF construction is restricted to the narrow interface band.

3.9.4 Density Projection

Density projection corrects particle positions to reduce local volume error. Let $\rho_h(\boldsymbol{x}_p)$ be the grid-interpolated density and ρ_0 the target density. A generic projection step moves particles along a density gradient direction:

$$\Delta \boldsymbol{x}_p = -\lambda \frac{\rho_h(\boldsymbol{x}_p) - \rho_0}{\|\nabla \rho_h(\boldsymbol{x}_p)\|^2 + \epsilon} \nabla \rho_h(\boldsymbol{x}_p), \tag{62}$$

with clamping controlled by `density_projection_error_clamp` and `density_projection_max_shift_ratio`. When `density_projection_interior_only=True`, the projection avoids free-surface particles and therefore does not need a full particle SDF rebuild every step.

3.9.5 Solid SDF Cut Cells

When `solid_sdf_cut_cell=True`, solid regions reduce the available fluid volume near boundaries. In finite-volume form, a pressure equation cell uses

$$\sum_{f \in \partial c} A_f \alpha_f \frac{p_c - p_{n(f)}}{\rho h_f} = \frac{V_c \alpha_c}{\Delta t} (\nabla \cdot \boldsymbol{u}^*)_c, \tag{63}$$

where α_c and α_f are cell and face open fractions. The keyword `solid_cut_cell_min_fraction` prevents singular tiny cut cells.

3.10 Double-Layer Semi-Implicit Two-Phase MPM

Related API. MPM.set_semi_implicit_solver_parameters, Porosity, Permeability.

The double-layer formulation represents the skeleton and pore fluid with disjoint particle sets

$$\mathcal{P} = \mathcal{P}_s \cup \mathcal{P}_f, \qquad \mathcal{P}_s \cap \mathcal{P}_f = \emptyset. \tag{64}$$

The skeleton is transferred to a collocated MPM grid, while fluid-velocity components occupy the faces of a staggered MAC grid. For cell volume V_c, the mapped solid fraction and porosity are

$$\phi_s = \frac{1}{V_c} \sum_{p \in \mathcal{P}_s} N_{cp} V_p (1 - n_p), \qquad \phi_f = n = 1 - \phi_s. \tag{65}$$

Solid nodal mass and momentum are

$$m_{s,i} = \sum_{p \in \mathcal{P}_s} N_{ip} m_{s,p}, \qquad \boldsymbol{p}_{s,i} = \sum_{p \in \mathcal{P}_s} N_{ip} m_{s,p} \boldsymbol{v}_{s,p}. \tag{66}$$

Let $\boldsymbol{w} = \boldsymbol{v}_f - \boldsymbol{v}_s$ be the relative phase velocity. A drag force density $\boldsymbol{d} = K_d \boldsymbol{w}$ produces equal-and-opposite phase accelerations

$$\boldsymbol{a}_s^d = \frac{K_d \boldsymbol{w}}{\rho_s (1-n)}, \qquad \boldsymbol{a}_f^d = -\frac{K_d \boldsymbol{w}}{\rho_f n}. \tag{67}$$

The supported coefficient laws include the permeability form

$$K_d = \frac{n^2 \gamma_f}{k}, \tag{68}$$

and the Ergun form

$$K_d = \frac{150 \mu_f (1-n)^2}{n d_g^2} + \frac{1.75 \rho_f (1-n)}{d_g} \|\boldsymbol{w}\|. \tag{69}$$

For Beetstra drag, define

$$Re = \frac{\rho_f d_g \|\boldsymbol{w}\|}{\mu_f}, \tag{70}$$

$$f_0 = \frac{10(1-n)}{n^2} + n^2 \left(1 + 1.5\sqrt{1-n}\right), \tag{71}$$

$$f_{Re} = \frac{0.413 Re}{24 n^2} \frac{n^{-1} + 3(1-n)n + 8.4 Re^{-0.343}}{1 + 10^{3(1-n)} Re^{-(5-4n)/2}}, \tag{72}$$

$$K_d = \frac{18 \mu_f n (1-n)}{d_g^2} (f_0 + f_{Re}). \tag{73}$$

Freezing K_d during one step and integrating drag by backward Euler gives

$$\boldsymbol{w}^{n+1} = \frac{\boldsymbol{w}^*}{1 + \Delta t K_d [\rho_s^{-1} (1-n)^{-1} + \rho_f^{-1} n^{-1}]}. \tag{74}$$

The fluid predictor also contains gravity and viscous diffusion,

$$\boldsymbol{v}_f^* = \boldsymbol{v}_f^n + \Delta t \left(\boldsymbol{g} + \boldsymbol{a}_f^d + \nu_f \nabla^2 \boldsymbol{v}_f^n \right). \tag{75}$$

Mixture incompressibility requires

$$\nabla \cdot [(1-n)\boldsymbol{v}_s + n \boldsymbol{v}_f] = 0, \tag{76}$$

or, after expanding the variable porosity,

$$(1-n) \nabla \cdot \boldsymbol{v}_s + n \nabla \cdot \boldsymbol{v}_f + \nabla n \cdot (\boldsymbol{v}_f - \boldsymbol{v}_s) = 0. \tag{77}$$

With pressure mobility

$$M_p = \frac{1-n}{\rho_s} + \frac{n}{\rho_f}, \tag{78}$$

the non-incremental fractional step yields

$$\Delta t \nabla \cdot (M_p \nabla p^{n+1}) = \nabla \cdot [(1-n)\boldsymbol{v}_s^* + n\boldsymbol{v}_f^*]. \tag{79}$$

Equivalently, with $k_1 = \Delta t M_p$,

$$\nabla \cdot (k_1 \nabla p^{n+1}) = f_{Poi}, \tag{80}$$

$$f_{Poi} = (1-n)\nabla \cdot \boldsymbol{v}_s^* + n\nabla \cdot \boldsymbol{v}_f^* + \nabla n \cdot (\boldsymbol{v}_f^* - \boldsymbol{v}_s^*). \tag{81}$$

Face averaging of k_1 produces a symmetric positive-definite pressure operator suitable for PCG/MGPCG. The assembled positive-diagonal matrix represents $-\nabla\cdot(k_1\nabla p)$, so its stored right-hand side is $-f_{Poi}$.

After the pressure solve, the continuum corrections are

$$\boldsymbol{v}_s^{n+1} = \boldsymbol{v}_s^* - \frac{\Delta t}{\rho_s}\nabla p^{n+1}, \qquad \boldsymbol{v}_f^{n+1} = \boldsymbol{v}_f^* - \frac{\Delta t}{\rho_f}\nabla p^{n+1}. \tag{82}$$

The solid correction is assembled weakly by sampling pressure to solid particles and scattering the pressure stress $-p\boldsymbol{I}$:

$$\boldsymbol{v}_{s,i}^{n+1} = \boldsymbol{v}_{s,i}^* + \frac{\Delta t}{m_{s,i}} \sum_{p \in \mathcal{P}_s} (1-n_p) V_p p_p^{n+1} \boldsymbol{I} \nabla N_{ip}. \tag{83}$$

At a fluid–air face, signed-distance samples $\varphi_F < 0$ and $\varphi_A > 0$ define the Ghost-Fluid fraction

$$\theta = \mathrm{clamp}\left(\frac{\varphi_F}{\varphi_F - \varphi_A}, 0.01, 1\right). \tag{84}$$

Eliminating the air-side ghost pressure from the zero-interface-pressure condition gives

$$p_g = -\frac{1-\theta}{\theta} p_F, \tag{85}$$

which is equivalent to a cut-face coefficient $k_1/(\theta h^2)$. Solid or domain-wall neighbors impose a Neumann condition by removing that neighbor from the pressure stencil. Fluid-particle volume remains consistent with mapped porosity through

$$V_p^f = \frac{m_p^f}{\rho_f n_p}. \tag{86}$$

This route supports two- and three-dimensional Cartesian coordinates, rejects axisymmetry, and requires `pressure_solver="MGPCG"`. Supported mapping choices are `USL`, `USF`, and `MUSL`. `Linear` and `SmoothLinear` use two MAC support nodes per direction, `GIMP` and `QuadBSpline` use three, and `CubicBSpline` uses four.

3.11 Adaptive MPM Refinement, Splitting, and Constraints

Related API. MPM.add_element, RefineThreshold, HangingConstraintMode.

Adaptive MPM is part of the MPM discretization. The implemented grid is a 2 : 1-balanced hierarchy with maximum level one or two above the background. A particle indicator η_p is projected to a coarse cell by

$$\eta_c = \max_{p \in c} \eta_p. \tag{87}$$

Cells with $\eta_c \geq \eta_{ref}$ are refined and then dilated by buffer cells. At each scheduled check a qualifying original coarse cell advances one level, capped by the configured maximum. Both levels use the same threshold: a second qualifying check promotes level one to level two. There is no independent second-level error estimator, child-cell indicator or coarsening step. Buffer dilation and subsequent 2 : 1 balancing may also promote neighboring cells. The user limiter `RefineRatio` is interpreted as the maximum fraction of original coarse cells that may be refined:

$$r_c = \frac{n_r}{N_c} \leq r_{\max}. \tag{88}$$

Here N_c is the number of coarse cells, n_r is the number of refined coarse cells, and $r_{\max}$ is the user value of `RefineRatio`. Internally the code converts this coarse-cell fraction into the capacity needed by the fine grid. For a two-level 2 : 1 hierarchy in d dimensions, each refined coarse cell replaces one coarse cell by 2^d fine cells, so the added-cell ratio is

$$\rho_{ref} = \frac{(2^d - 1)n_r}{N_c}. \tag{89}$$

Thus a 2D refined coarse cell contributes three added fine cells and a 3D refined coarse cell contributes seven added fine cells. A user value `RefineRatio`$= 0.5$ therefore still means that at most 50% of the coarse cells may refine, while the internal added-cell budget is 150% in 2D and 350% in 3D before boundary-node reserve and shape-function support are included.

Runtime level merging, neighbor balancing, node allocation and interface construction operate on device fields in both dimensions. For levels in $[0, L_{\max}]$, at most $L_{\max} - 1$ synchronous neighbor propagation passes produce the least non-coarsening level field with adjacent differences at most one. Support nodes are requested only for newly added refinement levels; unique requests append compact IDs without renumbering existing nodes. Device block scans order each appended node batch by logical index and preserve particle order within material groups, retaining locality in later transfers. Hanging constraints are expanded through strictly coarser levels to independent masters, while bridging weights use bounded device propagation from the coarse/fine interface. Unchanged topology retains its tables. Only scalar capacity/status counts and material launch bounds return to the host; particle and topology arrays remain on device until explicit inspection or recording.

A split particle creates 2^d children satisfying

$$\sum_a m_a = m_p, \qquad \sum_a m_a \boldsymbol{v}_a = m_p \boldsymbol{v}_p, \qquad \sum_a V_a = V_p. \tag{90}$$

On GPU the child index must be obtained with an atomic append counter. Only the active parent particle and its active state variables should be copied; a full preallocated field copy during every split step makes refinement memory bandwidth proportional to capacity rather than active particle count. For 3D split scratch storage, P parents satisfy $N_{old} + 7P \leq N_{cap}$ and $N_{old} \geq P$, hence $P \leq \lfloor N_{cap}/8 \rfloor$. Parent workspaces use this bound and the configured batch limit rather than reserving one entry for every particle. For free surfaces and slope faces, symmetric 2^d-child placement can put upper children outside the original material

domain. The 3D option `SplitInteriorOnly` therefore computes a coarse-cell particle-volume fraction and only splits cells satisfying

$$\frac{\sum_{p\in c} V_p}{V_c} \geq f_{int}. \tag{91}$$

Here f_{int} is the value of `SplitInteriorVolumeFraction`. Cells below this threshold may still refine the background grid, but their boundary particles remain unsplit so the split operation does not create an artificial surface protrusion.

Fine-grid hanging nodes obey

$$u_h = \sum_{a\in\mathcal{M}(h)} N_a(x_h) u_a. \tag{92}$$

Strong implicit enforcement writes $u = T\hat{u}$, so

$$\hat{K} = T^T K T, \qquad \hat{r} = T^T r. \tag{93}$$

Penalty enforcement instead adds

$$\Pi_c = \frac{1}{2} k_c \|Cu\|^2, \qquad r_c = C^T k_c C u, \qquad K_c = C^T k_c C. \tag{94}$$

For implicit adaptive solid MPM, strong shape-function constraints are preferred because they avoid artificial penalty stiffness and give cleaner reaction-force recovery.

3.12 Virtual Stress Boundary

Related API. MPM.add_virtual_stress_field, VirtualStress, VirtualForce.

Virtual stress belongs to MPM boundary loading. The boundary work $\int_{\Gamma_t} \delta\boldsymbol{u}\cdot\bar{\boldsymbol{t}}\,\mathrm{d}A$ is replaced by a volume integral over a virtual layer Ω_v. If $\boldsymbol{\sigma}^v\boldsymbol{n} = \bar{\boldsymbol{t}}$ and $\nabla\cdot\boldsymbol{\sigma}^v + \boldsymbol{b}^v = 0$, then

$$\int_{\Gamma_t} \delta\boldsymbol{u}\cdot\bar{\boldsymbol{t}}\,\mathrm{d}A = \int_{\Omega_v} \delta\boldsymbol{u}\cdot\boldsymbol{b}^v\,\mathrm{d}V - \int_{\Omega_v} \nabla\delta\boldsymbol{u} : \boldsymbol{\sigma}^v\,\mathrm{d}V. \tag{95}$$

Particle quadrature gives

$$\boldsymbol{f}^v_{i,p} = V_p N_i(\boldsymbol{x}_p)\boldsymbol{b}^v_p - V_p \boldsymbol{B}^T_{ip}\boldsymbol{\sigma}^v_p. \tag{96}$$

For pressure loading $\boldsymbol{\sigma}^v = -p\boldsymbol{I}$. The implementation marks occupied, neighboring empty, and boundary cells before applying this force so that the traction is not applied as an interior material stress.

3.13 Fluid Level Set Pressure Projection and Density Projection

Related API. fluid_level_set, fluid_level_set_function, density_projection, solid_sdf_cut_cell, nParticlesPerCell.

`fluid_level_set` belongs to the implicit incompressible-fluid FDM path. It is not only a visualization level set: it changes the pressure Poisson operator, the right-hand side, and

the post-solve velocity correction at a fluid-air interface. The cell-centered classification used by the kernels is

$$\chi_c = \begin{cases} 0, & \text{air}, \\ 1, & \text{fluid}, \\ 2, & \text{solid}. \end{cases}$$

The fluid signed-distance field ϕ_c follows the convention $\phi < 0$ in fluid and $\phi > 0$ in air. A particle rebuild approximates the surface by a narrow-band minimum over equivalent particle spheres/disks,

$$\phi_c = \min_p \left(\|\boldsymbol{x}_c - \boldsymbol{x}_p\| - r_p \right), \qquad r_p = \begin{cases} \frac{1}{2}\sqrt{2V_p}, & d = 2, \\ \frac{1}{2}\sqrt{3}\, V_p^{1/3}, & d = 3, \end{cases} \tag{97}$$

with a search band of $3h_{\min} + r_p$. Bounded and tagged variants update only the particle/interface band. The finalization step enforces a usable sign even when the particle band did not visit every cell,

$$\phi_c \leftarrow \begin{cases} \min(\phi_c, -\frac{1}{2}h_{\min}), & \chi_c = 1, \\ \max(\phi_c, \frac{1}{2}h_{\min}), & \chi_c \neq 1. \end{cases} \tag{98}$$

If `fluid_level_set_function` is supplied and the selected engine exposes `update_external_fluid_level_set`, that callable is used to update the same field and must use the same sign convention.

A cell is flagged as a free-surface cell when an axis-aligned neighbor changes between fluid and air. For a fluid cell f and neighboring air cell a, the ghost-fluid distance fraction used by the source kernels is

$$\theta_{fa} = \min\left(1, \max\left(0.01, \frac{\phi_f}{\phi_f - \phi_a}\right)\right), \qquad \phi_f < 0, \quad \phi_a > 0. \tag{99}$$

Thus θ is the fraction of the grid spacing from the fluid cell center to the interface, clamped away from zero to avoid singular matrix entries.

Let

$$\alpha_d = \frac{\Delta t}{\rho h_d^2}, \qquad p_{air} = p_{atm} + p_\sigma, \tag{100}$$

where p_σ is the stored surface-tension pressure contribution. For a fluid cell c, a fluid neighbor n contributes the usual symmetric pressure Laplacian entries

$$A_{cc} \mathrel{+}= \alpha_d, \qquad A_{cn} \mathrel{-}= \alpha_d. \tag{101}$$

An air neighbor has no pressure unknown. Instead the ghost-air pressure is inserted into the diagonal and right-hand side,

$$A_{cc} \mathrel{+}= \frac{\alpha_d}{\theta_{cn}}, \qquad b_c \mathrel{+}= \frac{\alpha_d}{\theta_{cn}} p_{air,n}. \tag{102}$$

A solid neighbor is treated as a no-through-pressure boundary; the right-hand side uses the boundary face velocity rather than adding an air pressure. The divergence term in the right-hand side is

$$b_c \mathrel{+}= \sum_d \frac{u^*_{d,c-\frac{1}{2}} - u^*_{d,c+\frac{1}{2}}}{h_d}, \tag{103}$$

with additional corrections where either side of a face is solid.

After the pressure solve, a face between two fluid cells uses the standard MAC correction

$$u_d^{n+1} = u_d^* - \frac{\Delta t}{\rho h_d}(p_R - p_L). \tag{104}$$

For a fluid-air face with fluid cell L and air cell R,

$$u_d^{n+1} = u_d^* - \frac{\Delta t}{\rho h_d\, \theta_{LR}} \left(p_{air,R} - p_L\right), \tag{105}$$

and the opposite orientation uses

$$u_d^{n+1} = u_d^* - \frac{\Delta t}{\rho h_d\, \theta_{RL}} \left(p_R - p_{air,L}\right).$$

If a solid cell touches the face, the non-cut-cell path sets the face velocity and force to zero. With `solid_sdf_cut_cell=True`, the open face fraction f_d multiplies pressure coefficients,

$$\alpha_d \to f_d \alpha_d, \tag{106}$$

and the divergence uses the blended face velocity

$$u_f = f_d u_{fluid} + (1 - f_d) u_{solid}.$$

Closed faces are assigned the solid face velocity.

Density projection uses the same cell classification and level-set band to move particles away from volume errors near the interface:

$$\Delta \boldsymbol{x}_p = -\lambda \frac{\rho_h(\boldsymbol{x}_p) - \rho_0}{\|\nabla \rho_h(\boldsymbol{x}_p)\|^2 + \epsilon} \nabla \rho_h(\boldsymbol{x}_p). \tag{107}$$

The implementation can also use the free-surface θ in pressure-driven face displacements. In 3D, global reconstruction costs $O(n_x n_y n_z)$; rebuilding the entire field during density projection is unnecessary when only the interface band changes. Sparse particle sampling can create an interior hole, mark a fluid cell as air, and impose a false pressure boundary; this is the usual reason for isolated pressure spikes after wave reflection.

4 SDF and Level Set Geometry

Related API. Geometry API addendum, fluid_level_set, solid_sdf_cut_cell, levelset_grid_number.

4.1 Signed-Distance Functions and Boolean Algebra

An exact signed-distance function $\phi(\boldsymbol{x})$ satisfies

$$\phi(\boldsymbol{x}) < 0 \text{ inside}, \qquad \phi(\boldsymbol{x}) = 0 \text{ on the surface}, \qquad \|\nabla \phi\| = 1 \tag{108}$$

where it is differentiable. The primitive constructors in `src/sdf` return callable geometry objects whose value is this signed distance or a close analytic approximation. Rigid transforms compose by inverse mapping:

$$\phi_{T\Omega}(\boldsymbol{x}) = \phi_{\Omega}(T^{-1}\boldsymbol{x}). \tag{109}$$

Uniform scaling by $s > 0$ preserves distances by

$$\phi_{s\Omega}(\boldsymbol{x}) = s\,\phi_{\Omega}(\boldsymbol{x}/s). \tag{110}$$

The standard Boolean operators are

$$\phi_{A\cup B} = \min(\phi_A, \phi_B), \tag{111}$$
$$\phi_{A\cap B} = \max(\phi_A, \phi_B), \tag{112}$$
$$\phi_{A\setminus B} = \max(\phi_A, -\phi_B). \tag{113}$$

The smooth blend operation replaces the hard min/max by a rounded transition over a user radius. Dilation and erosion shift the isosurface:

$$\phi_{\text{dilate}} = \phi - r, \qquad \phi_{\text{erode}} = \phi + r, \tag{114}$$

and a shell is the band $|\phi| - t \le 0$. These operations are used by `SDF2D` and `SDF3D`; 3D adds deformation operators such as twist, bend, wrap, elongation, radial transitions, slicing, and orientation to a selected axis. The result remains a useful implicit surface even when a deformation no longer preserves the exact Eikonal property.

4.2 Sampling, LocalGrid, and Interpolation

Analytic SDF objects can be sampled onto a regular grid for coupling with MPM, DEM, meshing, and visualization. Let a grid have origin $\boldsymbol{x}_0$, spacing h, and node index $\boldsymbol{i}$. The sampled field is

$$\phi_{\boldsymbol{i}} = \phi(\boldsymbol{x}_0 + h\boldsymbol{i}). \tag{115}$$

`LocalGrid.distance(p)` evaluates this table by bi-linear or tri-linear interpolation. In 3D, with cell-local coordinates $\xi, \eta, \zeta \in [0, 1]$,

$$\phi(\boldsymbol{x}) = \sum_{a,b,c\in\{0,1\}} (1-\xi)^{1-a}\xi^a(1-\eta)^{1-b}\eta^b(1-\zeta)^{1-c}\zeta^c\phi_{i+a,j+b,k+c}. \tag{116}$$

This interpolated field is the form consumed by cut-cell pressure operators and by level-set DEM/MPM geometry loaders. The sign convention must be kept consistent: changing inside/outside reverses wall normals, pressure cut fractions, and LSDEM overlap direction.

4.3 Surface Extraction and Mass Properties

The SDF wrapper can ray trace, construct point clouds, run marching cubes, and export mesh or particle files. Marching cubes treats every grid cell whose corner signs differ as an interface cell and linearly interpolates the crossing point on each edge:

$$\boldsymbol{x}_e = \frac{|\phi_b|\boldsymbol{x}_a + |\phi_a|\boldsymbol{x}_b}{|\phi_a| + |\phi_b|}. \tag{117}$$

The same interface samples are used to approximate volume, equivalent radius, center of mass, inertia, and surface node sets. Resolution controls both geometric accuracy and the cost of these derived properties. For contact, a coarse grid can underestimate curvature and create delayed contact; for a pressure cut cell, it can place the boundary on the wrong side of a face.

4.4 Fast Sweeping Redistance and Semi-Lagrangian Advection

After advection or noisy particle reconstruction, a level set is reinitialized to signed distance by solving the Eikonal equation

$$\|\nabla\phi\| = 1, \qquad \phi = 0 \text{ on } \Gamma. \tag{118}$$

`FastSweepingLevelSet` first marks target surface cells whose neighbors have opposite signs, estimates the interface distance by linear interpolation, and then repeatedly sweeps all coordinate directions. The local update solves the upwind discretization

$$\sum_{m=1}^{d} \left(\max(D_m^- \phi, -D_m^+ \phi, 0)\right)^2 = 1. \tag{119}$$

In sorted-neighbor form, if $a_1 \le a_2 \le a_3$ are accepted neighboring distances, the candidate with k active dimensions is the positive root of

$$\sum_{m=1}^{k} (\phi - a_m)^2 = h^2, \qquad \phi \ge a_k. \tag{120}$$

This is Zhao's fast sweeping method specialized to a Cartesian grid.

`SemiLagrangian` advects a field by tracing characteristics backward:

$$\phi^{n+1}(\boldsymbol{x}_i) = \phi^n(\boldsymbol{X}_i), \qquad \frac{\mathrm{d}\boldsymbol{X}}{\mathrm{d}t} = \boldsymbol{u}(\boldsymbol{X}, t), \qquad \boldsymbol{X}(t^{n+1}) = \boldsymbol{x}_i. \tag{121}$$

The code supports first- through fourth-order Runge-Kutta backtracing. Field interpolation is selected by `order`: linear interpolation for order 2, WENO3 for order 3, and WENO4 for order 4 in the current Python interface. Semi-Lagrangian advection is stable for large Courant numbers, but it is diffusive; redistance and optional smoothing are therefore part of the normal free-surface workflow.

This generic `SemiLagrangian` interface is distinct from the LSMPM soft-body scheme selector described in Level-Set MPM Soft Bodies. There, `SemiLagrangian` aliases the default `ReferenceMap` reconstruction; `MacCormack` and `WENO5` select dedicated incremental Taichi paths rather than an `order` value.

4.5 Curvature and Surface Tension

For fluid surfaces, `Fluid_Surface` stores a signed-distance field and builds surface particles or edge samples. Normals and curvature are computed from

$$\boldsymbol{n} = \frac{\nabla\phi}{\|\nabla\phi\| + \epsilon}, \qquad \kappa = \nabla \cdot \boldsymbol{n}. \tag{122}$$

On a regular grid, $\nabla\phi$ is evaluated by centered finite differences where neighbors are available, and $\nabla^2\phi$ or $\nabla\cdot\boldsymbol{n}$ gives the curvature proxy. The continuum surface-tension force has the standard form

$$\boldsymbol{f}_\sigma = \sigma_s \kappa \boldsymbol{n}\, \delta_\Gamma, \tag{123}$$

with δ_Γ represented by the narrow band or by sampled interface particles. This is why surface-resolution, smoothing, and redistance accuracy directly affect capillary forces.

4.6 How SDF and Level Sets Enter Solvers

Solver path	Geometry role
Implicit fluid MPM	`fluid_level_set` classifies fluid, air, and solid pressure cells; `solid_sdf_cut_cell` modifies face coefficients by the SDF boundary fraction.
DEM and LSDEM	level-set grids and SDF surfaces provide wall, rigid-body, and non-spherical particle geometry for overlap and normal evaluation.
MPDEM/LSMPM	SDF or level-set distance supplies the coupled contact gap between particles, material points, and level-set bodies.
Geometry generation	analytic SDFs generate surface meshes, particle samples, tetrahedral inputs, and diagnostic mass properties.

5 DEM

Related API. DEM.add_body, DEM.run, DEM.choose_contact_model.

5.1 Rigid-Body Equations

DEM particles store translational and rotational degrees of freedom. For a sphere or clump,

$$m\dot{\boldsymbol{v}} = \sum_c \boldsymbol{F}_c + m\boldsymbol{g}, \qquad \boldsymbol{I}\dot{\boldsymbol{\omega}} = \sum_c \boldsymbol{r}_c \times \boldsymbol{F}_c + \sum_c \boldsymbol{M}_c. \tag{124}$$

The explicit update advances velocity, angular velocity, position, and orientation from the accumulated contact, body, and coupling forces. Contact law details are collected in Contact Model; candidate generation is collected in Contact Search.

5.2 Level-Set MPM Soft Bodies

`scheme="LSMPM"` extends the level-set DEM path with deformable particles whose interior is discretized by material points and advanced by a total-Lagrangian MPM update. A soft

body still owns the same level-set template, bounding sphere, bounding box, and body id as a level-set rigid body, but it also stores a compact material-point range $[p_s^0, p_s^1)$, a compact local-grid range, and a list of surface material points. The reference configuration is sampled once from the template local grid. `MaterialPointsPerCell` or `ParticlePerCell` controls this sampling density; larger values give a smoother stress field and a denser contact surface, but increase TLMPM, level-set advection, and contact costs. The equal material-point reference volume is normalized to the integrated template volume,

$$V_p^0 = \frac{s_b^3 V_{\text{template}}}{N_b}, \tag{125}$$

so voxel-center inclusion error does not change body mass or bulk density when `MaterialPointsPerCell` is varied.

For a soft material point p, the code stores the reference coordinate $\boldsymbol{X}_p$, current coordinate $\boldsymbol{x}_p$, volume V_p^0, mass m_p, deformation gradient $\boldsymbol{F}_p$, and first Piola stress $\boldsymbol{P}_p$. Shape functions are precomputed in the body-local MPM grid at insertion time:

$$N_{ip} = N_i(\boldsymbol{X}_p), \qquad \nabla_0 N_{ip} = \nabla_{\boldsymbol{X}} N_i(\boldsymbol{X}_p). \tag{126}$$

The selectable soft shape functions are `Linear`, `QuadBSpline`, and `CubicBSpline`; their tensor-product support sizes are 2^3, 3^3, and 4^3 grid nodes, respectively. The explicit TLMPM nodal update is

$$m_i = \sum_p N_{ip} m_p, \tag{127}$$

$$m_i \boldsymbol{v}_i = \sum_p N_{ip} m_p \boldsymbol{v}_p, \tag{128}$$

$$\boldsymbol{f}_i = \sum_p N_{ip} m_p \boldsymbol{b} - \sum_p V_p^0 \, \boldsymbol{P}_p \nabla_0 N_{ip} + \sum_p N_{ip} \boldsymbol{f}_p^c. \tag{129}$$

After nodal acceleration is computed, material points receive the interpolated nodal velocity increment and their deformation gradient is advanced from the reference velocity gradient:

$$\dot{\boldsymbol{F}}_p = \left(\sum_i \boldsymbol{v}_i \otimes \nabla_0 N_{ip} \right), \qquad \boldsymbol{F}_p^{n+1} = \boldsymbol{F}_p^n + \Delta t \, \dot{\boldsymbol{F}}_p. \tag{130}$$

The soft constitutive model is selected by the DEM material attribute `ConstitutiveModel`. The finite-strain branch currently supports `NeoHookean`, `HenckyElastic`, `MooneyRivlin`, `Gent`, and `Hydrogel`. These models return $\boldsymbol{P} = \partial\psi/\partial\boldsymbol{F}$; the soft-body TLMPM force assembly uses this first Piola stress directly and does not convert it to Cauchy stress for the internal force.

5.2.0.1 Total reference-map reconstruction. The default explicit soft-body update is `ReferenceMap`. The initial template distance field ϕ_{init} is stored once when each body is inserted, including bodies generated after stepping starts. Every update reconstructs total motion from the current material-point state and samples this immutable field; the previous-step SDF is not a reconstruction input. This is a reference-map form of semi-Lagrangian pullback, adapted to MPM by grid projection rather than the source inverse-mapping method's nearest-node displacement interpolator.

Let $\boldsymbol{R}_0, \boldsymbol{R}$ denote initial/current body rotations, $\boldsymbol{c}_0, \boldsymbol{c}$ their centers, and s the constant template scale. The unscaled local coordinates and inverse-gradient particle patches are

$$\boldsymbol{X}_p = \boldsymbol{R}_0^T(\boldsymbol{x}_{p0} - \boldsymbol{c}_0)/s, \qquad \boldsymbol{y}_p = \boldsymbol{R}^T(\boldsymbol{x}_p - \boldsymbol{c})/s, \qquad \boldsymbol{B}_p = \boldsymbol{R}_0^T \boldsymbol{F}_p^{-1} \boldsymbol{R}. \tag{131}$$

At current local SDF node $\boldsymbol{y}_g$, use the existing linear or B-spline projection weights $\omega_{gp} = m_p N_{gp}$ to construct

$$\boldsymbol{X}_g = \frac{\sum_p \omega_{gp}[\boldsymbol{X}_p + \boldsymbol{B}_p(\boldsymbol{y}_g - \boldsymbol{y}_p)]}{\sum_p \omega_{gp}}, \tag{132}$$

$$\boldsymbol{B}_g = \frac{\sum_p \omega_{gp} \boldsymbol{B}_p}{\sum_p \omega_{gp}}. \tag{133}$$

The inverse displacement $\boldsymbol{d}_g = \boldsymbol{X}_g - \boldsymbol{y}_g$ and projected derivative $\boldsymbol{B}_g - \boldsymbol{I}$ are extended into neighboring unsupported cells by synchronous affine extrapolation. A wave only reads support from previous waves, avoiding traversal-order dependence. Trilinear interpolation of the initial field gives

$$\psi_g = \phi_{\text{init}}(\boldsymbol{X}_g), \qquad \boldsymbol{n}_0 = \frac{\nabla_X \phi_{\text{init}}}{\|\nabla_X \phi_{\text{init}}\|}, \qquad \phi_g = \frac{\psi_g}{\|\boldsymbol{B}_g^T \boldsymbol{n}_0(\boldsymbol{X}_g)\|}. \tag{134}$$

The denominator is inverse normal stretch, not $\det \boldsymbol{F}$. With an exact affine map and exact reference data it gives the exact distance to a plane. For curved boundaries or nonuniform deformation it is a local near-interface approximation: the image of a reference closest point need not be the current closest point. Projected $\boldsymbol{B}_g$ is an estimate, not necessarily the spatial derivative of the interpolated $\boldsymbol{X}_g$. When the reference gradient vanishes, the metric division is skipped. Downstream normals are obtained from the reconstructed grid field; exact surface-normal transport by $\boldsymbol{F}^{-T}$ does not make finite-band distance values exact.

Unsupported exterior nodes receive a positive cap, so a global Euclidean far-field SDF is not promised. Nonfinite/nonpositive particle determinants, nonpositive inverse Jacobians near the interface, and a support halo cutting the material interior are rejected. Near-interface reference queries outside the stored template domain enter the existing domain-departure audit. The initial-field cache requires one scalar per SDF slot; the selected reference-map route adds a nine-component inverse-derivative buffer and reuses the existing vector/weight projection buffers.

`ReferenceMap` is the default and `InverseMapping` plus `SemiLagrangian` are aliases for it. `MacCormack` explicitly selects the previous incremental RK2-backtraced, bounded MacCormack route. Its characteristic update has the form

$$\phi^{n+1}(\boldsymbol{y}_g) = \mathcal{I}[\phi^n](\boldsymbol{y}_{\text{departure}}), \qquad \dot{\boldsymbol{y}} = \boldsymbol{u}_{\text{local}}(\boldsymbol{y}), \tag{135}$$

where $\mathcal{I}$ is grid interpolation. Such repeated interpolation can accumulate error; reference-map reconstruction avoids that SDF history but still incurs material-motion, map-projection, and interpolation errors. After the material points move, body centers and bounds follow their current positions rather than the potentially larger mechanical background grid.

5.2.0.2 Soft level-set WENO5 advection.

The alternative `WENO5` path discretizes

$$\frac{\partial \phi}{\partial t} + \boldsymbol{u}_g \cdot \nabla \phi = 0 \tag{136}$$

with fifth-order Jiang–Shu one-sided reconstruction in the grid interior and the three-stage SSPRK(3,3) update. A seven-point directional halo is required; nodes within three cells of the compact-grid boundary fall back to one-sided first-order differences. Before MacCormack or WENO5 transport, soft material-point velocities are projected to the level-set grid and extended into uncovered near-interface nodes. WENO5 subcycles so that

$$\max_g \frac{\Delta t}{h_\phi}(|u_{gx}| + |u_{gy}| + |u_{gz}|) \le C_{\mathrm{WENO}}, \tag{137}$$

where `soft_levelset_advection_cfl` sets C_{WENO} and defaults to 0.20 when WENO5 is selected. The `soft_levelset_advection_interval` option accumulates the mechanical timestep and updates once after the requested number of TLMPM steps; its default is one. A reference-map update reconstructs total motion at that scheduled step; its CFL option does not subcycle the inverse map.

5.2.0.3 Soft level-set redistance.

Neither incremental transport nor locally corrected inverse-map reconstruction guarantees $\|\nabla\phi\| = 1$ throughout the field. Redistance is optional and disabled by default for `ReferenceMap`. Selecting `MacCormack` or `WENO5` without an explicit maintenance flag enables redistance; an explicit `enabled` value takes precedence. When enabled, the field is monitored after its update. In a narrow monitoring band,

$$E_\phi = \max_{|\phi| \le W_m} |\|\nabla\phi\| - 1| \tag{138}$$

is computed on compact soft grids. Local extrema are ignored in this monitor so that medial-axis points inside small particles do not trigger unnecessary redistance. If E_ϕ exceeds the user threshold, the current field is stored as ϕ_0 and the code solves the standard pseudo-time reinitialization equation in a wider band:

$$\frac{\partial \phi}{\partial \tau} + S(\phi_0)\,(\|\nabla\phi\| - 1) = 0, \qquad S(\phi_0) = \frac{\phi_0}{\sqrt{\phi_0^2 + \Delta x_\phi^2}}. \tag{139}$$

The implementation uses a first-order Godunov upwind norm and $\Delta\tau = c_\tau \Delta x_\phi$. The default monitoring and redistance bands are $W_m = 4\Delta x_\phi$ and $W_r = 8\Delta x_\phi$, with $c_\tau = 0.3$ and threshold 0.2. These bands are expressed in local level-set cells. For contact-heavy deformation, W_r should cover the contact band and a finite-difference halo:

$$W_r \gtrsim \max\left(8\Delta x_\phi, n_{cut} + 4\Delta x_\phi\right), \tag{140}$$

where n_{cut} is the largest normal cutoff in the active contact model. The template level-set extent must also be large enough to contain this redistance band; otherwise boundary clamping, not the redistance equation, limits accuracy.

5.2.0.4 Soft-soft and soft-rigid contact.

LSMPM contact uses a unified contact-node list. Rigid and soft level-set bodies both contribute template surface nodes. Soft

surface nodes are advected by the compact TLMPM grid using their precomputed total-Lagrangian shape functions. The surface trace uses eight-node trilinear support while bulk material points retain the configured MPM basis. For a directed contact node a from body i tested against body j, the narrow phase evaluates

$$\boldsymbol{y}_a = \boldsymbol{R}_j^T(\boldsymbol{x}_a - \boldsymbol{c}_j), \qquad g_a = \phi_j(\boldsymbol{y}_a). \tag{141}$$

For the linear, Hertz-Mindlin, and penalty LS contact models, $n_{cut} = 0$, so every candidate with $g_a < 0$ is active. Barrier contact uses a positive `NormalCutOff` and activates in the barrier band $g_a < n_{cut}$. The contact normal is

$$\boldsymbol{n} = \boldsymbol{R}_j \frac{\nabla\phi_j(\boldsymbol{y}_a)}{\|\nabla\phi_j(\boldsymbol{y}_a)\|}. \tag{142}$$

The same surface property object used by LSDEM contact supplies normal force, tangential force, damping, Coulomb truncation, and contact-history update. The LSMPM kernel dispatches by contact-model type: type-1 models use the normal and stiffness-parameter signature, while energy/barrier type-0 models use the SDF gradient signature, matching the rigid LSDEM wrappers.

Adjacent projected triangle areas are lumped to the template vertices and normalized so that $\sum_a \widehat{a}_a = 1$. The nodal coefficient is converted to physical area using the scaled reference template surface area,

$$a_a = s_b^2 A_{\mathrm{ref}} \widehat{a}_a. \tag{143}$$

If a legacy template has no positive A_{ref}, the code falls back to $4\pi r_{\mathrm{eq}}^2$. The same coefficient scales the assembled force and the contact tangent used by the timestep estimate. Rigid–rigid pairs retain one directed test. Whenever either side is soft, both directed surface-to-SDF tests are built and each carries one-half of its area coefficient, avoiding duplicate interface force while following both evolving surfaces. If the slave body is soft, its velocity and effective nodal mass are evaluated from the nearest advected soft surface node. The contact point is

$$\boldsymbol{x}_c = \boldsymbol{x}_a - \frac{1}{2} g_a \boldsymbol{n}. \tag{144}$$

For the LSMPM explicit contact time-step estimate, a rigid body contributes its body mass while a soft body contributes the conservative surface-node mass

$$m_a^{\mathrm{eff}} = \frac{m_b}{N_b^{\mathrm{surface}}}. \tag{145}$$

The minimum effective contact mass is paired with the largest area-scaled nodal tangent over all configured material pairs. Thus the stability check uses the same reference-area quadrature as the assembled force and does not mix normalized template weights with material-point masses.

Whenever either side is soft, its surface-node force enters the body-owned TLMPM grid through

$$\boldsymbol{f}_g^c \mathrel{+}= N_{ag} \boldsymbol{F}_a^c. \tag{146}$$

The surface shape functions satisfy partition of unity, so the transferred grid force sums to $\boldsymbol{F}_a^c$. A soft slave receives the opposite force at its nearest advected surface node and uses the same transfer. Rigid bodies still receive ordinary body force and torque. Soft surface-node forces are stored on `vertice.contact_force` and enter the next compact TLMPM grid update; no finite-radius soft material-point contact patch is used.

5.3 MPM Soft-Particle Backend and MPDEM Dispatch

Related API. MPM soft-particle API, DEM soft-rigid LSMPM workflow, MPDEM soft-affine IPC workflow.

The deformable part of `scheme="LSMPM"` is an MPM algorithm even when a legacy DEM example script drives the run. In the current source tree, the soft-particle implementation is collected under `src/mpm/soft_particle`: `Structs` defines the soft body, compact TLMPM grid, and material-point fields; `SoftBodyKernel` advances the explicit TLMPM state and advects the local level set; `LevelSet` samples, monitors, and redistances the deforming SDF; `NeighborKernel` builds directed body/contact-node lists; `ContactKernel` assembles explicit LSMPM contact forces; and `IPC`, `IPCFunction`, `IPCMPM`, `IPCTLMPM`, and `IPCULMPM` contain the implicit barrier/friction helpers for MPM soft-particle contact.

The DEM facade still owns the level-set rigid-body scene and the public `scheme="LSMPM"` compatibility path. The bridge in `src/mpm/soft_particle/DEMPMBridge.py` installs MPM soft-particle kernels on that scene so existing LSDEM generation, broad-phase search, level-set template storage, recorder output, and MPDEM examples keep working. This is a dispatch/ownership bridge, not a separate DEM constitutive model: soft-body stress is computed by MPM finite-strain material classes and soft-body kinematics are advanced by TLMPM. A pure `MPM(...,mode="SoftParticle")` configuration exists as a public option, but the current runnable level-set body/soft-point scene is the bridge path used by `DEM` or `MPDEM`.

Two broad contact paths share the same soft-particle fields:

1. the explicit LSMPM path uses DEM contact laws and applies the resulting contact force to soft material points before the next TLMPM P2G stage;
2. the implicit soft-particle IPC path builds an incremental potential, differentiates barrier/friction energies with respect to MPM grid displacements and affine control points, and solves the Newton/PCG system in one operator.

For search, both paths operate at the body level first. The selected DEM broad phase, `LinkedCell`, `HierarchicalLinkedCell`, or `BVH`, builds a soft/rigid body-pair list using body AABBs or bounding spheres enlarged by the contact margin. Only after a body pair is active are surface material points tested against the target body. This is intentionally different from building a global linked cell or BVH over every surface point, which would duplicate storage and lose the compact body-level structure of LSDEM/LSMPM.

Standalone Direct IPC-MPM follows the same device-residency rule. Its default `LinkedCell` search produces compact body pairs; a device prefix sum groups surface/body tasks, and each small per-surface segment is sorted by body id so contact ids remain stable across Newton assemblies. Swept CCD rebuilds body AABBs from both endpoints of the Newton direction and evaluates ACCD only for the resulting compact body pairs. Neither active-set path uses host `numpy.unique` or a surface-by-all-body scan.

5.4 Affine Body Dynamics

An AffineBody stores each body as an affine image of a template surface mesh. Instead of solving for every surface vertex independently, body b owns four world-space control points

$$\boldsymbol{y}_{b0}, \boldsymbol{y}_{b1}, \boldsymbol{y}_{b2}, \boldsymbol{y}_{b3} \in \mathbb{R}^3.$$

For a template vertex v, the code shifts the local coordinate by $(1/4, 1/4, 1/4)$ and evaluates the affine barycentric basis

$$N_{v0} = 1 - X_v - Y_v - Z_v, \qquad N_{v1} = X_v, \qquad N_{v2} = Y_v, \qquad N_{v3} = Z_v. \tag{147}$$

The world vertex position and velocity are therefore

$$\boldsymbol{x}_v = \sum_{\alpha=0}^{3} N_{v\alpha} \boldsymbol{y}_{b\alpha}, \qquad \boldsymbol{v}_v = \sum_{\alpha=0}^{3} N_{v\alpha} \dot{\boldsymbol{y}}_{b\alpha}. \tag{148}$$

This is equivalent to $\boldsymbol{x} = \boldsymbol{c} + \boldsymbol{A}\hat{\boldsymbol{x}}$, but the implementation assembles directly in the four-control-point basis. Initial control velocities are obtained from the user-provided translational velocity and angular velocity by

$$\dot{\boldsymbol{y}}_{b\alpha} = \boldsymbol{v}_b + \boldsymbol{\omega}_b \times (\boldsymbol{y}_{b\alpha} - \boldsymbol{c}_b). \tag{149}$$

The template surface is converted to a lumped vertex mass distribution and then to a 4×4 affine mass matrix

$$M^b_{\alpha\beta} = \sum_{v \in b} m_v N_{v\alpha} N_{v\beta}. \tag{150}$$

For one time step, with the explicit prediction $\tilde{\boldsymbol{y}} = \boldsymbol{y}^n + \Delta t\, \dot{\boldsymbol{y}}^n$, the inertial incremental potential is assembled componentwise as

$$\Pi^b_{in} = \frac{1}{2} \sum_{\alpha,\beta} M^b_{\alpha\beta} (\boldsymbol{y}_{b\alpha} - \tilde{\boldsymbol{y}}_{b\alpha}) \cdot (\boldsymbol{y}_{b\beta} - \tilde{\boldsymbol{y}}_{b\beta}). \tag{151}$$

Its gradient and Hessian are exactly $M^b(\boldsymbol{y} - \tilde{\boldsymbol{y}})$ and M^b. Body force contributes

$$\Pi^b_g = -\Delta t^2 \sum_{\alpha} m^b_\alpha\, \boldsymbol{g} \cdot \boldsymbol{y}_{b\alpha}, \tag{152}$$

where $m^b_\alpha = \sum_v m_v N_{v\alpha}$. The affine deformation gradient is

$$\boldsymbol{F}_b = \begin{bmatrix} \boldsymbol{y}_{b1} - \boldsymbol{y}_{b0} & \boldsymbol{y}_{b2} - \boldsymbol{y}_{b0} & \boldsymbol{y}_{b3} - \boldsymbol{y}_{b0} \end{bmatrix}, \tag{153}$$

and the current rigidity regularization is the analytic energy

$$\Pi^b_{rig} = \frac{\Delta t^2\, E_b\, V_b}{8} \left\| \boldsymbol{F}_b^T \boldsymbol{F}_b - \boldsymbol{I} \right\|_F^2. \tag{154}$$

The first variation uses $4c\, \boldsymbol{F}(\boldsymbol{F}^T\boldsymbol{F} - \boldsymbol{I})$ with $c = \Delta t^2 E_b V_b/8$, and the Hessian inserted into the linear system is assembled analytically in the same Taichi kernels, not by finite differences in the normal execution path.

5.4.0.1 Revolute-link energy.

For a joint between affine bodies (A) and (B), fixed affine weights map the initial world anchor and joint frame into both bodies. Let

$$\boldsymbol{r}_p = \boldsymbol{p}_A - \boldsymbol{p}_B, \qquad \boldsymbol{r}_a = \boldsymbol{a}_A - \boldsymbol{a}_B,$$

where the second side is a fixed world anchor/axis when (B=-1). With orthogonal material directors $\boldsymbol{u}_A, \boldsymbol{v}_A$ perpendicular to the axis, the motor residual for a body–body joint is

$$\boldsymbol{r}_m(\theta) = \boldsymbol{u}_B - (\cos\theta\, \boldsymbol{u}_A + \sin\theta\, \boldsymbol{v}_A).$$

The joint potential is

$$\Pi_j = \frac{\Delta t^2}{2}\left(k_p\|\boldsymbol{r}_p\|^2 + k_a\|\boldsymbol{r}_a\|^2 + k_m\|\boldsymbol{r}_m(\theta_m)\|^2\right).$$

The anchor and axis terms remove relative translation and tilt but do not penalize rotation about the axis. A violated angular limit reuses the director spring at the nearest lower or upper limit; inside the interval its energy is zero. Damping uses the previous-state angle gradient $\hat{\boldsymbol{J}}$ and the positive-semidefinite incremental potential

$$\Pi_d = \frac{\Delta t\, c_d}{2}\left[\hat{\boldsymbol{J}}^T(\boldsymbol{y} - \hat{\boldsymbol{y}})\right]^2.$$

Connected bodies are excluded from body–body IPC by default, while their contacts with other links and walls remain active. All joint terms and the connected-pair active-set filter are assembled on the Taichi device.

AffineBody contact follows the IPC workflow on the affine surface mesh. Body-body broad phase produces vertex-face and edge-edge candidates, while wall contact produces vertex-plane or vertex-facet candidates. The squared-distance barrier used for mesh primitives is

$$B_s(s) = -\kappa\left(\frac{s - \hat{d}^2}{\hat{d}^2}\right)^2 \ln\frac{s}{\hat{d}^2}, \qquad 0 < s < \hat{d}^2, \tag{155}$$

where s is a point-triangle or edge-edge squared distance. Plane-wall contact uses the gap form

$$B_g(g) = -\kappa(g - \hat{d})^2 \ln\frac{g}{\hat{d}}, \qquad 0 < g < \hat{d}. \tag{156}$$

The scalar barrier contribution is multiplied by Δt^2, the local surface measure, and $\hat{d}$. For a vertex-face candidate the code scatters the local gradient and Hessian through the vertex basis of the vertex body and the three face vertices. For an edge-edge candidate it scatters through the two edge vertices on each body and applies the standard IPC edge-edge distance type and mollifier; friction is assembled only after the edge-edge mollifier is fully active. Wall facet contact uses the same point-triangle barrier and the body vertex basis, with the wall degrees of freedom held fixed.

Friction uses the maximum of the two contacting material frictions, or the maximum of body and wall friction for body-wall contact. With normal $\boldsymbol{n}$, tangent projector $\boldsymbol{P} = \boldsymbol{I} - \boldsymbol{n}\otimes\boldsymbol{n}$, previous relative displacement $\hat{\boldsymbol{r}}$, and trial relative displacement $\boldsymbol{r}$, the regularized slip speed is

$$\bar{\boldsymbol{v}} = \boldsymbol{P}\frac{\boldsymbol{r} - \hat{\boldsymbol{r}}}{\Delta t}. \tag{157}$$

The friction energy uses the code function

$$f_0(\|\bar{\boldsymbol{v}}\|) = \begin{cases} \Delta t\,\|\bar{\boldsymbol{v}}\|, & \|\bar{\boldsymbol{v}}\| \ge \epsilon_v, \\ h^2(-h/3 + \epsilon_v\Delta t)/(\epsilon_v\Delta t)^2 + \epsilon_v\Delta t/3, & \|\bar{\boldsymbol{v}}\| < \epsilon_v, \end{cases} \quad h = \Delta t\,\|\bar{\boldsymbol{v}}\|. \tag{158}$$

Its gradient and Hessian are evaluated analytically and scattered with the same primitive weights as the normal barrier. If `contact_damping_stiffness`>0, the active contact Hessian is also reused to build a damping matrix K_{CD}, giving

$$\Pi_{CD} = \frac{1}{2}(\boldsymbol{y} - \hat{\boldsymbol{y}})^T K_{CD}(\boldsymbol{y} - \hat{\boldsymbol{y}}), \qquad \boldsymbol{r}_{CD} = K_{CD}(\boldsymbol{y} - \hat{\boldsymbol{y}}). \tag{159}$$

The nonlinear step is solved by Newton iterations. Each iteration assembles the energy, gradient, and Hessian; solves

$$\boldsymbol{K}\Delta\boldsymbol{y} = -\boldsymbol{r}; \tag{160}$$

optionally caps the step by `max_step`; applies CCD or ACCD; and then runs an Armijo line search. `COO` assembly inserts the Hessian into the existing coordinate sparse matrix and solves it by the project PCG solver. `HashTriplet` stores unsummed $(i, j, value)$ triplets and relies on the GPU triplet builder to sort/merge duplicate entries before PCG; it is restricted to Taichi CUDA. `MatrixFree` is accepted as an input keyword for compatibility, but in the current AffineBody implementation it is routed through the COO-backed PCG path rather than a separate matrix-free operator.

Because the surface mesh still provides the contact primitives, AffineBody neighbor search follows the same broad-phase principle as DEM: linked cells are appropriate for nearly uniform geometry, hierarchical cells for strong size variation, and BVH for mixed or highly nonuniform AABBs. The AffineBody neighbor helper updates primitive AABBs from the current affine mesh and uses $\hat{d}$ as the contact activation margin. The affine branch writes its own `AffineBody*.npz` state and `GraphicAffineBody*.vtu` surface output, so ordinary `DEM.choose_contact_model` force-law selection is bypassed for AffineBody IPC contact.

5.4.0.2 Level-set AffineBody alternative. When an affine template selects `ContactRepresentation="LevelSet"`, the surface mesh remains the quadrature, mass, and visualization proxy, but it no longer defines body–body closest features. For source surface point $\boldsymbol{x}_s$, target scale s_t, target affine matrix $\boldsymbol{A}_t$, and target SDF ϕ_t, the oriented gap is

$$\boldsymbol{q} = \boldsymbol{A}_t^{-1}(\boldsymbol{x}_s - \boldsymbol{y}_{t0}), \qquad g = s_t\phi_t\big((\boldsymbol{q} - \tfrac{1}{4}\mathbf{1})/s_t\big)\,. \tag{161}$$

Both source–target orientations are integrated with one-half of the source lumped surface measure. Trilinear interpolation supplies the field value, gradient, and cell-local Hessian. The exact fully implicit tangent includes both the pulled-back field Hessian and the second variation of $\boldsymbol{A}_t^{-1}$; the lagged tangent retains the positive barrier outer-product block. Dynamics uses the singular positive-gap IPC barrier. Optional initial overlap removal instead uses a C^2 Taylor continuation below a positive gap and stiffness continuation until the original barrier is strictly feasible. For a frozen level-set contact, let $\hat{\boldsymbol{r}}$ be the target material coordinate, $\hat{\boldsymbol{n}}$ the normalized pulled-back SDF gradient, and $\hat{\lambda}_n = \max[-b'(\hat{g}), 0]$. Its lagged friction potential is

$$\Pi_F^{ls} = c\,\mu\hat{\lambda}_n f_0\left(\left\|(\boldsymbol{I} - \hat{\boldsymbol{n}}\hat{\boldsymbol{n}}^T)\frac{[\boldsymbol{x}_s(\boldsymbol{y}) - \boldsymbol{x}_t(\boldsymbol{y};\hat{\boldsymbol{r}})] - [\boldsymbol{x}_s(\hat{\boldsymbol{y}}) - \boldsymbol{x}_t(\hat{\boldsymbol{y}};\hat{\boldsymbol{r}})]}{\Delta t}\right\|\right), \tag{162}$$

where c contains the time-step and symmetric surface measure. Freezing the material coordinate makes the slip a linear eight-control-point pullback; the regularized radial/tangential Hessian is projected positive semidefinite. Level-set particle–particle, particle–wall, and soft–affine paths use this lagged law. Fully implicit level-set friction and level-set contact damping remain unsupported.

For `ccd_type="ACCD"`, additive conservative advancement bounds the source sample displacement plus the largest target displacement over the eight corners of the target SDF grid box. Convexity of the norm of an affine velocity field makes the corner maximum a bound over that box. Each advance uses the current positive gap and the IPC safety fraction; an `accd_tolerance` offset keeps the accepted state away from zero.

5.4.0.3 Porosity-driven dense-packing projection.

`GenerateType="Distribute"` first samples affine-particle centers and sizes to the requested porosity without pairwise rejection. With random centers $\boldsymbol{p}$ and correction translations $\boldsymbol{t}$, the frictionless device solve minimizes

$$\frac{1}{2}\sum_b w_b\|\boldsymbol{t}_b\|^2 + \widetilde{E}_c(\boldsymbol{p}+\boldsymbol{t}) \tag{163}$$

subject to region translation bounds. Its matrix-free Hessian-vector product contains the complete $b''\nabla g\nabla g^T + b'\nabla^2 g$ term, and Newton, active-set classification, PCG, and Armijo trials remain in Taichi fields. At a converged free-set equilibrium the free coordinates satisfy the reduced KKT conditions for $\boldsymbol{H} = \boldsymbol{W} + \nabla^2 E_c$; bound-active coordinates remain fixed at their translation limits. Native level-set particles use their SDF continuation. Initially intersecting watertight, consistently wound TriangleMesh particles remain meshes. A fixed-bandwidth Gaussian surface repulsion is combined with a triangle-intersection-gated Minkowski separating- axis penalty and winding-number containment escape. Because reducing the original per-vertex deformation space to one translation per particle can make local triangle normals cancel, a continued bounding-sphere pair merit term supplies one coherent direction per rigid pair; actual mesh predicates, not that coarse term, certify success. After this translation- only untangling phase reports zero intersecting triangle pairs and zero contained surface vertices, strict point–triangle/edge–edge IPC with ACCD restores a positive-distance terminal state. Since per-particle translation preserves every internal edge, the ARAP term used by the general surface- untangling method is constant and has zero reduced gradient here. All untangling, terminal IPC, ACCD, and line-search iterations execute in Taichi; no mesh-to-SDF proxy is constructed. Discrete intersection and support-pair changes make this a piecewise-smooth feasibility projection.

5.5 Mixed LSMPM Soft-Affine IPC

Related API. DEM.set_affine_body_parameters, MPDEM soft-affine IPC workflow, dhat, contact_damping_stiffness, soft_background_damping.

The soft-affine IPC operator is the implicit branch used when LSMPM soft bodies and AffineBody rigid bodies are coupled with `soft_rigid_contact="IPC"`. Its unknown vector combines active soft MPM grid displacements and affine control-point positions:

$$\boldsymbol{y} = [\boldsymbol{y}_{10}, \ldots, \boldsymbol{y}_{b3}, \ldots, \Delta\boldsymbol{u}_1, \ldots, \Delta\boldsymbol{u}_{n_g}]^T. \tag{164}$$

The step potential is

$$\Pi(\boldsymbol{y}) = \Pi^{in}_{soft} + \Pi^{int}_{soft} + \Pi^{g}_{soft} + \Pi^{in}_{affine} + \Pi^{g}_{affine} + \Pi^{rig}_{affine} + \Pi^{ss}_{B} + \Pi^{sa}_{B} + \Pi^{aa}_{B} + \Pi_F + \Pi_D. \tag{165}$$

Here ss, sa, and aa denote soft-soft, soft-affine, and affine-affine contacts. The affine self terms are those in Affine Body Dynamics. The soft inertia and internal energy use TLMPM interpolation from the active soft grid:

$$\boldsymbol{x}_p(\boldsymbol{y}) = \boldsymbol{x}_p^n + \sum_i N_{ip}\Delta\boldsymbol{u}_i, \tag{166}$$

$$\boldsymbol{F}_p(\boldsymbol{y}) = \boldsymbol{F}_p^n + \sum_i \Delta\boldsymbol{u}_i \otimes \nabla_0 N_{ip}, \tag{167}$$

$$\Pi^{int}_{soft} = \Delta t^2 \sum_p V_p^0\, \psi(\boldsymbol{F}_p). \tag{168}$$

The first Piola stress and material tangent are taken directly from the selected finite-strain material class:

$$\boldsymbol{P}_p = \frac{\partial \psi}{\partial \boldsymbol{F}_p}, \qquad \mathbb{A}_p = \frac{\partial \boldsymbol{P}_p}{\partial \boldsymbol{F}_p}. \tag{169}$$

Therefore the soft material block is

$$\boldsymbol{K}^{mat}_{i\alpha,j\beta} = \Delta t^2 \sum_p V_p^0 \frac{\partial F_{p,mn}}{\partial u_{i\alpha}} \mathbb{A}_{p,mnrs} \frac{\partial F_{p,rs}}{\partial u_{j\beta}}, \tag{170}$$

and no duplicated $\boldsymbol{P}$ formula is embedded in the contact operator.

5.5.1 Soft-Soft Point-Level-Set Barrier

For soft-soft contact the source side is a stored soft surface material point and the target side is the other body's local SDF. With target rotation $\boldsymbol{R}_t$, center $\boldsymbol{c}_t$, and local query

$$\boldsymbol{q} = \boldsymbol{R}_t^T(\boldsymbol{x}_s - \boldsymbol{c}_t),$$

the active gap is

$$g = \phi_t(\boldsymbol{q}) - r_s, \qquad 0 < g < \hat{d}. \tag{171}$$

The barrier energy is the gap form

$$E_{ss} = a_s\, \hat{d}\, B_g(g), \qquad B_g(g) = -\kappa(g - \hat{d})^2 \ln(g/\hat{d}). \tag{172}$$

The trilinear level-set sample provides both the gradient and the local Hessian:

$$\boldsymbol{n}_x = \boldsymbol{R}_t \nabla_q \phi_t, \qquad \boldsymbol{H}_x = \boldsymbol{R}_t\, \nabla_q^2 \phi_t\, \boldsymbol{R}_t^T. \tag{173}$$

The source point depends on soft grid dofs through $\partial \boldsymbol{x}_s / \partial \boldsymbol{u}_i = N_{is}\boldsymbol{I}$. The contact gradient and Hessian contributed to source grid nodes are

$$\boldsymbol{g}_i^{ss} = a_s \hat{d}\, B_g'(g)\, N_{is} \boldsymbol{n}_x, \tag{174}$$

$$\boldsymbol{K}_{ij}^{ss} = a_s \hat{d}\, N_{is} N_{js} \left[B_g''(g) \boldsymbol{n}_x \otimes \boldsymbol{n}_x + B_g'(g) \boldsymbol{H}_x \right]. \tag{175}$$

In the current implementation the target level set is held fixed during this Newton linearization. The pair list is directed, so body A acting as source against body B and body B acting as source against body A are separate contributions. This gives a symmetric Hessian block on the active source-grid dofs of each directed term, while avoiding dense derivatives with respect to every target-grid SDF sample. It is a deliberate semi-implicit level-set contact approximation: the target body affects the residual through the sampled $\phi, \nabla\phi, \nabla^2\phi$, but the target SDF storage itself is not an unknown in that Newton block.

5.5.2 Soft-Affine Point-Triangle Barrier

For soft-affine contact, a soft surface material point is tested against a triangle on the affine surface mesh. The local point-triangle distance type is fixed during one narrow-phase assembly: vertex region, edge region, or face interior. Let

$$s = \|\boldsymbol{p} - \Pi_\triangle(\boldsymbol{p})\|^2$$

be the squared distance for that fixed closest-feature type. The local distance derivatives are assembled with respect to the point and the three triangle vertices:

$$\boldsymbol{z} = [\boldsymbol{p}, \boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}]^T, \qquad \boldsymbol{g}_s = \frac{\partial s}{\partial \boldsymbol{z}}, \qquad \boldsymbol{H}_s = \frac{\partial^2 s}{\partial \boldsymbol{z}^2}. \tag{176}$$

The barrier chain rule is

$$\nabla_{\boldsymbol{z}} E_{sa} = w_{sa} \frac{B'(d)}{2d} \boldsymbol{g}_s, \tag{177}$$

$$\nabla^2_{\boldsymbol{z}} E_{sa} = w_{sa} \left[\frac{1}{4} \left(\frac{B''(d)}{d^2} - \frac{B'(d)}{d^3} \right) \boldsymbol{g}_s \boldsymbol{g}_s^T + \frac{B'(d)}{2d} \boldsymbol{H}_s \right], \qquad d = \sqrt{s}, \tag{178}$$

with w_{sa} equal to the contact area proxy times $\hat{d}$ and any time-step scaling used by the incremental potential. The point block is mapped to soft grid dofs by $N_{is}\boldsymbol{I}$. Each triangle vertex block is mapped to the four affine body control points through the affine basis $N_{v\alpha}\boldsymbol{I}$. Consequently one active contact writes soft-soft grid blocks, soft-affine off-diagonal blocks, and affine-affine triangle blocks into the same sparse Hessian.

5.5.3 Soft-Affine Point-Level-Set Barrier

If the AffineBody target selects level-set contact, the mixed operator replaces the point–triangle stencil by the oriented implicit gap above. A soft surface point has trial position $\boldsymbol{x}_p = \boldsymbol{x}_p^n + \sum_i N_{ip} \Delta \boldsymbol{u}_i$, so its gap derivative pulls back to the MPM grid as

$$\frac{\partial g}{\partial \Delta \boldsymbol{u}_i} = N_{ip} \frac{\partial g}{\partial \boldsymbol{x}_p}. \tag{179}$$

The remaining twelve local columns differentiate the inverse affine map with respect to the target controls. Scattering one scalar potential through both pullbacks produces the soft–affine off-diagonal blocks and satisfies discrete translation action–reaction. The exact branch includes gap curvature; the lagged branch uses the positive barrier outer product. The level-set mixed branch also uses the frozen-material-coordinate friction potential above and pulls its tangent through the same soft and affine weights. Mixed CCD scans the relative path at a fraction of the target SDF spacing and bisects the first unsafe segment, including endpoint-feasible transient crossings.

5.5.4 Affine-Affine Contact and Contact Damping

Affine-affine contact uses the strict IPC mesh primitives: vertex-face point-triangle distance and edge-edge distance. Vertex-face assembly is identical to (178), except both the point and triangle vertices are affine template vertices and therefore map through affine basis weights. Edge-edge contact uses the fixed edge-edge distance type and edge mollifier before applying the same scalar barrier chain rule to $s = d^2$. Wall contact fixes the wall degrees of freedom and inserts only the affine body blocks.

Contact damping is not a surface-friction coefficient. It is a damping stiffness that reuses the active normal-contact Hessian to resist the current increment:

$$\Pi_D = \frac{1}{2} \Delta t \, k_D (\boldsymbol{y} - \hat{\boldsymbol{y}})^T \boldsymbol{K}_B (\boldsymbol{y} - \hat{\boldsymbol{y}}), \qquad \boldsymbol{r}_D = \Delta t \, k_D \boldsymbol{K}_B (\boldsymbol{y} - \hat{\boldsymbol{y}}). \tag{180}$$

For affine bodies this uses `contact_damping_stiffness` from `set_affine_body_parameters` or its aliases. For soft nodes, `soft_background_damping` is a separate background damping term analogous to ordinary MPM grid damping:

$$\boldsymbol{r}_i^{bd} = \Delta t\, c_{bg}\, m_i \boldsymbol{v}_i^{trial}, \qquad \boldsymbol{K}_{ii}^{bd} \approx c_{bg}\, m_i \boldsymbol{I}. \tag{181}$$

It damps soft-grid motion and is not a contact-pair property.

5.5.5 Friction, CCD, and Search

Mixed soft-affine friction is semi-implicit. After a normal contact is active, the operator stores the previous relative tangential displacement $\hat{\boldsymbol{r}}_t$, computes the trial relative displacement $\boldsymbol{r}_t$, and uses the same regularized potential as AffineBody friction:

$$E_F = \mu \lambda_n f_0 \left(\left\| (\boldsymbol{I} - \boldsymbol{n} \otimes \boldsymbol{n}) \frac{\boldsymbol{r}_t - \hat{\boldsymbol{r}}_t}{\Delta t} \right\| \right). \tag{182}$$

The tangent projector and active normal are frozen for the friction Hessian in one Newton assembly. The gradient and Hessian of f_0 are exact for the piecewise regularization, including the small-slip branch $\|\bar{\boldsymbol{v}}\| < \epsilon_v$. The resulting local friction block is mapped through the same soft grid shape weights and affine vertex/control weights as the normal point-triangle contact.

CCD uses the narrow-phase geometry matching each contact family:

Contact family	CCD geometry
soft-soft	point to target level-set gap, using a conservative bisection or ACCD advance on $g(\alpha) > 0$.
soft-affine	point-triangle distance for a TriangleMesh target; moving point-to-SDF gap with a grid-resolved conservative scan for a LevelSet target.
affine-affine	vertex-face point-triangle and edge-edge distances on affine surface meshes.

The level-set target is expanded by the IPC activation distance when building the broad-phase AABB. For soft-soft IPC this ensures that contact candidates are generated while the sampled level-set gap is still positive, which is required by a non-penetrating IPC barrier. The search route mirrors level-set DEM: body pairs are found by `LinkedCell`, `HierarchicalLinkedCell`, or `BVH`; then surface-point/body candidate lists are built inside each active body pair; finally the contact-type-specific narrow phase assembles energy, gradient, Hessian, friction, and CCD limits.

5.6 Strict Contact Derivatives

All IPC Hessians in the strict branches are analytic derivatives of the active distance expression, not finite-difference Hessians. For point-triangle contact, the closest feature determines the formula:

- vertex type: $s = \|\boldsymbol{p}-\boldsymbol{a}\|^2$ or the corresponding vertex, giving constant second derivatives between the point and that vertex;
- edge type: $s = \|\boldsymbol{p} - (1-t)\boldsymbol{a} - t\boldsymbol{b}\|^2$ with $t = ((\boldsymbol{p}-\boldsymbol{a})\cdot(\boldsymbol{b}-\boldsymbol{a}))/\|\boldsymbol{b}-\boldsymbol{a}\|^2$;
- face type: $s = (\boldsymbol{n}\cdot(\boldsymbol{p}-\boldsymbol{a}))^2$ with $\boldsymbol{n}$ from the normalized triangle normal and barycentric closest point in the triangle interior.

The implementation uses the full fixed-type gradient and Hessian before the barrier chain rule. The Hessian is piecewise smooth: it is exact inside a fixed closest-feature region, while feature changes are nonsmooth and are handled by rebuilding contact types between Newton assemblies. Edge-edge contact follows the same principle with fixed closest-edge parameters and the IPC edge-edge mollifier. Finite-difference tests compare the assembled barrier and friction gradients/Hessians for affine-affine, affine-soft, and soft-soft contact against numerical derivatives; they are a verification tool, not a runtime Hessian fallback.

5.7 Critical Timestep

For a linearized contact with stiffness k and reduced mass m^*, the contact eigenfrequency is

$$\omega = \sqrt{k/m^*}, \qquad m^* = \frac{m_i m_j}{m_i + m_j}. \tag{183}$$

The explicit timestep must satisfy

$$\Delta t < C_{DEM}\frac{2}{\omega_{max}}, \tag{184}$$

with a safety factor chosen by the DEM configuration. Hertz-Mindlin contacts use the current tangent normal stiffness in this estimate.

In the current LSMPM path the public estimate is deliberately more conservative than a pairwise reduced-mass analysis:

$$\Delta t_c = \sqrt{\frac{\min_a m_a^{\mathrm{eff}}}{\max_a k_a^{\tan}}}. \tag{185}$$

For linear contact, $k_a^{\tan}$ is the maximum normal or tangential surface-property stiffness multiplied by the physical nodal area. For the energy-conserving penalty law, the bound also includes the configured power `FreeParameter`, a penetration bound based on the larger of the point Verlet distance and the smallest soft mechanical-grid spacing, and the level-set gradient-error bound. `FreeParameter` must be at least two so the tangent remains finite at zero penetration. Barrier contact evaluates the barrier curvature at its normalized cutoff, while Hertz–Mindlin uses $2E^*\sqrt{R\delta}$ and $8G^*\sqrt{R\delta}$, each scaled by nodal area; the reported Hertz step is the minimum of this quadrature bound and the ordinary elastic-wave estimate.

6 Finite Element Method

Related API. FEM user API, FEM.create_mesh, FEM.set_solver, and FEM.add_contact.

6.1 Element, Solver, Material, and Contact Compatibility

Element topology does not determine the time integrator: elastic TET4 and HEX8 are both available in explicit and implicit FEM. The explicit-only restriction belongs to the incremental HEX8 elastoplastic formulation.

Formulation	Time integration	Constitutive support	Plasticity and contact limits
Elastic TET4 volume	Explicit or implicit	St. Venant–Kirchhoff or compressible Neo-Hookean	No incremental plasticity; standalone Barrier IPC is implicit
Elastic HEX8 volume	Explicit or implicit	St. Venant–Kirchhoff or compressible Neo-Hookean	No plasticity on the elastic route; standalone Barrier IPC is implicit
Incremental elastoplastic HEX8	Explicit only, with eight Gauss points	von Mises, Mohr–Coulomb, Drucker–Prager, state-dependent Mohr–Coulomb, Modified Cam–Clay, granular, SANISAND-MS, and NorSand	Standalone FEM IPC is rejected; explicit FEDEM surface contact remains available
Embedded cloth surface	Explicit or implicit	Cloth ARAP or cloth Neo-Hookean, with optional bending, stitch, target-spring, and SDF energies	No plasticity; standalone Barrier IPC is implicit, while cross-solver contact belongs to FEDEM

6.2 Rest Shape, Volume Elements, and Membranes

The FEM facade stores current nodal positions $\boldsymbol{x}_a$ separately from the optional rest shape $\boldsymbol{X}_a$. If the user does not supply `rest_shape`, preprocessing sets $\boldsymbol{X}_a = \boldsymbol{x}_a$. For a volume element at quadrature point q,

$$\boldsymbol{F}_q = \sum_a \boldsymbol{x}_a \otimes \nabla_X N_a(\boldsymbol{X}_q), \tag{186}$$

where the reference gradients and weighted volume are computed once from the rest shape. TET4 has constant gradients, while HEX8 evaluates this mapping at its Gauss points. Material stress and tangent come from the selected finite-strain model in `src/physics_model/consititutive_model`. For linear TET4, define

$$\boldsymbol{D}_m = [\boldsymbol{X}_1 - \boldsymbol{X}_0, \boldsymbol{X}_2 - \boldsymbol{X}_0, \boldsymbol{X}_3 - \boldsymbol{X}_0], \qquad \boldsymbol{D}_s = [\boldsymbol{x}_1 - \boldsymbol{x}_0, \boldsymbol{x}_2 - \boldsymbol{x}_0, \boldsymbol{x}_3 - \boldsymbol{x}_0].$$

The implementation evaluates the equivalent direct map $\boldsymbol{F} = \boldsymbol{D}_s \boldsymbol{D}_m^{-1}$. This is algebraically identical to the reference-gradient sum above and is constant throughout a linear tetrahedron; it removes repeated quadrature-point kinematics without changing the total-Lagrangian weak form. HEX8 cannot use this constant map because its trilinear gradients vary by quadrature point.

For a TRI3 membrane, define two material-space rest edges and two current edges,

$$\boldsymbol{D}_m = [\boldsymbol{U}_1 - \boldsymbol{U}_0, \boldsymbol{U}_2 - \boldsymbol{U}_0], \qquad \boldsymbol{D}_s = [\boldsymbol{x}_1 - \boldsymbol{x}_0, \boldsymbol{x}_2 - \boldsymbol{x}_0]. \tag{187}$$

When a cloth constitutive model is selected, the membrane deformation map is the graphics-style constant triangle map

$$\boldsymbol{F}_s = \boldsymbol{D}_s \boldsymbol{D}_m^{-1} \in \mathbb{R}^{3\times 2}. \tag{188}$$

This shortcut is selected by the cloth material, not by TRI3 topology alone; ordinary finite elements retain their standard element/quadrature mapping. The same rest shape also supplies reference dihedral angles or quadratic bending data. Optional garment stitch, target spring, and frozen-frame SDF energies add analytic gradients and Hessians to the same device assembly.

6.3 Updated-Lagrangian HEX8 Elastoplasticity

For a shared incremental elastoplastic model, each HEX8 owns eight persistent Gauss-point Cauchy stresses and the state-variable structure declared by that constitutive class. Preprocessing first accepts the configured rest-to-current deformation, including initial prestrain. Between two accepted configurations the assembler forms

$$\Delta \boldsymbol{F}_q = \boldsymbol{F}_q^n \left(\boldsymbol{F}_q^{n-1}\right)^{-1}, \qquad \boldsymbol{L}_q \simeq \frac{\Delta \boldsymbol{F}_q - \boldsymbol{I}}{\Delta t}. \tag{189}$$

The shared UL explicit constitutive function integrates $\boldsymbol{\sigma}_q^n = \mathcal{C}(\boldsymbol{\sigma}_q^{n-1}, \boldsymbol{L}_q, \Delta t, \boldsymbol{z}_q)$ and updates its history $\boldsymbol{z}_q$ in device fields. Force assembly converts the returned Cauchy stress to first Piola stress,

$$\boldsymbol{P}_q = J_q \boldsymbol{\sigma}_q \boldsymbol{F}_q^{-T}, \qquad \boldsymbol{f}_a^{\text{int}} \mathrel{+}= w_q \boldsymbol{P}_q \nabla_X N_a. \tag{190}$$

This route supports the shared von-Mises/J2, Mohr–Coulomb, Drucker–Prager, state-dependent Mohr–Coulomb, Modified Cam Clay, granular, SANISAND-MS, and NorSand models. It is intentionally explicit-only and is not combined with FEM IPC/AL; finite-strain elastic HEX8 remains available in both explicit and implicit solvers. A pressure-dependent model without one fixed preprocessing modulus, currently Modified Cam Clay, requires a finite user-specified Δt rather than an automatic wave-speed step.

6.4 Device Assembly and Nonlinear Solve

Element and contact energies, gradients, Hessians, time integration, and residual updates run in Taichi fields and kernels. `assemble_type="COO"` emits scalar coordinate triplets; `"HashTriplet"` accumulates device-side blocks. PCG requires local Hessian projection through `project_pd=True`; BiCGSTAB can consume an unprojected tangent, and `linear_solver="Scipy"` is the explicit host solve boundary rather than a NumPy FEM backend. Implicit dynamics uses Newmark inertia and Armijo line search, while explicit dynamics uses lumped mass and a symplectic update. A trial that misses strict Armijo decrease only by the configured precision's scaled energy roundoff is accepted only when the predicted directional decrease is itself below that roundoff; non-finite trials remain inadmissible.

6.5 FEM Collision Culling and CCD

FEM self-contact separates spatial broad phase from collision culling. The user selects `broad_phase="LinkedCell"` or `"BVH"`. Both backends produce conservative point–triangle (PT) and edge–edge (EE) AABB overlaps and reject incident primitives. The common culling layer then applies seam relations introduced by garment stitches, evaluates the exact closest-feature distance, marks

$$d_{PT/EE}^2 < (d_{min} + \hat{d})^2, \tag{191}$$

and prefix-sum compacts the active stencils and their integration measures. Thus the raw AABB list is never interpreted as the final barrier set.

The linked-cell grid is rebuilt for every query using count, prefix sum, and fill passes, so no fixed maximum number of deforming triangles is assigned to a cell. For EE queries each edge AABB is expanded by half the pairwise search radius; overlap of the two expanded boxes therefore represents one configured radius rather than twice that radius. The BVH keeps a balanced topology and refits current boxes, but passes its output through the same culling kernels.

For an IPC search direction $\boldsymbol{p}$, every admissible-step query rebuilds spatial candidates from swept bounds

$$\mathrm{AABB}_a = \mathrm{hull}(\boldsymbol{x}_a, \boldsymbol{x}_a + \boldsymbol{p}_a). \tag{192}$$

Conservative PT/EE CCD computes stencil step fractions α_c, compacts the stencils that can limit the step, and reduces

$$\alpha_{CCD} = \min\left(1, \min_c \alpha_c, \min_h \alpha_h\right), \tag{193}$$

where h denotes a one-sided plane constraint. This path intentionally has no Verlet multiplier because both Newton and CCD queries use freshly rebuilt current or swept bounds. Barrier IPC applies this bound before Armijo backtracking and uses d_{min} as the CCD clearance.

For multiple disconnected FEM bodies, each cell has a body ID and every surface primitive inherits that ID. A configured unordered pair $p = (b_i, b_j)$ owns its parameter set

$$\mathcal{P}_p = \{\hat{d}_p, d_{\min,p}, \kappa_p, \mu_p, \epsilon_{v,p}, s_{\mathrm{CCD},p}, \text{broad phase}_p\}. \tag{194}$$

Collision culling accepts a PT or EE stencil only when its two primitive body IDs form exactly p, then evaluates and scatters the corresponding barrier, friction, and Hessian terms into the shared global system. Once at least one pair is registered, unlisted body pairs are inactive. Independent pair-local culling buffers permit different activation radii and spatial backends without a host-side candidate partition or force copy.

7 FEM–DEM Coupling

Related API. FEM–DEM user API.

The maintained `FEDEM` path has three scheme-specific branches. Ordinary DEM spheres and rigid LSDEM bodies use explicit FEM and the selected DEM penalty law. AffineBody

uses elastic implicit FEM and a monolithic IPC potential. In every branch the FEM soft body remains a TET4/HEX8 volume or TRI3 membrane with a TRI3 contact boundary; the coupling does not create an FEM SDF.

The principal route restrictions are:

FEDEM route	Child solvers	Deformable material support	Contact restriction
Deformable–deformable soft particles	Explicit FEM	Elastic TET4/HEX8 or incremental elasto-plastic HEX8	Explicit Linear, Hertz–Mindlin, or energy-conserving contact
DEM sphere–FEM surface	Explicit FEM plus DEM	Volume, membrane, or cloth; plasticity only through explicit HEX8	Penalty contact, not standalone FEM IPC
Rigid LSDEM SDF–FEM surface	Explicit FEM plus LSDEM	Same explicit HEX8 plastic restriction	Only the rigid body owns an SDF
AffineBody–FEM mesh IPC	Implicit FEM plus AffineBody	Elastic TET4/HEX8 or cloth; no plasticity	Monolithic ordinary Barrier IPC with lagged friction

7.1 Explicit Deformable Soft-Particle Contact

A FEM soft particle is one disconnected TET4/HEX8 volume body. It uses the same volume stress integration as ordinary FEM and does not own an SDF. The contact representation is the extracted, currently deforming TRI3 boundary. After linked-cell or BVH AABB search, common collision culling rejects same-body candidates and compacts point–triangle (PT) and edge–edge (EE) stencils.

For a directed PT stencil (p, a, b, c), let $\boldsymbol{q}$ and $(\lambda_a, \lambda_b, \lambda_c)$ be the closest point and barycentric coordinates on the oriented face. With outward normal $\boldsymbol{n}$, the signed gap and overlap are

$$g_{PT} = (\boldsymbol{x}_p - \boldsymbol{q}) \cdot \boldsymbol{n}, \qquad \delta_{PT} = h - g_{PT}, \tag{195}$$

where h is `ContactThickness`. The relative velocity and force scatter use the same action–reaction weights

$$\boldsymbol{w} = (1, -\lambda_a, -\lambda_b, -\lambda_c), \qquad \boldsymbol{v}_{rel} = \sum_{i=0}^{3} w_i \boldsymbol{v}_i, \qquad \boldsymbol{f}_i \mathrel{+}= w_i(\boldsymbol{f}_n + \boldsymbol{f}_t). \tag{196}$$

The effective mass is $m^* = (\sum_i w_i^2/m_i)^{-1}$. The contact measure is one half of the query node's current lumped surface area, compensating for the two directed surface queries.

For an EE stencil, closest segment parameters $s, t \in [0, 1]$ give

$$\boldsymbol{w} = (1 - s, s, -(1 - t), -t), \qquad \delta_{EE} = h - \|(1 - s)\boldsymbol{x}_0 + s\boldsymbol{x}_1 - (1 - t)\boldsymbol{x}_2 - t\boldsymbol{x}_3\| . \tag{197}$$

The same effective-mass and action–reaction formulas apply. Since EE distance is unsigned, $h > 0$ is required for a pre-crossing force.

For the Linear model with surface measure A_c,

$$\boldsymbol{f}_n = \max\Big(A_c k_n \delta - 2\zeta_n \sqrt{m^* A_c k_n}\, v_n, 0\Big)\, \boldsymbol{n}, \tag{198}$$

$$\boldsymbol{f}_t^{trial} = -A_c k_t \boldsymbol{\xi} - 2\zeta_t \sqrt{m^* A_c k_t}\, \boldsymbol{v}_t, \tag{199}$$

followed by $\|\boldsymbol{f}_t\| \le \mu \|\boldsymbol{f}_n\|$. The Hertz–Mindlin branch uses the DEM damping and tangential-history conventions and defines the otherwise absent patch radius by $R_c = \sqrt{A_c/\pi}$. Tangential overlap is stored by contact kind and four-node stencil; Verlet lists are rebuilt after surface motion exceeds half the skin distance.

7.2 Ordinary DEM Sphere–Triangle Contact

The ordinary DEM branch migrates the executable `coupled_DEM_FEM` sphere–triangle laws into the package lifecycle used by `src/mpdem`. It is penalty contact, not FEM IPC.

For particle center $\boldsymbol{x}$, radius R, oriented face normal $\boldsymbol{n}$, and face centroid $\boldsymbol{c}$, define

$$d = (\boldsymbol{x} - \boldsymbol{c}) \cdot \boldsymbol{n}, \qquad \delta = R - d, \qquad r_c = \sqrt{R^2 - d^2}. \tag{200}$$

Only $0 < d < R$ produces a contact section. The source area correction is

$$\phi = \frac{A(\mathrm{disk}(\boldsymbol{x} - d\boldsymbol{n}, r_c) \cap \triangle_{123})}{\pi r_c^2}. \tag{201}$$

For the linear model, with relative contact-point velocity split into $v_n \boldsymbol{n} + \boldsymbol{v}_s$,

$$\boldsymbol{f}_n = \Big(k_n \delta - 2\zeta_n \sqrt{m k_n}\, v_n\Big)\, \boldsymbol{n}, \tag{202}$$

$$\boldsymbol{\xi}^{\mathrm{trial}} = \boldsymbol{v}_s \Delta t + \|\boldsymbol{\xi}^{n-1}\|\, \mathrm{normalize}\big(\boldsymbol{\xi}^{n-1} - (\boldsymbol{\xi}^{n-1} \cdot \boldsymbol{n})\boldsymbol{n}\big)\,, \tag{203}$$

$$\boldsymbol{f}_s^{\mathrm{trial}} = -k_s \boldsymbol{\xi}^{\mathrm{trial}}. \tag{204}$$

The trial tangent is Coulomb-capped by $\mu|f_n|$; below the cap the source tangential dashpot $-2\zeta_s\sqrt{m k_s}\boldsymbol{v}_s$ is added unchanged.

For Hertz–Mindlin contact,

$$a = \sqrt{R\delta}, \qquad k_n = 2E^* a, \qquad k_s = 8G^* a, \tag{205}$$

and the elastic normal magnitude is $\frac{2}{3}k_n\delta$. The source damping factor $\beta = -\log(e)/\sqrt{\pi^2 + \log^2(e)}$ multiplies $-1.8257\, v_n \sqrt{k_n m}$ and its tangential analogue. Both laws multiply the final force by ϕ.

The DEM particle receives $\boldsymbol{F} = \phi(\boldsymbol{f}_n + \boldsymbol{f}_s)$ and the source moment convention

$$\boldsymbol{M} = \boldsymbol{F} \times (\boldsymbol{x} - \boldsymbol{x}_c). \tag{206}$$

The FEM receives opposite-sign nodal contributions. With projected center $\boldsymbol{p}$, the source weights are the absolute subtriangle areas divided by the parent area, for example $w_1 = A(\boldsymbol{p}, \boldsymbol{x}_2, \boldsymbol{x}_3)/A_{123}$. They sum to one when $\boldsymbol{p}$ is inside the triangle. They are deliberately not renormalized when $\boldsymbol{p}$ is outside but the circular section still overlaps the face, preserving the source edge-case behavior rather than imposing a new exact action–reaction correction.

Because face size changes, broad search inserts each triangle's AABB expanded by the maximum sphere radius and skin into all overlapped cells. Every rebuild uses device count, prefix-sum, and fill passes, followed by per-particle exact distance culling and compact contact construction. Tangential history is inherited by particle ID and face ID. A shared step resets forces, rebuilds DEM/coupling lists when required, resolves DEM internal contact, resolves FEM–DEM action/reaction, advances DEM, and finally advances FEM once.

7.3 Rigid LSDEM SDF–FEM Boundary-Node Contact

The LSDEM branch uses the same FEM boundary mesh and the same `Linear`/`HertzMindlin` force law, but the target geometry is the rigid body's existing signed-distance grid. It does not construct an SDF for FEM. For each boundary node a, the narrow phase transforms $\boldsymbol{x}_a$ into the rigid level-set frame, queries signed distance g_a and gradient, and uses the normalized gradient as the contact normal. The contact measure is the current lumped nodal area

$$A_a = \frac{1}{3}\sum_{f \ni a} A_f(\boldsymbol{x}). \tag{207}$$

The nodal velocity and rigid contact-point velocity define the relative velocity. The resulting force is applied directly to FEM node a, and its exact opposite plus moment about the rigid center is applied to LSDEM. Each node–rigid contribution therefore satisfies action–reaction. The Linear law scales its stiffness by A_a. For the optional Hertz–Mindlin law, a surface node has no physical sphere radius, so GeoTaichi uses the explicit closure $R_a = \sqrt{A_a/\pi}$. This is a documented extension rather than a source-equivalent `coupled_DEM_FEM` equation.

Broad search uses the true enclosing radius rather than the equivalent-volume radius to select the linked-cell scale. Each rebuild forms the exact current world AABB of the rotated level-set box. Linked cells use rigid-AABB count, prefix-sum, and compact fill; BVH uses one current AABB leaf per rigid body. Both produce unique FEM boundary-node/rigid-body pairs, and the SDF query remains the exact narrow phase. A rigid SDF can rotate while its center is fixed, so this list is rebuilt every explicit step rather than guarded by a center-only Verlet test.

7.4 Elastic FEM–AffineBody IPC

The AffineBody branch combines affine controls $\boldsymbol{q}$ and FEM nodes $\boldsymbol{x}$ into one Newton vector. Affine surface vertices are reconstructed as

$$\boldsymbol{y}_v = \sum_{i=0}^{3} w_{vi}\boldsymbol{q}_{4b+i}.$$

Mixed collision culling retains only primitives whose endpoints belong to different subsystems. Point–triangle and edge–edge barrier gradients and analytic Hessians are first formed in 12 endpoint degrees of freedom, then pulled back through the affine weights and FEM identity map into the global block matrix. The local Hessian is spectrally projected to PSD.

The closest-feature classifier dispatches PP, PE, PT, and EE derivative kernels separately. This is mathematically the same analytic distance Hessian as the unified feature switch,

but prevents Taichi from statically inlining every large symbolic branch into one kernel. Current candidates are rebuilt during Newton assembly and swept candidates are rebuilt for every mixed CCD/ACCD line-search bound, so this branch has no Verlet multiplier. The Armijo step also enforces AffineBody self CCD and the FEM material Jacobian bound.

For lagged mixed friction, each fixed-point iteration freezes the closest feature normal $\hat{\boldsymbol{n}}$, signed closest-point weights

$$\hat{w}_i = \frac{\nabla_{\boldsymbol{x}_i} d^2 \cdot \hat{\boldsymbol{n}}}{2d}, \qquad \boldsymbol{P} = \boldsymbol{I} - \hat{\boldsymbol{n}} \otimes \hat{\boldsymbol{n}}, \tag{208}$$

and the barrier normal-force magnitude $\hat{\lambda}_n$. For an EE stencil the lagged law is active only after the IPC parallel-edge mollifier reaches one, matching the repository's standard IPC friction convention. With the beginning-of-step endpoint positions $\hat{\boldsymbol{x}}_i$ held fixed,

$$\bar{\boldsymbol{v}} = \frac{\boldsymbol{P}}{\Delta t} \sum_{i=0}^{3} \hat{w}_i (\boldsymbol{x}_i - \hat{\boldsymbol{x}}_i), \qquad E_f = \mu \hat{\lambda}_n f_0(\|\bar{\boldsymbol{v}}\|; \epsilon_v, \Delta t). \tag{209}$$

The gradient is pulled back through the same four endpoint weights. The three-dimensional relative Hessian is assembled analytically in the tangent plane: its transverse eigenvalue is $\mu \hat{\lambda}_n f_1(v)/v$, and its radial eigenvalue is $2\mu \hat{\lambda}_n (\epsilon_v - v)/\epsilon_v^2$ for $0 < v < \epsilon_v$ and zero in sliding. Both are non-negative, so the subsequent affine-control/FEM-node pullback remains symmetric PSD without a finite-difference Hessian. Pair properties select μ and ϵ_v, and positive `friction_iterations` refresh the frozen frame and normal force while preserving $\hat{\boldsymbol{x}}$.

The global matrix may use `COO` or `HashTriplet`; `PCG` is the device solve and `Scipy` is the explicit host solver option. Mixed contact currently supports the lagged potential above, not fully implicit friction. AffineBody self-contact also remains lagged and is restricted to its own single friction iteration by this monolithic coupling engine.

8 FEM–MPM Coupling

Related API. FEM–MPM user API.

The `FEMPM` path applies the same one-sided sphere–triangle section fraction, linear spring–dashpot law, or Hertz–Mindlin law as FEDEM, but the sphere represents an MPM material point. Its radius is the coupling radius R_p stored with the point and its relative velocity is

$$\boldsymbol{v}_{\mathrm{rel},p} = \boldsymbol{v}_p - \frac{\boldsymbol{v}_1^{\mathrm{FEM}} + \boldsymbol{v}_2^{\mathrm{FEM}} + \boldsymbol{v}_3^{\mathrm{FEM}}}{3}. \tag{210}$$

There is no MPM particle rotational degree of freedom. The resulting force is atomically accumulated in `ParticleCoupling.external_force`; the ordinary MPM P2G force operation therefore contributes it to the grid momentum equation without a host transfer. The opposite force is distributed to the three FEM nodes with the FEDEM absolute-subtriangle weights before explicit FEM update.

With `search="LinkedCell"`, every rebuild counts the expanded cells overlapped by each current triangle, prefix-sums the counts, and fills compact memberships. With `search="BVH"`, current triangle AABBs are Morton sorted and a linear BVH is rebuilt. A point queries the tree using its AABB expanded by $R_p + 2s$, where s is the coupling skin. Both paths

then apply the same exact point–triangle distance cull and compact history construction, so search choice changes acceleration and storage rather than contact physics.

The coupled step resets MPM and FEM force messages, updates child neighbor state, rebuilds the coupling list when either side moves more than $s/2$, evaluates DEM-style contact, performs MPM P2G/G2P and constitutive evolution, and advances FEM exactly once. The shared stability gate uses

$$\Delta t \le C_{\mathrm{CFL}} \min(\Delta t_{\mathrm{MPM}}, \Delta t_{\mathrm{FEM}}, \Delta t_{\mathrm{contact}}). \tag{211}$$

For `contact_model="IPC"`, a surface MPM sample is reconstructed from the compact active grid displacement,

$$\boldsymbol{x}_p(\boldsymbol{u}_g) = \boldsymbol{x}_p^n + \sum_a N_a(\boldsymbol{x}_p^n)\boldsymbol{u}_a, \tag{212}$$

and paired with a current FEM triangle. The IPC point–triangle squared-distance gradient $\boldsymbol{g} \in \mathbb{R}^{12}$ and analytic Hessian $\boldsymbol{H} \in \mathbb{R}^{12\times 12}$ are evaluated in geometric coordinates. The FEM internal energy remains Total Lagrangian: reference gradients and quadrature weights are fixed and $\boldsymbol{F} = \partial \boldsymbol{x}/\partial \boldsymbol{X}$. The Direct MPM side is Updated Lagrangian, while IPC geometry is evaluated in the current spatial configuration. Thus the monolithic system couples TL FEM, UL MPM, and spatial contact without imposing one shared deformation-gradient update on both discretizations. In Cartesian two dimensions the geometric primitive is instead a point and a FEM boundary edge. Its exact squared-distance derivatives have six entries and the zero-clearance CCD solves the moving point–line collinearity quadratic, then checks the finite edge coordinate. Finite-clearance ACCD advances the point and both edge endpoints in one bounded conservative loop. The same LinkedCell/BVH interface builds current and swept point–edge candidates.

For no-swirl axisymmetry, both children retain two meridional unknowns but evaluate a three-dimensional constitutive map,

$$\boldsymbol{F}_{\mathrm{axi}} = \begin{bmatrix} \partial r/\partial R & \partial r/\partial Z & 0 \\ \partial z/\partial R & \partial z/\partial Z & 0 \\ 0 & 0 & r/R \end{bmatrix}, \qquad \mathrm{d}V_0 = 2\pi R\,\mathrm{d}R\,\mathrm{d}Z. \tag{213}$$

The FEM TRI3 and Direct MPM tangent include the derivative $\partial F_{\theta\theta}/\partial u_{r,a} = N_a/R$. Contact remains a point–edge problem in the (r, z) meridian, but one MPM surface sample represents a full reference ring and therefore uses the fixed IPC weight $2\pi R\, w_p$. All radii are measured from the shared `axis_offset` and must be positive. The third stored FEM component is a constrained embedding DOF; it is not a swirl unknown. If $\boldsymbol{B}$ maps FEM nodal and active MPM-grid increments to the point and triangle positions, the coupled contributions are

$$\boldsymbol{r}_c = \boldsymbol{B}^T\boldsymbol{g}, \qquad \boldsymbol{K}_c = \boldsymbol{B}^T\,\mathrm{PSD}(\boldsymbol{H})\boldsymbol{B}. \tag{214}$$

For associated materials, the barrier, elastic FEM tangent, Direct MPM tangent, inertia, and lagged IPC friction Hessian share one projected symmetric Newton system using PCG. Nonassociated DP instead retains the physical material Jacobian and unprojected contact pullback $\boldsymbol{K}_c = \boldsymbol{B}^T\boldsymbol{H}\boldsymbol{B}$, stores both matrix triangles, and uses BiCGSTAB with residual Armijo. Both `COO` and `HashTriplet` assemble on the Taichi device; Scipy is only an explicit linear-solver option.

The MPM tangent may come from Neo-Hookean elasticity or the Direct ULMPM physical nonassociated Drucker–Prager response or associated Drucker–Prager/von-Mises incremental potentials in finite-strain associated plasticity. FEM remains elastic. The plastic material return, PK1 force, analytic spectral Hessian, determinant CCD, and accepted-state commit all run in Taichi kernels; selecting plasticity does not bypass swept IPC culling, contact CCD/ACCD, or Armijo. For Cartesian two-dimensional plasticity, Direct ULMPM embeds the incremental map in three-dimensional plane strain,

$$\boldsymbol{f}_{\mathrm{ps}} = \begin{bmatrix} 1+\partial u_x/\partial x_n & \partial u_x/\partial y_n & 0 \\ \partial u_y/\partial x_n & 1+\partial u_y/\partial y_n & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \boldsymbol{F}_{e,tr} = \boldsymbol{f}_{\mathrm{ps}}\boldsymbol{F}_{e,n}. \tag{215}$$

The grid/contact unknowns remain two-dimensional, but $\boldsymbol{F}_e$, the invariants, return mapping, and material tangent are three-dimensional. Intrinsic two-dimensional DP or J_2 invariants are not used. The coupled step snapshots FEM nodal state and Direct-MPM particle, grid, elastic-gradient, and plastic-history state on the device. Any nonlinear or post-solve failure restores that snapshot, so an exception cannot leave a partly committed coupled state.

The lagged friction frame and barrier normal force are frozen at the start of each friction fixed-point iteration. Its tangential relative increment includes both the interpolated MPM point motion and all three FEM vertex motions. Current proximity candidates are rebuilt for every Newton assembly and trial energy; swept candidates are rebuilt for point–triangle CCD/ACCD. Consequently the implicit branch does not use a Verlet multiplier. Armijo accepts only a finite-merit, non-inverted FEM/MPM state that also respects the conservative contact step; the merit is energy for potential materials and the physical residual norm for nonassociated DP. Continuum-solid MPM coupled to cloth FEM uses the same nonassociated DP and lagged-friction driver. Direct physical DP evaluation removes the material consistency outer loop; it does not remove the friction fixed point.

For nonassociated DP, `inexact_newton=True` adapts the Taichi Krylov forcing term using the same policy as IGA–MPM:

$$\eta_0 = 10^{-2}, \qquad \eta_k = \min\left(10^{-2}, \max\left(\eta_{min}, 0.9\left(\frac{\|\boldsymbol{R}_{free,k}\|}{\|\boldsymbol{R}_{free,k-1}\|}\right)^{3/2}\right)\right), \quad \eta_{min} = \max(10^{-7}, \eta_{configured}). \tag{216}$$

The true linear stopping threshold is $\max(\varepsilon_{\mathrm{lin,a}}, \eta_k\|\boldsymbol{R}_{free,k}\|)$. The force norm is measured before Dirichlet elimination; scaling by the eliminated right-hand side can overestimate the allowed error under prescribed loading. The forcing history resets at each lagged-friction outer iteration. Fixed node reactions are excluded from the force norm; prescribed displacement errors are checked separately. Force and physical-correction convergence are required together. The first force reference is retained throughout the attempted step's friction refreshes; only the Krylov forcing history restarts. A terminal inexact correction is checked at configured strict linear accuracy. It defaults to false because extra Newton assemblies may outweigh Krylov savings on small systems.

9 IGA

Related API. IGA.add_primitives, IGAMPM.build, and the IGAMPM monolithic barrier system.

9.1 NURBS Basis and Isoparametric Mapping

For knot-span basis $N_A(\xi)$ and weights w_A, the NURBS basis is

$$R_A(\xi) = \frac{N_A(\xi) w_A}{\sum_B N_B(\xi) w_B}. \tag{217}$$

The physical coordinate is

$$\boldsymbol{x}(\xi) = \sum_A R_A(\xi) \boldsymbol{P}_A, \tag{218}$$

with control points $\boldsymbol{P}_A$. Derivatives in physical space follow from the Jacobian

$$\boldsymbol{J} = \frac{\partial \boldsymbol{x}}{\partial \xi} = \sum_A \boldsymbol{P}_A \otimes \nabla_\xi R_A. \tag{219}$$

The code uses the same isoparametric idea as FEM. With reference control points $\boldsymbol{X}_A$, current control points $\boldsymbol{x}_A = \boldsymbol{X}_A + \boldsymbol{d}_A$, and quadrature point ξ_q,

$$\boldsymbol{X}_q = \sum_A R_A(\xi_q) \boldsymbol{X}_A, \qquad \boldsymbol{x}_q = \sum_A R_A(\xi_q) \boldsymbol{x}_A. \tag{220}$$

The reference Jacobian and the physical derivative of a basis function are

$$J_{0,ij} = \frac{\partial X_i}{\partial \xi_j}, \qquad \frac{\partial R_A}{\partial X_i} = \frac{\partial R_A}{\partial \xi_j} (J_0^{-1})_{ji}. \tag{221}$$

The quadrature weight stored by the element is

$$V_q = \det(J_0)\, j_2\, w_q, \tag{222}$$

where j_2 is the one-dimensional knot-span scaling used by the local Gauss-rule implementation. The reference configuration must satisfy $\det(J_0) > 0$ at every quadrature point. Device kernels record a failed finite-positive check and preprocessing raises an error after the kernel batch; an orientation-reversed or singular patch is not reinterpreted using $|\det(J_0)|$ and cannot contribute negative mass or energy.

9.2 NURBS Utility Library

The `src/nurbs` folder is not only an IGA backend. It also provides B-spline/NURBS primitives, knot operations, closest-point inversion, distance queries, and fitting utilities. The one-dimensional B-spline basis is computed from the Cox-de Boor recursion. For knot vector $\Xi = \{\xi_0, \ldots, \xi_m\}$,

$$N_{i,0}(\xi) = \begin{cases} 1, & \xi_i \le \xi < \xi_{i+1}, \\ 0, & \text{otherwise}, \end{cases} \tag{223}$$

$$N_{i,p}(\xi) = \frac{\xi - \xi_i}{\xi_{i+p} - \xi_i} N_{i,p-1}(\xi) + \frac{\xi_{i+p+1} - \xi}{\xi_{i+p+1} - \xi_{i+1}} N_{i+1,p-1}(\xi), \tag{224}$$

with zero contribution whenever a denominator vanishes. The derivative is

$$\frac{\mathrm{d}N_{i,p}}{\mathrm{d}\xi} = \frac{p}{\xi_{i+p} - \xi_i} N_{i,p-1} - \frac{p}{\xi_{i+p+1} - \xi_{i+1}} N_{i+1,p-1}. \tag{225}$$

`find_span`, `basis_functions`, and `basis_function_ders` implement these formulas; tensor-product 2D and 3D basis functions are outer products:

$$N_A(\xi, \eta, \zeta) = N_i^p(\xi) N_j^q(\eta) N_k^r(\zeta). \tag{226}$$

For rational NURBS, the basis is

$$R_A(\boldsymbol{\xi}) = \frac{w_A N_A(\boldsymbol{\xi})}{W(\boldsymbol{\xi})}, \qquad W(\boldsymbol{\xi}) = \sum_B w_B N_B(\boldsymbol{\xi}). \tag{227}$$

Its first derivative is

$$R_{A,\alpha} = \frac{w_A}{W} \left(N_{A,\alpha} - N_A \frac{W_{,\alpha}}{W} \right), \tag{228}$$

and second derivatives follow by differentiating this quotient. These derivatives are used by `SplineCurve.derivative`, `SplineSurface.derivative`, curvature evaluation, boundary normals, and the closest-point Hessians used by IGA-MPM contact.

9.2.1 Knot Insertion and Refinement

Knot insertion changes the control net but not the represented curve or surface. For a curve

$$\boldsymbol{C}(\xi) = \sum_i R_i(\xi) \boldsymbol{P}_i, \tag{229}$$

inserting a knot $\bar{\xi}$ updates the affected control points by affine combinations

$$\boldsymbol{P}_i^{new} = \alpha_i \boldsymbol{P}_i + (1 - \alpha_i) \boldsymbol{P}_{i-1}, \qquad \alpha_i = \frac{\bar{\xi} - \xi_i}{\xi_{i+p} - \xi_i}. \tag{230}$$

Tensor-product surfaces and volumes apply the same one-dimensional operation along the selected parametric direction. Repeated insertion up to a desired multiplicity lowers cross-element continuity; uniform refinement inserts new knots without changing geometry and is the NURBS analogue of mesh refinement.

9.2.2 Curves, Surfaces, Volumes, and Boundary Primitives

`SplineCurve`, `SplineSurface`, and `SplineVolume` expose the same geometry concepts: degree, knot vectors, control points, element spans, boundary entities, interpolation, derivatives with respect to knots and control points, and visualization/write helpers. B-spline primitives store only control points; NURBS primitives additionally store weights and expose weight-sensitivity routines such as $\partial \boldsymbol{x} / \partial w_A$:

$$\frac{\partial \boldsymbol{x}}{\partial w_A} = \frac{N_A}{W} (\boldsymbol{P}_A - \boldsymbol{x}). \tag{231}$$

Boundary activation constructs curve primitives from the edge of a surface or surface primitives from the face of a volume. This is the bridge used by IGA boundary conditions, IGA-MPM point-curve/point-surface contact, and geometric postprocessing.

9.2.3 Closest-Point Projection and Inversion

Curve inversion solves for a parameter whose tangent is orthogonal to the point residual:

$$f(u) = (\boldsymbol{C}(u) - \boldsymbol{x}) \cdot \boldsymbol{C}'(u) = 0. \tag{232}$$

Newton uses

$$f'(u) = \boldsymbol{C}'(u) \cdot \boldsymbol{C}'(u) + (\boldsymbol{C}(u) - \boldsymbol{x}) \cdot \boldsymbol{C}''(u), \tag{233}$$

with the parameter clamped to the valid knot interval or wrapped for closed curves. Surface inversion solves the two orthogonality equations

$$F_1 = (\boldsymbol{S}(u,v) - \boldsymbol{x}) \cdot \boldsymbol{S}_u = 0, \qquad F_2 = (\boldsymbol{S}(u,v) - \boldsymbol{x}) \cdot \boldsymbol{S}_v = 0. \tag{234}$$

The Newton matrix is

$$J_{\alpha\beta} = \boldsymbol{S}_\alpha \cdot \boldsymbol{S}_\beta + (\boldsymbol{S} - \boldsymbol{x}) \cdot \boldsymbol{S}_{\alpha\beta}. \tag{235}$$

The code uses a coarse initial search followed by Newton/Gauss iteration and boundary clamping. Boundary inversion is necessary because the closest point to a surface can lie on an edge or corner even when the unconstrained Newton iterate would leave the parametric domain.

9.2.4 Spline Distance Queries

Spline-spline distance routines minimize

$$d^2(u,v) = \|\boldsymbol{C}_1(u) - \boldsymbol{C}_2(v)\|^2 \tag{236}$$

or the surface/curve analogue. Lower bounds from convex-hull AABBs and capsule bounds reject pairs before local minimization. For a curve-curve pair, the stationarity equations are

$$(\boldsymbol{C}_1 - \boldsymbol{C}_2) \cdot \boldsymbol{C}_1' = 0, \qquad (\boldsymbol{C}_1 - \boldsymbol{C}_2) \cdot \boldsymbol{C}_2' = 0. \tag{237}$$

The Hessian contains tangent-tangent terms and residual-curvature terms, just as in the single-point inversion formula. The bisection and Newton variants in `Distance.py` trade robustness for speed; conservative broad-phase bounds are still required for contact search because local distance minimization is not by itself a global candidate generator.

9.2.5 Interpolation, Approximation, and Fitting

Global interpolation constructs a linear system

$$\sum_A N_A(\xi_j)\boldsymbol{P}_A = \boldsymbol{Q}_j \tag{238}$$

or its tensor-product surface version. Least-squares approximation minimizes

$$\min_{\boldsymbol{P}} \sum_j \left\| \sum_A N_A(\xi_j)\boldsymbol{P}_A - \boldsymbol{Q}_j \right\|^2 + \lambda_s \mathcal{R}(\boldsymbol{P}), \tag{239}$$

where $\mathcal{R}$ is an optional smoothing or regularization term in the helper routines. NURBS fitting also treats weights as unknowns and therefore uses Gauss-Newton or Newton-Raphson updates:

$$(J^T J + \lambda I)\Delta y = -J^T r. \tag{240}$$

Parameterization utilities choose chord-length, centripetal, grid, PCA, or boundary-based parameters depending on whether the input is a curve, gridded surface, or scattered point set. These routines are useful for converting measured or meshed geometry into the NURBS primitives consumed by IGA and IGA-MPM.

9.3 FEM/IGA Residual and Tangent Flow

Using the displacement approximation $\boldsymbol{u} = \sum_A R_A \boldsymbol{d}_A$, the small-strain stiffness matrix is

$$\boldsymbol{K}_{AB} = \int_\Omega \boldsymbol{B}_A^T \mathbb{C} \boldsymbol{B}_B \, \mathrm{d}\Omega, \tag{241}$$

and the mass matrix is

$$\boldsymbol{M}_{AB} = \int_\Omega \rho R_A R_B \boldsymbol{I} \, \mathrm{d}\Omega. \tag{242}$$

Because NURBS basis functions are usually smoother than C^0 finite elements, IGA can represent curved geometry exactly and achieve higher-order smooth stress fields for the same mesh topology.

The finite-strain IGA implementation assembles from an energy density $\Psi(\boldsymbol{F})$. At each quadrature point the deformation gradient is computed in code as

$$F_{ab} = \sum_A x_{A,a} \frac{\partial R_A}{\partial X_b}. \tag{243}$$

Its derivative with respect to a control-point coordinate is

$$\frac{\partial F_{ab}}{\partial x_{A,c}} = \delta_{ac} \frac{\partial R_A}{\partial X_b}. \tag{244}$$

Let $P_{ab} = \partial \Psi / \partial F_{ab}$ and $\mathbb{A}_{ab\,cd} = \partial^2 \Psi / \partial F_{ab} \partial F_{cd}$. The local internal force contribution follows by the chain rule:

$$r^{int}_{A,c} = \sum_q V_q \sum_{a,b} P_{ab} \frac{\partial F_{ab}}{\partial x_{A,c}} = \sum_q V_q \sum_b P_{cb} \frac{\partial R_A}{\partial X_b}. \tag{245}$$

The code inserts the negative of this local gradient into the right-hand side. The material tangent block is

$$K^{mat}_{Ac,Bd} = \sum_q V_q \sum_{a,b,e,f} \frac{\partial F_{ab}}{\partial x_{A,c}} \mathbb{A}_{ab\,ef} \frac{\partial F_{ef}}{\partial x_{B,d}}. \tag{246}$$

This is the standard FEM element loop, with NURBS replacing polynomial C^0 shape functions.

9.4 Material Energies in the IGA Backend

For the linear-elastic IGA material class, the strain measure is Green strain:

$$\boldsymbol{E} = \frac{1}{2}(\boldsymbol{F}^T\boldsymbol{F} - \boldsymbol{I}). \tag{247}$$

The implemented energy is

$$\Psi_{LE} = \frac{1}{2}\lambda \mathrm{tr}(\boldsymbol{E})^2 + \mu \boldsymbol{E} : \boldsymbol{E}. \tag{248}$$

For the Neo-Hookean material,

$$\Psi_{NH} = \frac{1}{2}\mu(I_2 - d) - \mu \ln J + \frac{1}{2}\lambda(\ln J)^2, \qquad I_2 = \mathrm{tr}(\boldsymbol{F}^T\boldsymbol{F}), \quad J = \det \boldsymbol{F}. \tag{249}$$

The gradient and Hessian are evaluated through invariant derivatives in the material class, then contracted with $\partial F/\partial x_A$ by the element assembly described above.

9.5 Static, Implicit Dynamic, and Explicit FEM/IGA Workflows

A static IGA solve minimizes the total potential

$$\Pi(\boldsymbol{x}) = \sum_e \sum_q \Psi(\boldsymbol{F}_q) V_q - \sum_i \boldsymbol{f}_i^{ext} \cdot \boldsymbol{u}_i. \tag{250}$$

Newton iteration solves

$$\boldsymbol{K}(\boldsymbol{x}^k)\Delta \boldsymbol{x} = -\boldsymbol{r}(\boldsymbol{x}^k), \qquad \boldsymbol{x}^{k+1} = \boldsymbol{x}^k + \alpha \Delta \boldsymbol{x}. \tag{251}$$

Dirichlet control-point degrees of freedom are imposed by row-column elimination: constrained columns are subtracted from the right-hand side, constrained rows and columns are zeroed, the constrained diagonal is set to one, and the constrained right-hand side is set to the prescribed increment. For constant prescribed velocity, the implicit boundary evaluates $\Delta u_c = \bar{v}_c \Delta t$ at every attempted timestep. A reduced retry must reduce this increment as well; retaining the original increment would increase the imposed speed and generate spurious acceleration.

For implicit dynamics, the code forms an incremental Newmark potential. With source variables a_N and b_N for the Newmark parameters, the inertia energy uses

$$c_1 = \frac{1}{2\Delta t^2 a_N b_N}, \qquad c_2 = c_1 \Delta t, \qquad c_3 = \frac{1}{2b_N} - 1. \tag{252}$$

For a control-point displacement increment $\Delta \boldsymbol{u}_i$,

$$\Pi_M = \sum_i \frac{1}{2} m_i \Delta \boldsymbol{u}_i \cdot (c_1 \Delta \boldsymbol{u}_i - 2c_2 \boldsymbol{v}_i^n - 2c_3 \boldsymbol{a}_i^n) - m_i \boldsymbol{g} \cdot \Delta \boldsymbol{u}_i. \tag{253}$$

Therefore the inertia residual and tangent are

$$\boldsymbol{r}_i^M = m_i(c_1 \Delta \boldsymbol{u}_i - c_2 \boldsymbol{v}_i^n - c_3 \boldsymbol{a}_i^n - \boldsymbol{g}), \qquad K_{ii}^M = c_1 m_i \boldsymbol{I}. \tag{254}$$

After the Newton step is accepted, velocity and acceleration are reconstructed from the Newmark formulas in the solver.

The explicit FEM/IGA flow is the mass-lumped counterpart:

$$m_i \boldsymbol{a}_i = \boldsymbol{f}_i^{ext} - \boldsymbol{f}_i^{int}, \qquad \boldsymbol{v}_i^{n+1} = \boldsymbol{v}_i^n + \Delta t\, \boldsymbol{a}_i^n, \qquad \boldsymbol{x}_i^{n+1} = \boldsymbol{x}_i^n + \Delta t\, \boldsymbol{v}_i^{n+1}. \tag{255}$$

It uses the same basis, quadrature, and internal-force formula, but avoids the global tangent solve and is limited by the element wave-speed CFL condition.

9.6 Line Search and Element Inversion Guard

The implicit IGA backend computes a trial step length with material CCD before energy backtracking. For each element,

$$\det(\boldsymbol{F} + \alpha \Delta \boldsymbol{F}) = a_3 \alpha^3 + a_2 \alpha^2 + a_1 \alpha + a_0 \tag{256}$$

in 3D, and the corresponding quadratic polynomial in 2D. The smallest admissible positive root bounds α so that the element does not invert during the step. This is the same continuous-detection idea described under Contact Search; after that bound is found, the line search accepts the first step with lower total incremental energy within tolerance.

10 MPM-DEM

Related API. DEMPM.add_body, DragForceModel, coupling_scheme.

10.1 MPM–AffineBody Route Selection

The common, Direct, and LSMPM labels denote three different coupled formulations.

Route	Coupling operator	Materials and plasticity	Restrictions
Native/common MPM–AffineBody	Incompressible FDM pressure projection with volume-fraction immersed-boundary forcing	Fluid response; plasticity is not applicable	Three-dimensional, fixed timestep, no DEM subcycling, no Barrier IPC; Native solid MPM–AffineBody is unsupported
LSMPM soft–AffineBody	Monolithic soft-grid displacement and affine-control Newton system with ordinary Barrier IPC	Hyperelastic soft particles only; multiple soft materials must all be Neo-Hookean	Owned by `scheme="LSMPM"`, not by the MPM facade; no plasticity

Route	Coupling operator	Materials and plasticity	Restrictions
Direct MPM–AffineBody	Monolithic active MPM-grid displacement and affine-control Newton system with ordinary Barrier IPC	Neo-Hookean, finite-strain Drucker–Prager, von Mises, or Modified Cam–Clay	Direct implicit ULMPM, three-dimensional, lagged friction; ordinary Direct self/multibody IPC remains unsupported

10.2 Contact Coupling

Direct MPM–AffineBody selects the physical nonassociated DP force and full nonsymmetric Jacobian with Taichi BiCGSTAB when friction and dilation angles differ. Accepted history is frozen while each Newton trial recomputes its return map. This removes the material consistency outer loop, while ABD, MPM, and mixed-contact lagged friction retain their separate fixed point. Residual-norm Armijo and the common material/contact CCD limit control each update. Associated materials retain projected symmetric PCG. Plastic history is committed only after the complete coupled solve converges. Optional `inexact_newton=True` applies the same capped Krylov forcing policy as FEM–MPM with floor 10^{-7}, resetting its history in every friction outer iteration. Its free force tolerance is `absolute_tolerance` plus `residual_tolerance` times the initial force norm (clamped below by one), with defaults 10^{-10} and 10^{-8}. Force convergence, represented-motion/strain correction, prescribed-motion feasibility, and refreshed lagged-friction convergence remain mandatory. The first free-force reference persists throughout the attempted step's friction loops; terminal inexact corrections are rechecked with strict linear accuracy. The public Direct route enables Inexact Newton by default for nonassociated DP; `inexact_newton=False` selects strict linear solves. Other materials retain their existing solver defaults. The LSMPM soft-particle–AffineBody route remains hyperelastic.

In contact-based MPDEM/DEMPM, DEM particles or LSDEM surfaces interact with MPM particles through a contact force $\boldsymbol{F}_{pm}$. Momentum conservation requires equal and opposite impulses:

$$\boldsymbol{p}_{DEM}^{n+1} = \boldsymbol{p}_{DEM}^{n} + \Delta t \boldsymbol{F}_{pm}, \tag{257}$$

$$\boldsymbol{p}_{MPM}^{n+1} = \boldsymbol{p}_{MPM}^{n} - \Delta t \boldsymbol{F}_{pm}. \tag{258}$$

The neighbor search builds candidate MPM-DEM pairs. The selected contact model then evaluates the same normal/tangential laws as DEM, or the special `FluidParticle` normal law for fluid contact.

MPM `sparse_grid` changes only the MPM background-grid storage and the MPM P2G/G2P node numbering. Contact-based MPDEM/DEMPM still builds DEM or coupled contact candidates with its own spatial search buffers, so the sparse MPM background grid can be used with Lagrangian coupling when the MPM branch itself supports BlockScan. CFDEM is different: the semi-resolved fluid-particle path samples a fluid grid and is intentionally kept on its non-sparse coupling layout.

10.3 Semi-Resolved Drag

For fluid-particle coupling, let

$$\boldsymbol{u}_r = \boldsymbol{u}_f(\boldsymbol{x}_p) - \boldsymbol{v}_p$$

be relative velocity, r_p particle radius, ϵ_f local fluid volume fraction, ρ_f fluid density, and μ_f viscosity. The code computes

$$Re_p = \frac{\epsilon_f \, 2\rho_f r_p \|\boldsymbol{u}_r\|}{\mu_f}, \tag{259}$$

unless `ParticleReynold` is supplied.

For the linear law, a Gidaspow-type coefficient β is used:

$$\boldsymbol{F}_d = \frac{V_p}{1-\epsilon_f}\beta \boldsymbol{u}_r. \tag{260}$$

For $\epsilon_f \le 0.8$,

$$\beta = \frac{150(1-\epsilon_f)^2\mu_f}{\epsilon_f d_p^2} + \frac{1.75(1-\epsilon_f)\rho_f\|\boldsymbol{u}_r\|}{d_p}. \tag{261}$$

For dilute cells, the Schiller-Naumann coefficient is used inside the same Gidaspow branch.

For the quadratic law, the implemented force is

$$\boldsymbol{F}_d = \frac{1}{2}\pi C_d \rho_f r_p^2 \epsilon_f^{2-\kappa}\|\boldsymbol{u}_r\|\boldsymbol{u}_r, \tag{262}$$

where

$$\kappa = 3.7 - 0.65\exp\left[-\frac{1}{2}\left(1.5 - \ln Re_p\right)^2\right]. \tag{263}$$

The drag on DEM particles is mapped back to fluid cells with opposite sign so the coupled momentum exchange is conservative up to interpolation and boundary handling.

10.4 Incompressible Semi-Resolved DEM Sphere Coupling

Related API. DragForceModel, ParticleFluid, DEMPM.add_property.

The incompressible 3D sphere path targets dense suspensions. DEM particle volume is mapped to pressure cells with a normalized Gaussian kernel, $\alpha_s = \sum_p W_{pc}V_p / \sum_c W_{pc}V_c$, and $\epsilon_f = 1 - \alpha_s$. The pressure projection enforces

$$\frac{\epsilon_f^{n+1} - \epsilon_f^n}{\Delta t} + \nabla\cdot(\epsilon_f \boldsymbol{u}_f^{n+1}) = 0. \tag{264}$$

With $\boldsymbol{u}_f^{n+1} = \boldsymbol{u}_f^* - \Delta t \nabla p/\rho_f$, the pressure matrix uses face-averaged ϵ_f/ρ_f, and its right-hand side includes the porosity-time term. A numerical floor $\epsilon_f \ge 0.05$ keeps packed cells conditioned.

The force set resolved by this path is

$$\boldsymbol{F}_{pf} = \boldsymbol{F}_d + \boldsymbol{F}_p, \qquad \boldsymbol{F}_p = -V_p \nabla p(\boldsymbol{x}_p). \tag{265}$$

Drag uses dependent-domain fluid averages. Its exact opposite is normalized over the influence domain, then each cell impulse is shared by its available MAC faces and divided by porosity-weighted fluid mass, so

$$\sum_i \epsilon_i m_i \Delta \boldsymbol{u}_i = -\Delta t \sum_p \boldsymbol{F}_{d,p}. \tag{266}$$

Zero slip returns zero before Reynolds-number logarithms or inverse powers are evaluated. Virtual mass, lubrication, lift, history, and rotation-induced hydrodynamic forces are not included in this branch.

10.5 Fully Resolved LSDEM Volume-Fraction IBM

The 3D LSDEM route follows the volume-fraction immersed-boundary formulation of Lai, Zhao et al. (CMAME 414, 116195, 2023). Moving particles do not become cut-cell solids: fictitious incompressible MPM fluid remains inside them. For the repository's negative-inside SDF convention,

$$\phi_s = \frac{\sum_{v:\Phi_v<0}(-\Phi_v)}{\sum_v |\Phi_v|}. \tag{267}$$

The mixture density and density-weighted solid fraction are

$$\rho = (1-\phi_s)\rho_f + \phi_s \rho_s, \qquad \varphi_s = \frac{\phi_s \rho_s}{\rho}. \tag{268}$$

The rigid velocity, corrected velocity, and explicit IBM force density are

$$\boldsymbol{u}_s(\boldsymbol{x}) = \boldsymbol{v}_p + \boldsymbol{\omega}_p \times (\boldsymbol{x} - \boldsymbol{x}_p), \tag{269}$$

$$\widehat{\boldsymbol{u}} = (1-\varphi_s)\boldsymbol{u}_f + \varphi_s \boldsymbol{u}_s, \tag{270}$$

$$\boldsymbol{f}_{IB} = \rho(\widehat{\boldsymbol{u}} - \boldsymbol{u}_f)/\Delta t. \tag{271}$$

The MAC source is applied before projection. Both PCG and the MGPCG level-zero operator use face-interpolated mixed density; coarser MG levels remain a geometric preconditioner. After pressure correction, DEM force and torque are

$$\boldsymbol{F}_f = \sum_c \left[\varphi_s(-\nabla p + \mu \nabla^2 \boldsymbol{u}_f) - (1-\varphi_s)\boldsymbol{f}_{IB}\right]_c V_c, \tag{272}$$

$$\boldsymbol{M}_f = \sum_c (\boldsymbol{x}_c - \boldsymbol{x}_p) \times \left[\varphi_s(-\nabla p + \mu \nabla^2 \boldsymbol{u}_f) - (1-\varphi_s)\boldsymbol{f}_{IB}\right]_c V_c. \tag{273}$$

Multiple-body cell resultants are partitioned by SDF volume fraction. User-selected `solid_sdf_cut_cell` continues to describe fixed walls only. Force convergence must be checked by refinement; about ten cells per particle diameter is the recommended starting point, not a universal convergence guarantee.

11 IGA-MPM

Related API. IGAMPM.build, IGAMPM.run, IGAMPM.monolithic_barrier_system.

11.1 Current IGA-MPM Facade and Solver Split

The current public IGA-MPM entry point is `geotaichi.IGAMPM`, implemented by `src/igampm/mainIGAMPM.py`. It owns an IGA facade or built IGA engine, an MPM facade or built MPM engine, a contact manager, and one coupling engine. The facade selects one of two solver-compatible branches. Implicit IGA plus implicit Direct MPM selects IPC. Explicit IGA plus native explicit MPM selects the DEM kwLinear or kwHertzMindlin point–NURBS law. Mixed solver pairs are rejected. The implicit contact system has the abstract block form

$$\begin{bmatrix} \boldsymbol{K}_{II} & 0 \\ 0 & \boldsymbol{K}_{MM} \end{bmatrix} + \begin{bmatrix} \boldsymbol{K}^c_{II} & \boldsymbol{K}^c_{IM} \\ \boldsymbol{K}^c_{MI} & \boldsymbol{K}^c_{MM} \end{bmatrix} \begin{bmatrix} \Delta \boldsymbol{q}_I \\ \Delta \boldsymbol{u}_M \end{bmatrix} = - \begin{bmatrix} \boldsymbol{R}_I + \boldsymbol{g}^c_I \\ \boldsymbol{R}_M + \boldsymbol{g}^c_M \end{bmatrix}. \tag{274}$$

$\boldsymbol{K}_{II}$ and $\boldsymbol{K}_{MM}$ come from the two sub-solvers. The contact blocks are generated from closest-point barrier and friction energies and are documented in IGA-MPM IPC Contact Model. The global tangent is assembled in user-selected device COO or HashTriplet storage.

The MPM block may use Direct implicit ULMPM with Neo-Hookean elasticity or the physical nonassociated Drucker–Prager response or associated Drucker–Prager/von-Mises incremental potentials described in Finite-Strain Associated Plasticity. At Newton iterate k, its elastic trial map is $\boldsymbol{F}_{e,\mathrm{tr}} = (\boldsymbol{I} + \nabla \boldsymbol{u}_k)\boldsymbol{F}_{e,n}$; the returned PK1 stress and analytic algorithmic tangent enter $\boldsymbol{R}_M$ and $\boldsymbol{K}_{MM}$ through the same device assembly as elasticity. Contact does not change this constitutive update: point–NURBS ACCD, IGA and MPM material CCD, and Armijo search still bound every trial step. Nonassociated DP uses its complete physical Jacobian, including the trial plastic-volume derivative, and BiCGSTAB with residual Armijo; it requires no material-consistency outer iteration. Associated DP retains the existing frozen-volume consistency loop and PCG. The IGA material remains elastic, and Direct TLMPM plasticity remains unsupported because its fixed reference basis needs explicit plastic-gradient storage. Plastic history is updated only by the accepted ULMPM advection kernel. IGA–MPM step preparation, nonlinear solution, and acceptance share one device-state transaction. The entry IGA control-point state, Direct-MPM particle/grid state, $\boldsymbol{F}_{e,n}$, plastic history, and displacement fields are restored if preparation, solution, or post-commit refresh fails.

Cartesian elastic IGA–MPM uses the ordinary two- or three-dimensional maps. Cartesian 2D DP/von-Mises MPM uses the three-dimensional plane-strain map defined in Finite-Strain Associated Plasticity. In the axisymmetric two-dimensional mode, both the IGA and Direct MPM material blocks use the no-swirl three-dimensional map

$$\boldsymbol{F}_{\mathrm{axi}} = \begin{bmatrix} F_{rR} & F_{rZ} & 0 \\ F_{zR} & F_{zZ} & 0 \\ 0 & 0 & r/R \end{bmatrix}, \tag{275}$$

and every reference material quadrature weight is multiplied by $2\pi R$. The NURBS contact closest-point problem, its implicit derivatives, ACCD, friction, and Armijo trials remain in the (r, z) meridian. A boundary MPM sample carries a fixed revolved reference measure $2\pi R w_p$. Both children and the coupling must use the same `axis_offset`; material/contact quadrature on the axis itself is rejected.

11.2 Explicit DEM-Law IGA–MPM Contact

The explicit branch treats a coupled MPM material point as a sphere with center $\boldsymbol{x}_p$, radius r_p, mass m_p, and velocity $\boldsymbol{v}_p$. For the closest point on an IGA surface,

$$\boldsymbol{x}_s(u,v) = \sum_A R_A(u,v)\boldsymbol{P}_A, \qquad g_n = \|\boldsymbol{x}_p - \boldsymbol{x}_s\| - r_p, \qquad \boldsymbol{n} = \frac{\boldsymbol{x}_p - \boldsymbol{x}_s}{\|\boldsymbol{x}_p - \boldsymbol{x}_s\|}. \tag{276}$$

A contact is active when $g_n < 0$. Positive NURBS weights place the surface inside its control hull, so a current control-point AABB expanded by r_p is a conservative cull before the all-span projected Gauss–Newton closest point solve.

The contact-point velocity and reaction distribution use the same rational basis,

$$\boldsymbol{v}_s = \sum_A R_A \boldsymbol{v}_A, \qquad \boldsymbol{f}_p = \boldsymbol{f}_n + \boldsymbol{f}_t, \qquad \boldsymbol{f}_A = -R_A \boldsymbol{f}_p. \tag{277}$$

Consequently $\boldsymbol{f}_p + \sum_A \boldsymbol{f}_A = \boldsymbol{0}$; because $\boldsymbol{x}_s = \sum_A R_A \boldsymbol{P}_A$, the normal-force moment also balances: $(\boldsymbol{x}_p - \boldsymbol{x}_s) \times \boldsymbol{f}_n = \boldsymbol{0}$. A finite-radius tangential force generally leaves the couple $(\boldsymbol{x}_p - \boldsymbol{x}_s) \times \boldsymbol{f}_t$, because neither the MPM point nor the IGA control point owns an independent contact-spin DOF. This is the same point-force convention as explicit MPM–DEM coupling and is not a claim of exact angular-momentum conservation under friction. The Linear law is the repository DEM spring–dashpot with static/dynamic Coulomb friction and persistent rotated tangential overlap. The Hertz–Mindlin law evaluates its penetration-dependent normal and shear stiffnesses through the same shared DEM constitutive function. IGA control points have no independent spin, so nonzero rolling friction is not admissible.

Explicit contact is currently three-dimensional Cartesian. It has no IPC barrier, CCD, ACCD, or line search. The coupled CFL gate applies the native MPM factor to the MPM material-wave and contact-law time scales; because the IGA backend has no independent spectral estimator, users must also verify IGA stability. Every explicit step resets subsystem forces, assembles IGA body/internal loads, places equal-and-opposite contact forces, advances MPM through P2G/grid/G2P, and advances IGA control points. The finite-radius sphere–NURBS geometry is intentionally different from the zero-radius sample plus `dmin` used by implicit IPC.

The static two-phase IGA-MPM idea can still be written abstractly as

$$\begin{bmatrix} \boldsymbol{K}_{ss} & \boldsymbol{K}_{si} \\ \boldsymbol{K}_{is} & \boldsymbol{K}_{ii} \end{bmatrix} \begin{bmatrix} \Delta \boldsymbol{u}_s \\ \Delta \boldsymbol{q}_i \end{bmatrix} = - \begin{bmatrix} \boldsymbol{R}_s \\ \boldsymbol{R}_i \end{bmatrix}, \tag{278}$$

where s denotes two-phase soil/MPM unknowns and i denotes IGA unknowns. Rigid-body adapters and old helper facades are not part of the current public `IGAMPM` workflow.

11.3 Static Two-Phase IGA-MPM Coupling

Related API. `IGAMPM.build`, `MPM.set_semi_implicit_solver_parameters`.

Static two-phase IGA-MPM uses MPM particles as quadrature points and IGA or background-grid functions as trial/test functions. With unknown vector $\boldsymbol{q}_I$ =

$[u_x, u_y, u_z, p]^T$, the effective stress residual is

$$R_{u,I} = \sum_p \left[-\nabla N_I : (\boldsymbol{\sigma}'_p - p_p \boldsymbol{I}) + N_I \rho_{mix,p} \boldsymbol{g} \right] V_p, \tag{279}$$

and the pressure residual is

$$R_{p,I} = \sum_p \left[N_I (\ln J_p^{n+1} - \ln J_p^n) + \Delta t \, k_p \nabla N_I \cdot \nabla p_p \right] V_p. \tag{280}$$

Differentiating gives K_{uu}, K_{up}, K_{pu}, and K_{pp}. The off-diagonal blocks express that pressure changes total stress and deformation changes porosity/storage. IGA-MPM IPC contact is documented in Contact Model: IGA-MPM IPC.

11.4 IGA-MPM IPC Contact CCD/ACCD

Related API. IGAMPM.monolithic_barrier_system, dhat, barrier_nnz, friction_nnz.

Finite positive NURBS weights place each boundary inside its control-hull axis-aligned bounding box. Let $\underline{d}$ be the point–box distance, rounded conservatively downward. Then $\underline{d} \le d$. If $\underline{d} \ge \hat{d}$, the pair has zero IPC barrier energy and derivatives, so closest-point projection can be skipped. Every pair remains in the ACCD table; inactive distance entries may be lower bounds. Positive SemiIPC multipliers always require the full projection. For relative-motion bound L, clearance d_{min}, and safety factor $0 < s < 1$, a trial segment is certified by

$$\alpha L \le s(\underline{d} - d_{min}) \quad \Longrightarrow \quad d(t) - d_{min} \ge (1-s)(\underline{d} - d_{min}) > 0, \qquad 0 \le t \le \alpha. \tag{281}$$

Otherwise the bounded ACCD loop performs the moving closest-point queries. Bounds are rebuilt from current control points at every contact query. For a three-dimensional surface, each knot-span control-hull box is computed once for that geometry and shared by all particle queries. Fixed-knot Greville coordinates are precomputed. A balanced span BVH is built from the reference control-hull centers, then refitted from the current leaf hulls after deformation. Stackless traversal skips entire subtrees only when their conservative lower bound cannot improve the current upper bound. The same hierarchy locates the exact nearest control point, retaining the smallest control index on distance ties. A previous closest parameter supplies an additional projected-Newton seed evaluated on the current surface. Its distance is an upper bound; a span is skipped only when its conservative lower bound cannot improve that bound, with the existing rounding guard. All potentially improving span seeds, boundary minima, and the nearest-control-point seed remain in the search. These hints require no physical-history rollback. Moving ACCD queries compute moving hulls and never substitute a stale stationary box. Before full projection, traversal may certify that every span lies outside the maximum of the barrier activation and strict-feasibility thresholds. The minimum lower bound over the pruned subtrees is then a conservative whole-surface distance bound. A potentially near leaf or positive SemiIPC multiplier retains full projection. An accepted Armijo trial retains its matching contact table and hulls while synchronizing trial vectors. Built-in Newton assembly reuses this query, and Armijo reuses the base query from assembly. Rejected or failed searches instead restore and requery the accepted geometry. External callbacks and SemiIPC multiplier changes retain explicit contact preparation.

CCD first constructs componentwise bounds enclosing both $\boldsymbol{x}_i$ and $\boldsymbol{x}_i + \alpha_{\max}\Delta\boldsymbol{x}_i$. Each particle's swept box queries a refitted surface BVH and, in 3D, the surface's swept knot-span BVH. Fixed positive NURBS weights make these control-hull boxes conservative over the entire linear trajectory. Clearance and a rounding guard expand the overlap test. Only overlapping candidates enter the motion-bound certificate and moving closest-point ACCD. The narrow phase remains necessary: AABB intersection alone does not determine time of impact. One device launch advances one ACCD iteration on compact candidates; the host reads the unfinished count between launches to avoid excessive Taichi compilation of nested closest-point loops.

Particle search directions are interpolated once per sample. If the surface control directions have componentwise bounds $\boldsymbol{v}^- \le \Delta\boldsymbol{P}_i \le \boldsymbol{v}^+$, define

$$L_{\text{box}} = \left\|\max\left(|\Delta\boldsymbol{x}_p - \boldsymbol{v}^-|, |\Delta\boldsymbol{x}_p - \boldsymbol{v}^+|\right)\right\|_2 \ge \max_i \|\Delta\boldsymbol{x}_p - \Delta\boldsymbol{P}_i\|_2. \tag{282}$$

Absolute values and the maximum are componentwise, and the implementation rounds the upper bound upward. Only whole segments certified by this cheap bound skip the control scan; other pairs retain the tighter original bound and ACCD progression. Lagged friction evaluates its normal at the closest parameters already found by the barrier query.

For lagged friction, residual-only assembly evaluates the same frozen-frame force as full assembly in both 2D and 3D. A static kernel flag suppresses Hessian evaluation and block scatter without resetting existing sparse values. Newton force convergence checks and residual line search use this path; full assembly retains the symmetric frozen-friction Hessian.

IGA-MPM IPC contact couples MPM surface particles to NURBS curves in 2D or NURBS surfaces in 3D. The current assembly updates the IGA control-point snapshot, maps the MPM grid displacement to temporary particle positions, and then finds the closest NURBS point. The MPM boundary sample is a point primitive; unlike the earlier research sphere–NURBS prototype, its storage radius is not subtracted from the gap. Finite clearance is represented consistently by IPC d_{min}:

$$\boldsymbol{x}_p(\alpha) = \boldsymbol{x}_p^n + \alpha\sum_i N_i(\boldsymbol{x}_p^n)\Delta\boldsymbol{u}_i, \qquad \boldsymbol{x}_s(\xi,\eta;\alpha) = \sum_A R_A(\xi,\eta)\left(\boldsymbol{P}_A^n + \alpha\Delta\boldsymbol{P}_A\right). \tag{283}$$

For a surface contact, the closest point satisfies

$$\boldsymbol{r}\cdot\boldsymbol{x}_{s,\xi} = 0, \qquad \boldsymbol{r}\cdot\boldsymbol{x}_{s,\eta} = 0, \qquad \boldsymbol{r} = \boldsymbol{x}_s - \boldsymbol{x}_p, \tag{284}$$

and for a curve only the ξ equation is solved. The active barrier set is then

$$d_{ps} = \|\boldsymbol{r}\| < \hat{d}. \tag{285}$$

This closest-point search is a DCD operation at the current trial state; it selects the contacts that will contribute barrier and friction blocks.

The CCD/ACCD step bound for IGA-MPM must be applied before accepting a Newton increment, otherwise a particle can pass through a moving NURBS surface between two DCD evaluations. For a fixed closest-point parameter over a short ACCD advance, the relative displacement is

$$\Delta\boldsymbol{r} = \sum_A R_A(\xi,\eta)\Delta\boldsymbol{P}_A - \sum_i N_i(\boldsymbol{x}_p)\Delta\boldsymbol{u}_i. \tag{286}$$

For a curved positive-weight NURBS surface the closest parameter changes with α, so one fixed-parameter point–point root is not a global CCD solution. The positive-weight convex-hull property instead gives

$$L = \max_A \left\| \sum_i N_i \Delta \boldsymbol{u}_i - \Delta \boldsymbol{P}_A \right\|, \tag{287}$$

because the relative motion to any NURBS point is a convex combination of these point–control-point relative directions. This bound includes motion on both sides and vanishes under a common translation. At ACCD iterate (k), the implemented increment is

$$\Delta\alpha_k = s\frac{d(\alpha_k) - d_{safe}}{L}, \qquad 0 < s < 1. \tag{288}$$

The target retained excess is

$$(1 - s)\,(d(0) - d_{safe})\,. \tag{289}$$

Each Taichi thread advances one ACCD iteration for an active MPM surface-sample/NURBS-surface pair. It evaluates $\boldsymbol{x}_p^0 + \alpha\Delta\boldsymbol{x}_p$ and $\boldsymbol{P}_A^0 + \alpha\Delta\boldsymbol{P}_A$ as virtual trial geometry, without mutating the shared control-point field, and rebuilds the closest-point DCD query after every increment. At each projected-Newton state, the homogeneous numerator and weight denominator and their first and second derivatives are accumulated over the local support, then converted by the quotient rule. This avoids support-sized rational derivative matrices inside the nested search. Projected Newton and backtracking share one state-evaluation site: accepted trial derivatives become the next Newton state. Rejected trials retain the base state and shrink the same direction; tolerances, limits and the projected-gradient fallback remain unchanged. Curves use the same single-site structure and dispatch span-midpoint then Greville seeds through one solver call, preserving their order and parameter bounds. Per-point seed searches are explicitly serial; contact kernels parallelize across particles. Each thread stores its pair TOC. After all bounded iterations, a separate device atomic-min reduction uses the final TOCs. A trial that crosses the retained gap or strict clearance is queried but not accepted under the retained-gap rule. Boundaries with the same NURBS degree signature share an immutable basis object, allowing closest-point, assembly, and ACCD kernels to reuse their degree specialization. Python dispatches these basis groups and reads a scalar active count between bounded iterations; all pair state and geometry stay on the device. This avoids excessive Taichi 1.7 CFG optimization of the nested moving closest-point search. This global bound can be more conservative than a local velocity estimate interpolated at one closest parameter.

The standalone contact-step entry point prepares the base DCD set and, by default, verifies the proposed trial before restoring that base state. During the production Armijo sequence the caller has already prepared the base DCD set and will evaluate the candidate trial immediately, so the internal ACCD path reuses that prepared state and omits the duplicate verification/restore query without changing the conservative step bound.

When applied in the monolithic IGA-MPM solve, the safe sequence is:

1. build or refresh DCD closest-point candidates with the current trial state;
2. compute $\alpha_{contact}$ from the device point-curve or point-surface positive-weight NURBS ACCD bound;
3. compute material CCD for the IGA and MPM subdomains if those increments may invert elements or particles;

4. start the energy line search from $\alpha_0 = \min(\alpha_{contact}, \alpha_{material}, 1)$;
5. assemble barrier/friction blocks only for the accepted trial state.

The existing `barrier_nnz` and `friction_nnz` capacities must be sized for the accepted DCD set after ACCD, not for a stale pre-line-search set.

12 Contact Search

Related API. DEM.choose_neighbor, BVH, bvh_rebuild_interval.

12.1 Broad-Phase Requirements

Contact search is separated from contact physics. The broad phase must be conservative: it may return false positives, but it must not discard a true overlap or a true IPC candidate. Every primitive receives an enlarged axis-aligned bounding box using the Verlet skin:

$$\mathrm{AABB}^{skin} = [x_{min} - s, x_{max} + s]. \tag{290}$$

The narrow phase then evaluates exact sphere-sphere, sphere-wall, point-point, point-ground, or IGA closest-point distances.

12.2 DCD and CCD Classification

GeoTaichi uses both discrete collision detection (DCD) and continuous collision detection (CCD). DCD classifies contacts at the current configuration:

$$d^n = d(\boldsymbol{x}^n), \qquad \text{active if } d^n < d_{act}. \tag{291}$$

DEM overlap, Verlet-list update, LSDEM surface tests, IPC barrier candidate creation, and closest-point contact initialization are DCD operations. DCD is not allowed to miss a candidate; exact contact models remove false positives.

CCD bounds a trial displacement before the configuration crosses a contact or inversion barrier. A primitive moves along

$$\boldsymbol{x}(\alpha) = \boldsymbol{x}^n + \alpha \Delta \boldsymbol{x}, \qquad 0 \le \alpha \le 1. \tag{292}$$

The CCD task is

$$\alpha_{ccd} = \min\{\alpha > 0 : \ d(\boldsymbol{x}(\alpha)) = d_{safe}\}. \tag{293}$$

For a point-plane pair,

$$d(\alpha) = \boldsymbol{n} \cdot (\boldsymbol{x}^n + \alpha \Delta \boldsymbol{x}) + b, \qquad \alpha = \frac{d_{safe} - d(0)}{\boldsymbol{n} \cdot \Delta \boldsymbol{x}}. \tag{294}$$

For point-point IPC contact, the squared distance is a quadratic polynomial in α:

$$\|\boldsymbol{r}^n + \alpha \Delta \boldsymbol{r}\|^2 = d_{safe}^2. \tag{295}$$

For IGA/FEM material inversion guard, the determinant is a polynomial:

$$\det(\boldsymbol{F} + \alpha\Delta\boldsymbol{F}) = 0 \tag{296}$$

or a slackened positive determinant threshold. The smallest admissible root is used to cap the step.

For moving triangle and segment primitives, exact CCD can be written as root-finding on a polynomial contact condition. For vertex-face (VF) CCD, with vertex $\boldsymbol{p}$ and triangle $(\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c})$,

$$f_{vf}(\alpha) = (\boldsymbol{p}(\alpha) - \boldsymbol{a}(\alpha)) \cdot [(\boldsymbol{b}(\alpha) - \boldsymbol{a}(\alpha)) \times (\boldsymbol{c}(\alpha) - \boldsymbol{a}(\alpha))] = 0. \tag{297}$$

The impact is valid only if the barycentric coordinates of $\boldsymbol{p}(\alpha)$ on the triangle are inside the triangle. For edge-edge (EE) CCD, with edges $(\boldsymbol{a}, \boldsymbol{b})$ and $(\boldsymbol{c}, \boldsymbol{d})$,

$$f_{ee}(\alpha) = (\boldsymbol{c}(\alpha) - \boldsymbol{a}(\alpha)) \cdot [(\boldsymbol{b}(\alpha) - \boldsymbol{a}(\alpha)) \times (\boldsymbol{d}(\alpha) - \boldsymbol{c}(\alpha))] = 0, \tag{298}$$

and the line parameters on both edges must lie in $[0, 1]$. These exact tests are appropriate for mesh/rigid geometry, but they are often more expensive than needed for MPM and IGA-MPM because the contact primitives are driven by grid or control-point increments.

In the shared production interface, ordinary PP/PT/EE CCD uses zero thickness: PP solves the quadratic, while PT and EE enumerate all real roots of the cubic coplanarity polynomial and validate the finite primitive at each root. An identically zero coplanarity polynomial uses a zero-clearance distance-based fallback. Here "analytic" describes algebraic candidate generation; it is not the rounding-error certificate provided by interval methods such as Tight Inclusion. Finite clearance uses the ACCD path below.

12.3 Additive Conservative Advancement CCD

ACCD, or additive conservative advancement CCD, avoids solving exact VF/EE or point-NURBS time-of-impact polynomials. It advances by a safe amount computed from the current distance and an upper bound on how fast that distance can decrease. For a primitive pair with distance $d(\boldsymbol{x})$, safe threshold d_{safe}, and vertex/control-point set $\mathcal{P}$, the distance speed is bounded by

$$-\frac{\mathrm{d}d}{\mathrm{d}\alpha} \le \left|\frac{\mathrm{d}d}{\mathrm{d}\alpha}\right| = \left|\sum_{m\in\mathcal{P}} \frac{\partial d}{\partial \boldsymbol{x}_m} \cdot \Delta\boldsymbol{x}_m\right| \le \sum_{m\in\mathcal{P}} \left\|\frac{\partial d}{\partial \boldsymbol{x}_m}\right\| \|\Delta\boldsymbol{x}_m\| = L. \tag{299}$$

If $d(\alpha_k) > d_{safe}$, the additive conservative step is

$$\Delta\alpha_k = s\frac{d(\alpha_k) - d_{safe}}{L_k + \epsilon}, \qquad \alpha_{k+1} = \min(1, \alpha_k + \Delta\alpha_k), \tag{300}$$

with conservative rescaling $0 < s < 1$. Define the retained initial excess

$$g = (1 - s)(d(0) - d_{safe}). \tag{301}$$

After the first accepted increment, the loop queries each next trial and stops at the preceding accumulated step if the trial excess would fall below g. It also stops when

$$\alpha_k \ge 1, \qquad \Delta\alpha_k \le \varepsilon_\alpha, \tag{302}$$

or when the iteration limit is reached. If the final $\alpha < 1$, the solver uses this value as the CCD cap for the Newton or explicit trial step.

The conservative bound L must include every moving primitive dof. For MPM, the moving point velocity is the shape-function interpolation of grid increments. For IGA-MPM, L includes both the MPM particle interpolation and the NURBS control-point increments. For rigid or DEM meshes, L includes vertex displacements induced by rigid translation and rotation.

The reusable device functions live in `src/contact_detection/continuous_contact_detection`. `CCD.py` provides ordinary zero-thickness point–point, point–triangle, edge–edge, and linear-gap CCD. `AdditiveCCD.py` provides their finite-clearance ACCD counterparts plus the point–NURBS bound. `PolynomialCCD.py` owns the point–point quadratic root and deformation-gradient determinant guard. The module rules are:

1. DCD/BVH builds a superset of candidate primitive pairs.
2. CCD or ACCD computes a trial fraction α for every candidate that is closing.
3. The global step cap is the minimum over contact CCD, material CCD, and solver-specific CFL or trust-region caps.
4. Contact forces/barrier Hessians are assembled only after the accepted trial configuration is chosen.

In the direct IPC MPM path, the accepted trial length is

$$\alpha_0 = \min(\alpha_{ground}, \alpha_{particle}, \alpha_{material}, \alpha_{CFL}), \tag{303}$$

where α_{ground} and $\alpha_{particle}$ come from contact CCD, $\alpha_{material}$ comes from the material determinant CCD, and α_{CFL} limits particle displacement relative to the barrier distance $\hat{d}$. Energy backtracking then operates on this CCD-limited step; the barrier force and friction formulas themselves belong to Contact Model.

12.4 Cell List Search

For near-uniform particle sizes, a cell list maps each AABB to overlapped bins. The search complexity is approximately

$$O(N + N\bar{n}_c), \tag{304}$$

where $\bar{n}_c$ is the average number of candidates in neighboring bins. This is efficient when the bin width is comparable to the largest interaction range. Strong size polydispersity or highly clustered particles increase $\bar{n}_c$ and favor a BVH.

12.5 BVH and LBVH

BVH/LBVH is the contact-search module used by large DEM and coupled contact problems. AABB centers are normalized to the simulation domain, encoded into Morton keys, radix sorted, and connected into a linear BVH from longest common prefixes. Leaf bounds are primitive AABBs; internal bounds are unions:

$$\mathrm{AABB}_I = \mathrm{AABB}_{left(I)} \cup \mathrm{AABB}_{right(I)}. \tag{305}$$

A rebuild recomputes Morton keys and topology; a refit only propagates bounds up the existing tree. During traversal, a node is rejected when its AABB does not overlap the query AABB. The search is conservative, so exact distance or wall tests remove false positives after broad-phase traversal.

For mixed DEM particles and patch/facet walls, the LBVH stores separate AABB batches and builds the prefix array from the current active counts, not from the maximum preallocated capacities. Particle and wall AABBs are therefore written and read through the same active prefix. When walls are marked static, the initial full build includes both particle and wall batches; later updates can rebuild or refit only the particle prefix while preserving the wall batch. Triangle and patch AABBs are enlarged by the Verlet skin and clamped to a small positive extent in every coordinate so that degenerate or nearly planar geometry does not create zero-width boxes.

DigitalElevation walls are not the same as patch/facet BVH walls. They are inserted through the elevation-grid wall path and use the DigitalElevation search data for particle-wall candidates, while particle-particle search may still use BVH when selected.

12.6 Implementation Selection and Narrow-Phase Geometry

The search method should be chosen from geometry distribution rather than from the contact law. `LinkedCell` is best when radii and interaction distances are nearly uniform; each particle visits only neighboring bins. The bin size must be at least the largest interaction range plus Verlet skin:

$$h_{cell} \geq 2R_{max} + s_{verlet}. \tag{306}$$

`HierarchicalLinkedCell` reduces the cost for polydisperse radii by placing different size classes on different grids. A candidate from a coarse level is accepted by the narrow phase only if the exact distance or overlap test passes. `BVH` is preferred for highly nonuniform particle sizes, clumps, level-set bodies, mesh facets, or mixed DEM-MPM scenes because the tree adapts to AABB size rather than a single bin width.

The narrow phase depends on primitive type:

Primitive pair	Narrow-phase quantity
Sphere-sphere	gap $g = \|\boldsymbol{x}_i - \boldsymbol{x}_j\| - R_i - R_j$ and normal $(\boldsymbol{x}_i - \boldsymbol{x}_j)/\|\boldsymbol{x}_i - \boldsymbol{x}_j\|$.
Sphere-wall or point-plane	signed plane distance $g = \boldsymbol{n} \cdot \boldsymbol{x} + b - R$.
Facet or mesh body	point-triangle, edge-edge, or point-plane distance after AABB overlap; exact barycentric tests remove false positives.
Patch wall	point-triangle or shell/patch facet tests after the patch AABB has been expanded by the Verlet skin and protected against zero extent.
DigitalElevation wall	elevation-grid cell lookup followed by the local facet/height narrow phase; this path is separate from patch/facet BVH wall batches.

Primitive pair	Narrow-phase quantity
Level-set body	signed distance $\phi(\boldsymbol{x})$, normal $\nabla\phi/\|\nabla\phi\|$, and optional grid interpolation from `LocalGrid`.
LSMPM soft body	Directed surface-node test against the slave SDF. Rigid and soft masters use level-set surface nodes; soft nodes are advected by the compact TLMPM grid. Any candidate with $g = \phi < 0$ for zero-cutoff models is passed to the same LS contact law; soft-side forces are stored on the surface node and transferred by its TLMPM shape functions.
NURBS curve or surface	closest-point inversion in parameter space followed by distance and derivative evaluation.
IPC particle-particle	point-point distance and the squared-distance Hessian used by the barrier energy.

For `scheme="LSMPM"`, broad body pairs are generated by the selected DEM neighbor class and then expanded into a contact-node table. The table is rebuilt from the current rigid and soft level-set surface nodes. If either body in a body pair is soft, the narrow phase evaluates both directed tests $i \to j$ and $j \to i$. This is necessary because a deforming surface node can penetrate a rigid or soft slave even when the opposite surface-node set does not penetrate the soft SDF. Pure rigid LSDEM pairs remain single-directional to avoid duplicating the same surface-node contact. The user-visible `point_coordination_number` controls the maximum number of slave bodies recorded for each LS contact node; undersizing triggers an assertion instead of silently dropping contacts.

The Verlet skin amortizes rebuild cost. If the maximum accumulated primitive displacement since the last rebuild is u_{max}, the list remains valid only while

$$2u_{max} < s_{verlet}. \tag{307}$$

Rigid rotation must be included through a bound on swept vertex displacement, not only center translation. For LSDEM and NURBS bodies, the search AABB must cover the deformed or moved surface plus the contact activation distance; otherwise the closest-point or level-set narrow phase cannot recover a missed broad-phase candidate.

13 Contact Model

Related API. DEM.choose_contact_model, kappa, dhat, friction, epsv.

13.1 DEM Normal and Tangential Contact

For two spheres, the normal gap is

$$g_n = \|\boldsymbol{x}_j - \boldsymbol{x}_i\| - (R_i + R_j), \tag{308}$$

and contact is active when $g_n < 0$. The penalty potential

$$\Pi_n = \frac{1}{2} k_n \langle -g_n \rangle^2 \tag{309}$$

gives

$$\boldsymbol{F}_n = k_n \delta_n \boldsymbol{n}, \qquad \delta_n = -g_n. \tag{310}$$

Normal damping is

$$\boldsymbol{F}_n^{damp} = -c_n [(\boldsymbol{v}_j - \boldsymbol{v}_i) \cdot \boldsymbol{n}] \boldsymbol{n}. \tag{311}$$

For restitution coefficient e,

$$\eta = -\frac{\ln e}{\sqrt{\pi^2 + (\ln e)^2}}, \qquad c_n = 2\eta \sqrt{k_n m^*}. \tag{312}$$

The tangential history is projected to the current contact plane:

$$\boldsymbol{\xi}_t^{n+1} = (\boldsymbol{I} - \boldsymbol{n} \otimes \boldsymbol{n})(\boldsymbol{\xi}_t^n + \Delta t \, \boldsymbol{v}_t). \tag{313}$$

The trial tangential force is returned to the Coulomb disk:

$$\boldsymbol{F}_t = \begin{cases} \boldsymbol{F}_t^{trial}, & \|\boldsymbol{F}_t^{trial}\| \le \mu \|\boldsymbol{F}_n\|, \\ \mu \|\boldsymbol{F}_n\| \dfrac{\boldsymbol{F}_t^{trial}}{\|\boldsymbol{F}_t^{trial}\|}, & \text{sliding.} \end{cases} \tag{314}$$

Rolling and twisting resistance apply the same elastic predictor and frictional return-mapping idea to rotational gaps.

13.2 Sphere-Triangle Area Fraction for Facet Walls

For a spherical particle touching a triangular wall facet, GeoTaichi does not assign the whole wall force to the facet unless the particle contact disk is fully covered by that triangle. Let $\boldsymbol{x}$ be the sphere center, R the sphere radius, $\boldsymbol{n}_f$ the oriented facet normal, and

$$d = (\boldsymbol{x} - \boldsymbol{a}_1) \cdot \boldsymbol{n}_f \tag{315}$$

the signed distance from the sphere center to the facet plane. If $0 < d < R$, the sphere-plane intersection is a disk with center

$$\boldsymbol{c} = \boldsymbol{x} - d\boldsymbol{n}_f \tag{316}$$

and radius

$$\rho = \sqrt{R^2 - d^2}, \qquad A_0 = \pi \rho^2. \tag{317}$$

The facet contribution is the area of the intersection between this disk and the triangle,

$$\chi_f = \frac{|A(\mathcal{T}_f \cap \mathcal{D}(\boldsymbol{c}, \rho))|}{A_0}, \tag{318}$$

with $\chi_f = 0$ outside the band $0 < d < R$. This is the quantity computed by `processCircleShape` for facet and patch walls. Infinite plane walls return the full fraction.

The triangle-disk area is evaluated as the signed sum of the three edge contributions. For an oriented edge from $\boldsymbol{a}$ to $\boldsymbol{b}$, define

$$L = \|\boldsymbol{b} - \boldsymbol{a}\|, \quad \boldsymbol{t} = (\boldsymbol{b} - \boldsymbol{a})/L, \quad l = (\boldsymbol{a} - \boldsymbol{c}) \cdot \boldsymbol{t}, \quad r = l + L,$$

and

$$y = \frac{[(\boldsymbol{b}-\boldsymbol{a})\times(\boldsymbol{a}-\boldsymbol{c})]\cdot\boldsymbol{n}_f}{L}. \tag{319}$$

The non-intersecting interval contribution over $[u, v]$ is the circular sector

$$S(u, v; y) = \frac{1}{2}\rho^2 \operatorname{wrap}_{[-\pi,\pi]}\left[\operatorname{atan2}(y, v) - \operatorname{atan2}(y, u)\right]. \tag{320}$$

If $|y| > \rho$, the whole edge contributes $S(l, r; y)$. Otherwise let $s = \sqrt{\rho^2 - y^2}$. The left outside interval $[l, \min(-s, r)]$ and right outside interval $[\max(s, l), r]$ contribute their sector terms, and the middle chord interval $[\max(-s, l), \min(s, r)]$ contributes

$$-\frac{1}{2}\left[\min(s, r) - \max(-s, l)\right]_+ y. \tag{321}$$

Summing these three edge areas gives the signed intersection area; the contact kernel uses its absolute fraction for activation and scaling.

For particle-wall contact lists, the normal gap test still uses the DEM contact law gap, but a facet contact is active only when

$$g_n < n_{cut} \quad \text{and} \quad \chi_f > \text{Threshold}. \tag{322}$$

The fraction scales the stored contact force and the equivalent stiffness used by the contact model, so a particle intersecting only a small part of a finite facet contributes only the covered area fraction. This prevents a small corner overlap with one triangle from being treated as a full infinite-plane contact.

13.3 Hertz-Mindlin Contact

Hertz contact replaces the constant normal stiffness by

$$F_n = \frac{4}{3}E^*\sqrt{R^*}\delta_n^{3/2}, \qquad k_n = 2E^*\sqrt{R^*\delta_n}. \tag{323}$$

Mindlin tangential stiffness is proportional to the contact radius:

$$k_t = 8G^*\sqrt{R^*\delta_n}. \tag{324}$$

The timestep estimate must use these tangent stiffnesses rather than the force secant.

13.4 IPC Barrier and Friction

Incremental potential contact uses an energy barrier for distances below $\hat{d}$:

$$B(d) = -\kappa(d - \hat{d})^2 \ln\frac{d}{\hat{d}}, \qquad 0 < d < \hat{d}. \tag{325}$$

Its first and second derivatives are

$$B'(d) = \kappa\left[-2(d - \hat{d})\ln\frac{d}{\hat{d}} - \frac{(d - \hat{d})^2}{d}\right], \tag{326}$$

$$B''(d) = \kappa\left[-2\ln\frac{d}{\hat{d}} - 4 + 4\frac{\hat{d}}{d} + \frac{(d - \hat{d})^2}{d^2}\right]. \tag{327}$$

When a kernel differentiates d^2, with $\boldsymbol{g} = \nabla d^2$ and $\boldsymbol{H}_{d^2} = \nabla^2 d^2$,

$$\nabla B = \frac{1}{2} B'(d) d^{-1} \boldsymbol{g}, \qquad \nabla^2 B = \frac{1}{4} \left(B'' d^{-2} - B' d^{-3} \right) \boldsymbol{g}\boldsymbol{g}^T + \frac{1}{2} B' d^{-1} \boldsymbol{H}_{d^2}. \tag{328}$$

Friction uses $\lambda_n = -B'(d)$, $\lambda_f = \mu \lambda_n$, a tangent projection $T = I - \boldsymbol{n} \otimes \boldsymbol{n}$, and a regularized slip norm. The Hessian is projected to positive semidefinite form before insertion.

13.5 IGA-MPM IPC Contact Model

Related API. `find_closest_surface_contacts`, `find_closest_curve_contacts`, `assemble_barrier_matrix_for_surface`, `assemble_barrier_matrix_for_curve`, and `monolithic_barrier_system`.

For an IGA surface,

$$\boldsymbol{x}_s(u, v) = \sum_A R_A(u, v) \boldsymbol{P}_A, \qquad R_A = \frac{N_A w_A}{\sum_B N_B w_B}. \tag{329}$$

The closest point to an MPM surface particle $\boldsymbol{x}_p$ satisfies

$$\boldsymbol{r} \cdot \boldsymbol{x}_{s,u} = 0, \qquad \boldsymbol{r} \cdot \boldsymbol{x}_{s,v} = 0, \qquad \boldsymbol{r} = \boldsymbol{x}_s - \boldsymbol{x}_p. \tag{330}$$

For curves, only the u orthogonality equation is used. The following derivation is the one represented in `src/igampm/Coupling/ipc_func.py`. It is needed because the closest NURBS parameter is not fixed when the IGA control points or the MPM grid displacement change.

13.5.1 Physical convergence criteria

Related API. IGAMPM, `IGAMPM.set_solver` coupling tolerances.

For a coupled correction, define particle motion and its gradient by

$$\delta \boldsymbol{u}_p = \sum_i N_{pi} \delta \boldsymbol{u}_i^M, \qquad \delta \boldsymbol{L}_p = \sum_i \delta \boldsymbol{u}_i^M \otimes \nabla N_{pi}. \tag{331}$$

Using the particle body's grid spacing ℓ_p, the common Newton and updated-friction probe measure is

$$\eta = \max \left\{ \frac{\|\delta \boldsymbol{u}^I\|_\infty}{\Delta t_I}, \max_p \frac{\|\delta \boldsymbol{u}_p\|_\infty}{\Delta t_M}, \max_p \frac{\ell_p \|\delta \boldsymbol{L}_p\|_{\max}}{\Delta t_M} \right\}. \tag{332}$$

Here $\|\boldsymbol{A}\|_{\max} = \max_{ij} |A_{ij}|$. Axisymmetry additionally includes $\max_p \ell_p |\delta u_{r,p}| / (r_p \Delta t_M)$, with radius measured from the configured axis. The strain term prevents cancelling particle translations from hiding a deformation increment. Almost-empty grid-support modes are measured by their physical particle influence; no nodal mass or transfer operator is changed.

13.5.1.1 Optional Direct ULMPM small-mass treatment. The separate option `MPM.set_solver.solver.small_mass_ratio`, $\alpha_m \in [0, 1)$, defaults to zero and leaves the transfer unchanged. When enabled, define $\bar{m} = N_g^{-1} \sum_{i=1}^{N_g} m_i$ over all background nodes and remove nodes with $m_i \le \max(10^{-12}, \alpha_m \bar{m})$ after P2G. Their mass and momentum accumulators are cleared, and their particle connections are removed from every force, tangent, contact and G2P operation for this step. No particle mass is deleted, but the retained interpolation

need not sum to one, and discarded grid momentum is not recovered. Diagnostics report the discarded mass fraction and momentum; threshold comparisons are required. A particle losing every support node rejects the operation.

The same particle measure is shared by standalone lagged MPM–IPC, FEM–MPM, IGA–MPM, and Direct MPM–ABD. FEM replaces the IGA control term with physical nodal displacement; ABD uses surface-vertex displacement $\delta\boldsymbol{x}_v = \sum_a B_{va}\delta\boldsymbol{y}_a$ rather than affine-control coefficients. SoftParticle–ABD uses its soft-point interpolation and physical mechanical-grid spacing; template rotations and scale transform the support gradients exactly as in force assembly. The inner correction and updated outer probe are measured before max-step truncation. An updated probe is never applied, and a small nonzero force alone cannot skip its correction solve. Force-based FEM, FEM–ABD, and FEM–MPM loops also retain the first force reference across friction refreshes and reset it per attempted step.

Nonassociated DP requires $\eta \le \varepsilon_N$, the prescribed-increment bound, and $R_k \le \varepsilon_a + \varepsilon_r \max(R_0, 1)$ simultaneously. The first residual R_0 is retained throughout the attempted step's lagged-friction solves and reset on retry. After refreshing friction, its unapplied probe must pass $\eta \le \varepsilon_f$ and the same force bound. An inexact terminal correction is re-solved at the configured linear accuracy before accepting it. Contact and material feasibility remain mandatory. Inexact IGA–MPM, FEM–MPM, and Direct MPM–ABD linear errors use the physical free-force reference before Dirichlet elimination: $\|\boldsymbol{e}_{\mathrm{lin}}\| \le \max(\varepsilon_{\mathrm{lin,a}}, \zeta_k R_k)$, with adaptive forcing $\zeta_k \le 10^{-2}$. This reference excludes the potentially much larger prescribed-increment contribution to the eliminated right-hand side.

For IGA–MPM nonassociated DP, residual Armijo uses the loading merit $\mathcal{M}_{\mathrm{D}} = \frac{1}{2}R_k^2 + \frac{1}{2}w_{\mathrm{D}}^2 d_{\mathrm{D}}^2$, where d_{D} is the maximum prescribed-displacement mismatch. When its initial value exceeds the Dirichlet tolerance, $w_{\mathrm{D}} = \max(R_0, 1)/d_{\mathrm{D},0}$; otherwise $w_{\mathrm{D}} = 0$. The weight is fixed within the Newton solve. Exact fixed corrections give $d_{\mathrm{D}}(\alpha) = (1-\alpha)d_{\mathrm{D}}$, so the additional slope is $-w_{\mathrm{D}}^2 d_{\mathrm{D}}^2$. This allows prescribed loading to proceed from a force-balanced state. The physical Jacobian and stopping tolerances are unchanged. Residual Armijo also permits a feasible trial within the original force band when both R_k and R_{trial} satisfy $\varepsilon_a + \varepsilon_r \max(R_0, 1)$. The physical-motion and Dirichlet tests still determine Newton acceptance, and the refreshed-friction probe remains mandatory. This avoids requiring strictly decreasing residual noise after force convergence; trials outside the band retain ordinary Armijo.

Standalone lagged MPM–IPC also selects the physical nonassociated DP map, full nonsymmetric HashTriplet storage, BiCGSTAB, and CCD-filtered residual Armijo. Its free-force target is $10^{-10} + 10^{-8}\max(R_0, 1)$, in addition to its configured correction-velocity and prescribed-motion bounds. It does not run the old frozen-volume material-consistency loop. Associated materials retain the projected symmetric path; fully implicit friction is a separate formulation and retains its existing configuration restrictions.

13.5.2 Closest-Point Implicit Differentiation

Use the parametric vector $\boldsymbol{\theta} = (u, v)^T$ for a surface and $\boldsymbol{\theta} = (u)$ for a curve. Let

$$\boldsymbol{t}_\alpha = \frac{\partial \boldsymbol{x}_s}{\partial \theta_\alpha}, \qquad \boldsymbol{c}_{\alpha\beta} = \frac{\partial^2 \boldsymbol{x}_s}{\partial \theta_\alpha \partial \theta_\beta}, \qquad F_\alpha(\boldsymbol{\theta}, \boldsymbol{y}) = \boldsymbol{r} \cdot \boldsymbol{t}_\alpha = 0, \tag{333}$$

where $\boldsymbol{y}$ denotes all active IGA and MPM dofs. Differentiating the orthogonality equations gives

$$\sum_\beta C_{\alpha\beta}\,\mathrm{d}\theta_\beta + \mathrm{d}_{\boldsymbol{y}} F_\alpha = 0, \qquad C_{\alpha\beta} = \frac{\partial F_\alpha}{\partial \theta_\beta} = \boldsymbol{t}_\alpha \cdot \boldsymbol{t}_\beta + \boldsymbol{r} \cdot \boldsymbol{c}_{\alpha\beta}. \tag{334}$$

The matrix $\boldsymbol{C}$ is the local closest-point Hessian in parameter space. When $\boldsymbol{C}$ is singular or nearly singular the closest point is not differentiable; the code therefore uses the active contact only after the closest-point search returns a usable distance.

For the particle position $\boldsymbol{x}_p$, $\mathrm{d}_{\boldsymbol{x}_p} F_\alpha = -\boldsymbol{t}_\alpha \cdot \mathrm{d}\boldsymbol{x}_p$. With

$$\boldsymbol{T} = \begin{bmatrix} \boldsymbol{t}_1^T \\ \cdots \\ \boldsymbol{t}_m^T \end{bmatrix}, \qquad m = 1 \text{ or } 2, \tag{335}$$

implicit differentiation gives

$$\frac{\partial \boldsymbol{\theta}}{\partial \boldsymbol{x}_p} = \boldsymbol{C}^{-1}\boldsymbol{T}. \tag{336}$$

For a control point $\boldsymbol{P}_A$, the direct variations are $\mathrm{d}\boldsymbol{x}_s = R_A\,\mathrm{d}\boldsymbol{P}_A$ and $\mathrm{d}\boldsymbol{t}_\alpha = R_{A,\alpha}\,\mathrm{d}\boldsymbol{P}_A$. Define

$$\boldsymbol{q}_{A\alpha} = R_A \boldsymbol{t}_\alpha + R_{A,\alpha}\boldsymbol{r}, \qquad \boldsymbol{Q}_A = \begin{bmatrix} \boldsymbol{q}_{A1}^T \\ \cdots \\ \boldsymbol{q}_{Am}^T \end{bmatrix}. \tag{337}$$

Then

$$\frac{\partial \boldsymbol{\theta}}{\partial \boldsymbol{P}_A} = -\boldsymbol{C}^{-1}\boldsymbol{Q}_A. \tag{338}$$

Equations (336) and (338) are the implicit-differentiation step. In the code they appear as `Dknot_div_Dpoint_Curve` and `Dknot_div_Dpoint_Surface`; the control-point version is inserted into the second-derivative formulas through $\boldsymbol{Q}_A^T\boldsymbol{C}^{-1}\boldsymbol{Q}_B$ and $\boldsymbol{Q}_A^T\boldsymbol{C}^{-1}\boldsymbol{T}$.

13.5.3 Reduced Distance Gradient and Hessian

Let

$$s(\boldsymbol{\theta}, \boldsymbol{y}) = \boldsymbol{r} \cdot \boldsymbol{r}, \qquad d = \sqrt{s}, \qquad \bar{s}(\boldsymbol{y}) = s(\boldsymbol{\theta}(\boldsymbol{y}), \boldsymbol{y}). \tag{339}$$

Because $F_\alpha = \boldsymbol{r} \cdot \boldsymbol{t}_\alpha = 0$, the first derivative of $\bar{s}$ does not contain an explicit $\partial\boldsymbol{\theta}/\partial\boldsymbol{y}$ term:

$$\frac{\partial \bar{s}}{\partial \boldsymbol{x}_p} = -2\boldsymbol{r}, \qquad \frac{\partial \bar{s}}{\partial \boldsymbol{P}_A} = 2R_A\boldsymbol{r}. \tag{340}$$

This is why `Ddistance2_div_Dpoint` returns $-2\boldsymbol{r}$ and `Ddistance2_div_Dctrlpt` returns $2R_A\boldsymbol{r}$.

The Hessian does contain the implicit closest-point correction. Eliminating $\mathrm{d}\boldsymbol{\theta}$ with $\boldsymbol{C}^{-1}$ gives the reduced Hessian blocks

$$\frac{\partial^2 \bar{s}}{\partial \boldsymbol{x}_p\,\partial \boldsymbol{x}_p} = 2\left(\boldsymbol{I} - \boldsymbol{T}^T\boldsymbol{C}^{-1}\boldsymbol{T}\right), \tag{341}$$

$$\frac{\partial^2 \bar{s}}{\partial \boldsymbol{P}_A\,\partial \boldsymbol{P}_B} = 2\left(R_A R_B \boldsymbol{I} - \boldsymbol{Q}_A^T\boldsymbol{C}^{-1}\boldsymbol{Q}_B\right), \tag{342}$$

$$\frac{\partial^2 \bar{s}}{\partial \boldsymbol{P}_A\,\partial \boldsymbol{x}_p} = 2\left(\boldsymbol{Q}_A^T\boldsymbol{C}^{-1}\boldsymbol{T} - R_A\boldsymbol{I}\right). \tag{343}$$

These three expressions are the mathematical form of `D2distance2_div_D2point_Surface`, `D2distance2_div_D2ctrlpt_Surface`, and `D2distance2_div_DctrlptDpoint_Surface`. For a curve, $m = 1$,

$$C = \boldsymbol{t} \cdot \boldsymbol{t} + \boldsymbol{r} \cdot \boldsymbol{c}, \qquad \boldsymbol{T} = \boldsymbol{t}^T, \qquad \boldsymbol{Q}_A = (R_A \boldsymbol{t} + R'_A \boldsymbol{r})^T,$$

so the same formulas reduce to

$$\bar{s}_{\boldsymbol{x}_p \boldsymbol{x}_p} = 2 \left(\boldsymbol{I} - \frac{\boldsymbol{t} \otimes \boldsymbol{t}}{C} \right), \tag{344}$$

$$\bar{s}_{\boldsymbol{P}_A \boldsymbol{P}_B} = 2 \left(R_A R_B \boldsymbol{I} - \frac{(R_A \boldsymbol{t} + R'_A \boldsymbol{r}) \otimes (R_B \boldsymbol{t} + R'_B \boldsymbol{r})}{C} \right), \tag{345}$$

$$\bar{s}_{\boldsymbol{P}_A \boldsymbol{x}_p} = 2 \left(\frac{(R_A \boldsymbol{t} + R'_A \boldsymbol{r}) \otimes \boldsymbol{t}}{C} - R_A \boldsymbol{I} \right). \tag{346}$$

These are the formulas used by `D2distance2_div_D2point_Curve`, `D2distance2_div_D2ctrlpt_Curve`, and `D2distance2_div_DctrlptDpoint_Curve`.

13.5.4 Barrier Chain Rule and Coupled Blocks

The barrier energy is evaluated as a scalar function of $d = \sqrt{\bar{s}}$:

$$E_B(\boldsymbol{y}) = B(d(\boldsymbol{y})). \tag{347}$$

For any two local variables $a, b \in \{\boldsymbol{P}_A, \boldsymbol{x}_p\}$,

$$\frac{\partial E_B}{\partial a} = \frac{B'(d)}{2d} \frac{\partial \bar{s}}{\partial a}, \tag{348}$$

$$\frac{\partial^2 E_B}{\partial a \, \partial b} = \frac{1}{4} \left(\frac{B''(d)}{d^2} - \frac{B'(d)}{d^3} \right) \frac{\partial \bar{s}}{\partial a} \left(\frac{\partial \bar{s}}{\partial b} \right)^T + \frac{B'(d)}{2d} \frac{\partial^2 \bar{s}}{\partial a \, \partial b}. \tag{349}$$

This is exactly the chain rule in `Barrier.gradient` and `Barrier.hessian`. The local dof vector is

$$\boldsymbol{y} = [\boldsymbol{P}_{A_1}, \ldots, \boldsymbol{P}_{A_m}, \boldsymbol{u}_{i_1}, \ldots, \boldsymbol{u}_{i_n}], \tag{350}$$

where $\partial \boldsymbol{x}_p / \partial \boldsymbol{u}_i = N_i(\boldsymbol{x}_p) \boldsymbol{I}$. The IGA unknowns are the active control-point increments themselves, while the MPM unknowns move the particle through the grid shape functions. Therefore the gradient contributions are

$$\boldsymbol{g}_{P_A} = \frac{B'}{2d} \, \bar{s}_{\boldsymbol{P}_A}, \qquad \boldsymbol{g}_{u_i} = N_i \frac{B'}{2d} \, \bar{s}_{\boldsymbol{x}_p}. \tag{351}$$

With

$$\boldsymbol{H}^B_{ab} = \frac{\partial^2 E_B}{\partial a \, \partial b},$$

a single active contact writes

$$\boldsymbol{K}_{P_A P_B} = \boldsymbol{H}^B_{\boldsymbol{P}_A \boldsymbol{P}_B}, \tag{352}$$

$$\boldsymbol{K}_{P_A u_i} = N_i \boldsymbol{H}^B_{\boldsymbol{P}_A \boldsymbol{x}_p}, \tag{353}$$

$$\boldsymbol{K}_{u_i P_A} = N_i \boldsymbol{H}^B_{\boldsymbol{x}_p \boldsymbol{P}_A}, \tag{354}$$

$$\boldsymbol{K}_{u_i u_j} = N_i N_j \boldsymbol{H}^B_{\boldsymbol{x}_p \boldsymbol{x}_p}. \tag{355}$$

Thus the same contact naturally produces IGA-IGA, IGA-MPM, MPM-IGA, and MPM-MPM blocks. The code inserts the negative barrier gradient into the right-hand side because the global Newton system is written as $K\Delta \boldsymbol{y} = -\nabla E$. Friction uses the same active closest point and normal $\boldsymbol{n} = \boldsymbol{r}/d$, then builds a regularized tangential Hessian on $\boldsymbol{I} - \boldsymbol{n} \otimes \boldsymbol{n}$; its block mapping uses the same R_A and N_i factors.

The monolithic system is

$$\begin{bmatrix} K_{iga} & 0 \\ 0 & K_{mpm} \end{bmatrix} + K_B + K_F, \tag{356}$$

with the negative barrier/friction gradients split into IGA and MPM right-hand sides. The capacities `barrier_nnz` and `friction_nnz` must cover the block writes from all active contacts.

13.6 MPM Self-Collision and Soft-Particle IPC

The MPM IPC path handles point-ground contact, legacy point-point contact between MPM surface particles, and the soft-particle point-to-level-set contact used by level-set soft bodies. For a plane ground,

$$d = \boldsymbol{n} \cdot \boldsymbol{x}_p + b, \qquad \frac{\partial d}{\partial \boldsymbol{u}_i} = N_i(\boldsymbol{x}_p)\boldsymbol{n}, \tag{357}$$

so

$$\boldsymbol{r}_i = B'(d)N_i\boldsymbol{n}, \qquad K_{ij} = B''(d)N_iN_j\,\boldsymbol{n} \otimes \boldsymbol{n}. \tag{358}$$

For two MPM surface particles a, b from different bodies,

$$\boldsymbol{r} = \boldsymbol{x}_a - \boldsymbol{x}_b, \qquad d = \|\boldsymbol{r}\|, \qquad \boldsymbol{n} = \boldsymbol{r}/d. \tag{359}$$

The point-coordinate Hessian is

$$H_{aa} = B''\boldsymbol{n} \otimes \boldsymbol{n} + \frac{B'}{d}(\boldsymbol{I} - \boldsymbol{n} \otimes \boldsymbol{n}), \tag{360}$$

$$H_{ab} = -H_{aa}, \qquad H_{bb} = H_{aa}. \tag{361}$$

Mapping to grid dofs gives

$$K_{ij}^{ab} = N_i(\boldsymbol{x}_a)H_{ab}N_j(\boldsymbol{x}_b). \tag{362}$$

Pairs with the same `bodyID` are skipped; otherwise a continuum body would penalize its own neighboring surface particles.

For level-set soft particles the preferred geometry is not point-point contact between two discretized surfaces. A source surface material point is tested against the other soft body's SDF,

$$g = \phi_j(\boldsymbol{R}_j^T(\boldsymbol{x}_s - \boldsymbol{c}_j)) - r_s,$$

and the gap barrier is differentiated through the source MPM shape functions. This gives a smoother and more geometry-consistent normal than selecting the nearest target surface point. The current strict soft-soft IPC implementation uses the trilinear SDF gradient and Hessian for the source-side block, as described in Soft-Soft Point-Level-Set Barrier. Older helper classes expose a zero SDF Hessian for simple point-level-set derivatives; those helpers are only an approximate legacy path and should not be used as the reference for strict IPC Hessian verification.

14 Constitutive Equations

Related API. MPM.add_material, MohrCoulomb, DruckerPrager, ModifiedCamClay, stress_integration.

14.1 Implemented Material Families

The Native material factory in `src/mpm/MaterialManager.py` selects a model from the MPM material type, solver type, strain measure, and user `model` keyword. The Direct facade separately selects its finite-strain class while building the Direct engine. The combined public inventory is:

Model	Regime	Main role
`LinearElastic`	small strain	isotropic Hooke law.
`ElasticPerfectlyPlastic`	small strain	von-Mises plasticity with optional softening.
`MohrCoulomb`	small strain	frictional soil with tensile cutoff, dilation, softening, and rate-dependent options.
`DruckerPrager`	small strain	smooth pressure-dependent cone with selectable MC matching.
`ModifiedCamClay`	small strain	critical-state clay with optional three-invariant M_θ, subloading, bonding, and an AL local solver.
`StateDependentMohrCoulomb`	small strain	void-ratio/state-parameter dependent friction and dilation.
`GranularMaterial`	small strain	pressure-aware granular elastic update.
`NorSand`	small strain	critical-state sand model with image stress and hardening parameters.
`SanisandMS`	small strain	multi-surface SANISAND-style sand plasticity and fabric.
`UserDefined`	small strain or fluid	user-supplied constitutive hook selected through the material factory.
`HenckyElastic`	finite strain	logarithmic strain hyperelasticity.
`NeoHookean`	finite strain	compressible Neo-Hookean hyperelasticity.
`DruckerPrager` (Direct implicit ULMPM)	finite-strain plasticity	multiplicative Hencky Drucker–Prager return with independent constant dilation and symmetric frozen-inner solves.

Model	Regime	Main role
`StateDependentDruckerPrager` (coupled Direct implicit ULMPM)	finite-strain plasticity	multiplicative Hencky DP with SDMC void-ratio/critical-state angle evolution and a nonsymmetric physical Jacobian.
`VonMises` (Direct implicit ULMPM)	finite-strain plasticity	associated multiplicative Hencky J_2 return with optional linear isotropic hardening.
`MooneyRivlin`	finite strain	two-invariant rubber model.
`Gent`	finite strain	limiting-chain rubber model.
`Hydrogel`	finite strain	soft finite-strain hydrogel response.
`Newtonian`	fluid	pressure plus viscous deviatoric stress.
`Bingham`	fluid	regularized yield-stress fluid.
`Rigid` or `RigidBody`	rigid/prescribed	mass and body identity without deformable stress evolution.

Some finite-strain hyperelastic models reject `B-BarMethod`; two-phase material selection also rejects combinations that the current solver path does not support, such as TLMPM and implicit two-phase material creation in the standard factory path.

14.2 Random-Field Material Parameters

The MPM random-field option is a per-particle material-parameter override. It does not generate a spatial random field internally. Instead, when `MPM.set_configuration(random_field=True)` is set, the material factory calls `random_field_initialize`, reads the `MaterialFile` path, and body insertion calls `read_random_field(start_particle,end_particle,stateVars)`. Rows in the file are consumed in insertion order. For a newly inserted block of particles $\mathcal{P} = [p_s, p_e)$, row $r = p - p_s$ supplies the state variables for particle p. The common elastic conversion is

$$G_p = \frac{E_p}{2(1+\nu_p)}, \qquad K_p = \frac{E_p}{3(1-2\nu_p)}, \tag{363}$$

and wave-speed estimates use the maximum over the loaded random field:

$$c_{max} = \max_p \sqrt{\frac{E_p(1-\nu_p)}{(1+\nu_p)(1-2\nu_p)\rho_p}}. \tag{364}$$

The currently implemented file readers are:

Model	File columns
`GranularMaterial`	`DensityYoungModulusPoissonRatio`.
`ElasticPerfectlyPlastic`	`DensityYoungModulusPoissonRatioYieldStress`.
`MohrCoulomb`	`DensityYoungModulusPoissonRatioCohesionFrictionDilationTensile`; angles are in degrees.

Model	File columns
`DruckerPrager`	`DensityYoungModulusPoissonRatioCohesionFrictionDilationTensile`; angles are in degrees.

`LinearElastic` reads random-field state in its stress kernel if `density`, `shear`, and `bulk` are present, but the current class does not define a file reader or random-field state variables. `ModifiedCamClay` defines random-field state branches for `density`, `poisson`, `lambda_`, `kappa`, and `m_theta`, but its active file reader is not implemented in this source snapshot. `NorSand` and `SanisandMS` explicitly reject random-field material parameters. Therefore a user should check the model-specific `read_random_field` method before assuming that `random_field=True` is supported.

14.3 Elasticity and Hyperelasticity

For small strain,

$$\boldsymbol{\varepsilon} = \frac{1}{2}(\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^T), \qquad \boldsymbol{\sigma} = \lambda \operatorname{tr}(\boldsymbol{\varepsilon})\boldsymbol{I} + 2G\boldsymbol{\varepsilon}. \tag{365}$$

The compressible Neo-Hookean energy is

$$\Psi(\boldsymbol{F}) = \frac{\mu}{2}(\operatorname{tr}(\boldsymbol{F}^T\boldsymbol{F}) - d) - \mu \ln J + \frac{\lambda}{2}(\ln J)^2, \qquad J = \det \boldsymbol{F}, \tag{366}$$

with first Piola stress

$$\boldsymbol{P} = \mu(\boldsymbol{F} - \boldsymbol{F}^{-T}) + \lambda \ln J\, \boldsymbol{F}^{-T}. \tag{367}$$

14.3.1 Finite-Strain Material Class Contract

The finite-strain implementation in `src/physics_model/consititutive_model/finite_strain` is class based. Each model derives from `FiniteStrainModel` and is responsible for its own energy, first Piola stress, and material tangent:

$$\psi = \psi_m(\boldsymbol{F}, \boldsymbol{q}), \qquad \boldsymbol{P}_m = \partial \psi_m / \partial \boldsymbol{F}, \qquad \mathbb{A}_m = \partial \boldsymbol{P}_m / \partial \boldsymbol{F}. \tag{368}$$

The MPM and soft-particle kernels should call these class methods rather than duplicating a local $\boldsymbol{P}$ formula in a contact or material kernel. This keeps the production PK1 stress, the internal force, and the implicit tangent consistent for `NeoHookean`, `HenckyElastic`, `MooneyRivlin`, `Gent`, `Hydrogel`, and the indexed associated-plastic models described in finite-strain associated plasticity. Shared invariant helpers, such as derivatives of I_1, I_2, and J, may remain in a kernel utility file, but the model-specific combination of those invariants belongs to the model class.

The soft-particle state-variable contract is intentionally small. The runtime material state stores the stress measure needed by TLMPM and exposes `estress` as the derived von-Mises equivalent stress for output:

$$\sigma_{vm} = \sqrt{\frac{3}{2}\boldsymbol{s} : \boldsymbol{s}}, \qquad \boldsymbol{s} = \boldsymbol{\sigma} - \frac{1}{3}\operatorname{tr}(\boldsymbol{\sigma})\boldsymbol{I}. \tag{369}$$

For finite-strain soft particles, $\boldsymbol{\sigma}$ used for `estress` is recovered from PK1 by

$$\boldsymbol{\sigma} = \frac{1}{J}\boldsymbol{P}\boldsymbol{F}^T. \tag{370}$$

`estress` is therefore an output/diagnostic state variable; it is not a replacement for the class-provided $\boldsymbol{P}$ or $\mathbb{A}$ used by the force and Hessian assembly.

14.4 Finite-Strain Drucker–Prager and Von Mises

Related API. MPM.add_material, DruckerPrager, and Direct finite-strain VonMises; the same Direct ULMPM material block is consumed by FEM–MPM IPC and IGA–MPM IPC.

The Direct implicit updated-Lagrangian path uses the multiplicative split

$$\boldsymbol{F} = \boldsymbol{F}_e\boldsymbol{F}_p, \qquad \boldsymbol{F}_{e,tr} = (\boldsymbol{I} + \nabla\boldsymbol{u})\boldsymbol{F}_{e,n}. \tag{371}$$

The engine stores the total map and the material stores $\boldsymbol{F}_p^{-1}$; thus the trial map also equals $(\boldsymbol{I}+\nabla\boldsymbol{u})\boldsymbol{F}_n\boldsymbol{F}_{p,n}^{-1}$. The DP and von-Mises implementations always use $d = 3$. In a Cartesian two-dimensional simulation, Direct ULMPM supplies the plane-strain incremental map

$$\boldsymbol{f}_{\mathrm{ps}} = \begin{bmatrix} 1+\partial u_x/\partial x_n & \partial u_x/\partial y_n & 0 \\ \partial u_y/\partial x_n & 1+\partial u_y/\partial y_n & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \boldsymbol{F}_{e,tr} = \boldsymbol{f}_{\mathrm{ps}}\boldsymbol{F}_{e,n}. \tag{372}$$

Thus the global unknown and contact geometry remain 2D while the elastic state, stress invariants, plastic return, and tangent are 3D. The `plane_strain=False` combination is rejected for these laws; finite-strain plane stress and an intrinsic-2D DP/J2 approximation are not implemented. Axisymmetric 2D instead supplies the no-swirl 3D map with $F_{\theta\theta} = r/R$. If

$$\boldsymbol{F}_{e,tr} = \boldsymbol{U}\,\mathrm{diag}(s_1,\ldots,s_d)\boldsymbol{V}^T, \tag{373}$$

the principal Hencky trial strain, trace, and deviator are

$$\boldsymbol{\epsilon}^{tr} = \log \boldsymbol{s}, \qquad t = \mathbf{1}^T\boldsymbol{\epsilon}^{tr}, \qquad \boldsymbol{e} = \boldsymbol{\epsilon}^{tr} - \frac{t}{d}\mathbf{1}, \qquad r = \|\boldsymbol{e}\|. \tag{374}$$

Numerically, $t = \log\det \boldsymbol{F}_{e,tr}$. The shared plastic response refines the initial SVD basis with two one-sided Jacobi sweeps on $\boldsymbol{B} = \boldsymbol{F}_{e,tr}\boldsymbol{V}$, then sets $s_j = \|\boldsymbol{B}_j\|$ and $\boldsymbol{U}_j = \boldsymbol{B}_j/s_j$. This reduces reconstruction roundoff without changing the constitutive equations; force, tangent and commit share this basis. The elastic energy and principal Kirchhoff stress are

$$\Psi_e = G\|\boldsymbol{e}\|^2 + \frac{1}{2}Kt^2, \qquad \boldsymbol{\tau} = 2G\boldsymbol{e} + Kt\mathbf{1}. \tag{375}$$

GeoTaichi's finite-strain Drucker–Prager model starts from the classical tension-positive cone

$$\sqrt{J_2} + q_\phi\frac{I_1}{3} - k_\phi \le 0. \tag{376}$$

The `dpType` formulas for q_ϕ, k_ϕ are the same circumscribed, middle-circumscribed, and inscribed Mohr–Coulomb mappings listed in the small-strain DP section. With

$$\alpha = \frac{\sqrt{2}}{3}q_\phi, \qquad \tau_c = \sqrt{2}k_\phi, \tag{377}$$

the trial yield function in principal Kirchhoff stress is

$$f^{tr} = 2Gr + \alpha dKt - \tau_c. \tag{378}$$

Let $\beta = \sqrt{2}q_\psi/3$, using the same cone mapping for the independent constant dilation angle. Omitted dilation defaults to the friction angle. On the smooth cone, the physical return is

$$\Delta\gamma = \frac{f^{tr}}{2G + d^2\alpha\beta K + H_\gamma}, \tag{379}$$

$$\boldsymbol{\epsilon}_e^{n+1} = \boldsymbol{\epsilon}^{tr} - \Delta\gamma\left(\frac{\boldsymbol{e}}{r} + \beta\mathbf{1}\right). \tag{380}$$

For perfect DP plasticity $H_\gamma = 0$. If the return reaches the cone apex, the projected trace and multiplier are

$$t_a = \frac{\tau_c}{\alpha dK}, \qquad \boldsymbol{\epsilon}_e^{n+1} = \frac{t_a}{d}\mathbf{1}, \qquad \Delta\gamma = \frac{t - t_a}{d\beta} \quad (\beta > 0). \tag{381}$$

This is a hydrostatic tensile apex, not an independent tensile-cutoff surface. For zero dilation the apex is a tensile cap with $\Delta\gamma = r$ and $\Delta\epsilon_v^p = t - t_a$; on the smooth cone, $\Delta\epsilon_v^p = d\beta\Delta\gamma$. Dilation does not evolve.

Nonassociated flow has a generally nonsymmetric physical tangent. Monolithic Standalone lagged MPM–IPC, FEM–MPM, IGA–MPM, and Direct MPM–ABD directly solve the physical residual with BiCGSTAB, without a material-consistency outer loop. At each trial, compute

$$J_p(\boldsymbol{F}) = \frac{\det \boldsymbol{F}}{\det \boldsymbol{F}_e^{returned}}, \qquad \boldsymbol{P} = J_p(\boldsymbol{F})\boldsymbol{P}_e(\boldsymbol{F}_e^{tr})(\boldsymbol{F}_{p,n})^{-T}. \tag{382}$$

The complete total tangent includes $\boldsymbol{P} \otimes \partial_{\boldsymbol{F}} \log J_p$ as well as the transformed local return tangent. Neither it nor the assembled nonsymmetric Jacobian is symmetrized or PSD projected. HashTriplet stores both off-diagonal triangles and uses node-block Jacobi preconditioning; COO uses scalar Jacobi. Line search decreases $\frac{1}{2}\|\boldsymbol{R}_{free}\|^2$, subject to conservative contact CCD and material feasibility bounds. Accepted history is committed only after convergence. Lagged friction retains its independent outer loop. Associated DP retains PCG and its existing frozen-volume consistency iteration.

Other Direct-MPM consumers retain their existing frozen-potential contract. That PCG route freezes the isotropic strain shift $\delta = \alpha\Delta\gamma - \Delta\epsilon_v^p/d$ and predicted plastic volume $\widehat{J}_p$ in an associated inner potential evaluated at $\boldsymbol{\epsilon}^{tr} + \delta\mathbf{1}$. For zero-dilation apex states use $\delta = \alpha r$, keeping the inner state inside its flat apex branch. The material outer loop refreshes both predictors without committing history and checks the maximum, over particles, of

$$\left|\log \frac{J_p^{candidate}}{\widehat{J}_p}\right|, \qquad \frac{\|\boldsymbol{\tau}_{physical} - \boldsymbol{\tau}_{inner}\|}{2G}. \tag{383}$$

This dimensionless criterion uses the existing material tolerance. Equivalent shifts within the tensile cap give zero stress mismatch, rather than forcing another equilibrium solve. The inner Hessian stays symmetric and admits PSD projection and energy Armijo; the physical nonassociated tangent does not.

Predictor updates use particle-local Aitken relaxation. Define $\boldsymbol{r}_k = (\log(J_p^{candidate}/\widehat{J}_{p,k}), \delta^{candidate} - \delta_k)^T$ and $\Delta\boldsymbol{r}_k = \boldsymbol{r}_k - \boldsymbol{r}_{k-1}$. When the previous resid-

ual and secant difference are nondegenerate, update

$$\omega_k = \mathrm{clip}_{[10^{-3},1]}\left(-\omega_{k-1}\frac{\boldsymbol{r}_{k-1}^T\Delta\boldsymbol{r}_k}{\|\Delta\boldsymbol{r}_k\|^2}\right), \tag{384}$$

$$\delta_{k+1} = \delta_k + \omega_k r_{k,2}, \tag{385}$$

$$\widehat{J}_{p,k+1} = \widehat{J}_{p,k}\exp(\omega_k r_{k,1}). \tag{386}$$

Otherwise retain the current relaxation. Each step or retry resets the previous residual to zero and ω to one. The convergence criterion above is evaluated before relaxation, so damping cannot conceal a physical stress or volume mismatch. This changes only the numerical iteration, not the constant-dilation DP law.

14.4.1 State-dependent finite-strain Drucker–Prager

Related API. StateDependentDruckerPrager.

The independent finite-strain model transfers the evolution law from `infinitesimal_strain/StateDependentMohrCoulomb.py`. Ordinary DP retains its constant angles. With accepted void ratio e_n, total Jacobian $J_n = \det \boldsymbol{F}_n$, and previous elastic Jacobian $J_{e,n}$, define

$$p_n = \max\left(-K\log J_{e,n}/J_{e,n},\, 1000\,\mathrm{Pa}\right), \tag{387}$$

$$e_c = e_{\mathrm{Tao}} - \lambda_c(p_n/(101000\,\mathrm{Pa}))^{\xi}, \tag{388}$$

$$e^{tr} = \mathrm{clip}_{[0.1,1.5]}\left((1+e_n)J^{tr}/J_n - 1\right), \qquad S = e^{tr} - e_c, \tag{389}$$

$$\phi = \max\left(\phi_c, \arctan[\tan\phi_c\exp(-n_f S)]\right), \tag{390}$$

$$\psi = \max\left(0, \arctan[-n_d S]\right). \tag{391}$$

Here p_n is compression-positive Cauchy pressure and is frozen for the entire step, including lagged-friction iterations. The finite volume formula reduces to SDMC's $\Delta e = (1+e)\Delta\epsilon_v$ for infinitesimal increments; axisymmetric J includes the hoop stretch. Trial evaluations never change accepted history. Elastic moduli and cohesion remain constant. The default `MiddleCircumscribed` cone uses the same $3+\sin\phi$ mapping as SDMC, with the other two DP mappings available. The return retains the finite-strain DP tensile apex, rather than an independent SDMC tensile cutoff.

The coefficients $\alpha(t), \beta(t), \tau_c(t)$ now depend on trial Hencky trace t, through $J^{tr} = \exp(t)/\det \boldsymbol{F}_{p,n}^{-1}$. Inside the void-ratio clamp, $de^{tr}/dt = (1+e_n)J^{tr}/J_n$; outside it this derivative is zero. In the dense branch $S < 0$,

$$\phi' = -n_f \sin\phi\cos\phi\,(e^{tr})', \qquad \psi' = \frac{-n_d(e^{tr})'}{1+(n_d S)^2}. \tag{392}$$

Both angle derivatives are zero on the loose branch. Let $m_i = e_i/r$, $D = 2G + 9K\alpha\beta$, and $D' = 9K(\alpha'\beta + \alpha\beta')$. The smooth-cone Jacobian is

$$\partial_j\gamma = \frac{2Gm_j + 3K(\alpha + t\alpha') - \tau_c' - \gamma D'}{D}, \tag{393}$$

$$\partial_j\epsilon_i^e = \delta_{ij} - (m_i+\beta)\partial_j\gamma - \gamma\beta' - \frac{\gamma}{r}(\delta_{ij} - 1/3 - m_i m_j). \tag{394}$$

At the apex,

$$t_a' = \frac{\tau_c'\alpha - \tau_c\alpha'}{3K\alpha^2}, \qquad \partial_j\epsilon_i^e = t_a'/3. \tag{395}$$

The hydrostatic apex stress $Kt_a\boldsymbol{F}_{e,tr}^{-T}$ therefore contributes the additional tangent term $K\,\mathrm{vec}(\boldsymbol{F}_{e,tr}^{-T})\otimes\partial_{\boldsymbol{F}_{e,tr}}t_a$. Together with the spectral return derivative and total plastic-volume derivative, these terms form the consistent physical PK1 Jacobian. Central finite differences validate the smooth elastic, cone, and apex branches; clamp and branch switches are piecewise differentiable.

This model follows the coupled physical residual/BiCGSTAB route and has no incremental energy potential or material-consistency outer iteration. Lagged friction retains its independent fixed point. Its 14-entry history appends void ratio, accepted total Jacobian, and previous-pressure cache to the original 11 DP entries; only accepted steps commit void ratio and volume, while rollback restores every entry. Supported owners are FEM–MPM, IGA–MPM, and 3D MPM–AffineBody IPC with Direct implicit ULMPM. Standalone stepping, Direct TLMPM, Native MPM, LSMPM soft materials, and parameter/history adjoints are unsupported.

14.4.2 Associated von Mises

For finite-strain `VonMises`, $\alpha = 0$ and the user stress σ_y follows the standard convention

$$\sqrt{3J_2} = \sigma_y, \qquad \tau_c(\bar{\epsilon}_p) = \sqrt{\frac{2}{3}}\,(\sigma_{y0} + H\bar{\epsilon}_p)\,. \tag{396}$$

The equivalent plastic increment and local hardening modulus are

$$\Delta\bar{\epsilon}_p = \sqrt{\frac{2}{3}}\Delta\gamma, \qquad H_\gamma = \frac{2}{3}H. \tag{397}$$

Thus `HardeningModulus=0` gives perfect J_2 plasticity, while a positive value gives linear isotropic hardening.

For associated flow, eliminating $\Delta\gamma$ gives the local incremental potential

$$\Psi_{inc} = G\|\operatorname{dev}\boldsymbol{\epsilon}_e^{n+1}\|^2 + \frac{1}{2}K\,\mathrm{tr}(\boldsymbol{\epsilon}_e^{n+1})^2 + \tau_c^n\Delta\gamma + \frac{1}{2}H_\gamma\Delta\gamma^2. \tag{398}$$

At the DP apex its continuous linear volumetric continuation has slope Kt_a. The code differentiates this potential analytically through the singular values to obtain PK1 and a symmetric spectral Hessian; no finite difference is used in production. Projected Newton may then project each particle Hessian to the PSD cone before sparse assembly. Material CCD bounds $\det(\boldsymbol{I} + \nabla\boldsymbol{u}) > 0$, while IPC swept culling/CCD and Armijo are unchanged by the material choice.

After an accepted step, the Taichi commit kernel stores

$$\boldsymbol{F}_{e,n+1} = \boldsymbol{U}\,\mathrm{diag}(\exp\boldsymbol{\epsilon}_e^{n+1})\boldsymbol{V}^T \tag{399}$$

and advances equivalent and volumetric plastic-strain diagnostics. Direct TLMPM currently rejects these plastic models because a fixed-reference formulation must retain $\boldsymbol{F}_p$ (or an equivalent evolving rest basis) explicitly. The implementation follows the force-equivalence and incremental potential construction of Li, Li, and Jiang (2022), but differs from their supplemental DP example by using physical cohesion and classical MC cone mapping, independent constant dilation, a hydrostatic apex, symmetric frozen-inner solves, and optional standard von Mises hardening; see `agent/FINITE_STRAIN_ASSOCIATED_PLASTICITY.md` for the complete deviation ledger.

14.5 Return Mapping and Explicit/Substepping Stress Integration

`ReturnMapping` and `ImplicitIntegration` both route to `PlasticMaterial.ImplicitIntegration` in `src/physics_model/consititutive_model/infinitesimal_strain/ElasPlasticity.py`. `SubStepping` routes to `PlasticMaterial.ExplicitIntegration`. A model may additionally expose `ImplicitIntegrationAL`; in the current source this special local solver is implemented for `ModifiedCamClay`. The velocity gradient gives

$$\boldsymbol{D}_p = \operatorname{sym}\left(\sum_i \boldsymbol{v}_i \otimes \nabla N_{ip}\right), \quad \boldsymbol{W}_p = \operatorname{skw}\left(\sum_i \boldsymbol{v}_i \otimes \nabla N_{ip}\right), \quad \Delta\boldsymbol{\varepsilon}_p = \Delta t\, \boldsymbol{D}_p. \tag{400}$$

In the small-strain material kernels the 3D Voigt increment is stored as

$$\Delta\boldsymbol{\varepsilon} = \Delta t \begin{bmatrix} L_{11} & L_{22} & L_{33} & \frac{1}{2}(L_{12}+L_{21}) & \frac{1}{2}(L_{23}+L_{32}) & \frac{1}{2}(L_{13}+L_{31}) \end{bmatrix}^T. \tag{401}$$

The objective increment includes

$$\Delta\boldsymbol{\sigma}^{rot} = \Delta t(\boldsymbol{\sigma}_n \boldsymbol{W}_p - \boldsymbol{W}_p \boldsymbol{\sigma}_n). \tag{402}$$

The elastic predictor is

$$\boldsymbol{\sigma}^{tr} = \boldsymbol{\sigma}_n + \mathbb{C}^e : \Delta\boldsymbol{\varepsilon}_p. \tag{403}$$

The helper `ElasticTensorMultiplyVector` applies

$$\mathbb{C}^e : \Delta\boldsymbol{\varepsilon} = \begin{bmatrix} a & b & b & 0 & 0 & 0 \\ b & a & b & 0 & 0 & 0 \\ b & b & a & 0 & 0 & 0 \\ 0 & 0 & 0 & 2G & 0 & 0 \\ 0 & 0 & 0 & 0 & 2G & 0 \\ 0 & 0 & 0 & 0 & 0 & 2G \end{bmatrix} \Delta\boldsymbol{\varepsilon}, \quad a = K + \frac{4}{3}G, \quad b = K - \frac{2}{3}G. \tag{404}$$

For $f(\boldsymbol{\sigma}, \alpha) \le 0$ and plastic potential g, backward Euler solves

$$\boldsymbol{\sigma}_{n+1} = \boldsymbol{\sigma}^{tr} - \Delta\lambda\, \mathbb{C}^e : \boldsymbol{n}_g, \tag{405}$$

$$\alpha_{n+1} = \alpha_n + \Delta\lambda h(\boldsymbol{\sigma}_{n+1}, \alpha_{n+1}), \tag{406}$$

$$f(\boldsymbol{\sigma}_{n+1}, \alpha_{n+1}) = 0, \qquad \Delta\lambda \ge 0. \tag{407}$$

The generic code evaluates $\boldsymbol{n}_f = \partial f/\partial\sigma$, $\boldsymbol{n}_g = \partial g/\partial\sigma$, and $\boldsymbol{a}_g = \mathbb{C}^e : \boldsymbol{n}_g$. Its scalar Newton denominator is

$$D_\lambda = \boldsymbol{n}_f : \boldsymbol{a}_g - H_{code}, \qquad \Delta\lambda = \frac{f}{D_\lambda}. \tag{408}$$

Here H_{code} is the value returned by `ComputePlasticModulus`; for softening it may be negative, so the formula is better read as the exact source denominator rather than a sign-convention-free textbook symbol. The stress and internal variables are then updated as

$$\boldsymbol{\sigma} \leftarrow \boldsymbol{\sigma} - \Delta\lambda\, \boldsymbol{a}_g, \qquad \boldsymbol{q} \leftarrow \boldsymbol{q} + \Delta\boldsymbol{q}(\Delta\lambda, \boldsymbol{n}_g). \tag{409}$$

If drift correction is enabled, the code repeats a local correction on the updated yield surface.

The consistent tangent is

$$\mathbb{C}^{alg} = \mathbb{C}^e - \frac{(\mathbb{C}^e : \boldsymbol{n}_g) \otimes (\boldsymbol{n}_f : \mathbb{C}^e)}{D_\lambda}. \tag{410}$$

It is symmetric only when the linearization is consistent and $\boldsymbol{n}_f = \boldsymbol{n}_g$.

14.5.1 Explicit Forward-Euler Substepping

The `SubStepping` branch first finds the elastic fraction α_e of the strain increment. If $f(\sigma_n) < 0$ and $f(\sigma_n + \mathbb{C}^e : \Delta\varepsilon) > 0$, a modified regula-falsi search finds the point where the increment crosses the yield surface:

$$f(\sigma_n + \alpha_e \mathbb{C}^e : \Delta\varepsilon) = 0, \qquad 0 \le \alpha_e \le 1. \tag{411}$$

The elastic part is applied directly. The remaining strain $\Delta\varepsilon_p^{pl} = (1 - \alpha_e)\Delta\varepsilon$ is integrated by the forward-Euler local substep

$$\Delta\sigma^e = \mathbb{C}^e : (\Delta T\, \Delta\varepsilon_p^{pl}), \tag{412}$$

$$D_\lambda = \boldsymbol{n}_f : \mathbb{C}^e : \boldsymbol{n}_g - H_{code}, \tag{413}$$

$$\Delta\lambda = \max\left(\frac{\boldsymbol{n}_f : \Delta\sigma^e}{D_\lambda}, 0\right), \tag{414}$$

$$\Delta\sigma = \Delta\sigma^e - \Delta\lambda\, \mathbb{C}^e : \boldsymbol{n}_g, \tag{415}$$

$$\Delta q = \Delta q(\Delta\lambda, \boldsymbol{n}_g). \tag{416}$$

This is a forward Euler plastic corrector because gradients and moduli are evaluated at the beginning of the substep.

The active explicit routine `NBURKDP2` uses an embedded two-stage estimate. For substep size ΔT and multiplier m,

$$(\Delta\sigma_1, \Delta q_1) = \Phi(\sigma_1, q_1, \Delta T), \tag{417}$$

$$\sigma_2 = \sigma_1 + m\Delta\sigma_1, \qquad q_2 = q_1 + m\Delta q_1, \tag{418}$$

$$(\Delta\sigma_2, \Delta q_2) = \Phi(\sigma_2, q_2, \Delta T), \tag{419}$$

$$\sigma^{temp} = \sigma_1 + \frac{1}{2}(m\Delta\sigma_1 + \Delta\sigma_2). \tag{420}$$

The stress error estimator is

$$e_\sigma = \frac{\left\| -\frac{1}{2}m\Delta\sigma_1 + \frac{1}{2}\Delta\sigma_2 \right\|}{\|\sigma^{temp}\|}, \tag{421}$$

and the next substep is scaled by a clamped factor

$$\Delta T_{new} = \Delta T \mathrm{clip}\left(\sqrt{\frac{10^{-6}}{e_\sigma}}, 0.1, 2\right). \tag{422}$$

The explicit path is useful for severe softening and yield-surface corners, but it supplies no global consistent tangent for implicit Newton assembly.

The source also contains `RKDP2` and `RKDP6` Runge-Kutta substepping helpers. They use embedded estimates to adapt ΔT from both stress and internal-variable errors:

$$e_q = \max_a \frac{|q_a^{high} - q_a^{low}|}{|q_a^{high}| + \epsilon}. \tag{423}$$

`RKDP2` uses a two-stage embedded rule. `RKDP6` uses a six-stage Dormand-Prince-style rule with a higher-order trial state and a lower-order error estimate. The active `ExplicitIntegration` path in this source calls `NBURKDP2`; the Runge-Kutta routines are part of the explicit integration library and are available to model implementations that choose to call them.

14.6 Von-Mises Elastic-Perfectly-Plastic Model

`ElasticPerfectlyPlastic` uses the equivalent deviatoric stress

$$q = \sqrt{3J_2}, \qquad J_2 = \frac{1}{2}\boldsymbol{s} : \boldsymbol{s}, \tag{424}$$

and yield function

$$f = q - \sigma_y(\bar{\varepsilon}_p). \tag{425}$$

The associated potential gives

$$\boldsymbol{n}_f = \boldsymbol{n}_g = \frac{\partial q}{\partial \sigma} = \frac{3}{2q}\boldsymbol{s}. \tag{426}$$

Without softening, the plastic modulus is zero. With softening, the internal variable is either scalar equivalent plastic strain or a stored plastic strain tensor. The code updates

$$\Delta\bar{\varepsilon}_p = \|\Delta\lambda\, \boldsymbol{n}_g\|_{eq}\,, \tag{427}$$

and inserts

$$H_{code} = \frac{\mathrm{d}\sigma_y}{\mathrm{d}\bar{\varepsilon}_p}\frac{\partial \bar{\varepsilon}_p}{\partial \varepsilon_p} : \boldsymbol{n}_g \tag{428}$$

into the generic return-mapping denominator.

14.7 Stationary MPM Wall Friction

Let v^* and a^* denote the free grid velocity and acceleration after the explicit update, and n the outward unit wall normal. For $w = \max(v^*\cdot n, 0) > 0$, define the tangent $t = v^* - (v^*\cdot n)n$ and project

$$v^+ = \max\left(0, 1 - \frac{\mu w}{\|t\|}\right) t, \qquad a^+ = a^* + \frac{v^+ - v^*}{\Delta t}. \tag{429}$$

For zero tangential speed the projected velocity is zero. For $w = 0$ the node separates or remains tangent and is unchanged. This enforces unilateral nonpenetration with a Coulomb impulse bound and cannot reverse slip or add kinetic energy. No second free acceleration increment is applied. The axis-aligned implementation is shared by ordinary and contact grid nodes.

14.8 Drucker-Prager and Mohr-Coulomb Softening and Rate Dependence

The sensitive-clay validation based on Zhang et al. (2021), section 4.3, uses a fixed 0.2 m grid in 2D plane strain, with no adaptive grid in either initialization or failure. The paper uses Tresca; the current control uses zero-friction, zero-dilation `DruckerPrager` with `MPM.add_material.material.dpType=Inscribed`. This gives $\sqrt{J_2} = c$, matching the paper's pure-shear strengths $c_p = 20$ kPa and $c_r = 4$ kPa, but not its full Tresca surface. The slope $dc/d\bar{\varepsilon}_p = -50$ kPa gives $\bar{\varepsilon}_{pr} = 0.32$. DP's accumulated shear history at zero dilation uses the standard invariant $\bar{\varepsilon}_p = \sqrt{2e_p : e_p/3}$ and therefore takes endpoint 0.32. For comparison, the separate internal soft-MC helper uses $\sqrt{2J_2(e_p)/3} = \bar{\varepsilon}_p/\sqrt{2}$, while its

recorded scalar `strain` already uses the standard normalization. That MC conversion must not be applied to the DP endpoint. The same self-contained example exposes `--materialmc` for the corrected MC comparison on identical 0.2 m uniform grids, boundaries and timesteps. Its shear yield is Tresca at zero friction, while the native smoothed deviatoric potential is retained. Both material choices disable adaptive grids. In general DP accumulates $\sqrt{2\,\mathrm{d}\varepsilon_p : \mathrm{d}\varepsilon_p/3}$, including any volumetric tensile-cap flow; the paper defines its invariant using deviatoric plastic strain. Their agreement in zero-dilation pure shear does not make the complete models identical. For the native tensor-history invariant $\kappa = \sqrt{e_p : e_p/3}$, $\partial\kappa/\partial\varepsilon_p = e_p/(3\kappa)$ under tensor contraction (double weight on Voigt shear entries). MC and softening Mises use this derivative without changing their persisted history normalization. Both models refresh softened strength from local history. Generic implicit returns and all explicit substep schemes refresh parameters before testing the updated yield residual, including after crossing the residual endpoint.

With compression-positive $p = -\mathrm{tr}(\sigma)/3$ and $J_2 = s : s/2$, Drucker-Prager uses

$$f_s = \sqrt{J_2} + q_\phi \sigma_m - k_\phi, \qquad f_t = \sigma_m - T, \qquad \sigma_m = -p. \tag{430}$$

The three `dpType` cone mappings are

$$q_\phi^{circ} = \frac{6\sin\phi}{\sqrt{3}(3-\sin\phi)}, \qquad k_\phi^{circ} = \frac{6c\cos\phi}{\sqrt{3}(3-\sin\phi)}, \tag{431}$$

$$q_\phi^{mid} = \frac{6\sin\phi}{\sqrt{3}(3+\sin\phi)}, \qquad k_\phi^{mid} = \frac{6c\cos\phi}{\sqrt{3}(3+\sin\phi)}, \tag{432}$$

$$q_\phi^{ins} = \frac{3\tan\phi}{\sqrt{9+12\tan^2\phi}}, \qquad k_\phi^{ins} = \frac{3c}{\sqrt{9+12\tan^2\phi}}. \tag{433}$$

The plastic potential replaces q_ϕ by q_ψ. Exactly zero friction is retained without an artificial positive angle. The substepping path preserves signed negative yield values inside the elastic domain for elastic/plastic intersection searches. During local plastic integration it recomputes softened strength from the current local history before committing the state. The analytical pure-shear check spans elastic loading, linear softening and the residual plateau. The classical closed-form path below still freezes strength within its step; the clay validation explicitly selects `MPM.set_configuration.stress_integration=SubStepping`. The source uses the sign convention $\sigma_m = \mathrm{tr}(\sigma)/3$, so compression has negative σ_m. The implemented shear and tensile branch values are

$$f_s = \sqrt{J_2} + q_\phi \sigma_m - k_\phi, \qquad f_t = \sigma_m - T. \tag{434}$$

For the shear branch, DP overrides the generic return mapping by a closed form. With trial mean stress σ_m^{tr}, trial deviatoric stress s^{tr}, and $r^{tr} = \sqrt{J_2^{tr}}$,

$$\Delta\lambda = \frac{r^{tr} + q_\phi \sigma_m^{tr} - k_\phi}{G + K q_\phi q_\psi - H_{code}}. \tag{435}$$

The updated mean stress and deviatoric scaling are

$$\sigma_m = \sigma_m^{tr} - K q_\psi \Delta\lambda, \qquad s = s^{tr}\frac{k_\phi - q_\phi \sigma_m}{r^{tr}}. \tag{436}$$

For the tensile branch,

$$\Delta\lambda = \frac{\sigma_m^{tr} - T}{K}, \qquad \sigma_m = T, \qquad s = s^{tr}. \tag{437}$$

The equivalent plastic strain increments stored by the code are

$$\Delta\bar{\varepsilon}_p^{shear} = \Delta\lambda\sqrt{\frac{1}{3} + \frac{2q_\psi^2}{9}}, \qquad \Delta\bar{\varepsilon}_p^{tension} = \Delta\lambda\frac{\sqrt{2}}{3}. \tag{438}$$

Mohr-Coulomb uses $\epsilon = \sqrt{3}\sigma_m$, $\sqrt{2J_2}$, and Lode angle θ:

$$f_s = \sqrt{\frac{3}{2}}\sqrt{2J_2}\left[\frac{\sin(\theta+\pi/3)}{\sqrt{3}\cos\phi} + \frac{\cos(\theta+\pi/3)\tan\phi}{3}\right] + \sqrt{\frac{1}{3}}\epsilon\tan\phi - c. \tag{439}$$

The tensile cutoff is

$$f_t = \sqrt{\frac{2}{3}}\cos\theta\sqrt{2J_2} + \sqrt{\frac{1}{3}}\epsilon - T. \tag{440}$$

The MC gradient is assembled by differentiating with respect to ϵ, $\sqrt{2J_2}$, and θ:

$$\frac{\partial f}{\partial\sigma} = \frac{\partial f}{\partial\epsilon}\frac{\partial\epsilon}{\partial\sigma} + \frac{\partial f}{\partial\sqrt{2J_2}}\frac{\partial\sqrt{2J_2}}{\partial\sigma} + \frac{\partial f}{\partial\theta}\frac{\partial\theta}{\partial\sigma}. \tag{441}$$

The potential derivative uses a smoothed Menetrey-Willam-type shape in the deviatoric plane. Its eccentricity is clamped in code and the tensile branch uses separate constants for a smooth cutoff potential. Therefore MC is generally non-associated and its tangent should be treated as nonsymmetric unless the selected parameters make $g = f$.

Softening is controlled by `SoftType`, peak/residual strengths, `PlasticDevStrain`, and `ResidualPlasticDevStrain`. Linear softening is

$$a(\bar{\varepsilon}_p) = a_p + (a_r - a_p)\,\mathrm{clip}\left(\frac{\bar{\varepsilon}_p - \bar{\varepsilon}_p^{peak}}{\bar{\varepsilon}_p^{res} - \bar{\varepsilon}_p^{peak}}, 0, 1\right), \tag{442}$$

for distinct ordered strain endpoints, where $a \in \{c, \phi, \psi\}$. The peak is returned before the start and the residual at/after the end. The strength derivative is $(a_r - a_p)/(\bar{\varepsilon}_p^{res} - \bar{\varepsilon}_p^{peak})$ inside the interval and zero on the plateaus and at the endpoints. The shared linear law takes no scale parameter. To preserve a formerly clipped factor $\eta \geq 1$, replace the end strain by $\bar{\varepsilon}_p^{peak} + (\bar{\varepsilon}_p^{res} - \bar{\varepsilon}_p^{peak})/\eta$. Thus endpoints 0/0.02 reproduce the previous clipped factor-5 law over 0/0.1. See `material.SoftType`. Exponential softening is

$$a = a_r + (a_p - a_r)\alpha\exp[-\beta(\bar{\varepsilon}_p - \bar{\varepsilon}_p^{peak})]. \tag{443}$$

The source DP plastic-modulus convention is

$$H_{code} = \left(\frac{\partial f}{\partial c}\frac{\mathrm{d}c}{\mathrm{d}\bar{\varepsilon}_p} + \frac{\partial f}{\partial\phi}\frac{\mathrm{d}\phi}{\mathrm{d}\bar{\varepsilon}_p}\right)\frac{\partial\bar{\varepsilon}_p}{\partial\lambda}. \tag{444}$$

The return denominator subtracts H_{code} from the elastic directional modulus; softening therefore reduces this denominator. DP uses the scalar history rate $\partial\bar{\varepsilon}_p/\partial\lambda = \sqrt{1/3 + 2q_\psi^2/9}$ on the shear branch, not its negative or unity. For a tensor-history formulation, the chain rule instead reads

$$\frac{\partial\bar{\varepsilon}_p}{\partial\lambda} = \frac{\partial\bar{\varepsilon}_p}{\partial\varepsilon_p} : \boldsymbol{n}_g, \tag{445}$$

where $\bar{\varepsilon}_p$ is the chosen equivalent plastic-strain invariant. Strength derivatives and history normalization must both match that invariant. The DP history and modulus are checked

by finite differences along the plastic multiplier and by the analytical shear loading curve. For arbitrary DP flow the history rate is $\sqrt{2\boldsymbol{n}_g : \boldsymbol{n}_g/3}$; hydrostatic tensile flow therefore gives $\sqrt{2}/3$, distinct from the zero-dilation shear rate $1/\sqrt{3}$. When DP or MC tensile strength is limited by the softened apex $T = c\cot\phi$, the tensile branch includes $\partial f_t/\partial\kappa = -c'\cot\phi + c\csc^2\phi\,\phi'$. If the prescribed cap remains below the apex, its derivative is zero. Random-field derivatives use particle-specific peak/residual strengths, matching the local material-parameter update.

Rate dependence uses the inertial number

$$I = \dot{\gamma} d\sqrt{\frac{\rho_s}{p}}, \tag{446}$$

and the implemented friction coefficient

$$\mu(I) = \mu_s + \frac{\dot{\gamma}(\mu_d - \mu_s)}{I_0\sqrt{p/(\rho_s d^2)} + \sqrt{\dot{\gamma}^2 + \epsilon^2}}. \tag{447}$$

Drucker-Prager recomputes q_ϕ and k_ϕ from this friction. The standalone rate-dependent solver also solves a scalar plastic shear rate $\dot{\gamma}_p$. With $C = I_0\sqrt{p/(\rho_s d^2)}$, $\tau^{tr} = \sqrt{J_2^{tr}}$, and $A = G\Delta t$, the residual is

$$R(\dot{\gamma}_p) = \tau^{tr} - A\dot{\gamma}_p - p\mu_s - p(\mu_d - \mu_s)\frac{\dot{\gamma}_p}{C + \sqrt{\dot{\gamma}_p^2 + \epsilon^2}}. \tag{448}$$

Newton iteration updates

$$\dot{\gamma}_p \leftarrow \max\left(0, \dot{\gamma}_p - \frac{R}{R'}\right), \tag{449}$$

then scales the trial deviatoric stress by

$$s = s^{tr}\max\left(0, 1 - \frac{G\Delta t\,\dot{\gamma}_p}{\tau^{tr}}\right). \tag{450}$$

14.9 State-Dependent and Critical-State Sand Models

`StateDependentMohrCoulomb`, `NorSand`, and `SanisandMS` are all critical-state or state-parameter models: strength and dilatancy depend on the current density relative to a critical-state line. A common measure is

$$\psi = e - e_c(p), \qquad e_c(p) = e_\Gamma - \lambda_c\left(\frac{p}{p_{ref}}\right)^\xi \quad \text{or} \quad v_c(p) = N - \lambda\ln\frac{p}{p_{ref}}, \tag{451}$$

depending on the model family. Positive ψ denotes a looser state and negative ψ a denser state. State-dependent Mohr-Coulomb modifies the friction and dilation angles from this state variable before using the MC surface described above:

$$\phi = \phi_c + \Delta\phi(\psi, p), \qquad \psi_d = \psi_d(\psi, p, \varepsilon_p). \tag{452}$$

NorSand represents the stress state by stress ratio $\eta = q/p$, critical stress ratio M, and an image pressure or image stress on the yield surface. In compact critical-state form, the plastic loading surface can be viewed as

$$f = \eta - \eta_i = 0, \qquad \eta_i = M\left[1 - \ln\left(\frac{p_i}{p}\right)\right], \tag{453}$$

with hardening controlled by the distance between the current state and the image state. The source parameters `G0`, `kappa`, `lambda`, `M`, `N`, `beta`, `vc0`, `v0`, `h`, and `p_ref` set pressure elasticity, critical-state line, dilatancy, initial specific volume, and hardening.

`SanisandMS` is a multi-surface SANISAND-style model. It uses pressure-dependent elasticity

$$G = G_0 p_{atm} \frac{(2.97 - e)^2}{1 + e} \sqrt{\frac{p}{p_{atm}}}, \qquad K = \frac{2G(1+\nu)}{3(1-2\nu)} \tag{454}$$

in the usual sand-model form, then evolves fabric and bounding-surface variables. A typical stress-ratio plasticity statement is

$$\dot{\alpha} = h(\alpha^b - \alpha)\dot{\lambda}, \qquad D = A(\alpha^d - \alpha) : \boldsymbol{n}, \tag{455}$$

where α is a back-stress or fabric-like tensor, α^b and α^d are bounding/dilatancy targets, and D controls volumetric plastic strain. The implementation stores the SANISAND material constants `Mc`, `c`, `lambda_c`, `e0`, `ksi`, `m`, `h0`, `ch`, `nb`, `A0`, `nd`, `zeta`, `mu0`, and `beta`. Its local update contains a Runge-Kutta style integration path for the state variables.

`GranularMaterial` is not a critical-state plasticity model. It is a pressure-aware granular solid update used when a simple granular continuum is needed without the full MC/DP return mapping. It should be read as a separate constitutive choice rather than a softening option of Mohr-Coulomb.

14.10 Modified Cam-Clay Return Mapping and Augmented Lagrangian View

MCC uses compression-positive pressure

$$p = -\frac{1}{3}\mathrm{tr}(\sigma), \qquad q = \sqrt{3J_2}, \tag{456}$$

preconsolidation pressure p_c, void ratio e, and the hardening factor

$$\upsilon = \frac{1+e}{\lambda - \kappa}. \tag{457}$$

The elastic moduli are pressure dependent:

$$K = \frac{1+e}{\kappa}\max(p, 100), \qquad G = \frac{3K(1-2\nu)}{2(1+\nu)}. \tag{458}$$

If bonding is active, the code adds a hydrate/bond shear contribution

$$G \leftarrow G + m_s \chi s_h, \qquad K \leftarrow \frac{2G(1+\nu)}{3(1-2\nu)}. \tag{459}$$

The current yield function is

$$f(p, q, p_c) = \frac{q^2}{M_\theta^2} + (p + p_{cc})\left[p - r(p_c + p_{cd} + p_{cc})\right] \le 0. \tag{460}$$

Here $r \in (0, 1]$ is the subloading ratio, p_{cd} and p_{cc} are bonding shifts, and M_θ is either the constant `StressRatio` or a Lode-angle dependent value. With `three_invariants=True`, the source uses

$$M_\theta = M - \frac{M^2}{3+M}\cos\left(\frac{3}{2}\theta\right). \tag{461}$$

For the unbonded, full-loading case ($r = 1, p_{cd} = p_{cc} = 0$), (460) reduces to the usual ellipse $q^2/M^2 + p(p - p_c) = 0$.

The plastic potential derivative used for flow is

$$\frac{\partial g}{\partial p} = 2p - p_c - p_{cd}, \qquad \frac{\partial g}{\partial q} = \frac{2q}{M_\theta^2}. \tag{462}$$

The yield gradient is

$$\frac{\partial f}{\partial p} = 2p + p_{cc} - r(p_c + p_{cd} + p_{cc}), \qquad \frac{\partial f}{\partial q} = \frac{2q}{M_\theta^2}, \tag{463}$$

plus the $M_\theta(\theta)$ term when the three-invariant option is enabled. Because f and g differ when bonding or subloading is active, the algorithmic tangent is generally nonsymmetric.

14.10.1 Nested Scalar Return Mapping

The default `ImplicitIntegration` override in `ModifiedCamClay.py` solves for a scalar plastic multiplier $\Delta\lambda$. For a trial state (p^{tr}, q^{tr}), the updated pressure and stress ratio are

$$p = \frac{p^{tr} + K\Delta\lambda p_c}{1 + 2K\Delta\lambda}, \tag{464}$$

$$q = \frac{q^{tr}}{1 + 6G\Delta\lambda/M_\theta^2}. \tag{465}$$

The preconsolidation pressure is solved by an inner Newton iteration from

$$p_c = p_c^n \exp\left[\upsilon\Delta\lambda\frac{2p^{tr} - p_c - p_{cd}}{1 + 2K\Delta\lambda}\right]. \tag{466}$$

The outer Newton equation is $f(p, q, p_c) = 0$. The derivative used by the source has the structure

$$\frac{\mathrm{d}f}{\mathrm{d}\Delta\lambda} = f_p\frac{\mathrm{d}p}{\mathrm{d}\Delta\lambda} + f_q\frac{\mathrm{d}q}{\mathrm{d}\Delta\lambda} + f_{p_c}\frac{\mathrm{d}p_c}{\mathrm{d}\Delta\lambda} + \text{bonding and subloading terms}. \tag{467}$$

The leading unbonded terms are

$$\frac{\mathrm{d}p}{\mathrm{d}\Delta\lambda} = -\frac{K(2p - p_c - p_{cd})}{1 + (2K + \upsilon(p_c + p_{cd}))\Delta\lambda}, \tag{468}$$

$$\frac{\mathrm{d}p_c}{\mathrm{d}\Delta\lambda} = \frac{\upsilon(p_c + p_{cd})(2p - p_c - p_{cd})}{1 + (2K + \upsilon(p_c + p_{cd}))\Delta\lambda}, \tag{469}$$

$$\frac{\mathrm{d}q}{\mathrm{d}\Delta\lambda} = -\frac{q}{\Delta\lambda + M_\theta^2/(6G)}. \tag{470}$$

After convergence, the deviatoric trial direction is kept and the stress is assembled as

$$\boldsymbol{\sigma}_{n+1} = \frac{q}{q^{tr}}\boldsymbol{s}^{tr} - p\boldsymbol{I}. \tag{471}$$

The void-ratio update follows the volumetric strain increment in the source:

$$e_{n+1} = e_n + (1 + e_{ref})\Delta\varepsilon_v. \tag{472}$$

14.10.2 Subloading and Bonding Variables

The subloading ratio r is clamped to $[10^{-5}, 1]$. Explicit initialization of the local return problem may set a ratio near one from the pressure ratio. Plastic history increments use the evolution law continuously, so a zero increment leaves r unchanged and $r = 1$ has zero rate:

$$r_{n+1} = r_n - u_s \left(1 + \frac{p_{cd} + p_{cc}}{p_c}\right) \ln(r_n) \sqrt{(\Delta\varepsilon_v^p)^2 + (\Delta\varepsilon_d^p)^2}. \tag{473}$$

Bonding uses

$$p_{cd} = a(\chi s_h)^b, \qquad p_{cc} = c(\chi s_h)^d, \tag{474}$$

and degrades the bond variable by accumulated plastic distortional strain:

$$\chi_{n+1} = \chi_n - m_{deg}\chi_n \Delta\varepsilon_d^p, \qquad 0 \le \chi_{n+1} \le 1. \tag{475}$$

These terms enter both the yield surface and the plastic-modulus calculation. For the generic history integrator, set $a_v = -\mathrm{tr}(\boldsymbol{n}_g)$ and $a_d = \sqrt{2\boldsymbol{n}_g : \boldsymbol{n}_g/3}$. Its hardening chain uses

$$p_c' = \upsilon(p_c + p_{cd})a_v, \qquad p_{cd}' = -m_{deg} b p_{cd} a_d, \qquad p_{cc}' = -m_{deg} d p_{cc} a_d, \tag{476}$$

$$f_{p_c} = f_{p_{cd}} = -r(p + p_{cc}), \quad f_{p_{cc}} = (1 - r)p - r(p_c + p_{cd} + 2p_{cc}), \quad f_r = -(p + p_{cc})(p_c + p_{cd} + p_{cc}). \tag{477}$$

Here primes denote rates per plastic multiplier. The modulus is $H_{code} = f_{p_c} p_c' + f_{p_{cd}} p_{cd}' + f_{p_{cc}} p_{cc}' + f_r r'$, using the current flow rates in r', not stored increments from a previous step. One-sided finite differences check the unbonded, bonded, subloading and combined cases, including the normal surface $r = 1$.

14.10.3 Augmented-Lagrangian Local Solver

`ImplicitIntegrationAL` is implemented for MCC in the current source. It solves the four unknowns

$$\boldsymbol{y} = [p,\ q,\ p_c,\ \Delta\lambda]^T \tag{478}$$

with residuals

$$R_0 = p - p^{tr} - K\Delta\lambda(-2p + p_c), \tag{479}$$

$$R_1 = q\left(1 + \frac{6G\Delta\lambda}{M_\theta^2}\right) - q^{tr}, \tag{480}$$

$$R_2 = \ln\frac{p_c}{p_c^{tr}} - \upsilon\Delta\lambda\frac{2p^{tr} - p_c - p_{cd}}{1 + 2K\Delta\lambda}, \tag{481}$$

$$R_3 = f(p, q, p_c). \tag{482}$$

The Jacobian is evaluated by finite differences, and a backtracking safeguard keeps

$$p \ge 100, \qquad q \ge 0, \qquad p_c \ge 100, \qquad \Delta\lambda \ge 0. \tag{483}$$

If the AL solve fails to reach the yield tolerance, the source applies drift correction before updating particle state. This path is useful when a coupled local solve is preferred over the nested scalar return, but it is currently a model-specific integration option rather than the generic `ElasPlasticity.ImplicitIntegration` path.

14.11 Newtonian and Bingham Fluids

For a Newtonian fluid,

$$\boldsymbol{\sigma} = -p\boldsymbol{I} + 2\mu\boldsymbol{D}. \tag{484}$$

Bingham regularization uses

$$\boldsymbol{\tau} = 2\left(\mu + \frac{\tau_y}{2\|\boldsymbol{D}\| + \epsilon}\right)\boldsymbol{D}. \tag{485}$$

15 Linear Solver

Related API. assemble_type, matrix_symmetric, multilevel, calculate_reaction_force.

15.1 Discrete Systems and Assembly Choices

Implicit MPM, IGA, pressure projection, and coupled solvers all reduce to

$$Ax = b \tag{486}$$

or to the Newton increment form

$$K_{eff}\Delta u = -R. \tag{487}$$

`MatrixFree` evaluates Ap or Kp by re-running element or particle-grid loops and does not store tangent entries. It is memory efficient but supports only operations that can be expressed as matrix-vector products. `COO` stores triplets (i, j, a_{ij}), which makes row/column constraints, host conversion, and debugging easier. `CSR` stores compressed row structure for faster repeated sparse matrix-vector products. `HashTriplet` reduces duplicate device writes before conversion to an assembled matrix.

A particle with n_s supporting nodes in d dimensions contributes at most

$$(dn_s)^2 \tag{488}$$

scalar entries for a full tangent and $dn_s(dn_s + 1)/2$ for a symmetric tangent. Symmetric storage is valid only when the material, boundary, and contact tangents are symmetric. Non-associated plasticity, friction, and most monolithic contact blocks should be treated as nonsymmetric unless a symmetric approximation is explicitly derived.

For monolithic FEM–MPM and IGA–MPM, Newton first evaluates the residual. If it is already converged, no tangent is assembled. Otherwise the same trial uses one prepared ULMPM deformation/plastic spectral response for force and tangent. The cache is refreshed after any change of trial or lagged material state, so it does not freeze the physical constitutive response across steps.

Classical FEM reference gradients and weights, and implicit IGA reference basis/gradient/weight/radius data, are independent of the current iterate. IGA HashTriplet sums quadrature contributions into one element/node-pair raw slot. In the coupled hash path, immutable FEM/IGA material connectivity instead maps directly into permanent reduced

slots. Dynamic MPM/contact contributions can overlap those slots; resetting clears values, while topology rebuilds retain the permanent coordinates. The upper-triangle representation and mirrored Dirichlet elimination preserve the same symmetric operator. These changes do not alter PSD projection, line search, or the non-associated material outer solve. For residual-merit line search with prescribed correction p_c, strong elimination gives $b_{\mathrm{DBC},f} = b_f - K_{fc}p_c$. Hence the free physical product is recovered without a duplicate raw matrix:

$$(Kp)_f = (A_{\mathrm{DBC}}p)_f + b_f - b_{\mathrm{DBC},f}.$$

The prescribed components of the direction must equal p_c; fixed rows are excluded from the merit norm. This identity also applies to nonsymmetric friction Jacobians.

15.2 Krylov Solvers

15.2.1 Conjugate Gradient

For a symmetric positive definite matrix, CG minimizes the quadratic energy in a Krylov subspace. Starting from x_0,

$$r_0 = b - Ax_0, \qquad p_0 = r_0, \tag{489}$$

$$\alpha_k = \frac{r_k^T r_k}{p_k^T A p_k}, \tag{490}$$

$$x_{k+1} = x_k + \alpha_k p_k, \tag{491}$$

$$r_{k+1} = r_k - \alpha_k A p_k, \tag{492}$$

$$\beta_k = \frac{r_{k+1}^T r_{k+1}}{r_k^T r_k}, \tag{493}$$

$$p_{k+1} = r_{k+1} + \beta_k p_k. \tag{494}$$

This is the structure used by `MatrixFreeCG`. CG is the right choice for elastic SPD tangents, pressure Poisson systems, and symmetric constrained systems. It is not robust for nonsymmetric tangents from non-associated plasticity or frictional contact.

15.2.2 Preconditioned CG

PCG applies a preconditioner $M^{-1} \approx A^{-1}$:

$$z_k = M^{-1} r_k, \tag{495}$$

$$\alpha_k = \frac{r_k^T z_k}{p_k^T A p_k}, \tag{496}$$

$$\beta_k = \frac{r_{k+1}^T z_{k+1}}{r_k^T z_k}, \tag{497}$$

$$p_{k+1} = z_{k+1} + \beta_k p_k. \tag{498}$$

`MatrixFreePCG` uses a diagonal preconditioner assembled from local diagonal contributions. The preconditioner must be positive; zero or negative diagonal entries indicate an indefinite tangent, bad constraints, or an incompatible solver choice.

15.2.3 BiCGSTAB

For nonsymmetric systems, BiCGSTAB keeps a fixed shadow residual $\hat{r}$. One iteration is

$$\rho_k = \hat{r}^T r_k, \tag{499}$$

$$\beta_k = \frac{\rho_k}{\rho_{k-1}} \frac{\alpha_{k-1}}{\omega_{k-1}}, \tag{500}$$

$$p_k = r_k + \beta_k (p_{k-1} - \omega_{k-1} v_{k-1}), \tag{501}$$

$$v_k = A p_k, \qquad \alpha_k = \frac{\rho_k}{\hat{r}^T v_k}, \tag{502}$$

$$s_k = r_k - \alpha_k v_k, \qquad t_k = A s_k, \tag{503}$$

$$\omega_k = \frac{t_k^T s_k}{t_k^T t_k}, \tag{504}$$

$$x_{k+1} = x_k + \alpha_k p_k + \omega_k s_k, \tag{505}$$

$$r_{k+1} = s_k - \omega_k t_k. \tag{506}$$

The block-triplet implementation uses right block-Jacobi preconditioning, applying M^{-1} to p_k and s_k before their matrix products and solution updates. It fuses vector updates and reductions, with four scalar returns per ordinary iteration. The shadow residual and recurrence remain fixed until prospective convergence. Verification then computes $r = b - Ax$; if this true residual fails the tolerance, it resets $\hat{r} = r$ and restarts with $p = r$. Termination also reports the true residual. This avoids repeatedly discarding the recurrence on strongly scaled nonsymmetric systems. A cached row adjacency references the reduced blocks without copying their values; CUDA reduces each block row within one warp. Pattern changes, active prefix changes and transpose mode invalidate the adjacency.

This is the solver family used when the matrix is nonsymmetric. Breakdown can occur if ρ_k, ω_k, or a denominator becomes too small; in that case the practical fix is usually not a looser tolerance but a better preconditioner, a smaller Newton step, or a corrected tangent/constraint.

15.3 COO, CSR, and Hash Reduction

COO assembly first emits all local triplets:

$$A_{ij} = \sum_e A_{ij}^{(e)} \quad \Rightarrow \quad \{(i, j, A_{ij}^{(e)})\}_e. \tag{507}$$

Because many elements or particles write the same global entry, duplicate triplets must be summed. Host-side sparse conversion can do this through SciPy. Device-side `HashReduction` hashes the pair (i, j), reduces all values with the same key, and emits the unique triplet set:

$$h(i, j) = (i\, n_{dof} + j) \bmod n_{bucket}. \tag{508}$$

The hash table must resolve collisions by comparing the full key, not just the bucket id. Precision also matters: CUDA shuffle reductions for float32 and float64 values are different operations.

CSR conversion stores row offsets o_i, column indices c_k, and values a_k. Sparse matrix-vector multiplication is

$$y_i = \sum_{k=o_i}^{o_{i+1}-1} a_k x_{c_k}. \tag{509}$$

CSR is preferred when the sparsity pattern is reused over many Krylov iterations; COO is easier when constraints, duplicates, or debug export are the dominant concern.

15.4 Dirichlet Conditions and Reaction Force

MatrixFree currently uses penalty Dirichlet conditions:

$$\Pi_D = \frac{1}{2} k_D \|u_c - \bar{u}_c\|^2, \qquad r_D = k_D(u_c - \bar{u}_c), \qquad K_D = k_D I. \tag{510}$$

COO and HashTriplet can use strong row-column elimination:

$$b_i \leftarrow b_i - K_{ic}\bar{u}_c, \quad i \neq c, \tag{511}$$

$$K_{ic} \leftarrow 0, \quad K_{ci} \leftarrow 0, \quad K_{cc} \leftarrow 1, \quad b_c \leftarrow \bar{u}_c. \tag{512}$$

The physical support reaction must be recovered from the original residual:

$$R_c = r_c + K_{cf}\Delta u_f + K_{cc}\Delta u_c. \tag{513}$$

Penalty MatrixFree reactions are approximate penalty forces and should be compared to strong-constraint reactions only after considering the selected penalty stiffness.

15.5 MGPCG Pressure Solver

The incompressible pressure projection solves

$$\nabla \cdot \left(\frac{\Delta t}{\rho} \nabla p \right) = \nabla \cdot u^*. \tag{514}$$

Discretely, $A = \Delta t\, D \rho^{-1} G$. Cell-centered and staggered-grid variants store diagonal and face coefficients. A representative cell equation is

$$A_{cc} p_c - \sum_{n \in N(c)} A_{cn} p_n = b_c, \tag{515}$$

with face coefficients removed or modified by air and solid boundary conditions. Fluid cells are unknowns, solid cells impose no-through conditions, and air cells impose free-surface pressure. Incorrectly marking an interior fluid cell as air creates a false boundary and can produce pressure spikes.

MGPCG uses a multigrid V-cycle as the PCG preconditioner. A standard V-cycle on level ℓ is

1. pre-smooth $A_\ell z_\ell = r_\ell$ by red-black Gauss-Seidel or damped Jacobi;
2. restrict residual $r_{\ell+1} = R(r_\ell - A_\ell z_\ell)$;
3. solve or smooth on the coarse level;

4. prolongate and correct $z_\ell \leftarrow z_\ell + P z_{\ell+1}$;
5. post-smooth.

Damped Jacobi uses

$$z_i^{new} = z_i + \omega \frac{r_i - (Az)_i}{A_{ii}}, \qquad \omega \approx \frac{2}{3}. \tag{516}$$

Red-black Gauss-Seidel updates checkerboard colors in sequence so the newest neighbor values are used within the same sweep. Full-weighting restriction and injection restriction are both present in the multigrid pressure paths; the right choice depends on grid staggering and boundary treatment.

The PCG outer loop with a V-cycle preconditioner is

$$M^{-1} r = \mathrm{VCycle}(r). \tag{517}$$

The tolerance combines an absolute floor and a relative reduction from the initial residual:

$$\|r_k\|^2 \le \max(\mathrm{abs_tol},\ \mathrm{rel_tol}\|r_0\|^2). \tag{518}$$

If the V-cycle does not reduce residuals, inspect cell labels, face coefficients, pressure boundary conditions, and the null-space constraint before changing Krylov tolerances.

16 Optimization Methods

Related API. max_iteration_number, line_search_work_tol, IGAMPM.monolithic_barrier_system.

16.1 Newton Linearization Used by Global and Local Solvers

Return mapping, MCC, closest-point projection, and implicit global equilibrium all use Newton linearization. For residual $R(x) = 0$,

$$J(x^k)\Delta x = -R(x^k), \qquad x^{k+1} = x^k + \alpha \Delta x. \tag{519}$$

In implicit MPM and IGA, x is a vector of nodal/control-point increments and J is the assembled or matrix-free tangent. In constitutive updates, x may be a scalar plastic multiplier or the MCC vector $[p, q, p_c, \Delta\lambda]^T$. The stopping metrics are residual norm, displacement increment, or local yield residual depending on the caller.

16.2 Newton Optimizer Classes in `src/optimizer`

The `Newton` helper minimizes an objective $f(x)$ with known gradient and Hessian. At each iteration it computes

$$\boldsymbol{g} = \nabla f(\boldsymbol{x}), \qquad \boldsymbol{H} = \nabla^2 f(\boldsymbol{x}). \tag{520}$$

The step is

$$\boldsymbol{p} = \begin{cases} -g/H, & n = 1,\ |H| > 10^{-12}, \\ -\boldsymbol{H}^{-1}\boldsymbol{g}, & 2 \le n \le 4,\ \|\boldsymbol{H}\|^2 > 10^{-12}, \\ \text{LU-solve}(\boldsymbol{H}, -\boldsymbol{g}), & 5 \le n \le 20, \\ -0.1\boldsymbol{g}, & \text{near-singular Hessian.} \end{cases} \tag{521}$$

The Armijo backtracking condition is

$$f(\boldsymbol{x} + \alpha \boldsymbol{p}) \le f(\boldsymbol{x}) + 10^{-4} \alpha\, \boldsymbol{g} \cdot \boldsymbol{p}, \tag{522}$$

with repeated halving down to a small step safeguard.

16.3 Augmented-Lagrangian Newton

`AugmentedLagrangianNewton` solves

$$\min_{\boldsymbol{x}} f(\boldsymbol{x}) \quad \text{s.t.} \quad g_i(\boldsymbol{x}) \le 0. \tag{523}$$

The implemented augmented Lagrangian is

$$\mathcal{L}_\rho(\boldsymbol{x}, \lambda) = f(\boldsymbol{x}) + \sum_i \frac{\rho}{2} \left\langle g_i(\boldsymbol{x}) + \frac{\lambda_i}{\rho} \right\rangle_+^2. \tag{524}$$

Its gradient and Hessian are

$$\nabla \mathcal{L}_\rho = \nabla f + \sum_{i \in A} \rho \left(g_i + \frac{\lambda_i}{\rho} \right) \nabla g_i, \tag{525}$$

$$\nabla^2 \mathcal{L}_\rho = \nabla^2 f + \sum_{i \in A} \rho \left[\nabla g_i \nabla g_i^T + \left(g_i + \frac{\lambda_i}{\rho} \right) \nabla^2 g_i \right], \tag{526}$$

where $A = \{ i : g_i + \lambda_i/\rho > 0 \}$. The inner problem is solved by the same Newton/Armijo procedure. The multiplier and penalty update are

$$\lambda_i^{k+1} = \max(0, \lambda_i^k + \rho g_i(\boldsymbol{x}^{k+1})), \qquad \rho \leftarrow \min(10^5, \rho\, \eta_\rho) \tag{527}$$

when constraint violation does not decrease sufficiently.

16.4 Energy Line Search and CCD-Limited IPC Steps

IPC Newton steps are first capped by the DCD/CCD procedure in Contact Search. The direct IPC MPM line search computes

$$g_0 = -\boldsymbol{r}^T \Delta \boldsymbol{u}. \tag{528}$$

If g_0 is not a descent quantity, the optional `line_search_descent_fallback` replaces the trial direction by the current right-hand side. The initial step is

$$\alpha_0 = \mathrm{ccd}(\Delta \boldsymbol{u}), \tag{529}$$

which includes ground contact CCD, particle-particle CCD, material determinant CCD, and a CFL-like displacement cap. Energy backtracking then accepts the first step satisfying

$$E(\boldsymbol{u} + \alpha \Delta \boldsymbol{u}) \le E(\boldsymbol{u}) + \mathrm{atol} + \mathrm{rtol} \max(1, |E(\boldsymbol{u})|), \tag{530}$$

halving α until the condition, the minimum-alpha threshold, or the maximum-backtracking limit is reached. The IGA backend uses the same pattern: material CCD first, then energy decrease.

For standalone IGA with prescribed displacement increments $\boldsymbol{b}$, split the increment into constrained and free components, $\boldsymbol{u} = (\boldsymbol{u}_c, \boldsymbol{u}_f)$. Initialize $\boldsymbol{u}_c = \boldsymbol{b}$ and $\boldsymbol{u}_f = \boldsymbol{0}$, then minimize $E(\boldsymbol{b}, \boldsymbol{u}_f)$. Newton corrections satisfy $\Delta \boldsymbol{u}_c = \boldsymbol{0}$, so every line-search trial retains the same prescribed boundary. Energy need only decrease within this constrained space; the change from the previous boundary position may increase stored energy because the moving boundary does work. Explicit IGA instead prescribes total displacements relative to the initial control points.

16.5 Scalar Root Finders

`utils/Root.py` provides Newton, Halley, and secant methods. With first derivative f', Newton uses

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}. \tag{531}$$

If f'' is supplied, Halley's correction is

$$\Delta x_H = \frac{f/f'}{1 - \frac{1}{2}(f/f')f''/f'}, \qquad x_{k+1} = x_k - \Delta x_H, \tag{532}$$

provided the denominator remains well behaved. Without derivatives, the secant update uses two previous evaluations:

$$x_{k+1} = x_k - f(x_k)\frac{x_k - x_{k-1}}{f(x_k) - f(x_{k-1})}. \tag{533}$$

The vectorized helper applies the same formulas componentwise and reports zero-derivative or convergence flags.

16.6 BFGS Status in the Current Source

The current source snapshot contains Newton, augmented-Lagrangian Newton, backtracking line search, CCD-limited line search, and scalar Newton/Halley/secant root finders. A search of the source tree does not show a BFGS or L-BFGS implementation. Therefore this manual does not list BFGS as a supported GeoTaichi optimizer unless such code is added later.

Part II

API Reference

17 API Reference Reading Rules

This part is optimized for lookup. It first defines the public surface, then provides function-to-dictionary links, expanded dictionary schemas, detailed facade entries, and one vertical entry per keyword or model. Each user-facing function links to the dictionary schema

it consumes, and each dictionary schema links to the detailed keyword entries and related theory. Large reference sections are hierarchical rather than flat: facade entries are grouped by module, keyword entries are grouped by letter range, and model entries are grouped by physical role before the individual entry title.

17.1 Public API Definition

Public API means the facade methods in `geotaichi`, `src/mpm/mainMPM.py`, `src/dem/mainDEM.py`, `src/mpdem/mainDEMPM.py`, `src/fem/mainFEM.py`, `src/fedem/mainFEDEM.py`, `src/fempm/mainFEMPM.py`, `src/iga/mainIGA.py`, `src/igampm/mainIGAMPM.py`, and the user-visible IGA–MPM contact helpers. Internal Taichi kernels are documented when they define a keyword, compatibility rule, or mathematical operator exposed to users.

All public solver facades inherit `diagnostics_snapshot()`. It returns a JSON-friendly common time/step/timestep/target view and delegates to a specialized engine snapshot when nonlinear, linear, contact, CCD, friction, retry, or failure diagnostics are available.

The solver facades also expose two console diagnostics with a common contract. `print_memory_info()` reports the configured capacities and, where the engine is already built, mesh, degree-of-freedom, and sparse-matrix sizes. `print_neighbor_search_info()` reports the active search method, runtime search object, Verlet/contact capacities, and current pair counts when those quantities apply. A facade without neighbor search returns without printing a search section.

17.2 Dictionary and Nested-Dictionary Reading Rules

Dictionary arguments are documented by call site and nesting path. A path such as `MPM.add_element.element.AdaptiveGrid.RefineThreshold` means: `add_element` receives an `element` dict, that dict may contain an `AdaptiveGrid` dict, and `RefineThreshold` is a key inside the nested `AdaptiveGrid` dict. The expanded schema tables use one row per keyword; rows are not grouped when the keys have different meanings.

The **Path** column identifies the exact dictionary location. The **Status** column states whether the key is required or optional. The **Keyword entry** column jumps to the detailed keyword reference when a global keyword entry exists. The **Related theory** links identify the formula or algorithm activated by that key.

17.3 Public Function to Dictionary Links

Public function	Expanded dictionary schema	Detailed keyword/model entries
`MPM.set_configuration`	`MPM.set_configuration` schema	solver_type, material_type, random_field
`MPM.set_solver`	`MPM.set_solver` schema	Timestep, SimulationTime, SaveInterval

Public function	Expanded dictionary schema	Detailed keyword/model entries
`MPM.memory_allocate`	`MPM.memory_allocate` schema, max_constraint_number schema	`max_constraint_number`, `max_particle_number`
`MPM.set_implicit_solver_parameters`	implicit solver schema	assemble_type, matrix_symmetric, max_iteration_number
`MPM.set_semi_implicit_solver_parameters`	semi-implicit solver schema	pressure_beta, pressure_solver, multilevel
`MPM.add_material`	MPM material schema, random-field schema	MohrCoulomb, DruckerPrager, ModifiedCamClay
`MPM.add_element`	`MPM.add_element` schema, AdaptiveGrid schema	AdaptiveGrid, RefineThreshold, HangingConstraintMode
`MPM.add_region`	`MPM.add_region` schema	RegionName, RegionFunction
`MPM.add_body` and `MPM.add_body_from_file`	`MPM.add_body` schema, body Template schema, body-from-file schema	Template, Period, Traction
`MPM.add_boundary_condition`	MPM boundary schema	BoundaryType, TractionConstraint
`DEM.set_configuration`, `DEM.memory_allocate`, `DEM.add_body`	DEM expanded dictionary schemas	BVH, bvh_rebuild_interval
`DEM.set_configuration`, `DEM.set_affine_body_parameters`, `DEM.add_template`, `DEM.add_body`	DEM expanded dictionary schemas	`AffineBody`, `affine_body_parameters`, `max_affine_body_number`
`DEMPM.set_configuration`, `DEMPM.memory_allocate`, `DEMPM.add_property`	MPDEM/CFDEM schemas	DragForceModel, FluidParticle
`FEM.create_mesh`, `FEM.set_solver`, `FEM.add_contact`	FEM lifecycle and keyword tables	`assemble_type`, `linear_solver`, `project_pd`, `broad_phase`, `dhat`, `dmin`
`FEDEM.set_configuration`, `FEDEM.memory_allocate`, `FEDEM.add_property`	FEM–DEM lifecycle and contact-property tables	`contact_coordination_number`, `max_contact_pairs`, `max_facet_cell_pairs`, `verlet_distance_multiplier`, `LinkedCell`, `BVH`, `IPC`
`FEMPM.set_configuration`, `FEMPM.memory_allocate`, `FEMPM.add_property`	FEM–MPM lifecycle and contact-property tables	`LinkedCell`, `BVH`, `contact_coordination_number`, `max_contact_pairs`

Public function	Expanded dictionary schema	Detailed keyword/model entries
IGA.add_primitives, IGAMPM.build, IGAMPM IPC helpers	IGA/IGAMPM schemas	line_search_work_tol, dhat, kappa

18 Quick API Locator

Workflow	Main API entries	Related theory
Build an MPM simulation	mpm-set-configuration, mpm-memory-allocate, mpm-add-material, mpm-add-element, mpm-add-body, mpm-dict-schemas, mpm-run	MPM weak form and transfers
Use implicit solid MPM	mpm-set-implicit-solver-parameters, key-assemble-type, key-matrix-symmetric, key-calculate-reaction-force	Implicit solid MPM, MPM IPC CCD/ACCD, Implicit Dirichlet constraints
Run incompressible fluid MPM	key-fluid-level-set, key-density-projection, key-multilevel, mpm-set-solver	Incompressible fluid MPM, Fluid level set and density projection, MGPCG multilevel solver
Use adaptive MPM	key-refinethreshold, key-refinecriterion, key-hangingconstraintmode, mpm-add-element	Adaptive MPM
Apply virtual stress boundary	mpm-add-virtual-stress-field, key-virtualstress, key-virtualforce	Virtual stress boundary
Build DEM contact simulation	dem-set-configuration, dem-choose-neighbor, dem-choose-contact-model, dem-add-property, dem-dict-schemas, dem-run	DEM, Contact Search, Contact Model
Use DEM Affine-Body	dem-set-configuration, dem-set-affine-body-parameters, dem-add-template, dem-add-body, dem-add-property	Affine body dynamics, Contact Search
Couple DEM and MPM/CFDEM	dempm-cfdem-set-configuration, dempm-cfdem-add-body, dempm-cfdem-add-property, mpdem-cfdem-dict-schemas, key-dragforcemodel	MPM-DEM, Incompressible DEM-sphere coupling

Workflow	Main API entries	Related theory
Build volume or cloth FEM	fem-create-mesh, fem-add-material, fem-set-solver, fem-add-contact, fem-run	Finite Element Method, FEM collision culling and CCD
Couple DEM/LS-DEM/AffineBody to deforming FEM surfaces	fedem-user-api	FEM–DEM coupling
Couple MPM points to deforming FEM surfaces	fempm-user-api	FEM–MPM coupling
Use IGA or IGA-MPM IPC	iga-add-primitives, igampm-build, igampm-run, iga-igampm-dict-schemas, iga-mpm-ipc-coupling-monolithic-barrier-system	IGA, IGA-MPM, IGA-MPM CCD/ACCD, IGA-MPM IPC contact model

19 Global Entry Points

19.1 Runtime Initialization

Most scripts start with:

```
import geotaichi as gt

gt.init(dim=3, arch="gpu", default_fp="float64")
```

Function or symbol	User meaning
`gt.init(...)`	Initializes Taichi, sets global dimension, backend, default float and integer precision, random seed, GPU memory policy, optional kernel profiler, and logging.
`gt.make_print_to_file(path)`	Redirects terminal output to a date-stamped log file under `path`.
`gt.DEM`, `gt.MPM`, `gt.DEMPM`, `gt.MPDEM`, `gt.FEM`, `gt.FEDEM`, `gt.FEMPM`, `gt.IGA`, `gt.IGAMPM`	Main user facades imported from `src`. `MPDEM` is an alias-style constructor around the same coupled `DEMPM` class with MPM/DEM ordering semantics.
`gt.sphere`, `gt.box`, `gt.cylinder`, `gt.rectangle`, `gt.circle`, etc.	Signed-distance helper functions imported for geometry, level-set, and user-defined region workflows.

gt.init keyword	Common values	Meaning
dim	2, 3	Global spatial dimension used by MPM and IGA-style modules.
arch	cpu, gpu	Taichi backend. On Apple Silicon, cpu requests are redirected to Metal in the current code.
cpu_max_num_threads	integer	Optional CPU thread cap.
offline_cache	boolean	Enables Taichi offline compilation cache.
debug	boolean	Enables Taichi debug runtime checks.
default_fp	float32, float64	Global floating point type.
default_ip	int32, int64	Global integer type.
device_memory_GB	number or None	Explicit GPU memory allocation.
device_memory_fraction	number or None	Fractional GPU memory allocation.
kernel_profiler	boolean	Enables Taichi kernel profiling.
print_ir	boolean	Prints Taichi IR during compilation.
log	boolean	Enables the date-stamped log redirect.
random_seed	integer	Taichi random seed.

20 Common Simulation Lifecycle

The high-level modules follow the same lifecycle:

```
solver = gt.MPM()
solver.set_configuration(...)
solver.set_solver({...})
solver.memory_allocate({...})
solver.add_material(model="LinearElastic", material={...})
solver.add_element(element={...})
solver.add_region(region={...})
solver.add_body(body={...})
solver.add_boundary_condition(boundary=[...])
solver.select_save_data(...)
solver.run()
```

DEM and coupled DEM-MPM scripts add contact-model calls:

```
dem.choose_contact_model(
    particle_particle_contact_model="Linear␣Model",
    particle_wall_contact_model="Linear␣Model",
)
dem.add_property(0, 0, {"NormalStiffness": 1e5, "Friction": 0.4})
```

Coupled DEMPM scripts configure the sub-solvers first, then the coupling facade:

```
dem = gt.DEM()
```

```
mpm = gt.MPM()
dempm = gt.DEMPM(dem, mpm)
dempm.set_configuration(coupling_scheme="CFDEM", domain=[1, 1, 1])
dempm.set_solver({...})
dempm.memory_allocate(memory={...}, dem_memory={...}, mpm_memory={...})
dempm.add_body(...)
dempm.choose_contact_model(...)
dempm.run(drag_model={"DragForceModel": "AbrahamModel"})
```

21 MPM User API

21.1 MPM Public Functions

Function	Purpose
`MPM(title,log=True)`	Creates the MPM facade, simulation state, scene, generator, engine, recorder, and solver handles.
`set_configuration(**kwargs)`	Sets dimension-dependent domain, boundary, gravity, solver type, material type, mapping, shape function, stabilization, fluid options, adaptive/THB/random-field flags, direct/backend flags, `ipc_contact`, and `soft_particle` mode.
`set_implicit_solver_parameters(**kwargs)`	Sets implicit solid/fluid solver controls: linear solver, Newmark parameters, tolerances, assembly type, multigrid options, and reaction-force output.
`set_semi_implicit_solver_parameters(dict)`	Sets the documented double-layer semi-implicit pressure-solver parameters, pressure beta, assembly type, and multigrid controls.
`set_fbar_parameters(**kwargs)`	Configures F-bar Jacobian averaging and Jacobian clamp.
`set_solver(dict)`	Sets timestep, final simulation time, CFL factor, adaptive timestep flag, save interval, and save path.
`memory_allocate(dict)`	Allocates material, particle, constraint, neighbor, and implicit degree-of-freedom capacities.
`add_contact(contact_type,**contact_phys)`	Activates MPM contact detection such as `MPMContact`, `GeoContact`, or `DEMContact`. Passing `"IPC"`, `"IPCContact"`, `"Barrier"`, or `"BarrierContact"` sets the direct/soft-particle IPC parameter dict instead of ordinary contact detection. Its `contact_search` defaults to device `Linked Cell`; explicit `"Brust"` remains available for very small body counts.
`add_material(model,material)`	Adds one or more MPM materials. Material ID 0 is reserved internally for rigid/default use; user deformable materials normally start at 1.

Function	Purpose
`add_element(element)`	Creates background elements and nodes; accepts normal, staggered, and adaptive element dictionaries.
`add_region(region)`	Registers named geometric regions used by body generation and boundary selection.
`add_body(body)`	Generates material points from registered regions.
`create_body()`, `create_ground()`, `add_ground(ground)`	Direct backend only. The factory methods create direct-engine body/ground containers; `add_body` accepts the populated body container and `add_ground` stores the ground object. Native MPM rejects these three direct-only operations.
`add_body_from_file(body)`	Reads particle data from files, including restart NPZ workflows.
`add_polygons(body)`	Adds polygonal contact geometry for DEM-style contact coupling.
`read_restart(file_number, file_path,is_continue=True)`	Loads MPM particles from saved NPZ output and advances print counters for continuation.
`add_boundary_condition(boundary)`	Adds grid velocity/reflection/friction, absorbing, traction, displacement, particle traction, and solid-cell boundary conditions.
`add_particle_traction(traction)`	Adds particle load constraints.
`add_virtual_stress_field(field,region_name=None,function=None)`	Adds a virtual stress field using a registered region or callable function.
`clean_boundary_condition(boundary)`	Removes matching boundary entries.
`write_boundary_condition(output_path)`	Writes current boundary constraints to disk.
`select_save_data(particle=True,grid=False,object=True)`	Selects recorder channels.
`choose_coupling_region(region_name=None,function=None)`	Marks MPM particles participating in DEM-MPM coupling.
`modify_parameters(**kwargs)`	Changes runtime parameters such as `SimulationTime`, `Timestep`, `CFL`, `SaveInterval`, `SavePath`, `gravity`, and damping.
`add_spatial_grid()`	Builds MPM spatial hashing for neighbor/coupling queries.
`add_engine()`, `add_recorder()`, `add_solver(**kwargs)`	Lazily builds the engine, output recorder, and solver implementation.
`add_postfunctions(**functions)`	Registers callback functions for post-step/post-output workflows.

Function	Purpose
`set_window(window)`	Sets visualization camera and display options.
`add_essentials(**kwargs)`	Finalizes the scene before running. It handles initial gravity fields and optional `fluid_level_set_function`, `solid_sdf_function`, and `ibm_source_function` callbacks.
`run(visualize=False,**kwargs)`	Runs the simulation. The same build-time callbacks may be supplied through `run`; `ibm_function` remains an alias of `ibm_source_function`.
`check_critical_timestep()`	Computes and applies a CFL-based stable timestep if needed.
`update_particle_properties(...)`	Updates selected particle field values.
`delete_particles(...)`	Deletes particles by body ID, region, or predicate.
`save_data()`	Writes selected output channels.
`postprocessing(...)`	Converts saved output to visualization files.

21.2 MPM Dictionary Schema Locator

The old compact keyword tables have been replaced by expanded schemas. Use the links below when looking up a public function or a nested dictionary; each target section lists one keyword per row.

Public call	Expanded schema	Related theory
`MPM.set_configuration`	configuration keyword schema	MPM
`MPM.set_solver`	solver dict schema	MPM
`MPM.memory_allocate`	memory dict schema and max_constraint_number nested schema	MPM
`MPM.set_implicit_solver_parameters`	implicit solver keyword schema	Implicit MPM
`MPM.set_semi_implicit_solver_parameters`	semi-implicit solver keyword schema	Two-phase MPM
`MPM.add_element`	element dict schema and AdaptiveGrid nested schema	Adaptive MPM
`MPM.add_region`	region dict schema	MPM
`MPM.add_body`	body dict schema, Template nested schema, and Traction nested schema	MPM
`MPM.add_body_from_file`	body-from-file schema	MPM
`MPM.add_material`	material keyword schema and random-field file schema	Constitutive equations

Public call	Expanded schema	Related theory
`MPM.add_boundary_condition`	boundary/output dict schema	Dirichlet constraints

21.3 MPM Dictionary Schemas and Random Field

Related theory. MPM, adaptive MPM, incompressible fluid MPM, virtual stress boundary, and constitutive equations.

The public MPM API often accepts a Python `dict`. The tables below expand the dict keys that are consumed directly by the current code. Keys marked "required" are read through `DictIO.GetEssential`; keys marked "optional" are read through `DictIO.GetAlternative`.

21.3.1 `MPM.set_configuration(**kwargs)`

Path	Status	Meaning and links
`MPM.set_configuration.domain`	required if unset	Simulation domain size in the current global dimension.
`MPM.set_configuration.is_axisymmetric`	optional, `False`	Canonical boolean alias for axisymmetric 2D kinematics; use it together with `dimension=2`.
`MPM.set_configuration.is_2DAxisy`	optional, `False`	Enables axisymmetric 2D volume and inertia terms.
`MPM.set_configuration.mode`	optional, `Normal`	High-level run mode; `Lightweight` is restricted to explicit ULMPM.
`MPM.set_configuration.mpm_backend`, `solver_backend`, `backend`	optional, `Native`	Backend selector. Accepted values normalize to `Native` or `Direct`; `Direct` builds the standalone engine classes under `src/mpm/engines/direct` and `src/mpm/soft_particle`.
`MPM.set_configuration.ipc`, `ipc_contact`	optional, `False`	Enables the direct MPM IPC branch and stores IPC parameters supplied later by `MPM.add_contact("IPC",...)`. See MPM IPC.
`MPM.set_configuration.soft_particle`, `soft_particle_mode`	optional, `False`	Enables `mode="SoftParticle"`. A boolean turns on the default `LevelSetTLMPM` backend; a dict accepts `Enabled` and `Backend`. See MPM soft-particle API.

Path	Status	Meaning and links
MPM.set_configuration.soft_particle_backend	optional, LevelSetTLMPM	Backend name stored with soft-particle mode. Current level-set runnable workflows use the DEM/MPDEM bridge to install the src/mpm/soft_particle kernels on the level-set scene.
MPM.set_configuration.boundary	optional, [None,None,None]	Domain boundary behavior per axis.
MPM.set_configuration.gravity	optional	Body acceleration vector.
MPM.set_configuration.background_damping	optional, 0	Nodal damping coefficient used by supported engines.
MPM.set_configuration.alphaPIC	optional, 0	PIC/FLIP blend; PIC sets it to 1 and FLIP sets it to 0.
MPM.set_configuration.mapping	optional, MUSL	Particle-grid update order: USL, USF, MUSL, or G2P2G.
MPM.set_configuration.stabilize	optional, None	Stabilization method such as B-Bar Method or F-BarMethod. See MPM.
MPM.set_configuration.gauss_number	optional, 0	Positive value activates Gauss integration for supported FEM-style MPM branches.
MPM.set_configuration.boundary_direction_detection	optional, False	Stores particle boundary-normal direction flags.
MPM.set_configuration.free_surface_detection	optional, False	Enables free-surface flags and neighbor state.
MPM.set_configuration.fluid_level_set	optional, False	Enables fluid level-set classification. Keyword entry: fluid_level_set.
MPM.set_configuration.fluid_domain_volume_fraction	optional, 0.1	Fluid-cell occupancy threshold in the incompressible FDM domain classifier. The accepted range is $[0, 1]$.
MPM.set_configuration.velocity_projection	optional, PIC/FLIP	Selects PIC, FLIP, PIC/FLIP, Affine, or Taylor G2P update.
MPM.set_configuration.moving_least_square	optional, False	Activates MLS/APIC-style transfer support.
MPM.set_configuration.shape_function	optional, Linear	Shape function: Linear, SmoothLinear, GIMP, QuadBSpline, or CubicBSpline.
MPM.set_configuration.solver_type	optional, Explicit	Selects the documented Explicit, Implicit, or SemiImplicit families. Keyword entry: solver_type.
MPM.set_configuration.shape_smooth	optional	Shape-function smoothing selector.
MPM.set_configuration.pressure_smoothing	optional	Pressure smoothing switch for supported fluid/two-phase branches.

Path	Status	Meaning and links
`MPM.set_configuration.strain_smoothing`	optional	Strain smoothing switch for supported solid branches.
`MPM.set_configuration.configuration`	optional, `ULMPM`	Updated or total Lagrangian branch: `ULMPM` or `TLMPM`.
`MPM.set_configuration.material_type`	optional, `Solid`	Selects `Solid`, `Fluid`, `TwoPhaseSingleLayer`, or `TwoPhaseDoubleLayer`. Keyword entry: material_type.
`MPM.set_configuration.visualize`	optional	Enables visualization setup.
`MPM.set_configuration.sparse_grid`	optional, `None`	Sparse-grid storage selector. A dict enables BlockScan for supported MPM branches; contact-based MPDEM/DEMPM may use it through Lagrangian coupling, while CFDEM still rejects it.
`MPM.set_configuration.particle_shifting`	optional, `False`	Enables particle shifting fields.
`MPM.set_configuration.density_projection`	optional, `False`	Enables density-based particle correction. Keyword entry: density_projection.
`MPM.set_configuration.density_projection_tolerance`	optional, `0.01`	Density projection convergence tolerance.
`MPM.set_configuration.density_projection_error_clamp`	optional, `0.01`	Limits density error used to compute particle shift.
`MPM.set_configuration.density_projection_max_shift_ratio`	optional, `0.05`	Maximum particle shift as a fraction of cell size.
`MPM.set_configuration.density_projection_interior_only`	optional, `True`	Restricts density projection to interior fluid particles.
`MPM.set_configuration.delayed_fluid_advection`	optional, `True`	Delays particle advection in the incompressible fluid branch.
`MPM.set_configuration.solid_sdf_cut_cell`	optional, `False`	Enables solid SDF cut-cell pressure coefficients.
`MPM.set_configuration.solid_cut_cell_min_fraction`	optional, `0.01`	Minimum open fraction for cut-cell faces/cells.
`MPM.set_configuration.stress_integration`	optional, `ReturnMapping`	Selects `ReturnMapping`/`ImplicitIntegration`, `SubStepping`, or model-specific `ImplicitIntegrationAL`. See stress integration.
`MPM.set_configuration.discretization`	optional, `FEM`	Top-level MPM discretization selector. Explicit/implicit solid and double-layer semi-implicit two-phase use `FEM`; implicit incompressible fluid with staggered elements uses `FDM`.

Path	Status	Meaning and links
`MPM.set_configuration.set_THB`	optional, `False`	Enables THB/adaptive hierarchy support.
`MPM.set_configuration.particle_traction_method`	optional, `Stable`	Particle traction algorithm: `Stable`, `Virtual`, or `None`.
`MPM.set_configuration.AOSOA`	optional, `False`	AOSOA storage flag.
`MPM.set_configuration.random_field`	optional, `False`	Enables per-particle material parameters from `MaterialFile`. See random-field schema.
`MPM.set_configuration.drift_correct`	optional, `True`	Enables post-integration yield drift correction.
`MPM.set_configuration.track_energy`	optional	Stores energy terms where supported.

21.3.2 `MPM.set_solver(solver)`

Path	Status	Meaning and links
`MPM.set_solver.solver.Timestep`	required	Initial timestep. Keyword entry: Timestep.
`MPM.set_solver.solver.SimulationTime`	required	Physical duration advanced by the run. Keyword entry: Simulation-Time.
`MPM.set_solver.solver.CFL`	optional, `0.5`	Safety factor for critical timestep checks.
`MPM.set_solver.solver.AdaptiveStep`	optional, `0`	Enables adaptive timestep logic.
`MPM.set_solver.solver.SaveInterval`	optional	Output interval; default is final time divided by 20.
`MPM.set_solver.solver.SavePath`	optional, `OutputData`	Output directory.
`MPM.set_solver.solver.GridSize`	Direct backend alias	Background-grid spacing; the canonical Direct spelling is `dx`, while `MPM.add_element.ElementSize` may also supply it.
`MPM.set_solver.solver.OutputInterval`	Direct backend alias	Positive integer number of steps between outputs; `interval` is the lower-level alias.

Path	Status	Meaning and links
`MPM.set_solver.solver.small_mass_ratio`	Direct implicit ULMPM; `0.0`	Finite real in $[0, 1)$. Zero disables removal. A positive ratio cuts grid nodes at or below $\max(10^{-12}, \alpha_m \bar{m})$ after P2G, using the mean over all background nodes; every particle stencil is compacted consistently. Losing all support is an error. Diagnostics report discarded grid mass and momentum; nonzero ratios are rejected for Native, explicit or TLMPM solvers. See small-mass treatment.
`MPM.set_solver.solver.plane_strain`	Direct Cartesian 2D only	Selects the 3D plane-strain material embedding. It defaults to `True` for Direct finite-strain DP/von Mises and `False` for ordinary 2D hyperelasticity; `False` is rejected for Cartesian 2D DP/von Mises.
`MPM.set_solver.solver.enable_step_retry`	optional, `False`	Enables bounded accepted-state retry for Direct implicit MPM and Direct soft-particle IPC only. Explicit and native MPM reject `True`.
`MPM.set_solver.solver.step_retry_max_retries`	optional, `2`	Maximum additional nonlinear attempts after the original attempt.
`MPM.set_solver.solver.step_retry_reduction`	optional, `0.5`	Strict geometric timestep reduction in $(0, 1)$.
`MPM.set_solver.solver.step_retry_minimum_timestep`	optional, `0`	Optional nonnegative lower timestep bound; zero disables the bound.

21.3.3 `MPM.memory_allocate(memory)`

Path	Status	Meaning and links
`MPM.memory_allocate.memory.max_material_number`	optional, `0`	Material slot capacity.
`MPM.memory_allocate.memory.max_particle_number`	optional, `0`	Particle capacity; adaptive splitting must fit inside this value.
`MPM.memory_allocate.memory.max_constraint_number`	optional, `\{\}`	Nested boundary-capacity dict; see max_constraint_number schema.
`MPM.memory_allocate.memory.verlet_distance_multiplier`	optional, `0`	Neighbor/contact-list buffer multiplier.

Path	Status	Meaning and links
`MPM.memory_allocate.memory.dof_multiplier`	optional, `2`	Extra implicit active degree-of-freedom capacity.

21.3.4 `memory.max_constraint_number` nested dict

Path	Status	Meaning
`memory.max_constraint_number.max_velocity_constraint`	optional	Capacity for `VelocityConstraint` entries.
`memory.max_constraint_number.max_reflection_constraint`	optional	Capacity for `ReflectionConstraint` entries.
`memory.max_constraint_number.max_friction_constraint`	optional	Capacity for `FrictionConstraint` entries.
`memory.max_constraint_number.max_absorbing_constraint`	optional	Capacity for `AbsorbingConstraint` entries.
`memory.max_constraint_number.max_traction_constraint`	optional	Capacity for grid traction constraints.
`memory.max_constraint_number.max_particle_traction_constraint`	optional	Capacity for particle traction constraints.
`memory.max_constraint_number.max_displacement_constraint`	optional	Capacity for implicit displacement constraints.
`memory.max_constraint_number.max_node_pressure_constraint`	optional	Capacity for pressure-node constraints in fluid/two-phase branches.

21.3.5 `MPM.set_implicit_solver_parameters(**kwargs)`

Path	Status	Meaning and links
`implicit.linear_solver`	optional, `PCG`	Linear solver: `CG`, `PCG`, `BiCG`, or `MGPCG`.
`implicit.calculate_reaction_force`	optional, `False`	Stores support reactions; interpretation depends on penalty vs strong Dirichlet.
`implicit.integration_scheme`	optional, `Newmark`	Time integration scheme for implicit solid MPM.
`implicit.displacement_tolerance`	optional, `1e-4`	Newton displacement tolerance.
`implicit.residual_tolerance`	optional	Nonlinear or pressure residual tolerance.

Path	Status	Meaning and links
`implicit.linear_solver_relative_tolerance`	optional, `0`	Relative stopping tolerance forwarded to the implicit Krylov solve; zero leaves the relative test disabled and uses the established absolute criterion.
`implicit.symmetrize_matrix_free_tangent`	optional, `False`	Requests the supported matrix-free solid path to symmetrize its tangent action. Do not use it to hide a genuinely nonsymmetric material/contact tangent.
`implicit.use_elastic_matrix_free_tangent`	optional, `False`	Uses the supported elastic approximation for the matrix-free tangent instead of the full constitutive tangent.
`implicit.quasi_static`	optional, `False`	Removes inertial terms where the selected solver branch supports it.
`implicit.newmark_parameter`	optional, `[0.5,0.25]`	Newmark $[\gamma, \beta]$. Keyword entry: newmark_parameter.
`implicit.max_iteration_number`	optional	Maximum nonlinear/pressure iterations. Keyword entry: max_iteration_number.
`implicit.assemble_type`	optional	`MatrixFree`, `COO`, or `HashTriplet`. Keyword entry: assemble_type.
`implicit.matrix_symmetric`	optional, `None`	HashTriplet symmetry hint; it must match the material/contact tangent.
`implicit.multilevel`	optional	Number of MGPCG levels.
`implicit.pre_and_post_smoothing`	optional	Multigrid smoothing iterations before and after restriction/prolongation.
`implicit.bottom_smoothing`	optional	Bottom-grid smoothing iterations.

21.3.6 `MPM.set_semi_implicit_solver_parameters(dict)`

This manual documents this call only for `material_type="TwoPhaseDoubleLayer"`. That route requires the MGPCG pressure solve described in Double-Layer Semi-Implicit Two-Phase MPM.

Path	Status	Meaning and links
`semi_implicit.pressure_beta`	optional, `1.0`	Pressure relaxation parameter. Keyword entry: pressure_beta.
`semi_implicit.pressure_bate`	optional	Backward-compatible misspelling of `pressure_beta`.

Path	Status	Meaning and links
`semi_implicit.pressure_solver`	required	`MGPCG` for the double-layer pressure projection.
`semi_implicit.assemble_type`	optional, `MatrixFree`	Pressure operator storage selector; `MatrixFreeMGPCG` is normalized to `MatrixFree` plus `pressure_solver="MGPCG"`.
`semi_implicit.multilevel`	optional	Number of MGPCG levels.
`semi_implicit.pre_and_post_smoothing`	optional	Multigrid smoothing iterations around transfer operators.
`semi_implicit.bottom_smoothing`	optional	Bottom-grid smoothing iterations.

21.3.7 `MPM.add_element(element)`

Path	Status	Meaning and links
`MPM.add_element.element.ElementType`	required	Element family: commonly `Q4N2D`, `R8N3D`, or `Staggered`.
`MPM.add_element.element.ElementSize`	required	Background grid spacing vector.
`MPM.add_element.element.ShapeFunction`	optional on Direct MPM	`QuadBSpline` selects the quadratic tensor-product basis with three nodes per axis and finite-grid boundary polynomials; every grid axis requires at least four nodes. Values, gradients, and IPC Hessians share the boundary correction. See shape functions.
`MPM.add_element.element.GhostCell`	optional	Number of ghost-cell layers.
`MPM.add_element.element.AdaptiveGrid`	optional nested dict	Adaptive-grid controls; see AdaptiveGrid schema.

21.3.8 `element.AdaptiveGrid` nested dict

Path	Status	Meaning and links
`element.AdaptiveGrid.RefineInterval`	optional	Steps between refinement checks.
`element.AdaptiveGrid.ParticleSplitInterval`	optional	Steps between split checks; defaults to `RefineInterval`.

Path	Status	Meaning and links
`element.AdaptiveGrid.RefineThreshold`	optional	Indicator threshold for refinement. Keyword entry: RefineThreshold.
`element.AdaptiveGrid.RefineCriterion`	optional, `Auto`	Indicator type; `Auto` selects the code default.
`element.AdaptiveGrid.RefineRatio`	optional	Maximum fraction of original coarse cells that may be refined; internally this reserves $2^d - 1$ added fine cells per refined coarse cell. Keyword entry: RefineRatio.
`element.AdaptiveGrid.RefineParticles`	optional	Splits particles when cells refine.
`element.AdaptiveGrid.ParticleSplitBatch`	optional	3D split batch size used to limit peak GPU memory.
`element.AdaptiveGrid.SplitInteriorOnly`	optional	In 3D, split particles only in cells whose particle volume fraction marks them as interior cells.
`element.AdaptiveGrid.SplitInteriorVolumeFraction`	optional	Minimum coarse-cell particle-volume fraction used by `SplitInteriorOnly`.
`element.AdaptiveGrid.BufferCells`	optional	Cells used to dilate refined regions.
`element.AdaptiveGrid.Penalty`	optional	Enables explicit hanging-node penalty mode.
`element.AdaptiveGrid.PenaltyIterations`	optional	Number of hanging-penalty correction iterations.
`element.AdaptiveGrid.PenaltyBeta`	optional	Penalty stiffness scale; the Young modulus is inferred from the active material.
`element.AdaptiveGrid.BridgingDomain`	optional	Explicit-only bridging domain; implicit solid MPM rejects the current penalty path.
`element.AdaptiveGrid.BridgingCells`	optional	Number of cells used for the bridging transition.
`element.AdaptiveGrid.HangingConstraintMode`	optional, `Auto`	`Auto`, `Penalty`, `Shape`, or `ShapeFunction`; implicit solid uses shape-function constraints.
`element.AdaptiveGrid.HangingShapeCapacityFactor`	optional	Extra allocation factor for strong hanging constraints.

Runtime adaptive topology, including compact-node allocation, hanging-master resolution and bridging weights, is constructed in device fields for both 2D and 3D. Existing node IDs remain stable as the grid grows. Particle splitting regroups material indices on device; only per-material launch bounds and scalar capacity/status counts are returned to Python.

NumPy node-map snapshots are explicit setup/inspection interfaces, not part of the refinement update. Node allocation and constraint workspaces respect the configured capacity; overflow or an unresolved constraint chain raises an error.

21.3.9 `MPM.add_region(region)`

Path	Status	Meaning and links
`MPM.add_region.region.Name`	required	Region identifier referenced by body and boundary dictionaries.
`MPM.add_region.region.Type`	required	Region shape: `Rectangle`, `Spheroid`, `Cylinder`, `Rectangle2D`, `Triangle2D`, `Polygon2D`, `UserDefined`, and related region types.
`MPM.add_region.region.BoundingBoxPoint`	required for bounded shapes	Local minimum/basis point of the bounding box.
`MPM.add_region.region.BoundingBoxSize`	required for bounded shapes	Local bounding-box size.
`MPM.add_region.region.rotate`	optional	3D rotation control.
`MPM.add_region.region.rotate2D`	optional	2D rotation angle.
`MPM.add_region.region.RotateCenter`	optional	Rotation center.
`MPM.add_region.region.Vertices`	required for `Polygon2D`	Polygon vertices.
`MPM.add_region.region.RegionFunction`	required for `UserDefined`	Callable that returns whether a point lies in the region.
`MPM.add_region.region.RegionVolume`	optional	Callable or value for user-defined volume estimation.
`MPM.add_region.region.all_in`	optional, `True`	Inclusion rule used by shared region helpers.

21.3.10 `MPM.add_body(body)` and body templates

Path	Status	Meaning and links
`MPM.add_body.body.Period`	optional	Insertion time window $[start, end, interval]$.
`MPM.add_body.body.Template`	required	One nested template dict or a list of template dicts. See body Template schema.
`MPM.add_body.body.WriteFile`	optional	Writes generated particles instead of immediately inserting them.

Path	Status	Meaning and links
`MPM.add_body.body.WritePath`	optional	Output path for generated particle files.
`MPM.add_body.body.Visualize`	optional	Writes particle VTK for inspection.
`MPM.add_body.body.CheckHistory`	optional	Deletes/replaces particles in the region before insertion.

21.3.11 `body.Template` nested dict

Path	Status	Meaning and links
`body.Template.RegionName`	required	Registered region name used for particle generation.
`body.Template.nParticlesPerCell`	required	Number of particles along each cell axis, not total particles in a cell. One fully filled cell contains `nParticlesPerCell**dimension` particles; fluid pressure-domain quality is sensitive to this value.
`body.Template.BodyID`	required	Body identifier stored on generated particles.
`body.Template.MaterialID`	required for deformable bodies	Material slot assigned to generated particles.
`body.Template.RigidBody`	optional, `False`	Generates rigid material points when enabled.
`body.Template.Density`	optional for rigid bodies	Rigid density used for mass generation.
`body.Template.Phase`	required for two-phase double layer	Phase index of the generated material points.
`body.Template.ParticleStress`	optional	Initial stress dictionary or vector; default is zero stress.
`body.Template.InitialVelocity`	optional	Initial particle velocity.
`body.Template.FixVelocity`	optional	Per-component `Free`/fixed flags.
`body.Template.Traction`	optional nested dict	Particle traction settings; see body Template Traction schema.

21.3.12 `body.Template.Traction` nested dict

Path	Status	Meaning
`body.Template.Traction.Pressure`	optional	Scalar pressure load converted to particle traction.
`body.Template.Traction.OuterNormal`	optional	User-supplied outward normal for the traction.
`body.Template.Traction.FluidPressure`	optional	Fluid pressure term for two-phase particle traction.
`body.Template.Traction.ConfiningPressure`	optional	Confining pressure used by virtual/particle traction workflows.

21.3.13 `MPM.add_body_from_file(body)`

Path	Status	Meaning
`MPM.add_body_from_file.body.FileType`	required	`TXT`, `NPZ`, or `OBJ`.
`MPM.add_body_from_file.body.Template`	required	Import template or list of templates.
`MPM.add_body_from_file.body.ParticleFile`	required for TXT/OBJ	Input particle or geometry file path.
`MPM.add_body_from_file.body.File`	required for NPZ	NPZ input path.
`MPM.add_body_from_file.body.Restart`	optional	Loads restart state from NPZ.
`MPM.add_body_from_file.body.ParticleNumber`	optional, -1	Number of TXT particles to read; -1 means all.
`MPM.add_body_from_file.body.Orientation`	optional	File geometry rotation/orientation.
`MPM.add_body_from_file.body.ScaleFactor`	optional	File geometry scale.
`MPM.add_body_from_file.body.Offset`	optional	File geometry translation.

21.3.14 `MPM.add_material(model,material)`

Path	Status	Meaning and links
`material.MaterialID`	required	Material slot. User deformable material IDs normally start at 1.
`material.MaterialFile`	optional	Random-field text file; see random-field schema.
`material.Density`	model-dependent	Mass density for solid/fluid models.

Path	Status	Meaning and links
material.YoungModulus, material.ElasticModulus	required for elastic/plastic solids	Elastic Young modulus. ElasticModulus is an LSMPM soft-material compatibility alias normalized to YoungModulus.
material.ConstitutiveModel, material.MaterialModel, material.SoftConstitutiveModel	LSMPM soft material selector	Selects the soft finite-strain material; SoftConstitutiveModel is a compatibility alias and new inputs should use ConstitutiveModel.
material.PoissonRatio	optional, often 0.3	Elastic Poisson ratio.
material.YieldStress	required for ElasticPerfectlyPlastic and Direct VonMises	Standard von-Mises equivalent yield stress; it has stress units.
material.HardeningModulus	optional, 0, for Direct VonMises	Linear isotropic hardening modulus; zero gives perfect plasticity.
material.ResidualYieldStress	optional	Residual yield stress for softening.
material.Cohesion	model-dependent	Peak cohesion for MC/DP models.
material.Friction	model-dependent	Peak friction angle or coefficient depending on model.
material.Dilation	model-dependent	Dilation angle for non-associated plastic flow.
material.FrictionAngle	required for Direct DruckerPrager	Friction angle in degrees.
material.fai_c	required for Direct StateDependentDruckerPrager	Critical friction angle in degrees; FrictionAngle is an alias. See the finite-strain state-dependent model.
material.e0,material.e_Tao,material.lambda_c,material.ksi,material.nd,material.nf	each defaults to 0.3	Dimensionless state-law parameters for Direct StateDependentDruckerPrager: initial void ratio, critical-state intercept, pressure coefficient/exponent, dilation/friction evolution coefficients. Requires $0.1 \leq e0 \leq 1.5$, positive e_Tao,ksi, and nonnegative lambda_c,nd,nf. All must be finite.

Path	Status	Meaning and links
material.DilationAngle	optional for Direct DruckerPrager	Independent constant dilation angle in degrees; defaults to FrictionAngle. Standalone lagged MPM–IPC, FEM–MPM, IGA–MPM, and Direct MPM–ABD use direct nonsymmetric Newton/BiCGSTAB with residual Armijo and no material outer loop; other Direct-MPM consumers retain frozen symmetric inner solves.
material.Tensile	optional	Tensile cutoff strength.
material.ResidualCohesion	optional	Residual cohesion after softening.
material.ResidualFriction	optional	Residual friction after softening.
material.ResidualDilation	optional	Residual dilation after softening.
material.PlasticDevStrain	optional	Plastic strain at the start/peak of softening.
material.ResidualPlasticDevStrain	optional	Plastic strain at residual strength.
material.SoftType	optional	Linear or Exponential softening where supported.
material.dpType	optional	DP-to-MC cone mapping: Circumscribed (ordinary Direct DP default), MiddleCircumscribed (state-dependent Direct DP default), or Inscribed.
material.RateDependent	optional	Enables the DP inertial-number rate-dependent friction branch.
material.StaticFriction	optional	Static friction used by rate-dependent or granular models.
material.DynamicFriction	optional	Dynamic friction used by rate-dependent or granular models.
material.AverageDiameter	model-dependent	Grain diameter for granular/rate-dependent closures.
material.InertialNumber	model-dependent	Reference inertial number I_0.
material.GrainDensity	model-dependent	Grain density for inertial-number models.
material.eps	optional	Small regularization value in rate-dependent friction.
material.StressRatio	required for ModifiedCamClay	MCC critical-state stress ratio M.

Path	Status	Meaning and links
material.lambda	required for ModifiedCamClay	MCC compression index.
material.kappa	required for ModifiedCamClay	MCC swelling/recompression index.
material.void_ratio_ref	required for ModifiedCamClay	Reference void ratio.
material.OverConsolidationRatio	optional	OCR used to initialize p_c.
material.pc	optional	Initial preconsolidation pressure.
material.pressure_ref	optional	Reference pressure in the MCC void-ratio relation.
material.beta	optional	Shape modifier in the implemented MCC yield surface.
material.ThreeInvariants	optional	Enables the three-invariant MCC $M(\theta)$ branch.
material.Porosity	required for two-phase materials	Initial porosity.
material.FluidBulkModulus	required for two-phase materials	Fluid bulk modulus.
material.Permeability	required for two-phase materials	Hydraulic permeability.
material.SolidDensity	optional	Solid phase density in two-phase materials.
material.FluidDensity	optional	Fluid phase density.
material.FluidViscosity	optional	Fluid viscosity.
material.Viscosity	optional	Viscosity for fluid or two-phase closures.
material.GrainDiameter	optional	Grain diameter for drag/permeability closures.
material.FluidUnitWeight	optional	Fluid unit weight.
material.DragModel	optional	Two-phase drag model selector.
material.BulkModulus	fluid-specific	Fluid bulk/modulus parameter.
material.cl	optional	Linear artificial viscosity coefficient for fluid models.
material.cq	optional	Quadratic artificial viscosity coefficient for fluid models.
material.atmospheric_pressure	optional	Reference pressure for fluid models.

Path	Status	Meaning and links
material.surface_tension	optional	Surface tension coefficient where supported.
material.gamma	optional	Fluid model parameter used by selected branches.
material.YieldStress	required for Bingham	Bingham yield stress.
material.CriticalShearRate	optional	Bingham regularization strain-rate scale.

21.3.15 Random-field material files

Item/path	Required format	Meaning
MPM.set_configuration.random_field	True	Turns on global random-field mode before add_material and body insertion.
material.MaterialFile	text file path	File read by np.loadtxt; hash-prefixed lines are comments.
MaterialFilerowcount	at least inserted particle count	Rows are consumed in particle order between start_particle and end_particle.
GranularMaterialcolumns	DensityYoungModulusPoissonRatio	Code converts Young/Poisson to per-particle shear and bulk.
ElasticPerfectlyPlasticcolumns	DensityYoungModulusPoissonRatioYieldStress	Residual/softening keys still come from the material dict.
MohrCoulombcolumns	DensityYoungModulusPoissonRatioCohesionFrictionDilationTensile	Friction and dilation are in degrees.
DruckerPragercolumns	DensityYoungModulusPoissonRatioCohesionFrictionDilationTensile	Residual strengths and dpType still come from the material dict.
LinearElasticstatus	not implemented as file reader	The current material class does not override read_random_field.
ModifiedCamClaystatus	not implemented as file reader	MCC has random-field state branches but no active file reader in this source snapshot.
NorSand/SanisandMSstatus	unsupported	These classes explicitly reject random-field mode in the current source snapshot.

21.3.16 `MPM.add_boundary_condition(boundary)` and output dicts

Path	Status	Meaning and links
`boundary.BoundaryType`	required	`VelocityConstraint`, `ReflectionConstraint`, `FrictionConstraint`, `AbsorbingConstraint`, `TractionConstraint`, `DisplacementConstraint`, `ParticleTractionConstraint`, or `SolidCell`.
`boundary.StartPoint`	required	Lower corner of the selected boundary region.
`boundary.EndPoint`	required	Upper corner of the selected boundary region.
`boundary.NLevel`	optional, `All`	Grid level selector for adaptive grids.
`boundary.CellThickness`	optional	Solid-cell thickness for staggered fluid boundaries.
`boundary.SolidCellThickness`	optional	Alias/alternative solid-cell thickness.
`boundary.Norm`	type-dependent	Axis index, signed axis, or axis-aligned vector.
`boundary.Sign`	optional	Reflection/friction side selector.
`boundary.Friction`	required for friction constraints	Boundary Coulomb friction coefficient.
`boundary.Velocity`	optional	Vector prescribed velocity; components may be `None`.
`boundary.VelocityX`	optional	Prescribed x velocity component.
`boundary.VelocityY`	optional	Prescribed y velocity component.
`boundary.VelocityZ`	optional	Prescribed z velocity component.
`boundary.Displacement`	optional	Vector prescribed displacement for implicit MPM.
`boundary.DisplacementX`	optional	Prescribed x displacement component.
`boundary.DisplacementY`	optional	Prescribed y displacement component.
`boundary.DisplacementZ`	optional	Prescribed z displacement component.
`boundary.ExternalForce`	optional	Vector grid force for traction constraints.
`boundary.ExternalForceX`	optional	x component of grid traction force.
`boundary.ExternalForceY`	optional	y component of grid traction force.

Path	Status	Meaning and links
boundary.ExternalForceZ	optional	z component of grid traction force.
particle_traction.Pressure	optional	Particle pressure traction.
particle_traction.OuterNormal	optional	Particle traction outward normal.
particle_traction.RegionName	optional	Region selector for particle traction.
particle_traction.RegionFunction	optional	Callable particle selector.
particle_traction.FluidPressure	optional	Two-phase fluid pressure contribution.
virtual_stress.ConfiningPressure	optional	Scalar confining pressure for virtual stress.
virtual_stress.VirtualStress	optional	Callable returning a 6-component stress vector.
virtual_stress.VirtualForce	optional	Additional virtual force field.
select_save_data.particle	optional, True	Enables particle output family.
select_save_data.grid	optional, False	Enables grid output family.
select_save_data.object	optional, True	Enables object/geometry output family.
recorder.write_bodyID	optional	Writes particle body IDs.
recorder.write_materialID	optional	Writes particle material IDs.
recorder.write_volume	optional	Writes particle volume.
recorder.write_displacement	optional	Writes displacement.
recorder.write_velocity	optional	Writes velocity.
recorder.write_pressure	optional	Writes pressure.
recorder.write_state_variables	optional	Writes constitutive state variables.
recorder.write_background_grid	optional	Writes background-grid data.

21.4 MPM Soft-Particle API

Related theory. MPM soft-particle backend, Level-set MPM soft bodies, and Mixed LSMPM soft-affine IPC.

The soft-particle implementation belongs to the MPM module and is stored in `src/mpm/soft_particle`. Current runnable level-set soft-particle examples use `DEM` or `MPDEM` to provide LSDEM-style templates, generation, neighbor search, and output, then install the MPM soft-particle backend through the bridge. The pure `MPM(mode="SoftParticle")` switch is exposed and validated, but `MPM.run()` currently rejects it until level-set scene ownership is fully owned by MPM.

Entry or key	Meaning
`MPM.set_configuration(mode="SoftParticle")`	Enables the soft-particle mode alias. Equivalent aliases include `softparticle`, `lsmpm`, and `levelsetmpm`.
`MPM.set_configuration(soft_particle=True)`	Explicit soft-particle mode switch. A dict may provide `Enabled` and `Backend`.
`soft_particle_backend="LevelSetTLMPM"`	Stored backend name for the level-set total-Lagrangian MPM implementation.
`shape_function`	In soft-particle mode this is the TLMPM interpolation shape; accepted values are `Linear`, `QuadBSpline`, and `CubicBSpline`.
`DEM.set_configuration(soft_levelset_reinitialization=...)`	Configures SDF advection, redistance, volume correction, and domain checks for the current runnable `scheme="LSMPM"` bridge. These keys are not forwarded by `MPM.set_configuration`; use the DEM/MPDEM LSMPM facade until MPM owns the complete level-set scene lifecycle.
`MPM.add_contact("IPC",kappa=...,dhat=...,mu=...,epsv=...)`	Enables the direct IPC branch and stores barrier/friction parameters used by direct implicit MPM or soft-particle IPC engines.
`src/mpm/soft_particle/IPC.py`	Defines reusable `Barrier` and `Friction` objects with `kappa`, `dhat`, `mu`, and `epsv`.
`src/mpm/soft_particle/IPCFunction.py`	Defines point-point, point-ground, and point-level-set derivative helpers. Strict soft-affine IPC uses the full SDF and point-triangle Hessians documented in theory sections.
`src/mpm/soft_particle/DEMPMBridge.py`	Installs MPM soft-particle kernels on the DEM/MPDEM level-set scene for the current runnable LSMPM workflows.

Soft-particle validation requires `dimension=3`, `configuration="TLMPM"`, `solver_type="Explicit"`, `material_type="Solid"`, `discretization="FEM"`, `stabilize=None`, and `gauss_number=0`. Those restrictions describe the explicit TLMPM soft-body integration path. Implicit soft-affine IPC is reached through the MPDEM/DEM soft-rigid IPC workflow, where the mixed operator owns the Newton/PCG system for soft grid dofs and affine control points.

22 DEM User API

22.1 DEM Public Functions

Function	Purpose
`DEM(title,log=True)`	Creates a DEM simulation facade.

Function	Purpose
`set_configuration(**kwargs)`	Sets domain, boundary, gravity, integration engine, neighbor search, DEM scheme, shell support, contact-point iterative model, visualization, energy tracking, and AffineBody switches.
`set_affine_body_parameters (**kwargs)`	Sets AffineBody assembly, barrier, friction, Newton, linear-solver, line-search, and CCD/ACCD parameters.
`set_solver(dict)`	Sets timestep, final time, CFL factor, adaptive step, save interval, and save path.
`memory_allocate(dict)`	Allocates materials, particles, spheres, clumps, LS-DEM bodies, walls, servo walls, BVH/search capacities, and contact-list sizes.
`add_region(region)`	Registers insertion regions.
`add_attribute(materialID,a ttribute)`	Adds material attributes such as density to a DEM material slot.
`add_template(template)`	Adds clump, level-set, general, or AffineBody surface templates.
`preprocess_soft_grid_templ ate(...)`	For a registered LSMPM template, precomputes material points, mechanical-grid topology and interpolation support, caches the result, and returns exact capacity/spacing diagnostics for `memory_allocate`.
`create_body(body)`, `add_body (body)`, `add_body_from_file (body)`	Generates, inserts, or reads DEM particles and rigid bodies; in `scheme="LSMPM"` this also inserts or generates `BodyType="SoftBody"` level-set MPM particles. In `scheme="AffineBody"`, `GenerateType="Distribute"` performs porosity-driven overlapping placement and a frictionless IPC nonpenetration projection.
`create_body_batch(body)`	Inserts a batch through the generator's batch path and returns its generator result. Use the same body schema and call-order requirements as `create_body`.
`add_joint(joint)`	Adds an AffineBody revolute joint before the first run; the same API is available to the mixed LSMPM soft–AffineBody IPC route.
`set_joint_target_angle(join tID,target_angle)`	Updates a revolute joint motor target in degrees before or after AffineBody engine initialization.
`add_wall(body)`, `add_wall_fro m_file(body)`	Adds wall geometry from dictionaries or files.
`choose_neighbor()`	Builds the selected neighbor-search backend. In `Af fineBody`, the affine engine owns its vertex-face and edge-edge neighbor helper.
`choose_contact_model(pp,pw)`	Selects particle-particle and particle-wall contact models. `AffineBody` uses its IPC-style contact branch instead.

Function	Purpose
`add_property(materialID1,materialID2,property,dType="all")`	Adds surface contact properties; in `AffineBody`, stores friction properties for `particle-particle` and `particle-wall` affine contact.
`load_history_contact()`, `read_restart(...)`	Restart history contact state and saved particles/walls.
`select_save_data(...)`	Selects DEM output channels.
`modify_parameters(**kwargs)`	Modifies timestep, final time, CFL, save/output, and gravity.
`add_engine(callback)`, `add_recorder()`, `add_solver(**kwargs)`	Builds runtime engine, recorder, and solver.
`add_postfunctions(**functions)`	Registers post-step callback functions.
`add_essentials(**kwargs)`	Finalizes contacts, walls, servo controls, and solver essentials.
`set_static_wall(static_wall=True)`, `servo_switch(status)`	Controls wall status and servo wall activation.
`set_window(window)`	Sets visualization parameters.
`run(visualize=False,**kwargs)`	Runs DEM.
`is_lsmpm_soft_affine_ipc()`	Reports whether the configured scene is the mixed LSMPM soft–AffineBody IPC route. This is primarily an orchestration query used by the coupled facade.
`check_critical_timestep()`, `get_critical_timestep()`	Computes stable DEM timestep estimates.
`update_material_properties`, `update_particle_properties`, `update_wall_status`, `update_contact_properties`	Runtime property updates.
`delete_particles`, `delete_walls`, `save_data`, `postprocessing`	Data management and visualization conversion.

22.2 DEM Keywords

Call site	Keyword	Meaning and accepted values
`set_configuration`	`domain`, `boundary`, `gravity`	Domain, per-axis boundary, and body force. Boundary values are `None`, `Reflect`, `Destroy`, and `Period`.
	`engine`	`SymplecticEuler`, `VelocityVerlet`, or `PredictCorrector`.

Call site	Keyword	Meaning and accepted values
	search	Brust, LinkedCell, HierarchicalLinkedCell, or BVH.
	search_direction	Up or Down.
	scheme	DEM, LSDEM, LSMPM, PolySuperEllipsoid, PolySuperQuadrics, or AffineBody.
	soft_rigid_contact	Contact route for scheme="LSMPM" soft-rigid workflows. "DEM" uses explicit DEM/LSDEM contact laws; "IPC" uses the mixed soft-affine IPC operator when AffineBody rigid particles are present. Alias: lsmpm_soft_rigid_contact.
	affine_body, affine_body_parameters, affine_assemble_type	AffineBody shortcut and parameter overrides. affine_body=True selects scheme="AffineBody"; affine_assemble_type forwards to the affine assemble_type parameter.
	iterative_model	LagrangianMultiplier, PCN, or GJK.
	track_energy, visualize, enable_shell	Output/physics flags.
memory_allocate	max_material_number, max_particle_number, max_sphere_number, max_clump_number	Core DEM material and particle capacities.
	max_rigid_body_number, max_affine_body_number, max_rigid_template_number, levelset_grid_number, surface_node_number, point_coordination_number, max_soft_body_number, max_material_point_number	LSDEM/LSMPM and AffineBody capacities. max_affine_body_number is preferred for AffineBody; if it is absent, max_rigid_body_number is accepted for compatibility.
	max_patch_number, max_facet_number, max_servo_wall_number, max_plane_number, max_digital_elevation_facet_number	Wall and boundary geometry capacities.

Call site	Keyword	Meaning and accepted values
	`body_coordination_number`, `wall_coordination_number`, `wall_per_cell`, `compaction_ratio`, `verlet_distance_multiplier`, `bvh_rebuild_interval`	Contact-list and neighbor-search capacities.
	`hierarchical_level`, `hierarchical_size`	Hierarchical linked-cell settings.
`add_region`	`Name`, `Type`, `BoundingBoxPoint`, `BoundingBoxSize`, `rotate`, `rotate2D`, `RotateCenter`, `Vertices`, `RegionFunction`, `RegionVolume`, `all_in`	Same shared region schema as MPM, with additional fill-count/fill-volume semantics inside DEM insertion.
`add_body/create_body`	`GenerateType`, `BodyType`, `RegionName`, `Template`, `Period`, `CheckHistory`, `WriteFile`, `WritePath`, `Visualize`, `TryNumber`, `PoissonSampling`, `Porosity`, `Fraction`	Body generation and insertion controls.
`BodyTemplate`	`Radius`, `BodyPoint`, `GroupID`, `MaterialID`, `InitialVelocity`, `InitialAngularVelocity`, `FixVelocity`, `FixAngularVelocity`, `FixMotion`, `BodyOrientation`, `Name`, `ScaleFactor`, `BoundingRadius`, `MinRadius`, `MaxRadius`, `BodyNumber`, `TotalParticles`, `RadiusDistribution`	Sphere, clump, level-set, and template-body parameters.
`add_wall`	`WallType`, `WallShape`, `WallID`, `WallVertice`, `WallCenter`, `WallRadius`, `WallHeight`, `OuterNormal`, `ControlType`, `TargetStress`, `LimitVelocity`, `Gain`	Wall and servo wall geometry/control parameters.
`set_solver`	`Timestep`, `SimulationTime`, `SaveInterval`, `SavePath`, `CFL`, `AdaptiveStep`	Time integration and output.

22.3 DEM and MPDEM Contact Models

Contact model	Surface property keywords
`LinearModel`	Either `NormalStiffness`/`TangentialStiffness` or `EffectiveModulus`/`NormalToShearRatio`; `StaticFriction` or `Friction`; `DynamicFriction` or `Friction`; `RollingFriction`; `NormalViscousDamping`; `TangentialViscousDamping`.
`HertzMindlinModel`	`ShearModulus`, `Poisson`, `StaticFriction` or `Friction`, `DynamicFriction` or `Friction`, `RollingFriction`, `Restitution`.
`LinearRollingModel`	Linear stiffness parameters plus `RollingStiffness`, `TwistingStiffness`, `RollingFriction`, `TwistingFriction`, `RollingViscousDamping`, `TwistingViscousDamping`.
`JiangRollingModel`	Rolling-resistance contact model for DEM particles; shares stiffness/friction concepts with the rolling base class.
`EnergyConservingModel`	LSDEM/LSMPM penalty contact. LSMPM scales the force and tangent by reference surface quadrature; `FreeParameter` must be at least 2 so the zero-gap explicit tangent is finite.
`BarrierModel` or `BarrierFunctionModel`	LSDEM/LSMPM barrier contact; the LSMPM critical-step bound includes the area-scaled barrier curvature and monitored SDF-gradient error.
`FluidParticle`	MPDEM fluid-particle normal contact: `NormalStiffness`, `NormalViscousDamping`, optional `NonSlipFraction`.
`UserDefined`	Placeholder for user implementation in contact managers.

22.4 DEM Dictionary Schemas

Related theory. DEM, contact search, and contact models.

DEM dict	Key	Meaning
`set_configuration`	`domain`	Required if the simulation domain has not been set.
	`boundary`	Per-axis boundary mode, default `[None,None,None]`.
	`gravity`	Gravity vector, default $[0, 0, -9.8]$.
	`engine`	`SymplecticEuler`, `VelocityVerlet`, or `PredictCorrector`.
	`search`	`Brust`, `LinkedCell`, `HierarchicalLinkedCell`, or `BVH`.
	`search_direction`	`Up` or `Down`.
	`digital_elevation_contact`	Digital-elevation walls currently force the `heightfield` contact mode; another requested value is ignored with a warning.

DEM dict	Key	Meaning
	`scheme`	`DEM`, `LSDEM`, `LSMPM`, `PolySuperEllipsoid`, `PolySuperQuadrics`, or `AffineBody`.
	`soft_shape_function`	`LSMPM` mechanical-grid shape function. Accepted values are `Linear`, `QuadBSpline`, and `CubicBSpline`; aliases `lsmpm_shape_function` and `shape_function` are accepted.
	`soft_grid_storage`	`Dense` or contiguous support-`Compact` storage. Compatibility values `Sparse`, `BlockSparse`, and `FixedSparse` normalize to `Compact`; they do not allocate pointer SNodes.
	`soft_grid_type`	`Hexahedron` or the normalized tetrahedral option. Tetrahedral support forces the linear basis and four-node support.
	`soft_mechanical_grid_spacing_ratio`	Positive mechanical-grid spacing divided by template equivalent radius; default 0.15.
	`soft_pic_fraction`	Soft TLMPM PIC fraction in $[0, 1]$; default 0 is the established FLIP-style update.
	`soft_rigid_contact`, `lsmpm_soft_rigid_contact`	Selects `DEM` explicit contact or `IPC` mixed soft-affine implicit contact for LSMPM soft-rigid workflows.
	`soft_levelset_reinitialization`	`LSMPM` soft-level-set control. A Boolean only enables/disables redistance. A nested dict may set `enabled`, `advection_scheme`, `advection_cfl`, `advection_interval`, `grad_threshold`/`threshold`, `check_interval`, `monitor_band`/`monitor_band_cells`, `reinit_band`/`reinit_band_cells`, `cfl`/`pseudo_time_cfl`, `iterations`, `volume_correction`/`preserve_volume`, `volume_epsilon_cells`, `volume_tolerance`, `volume_max_shift_cells`, `volume_iterations`, `volume_interval`, `domain_check`, and `domain_tolerance_cells`.
	`soft_levelset_advection_scheme`	Default `ReferenceMap`; `InverseMapping` and `SemiLagrangian` alias it. `MacCormack` selects the former incremental route; `WENO5` selects finite-difference transport.
	`soft_levelset_advection_cfl`	Positive WENO5 subcycling limit; selecting WENO5 without an explicit value sets 0.20. Other schemes default to 1.0; reference-map reconstruction does not use CFL subcycling.
	`soft_levelset_advection_interval`	Positive number of mechanical steps accumulated between level-set advections; default 1.

DEM dict	Key	Meaning
	`soft_levelset_reinit_*`	Flat aliases for the same LSMPM redistance parameters: `soft_levelset_reinit_grad_threshold`, `soft_levelset_reinit_check_interval`, `soft_levelset_reinit_monitor_band`, `soft_levelset_reinit_band`, `soft_levelset_reinit_cfl`, and `soft_levelset_reinit_iterations`.
	`soft_levelset_domain_check`	Enables the compact SDF-domain departure audit; default `True`.
	`soft_levelset_domain_tolerance_cells`	Non-negative tolerated departure beyond the stored local level-set domain, in local cells; default 10^{-5}.
	`affine_body`	Boolean shortcut. When true, the DEM scheme is set to `AffineBody`.
	`affine_body_parameters`	Nested dict passed to `set_affine_body_parameters`.
	`affine_assemble_type`	Shortcut forwarded as AffineBody `assemble_type`.
	`track_energy`, `visualize`, `enable_shell`	Energy/output/shell flags.
	`iterative_model`	`LagrangianMultiplier`, `PCN`, or `GJK`.
`set_solver`	`Timestep`, `SimulationTime`, `SaveInterval`	Required time and output controls.
	`CFL`, `AdaptiveStep`, `SavePath`	Optional timestep safety, adaptive stepping, and output path.
`memory_allocate`	`max_material_number`	Required material capacity.
	`max_particle_number`	Required for `scheme="DEM"`.
	`max_sphere_number`, `max_clump_number`	Optional DEM body capacities.
	`max_rigid_body_number`	Required for level-set or implicit-surface schemes; accepted as a compatibility fallback for `AffineBody`.
	`max_affine_body_number`	Preferred AffineBody body capacity.
	`max_rigid_template_number`	Template capacity, default 1.
	`levelset_grid_number`, `surface_node_number`	Required for `LSDEM`/`LSMPM`.

DEM dict	Key	Meaning
	`point_coordination_number`	LSDEM/LSMPM point contact capacity, default $[4, 2]$. The first value bounds point/body candidates and the second bounds point/wall candidates; LSMPM undersizing raises an assertion rather than dropping soft contacts.
	`max_soft_body_number`, `max_material_point_number`	Required for `LSMPM`.
	`soft_grid_number`	LSMPM mechanical-grid capacity. It defaults to `levelset_grid_number`, but compact support and a separately refined mechanical grid may require a distinct value.
	`max_soft_velocity_constraint`	Non-negative capacity for soft mechanical-grid velocity constraints; default 0.
	`soft_template_material_point_number`	Cached template material-point support capacity. If absent, registered preprocessed templates are summed, then the runtime material-point capacity is used as fallback.
	`soft_template_surface_node_number`	Cached template surface-support capacity, inferred from preprocessed templates or from surface nodes times template count.
	`soft_template_levelset_node_number`	Cached template SDF-support capacity, inferred from preprocessed templates or from level-set nodes times template count.
	`max_patch_number`, `max_facet_number`, `max_servo_wall_number`, `max_plane_number`, `max_digital_elevation_facet_number`	Wall geometry capacities.
	`compaction_ratio`, `hierarchical_level`, `bvh_rebuild_interval`	Search and allocation controls.
	`hierarchical_size`	Required when `search="HierarchicalLinkedCell"`.
	`body_coordination_number`, `wall_coordination_number`, `verlet_distance_multiplier`, `wall_per_cell`	Contact-list and Verlet buffer capacities.
`add_attribute`	`Density`	Required material density.

DEM dict	Key	Meaning
	`ForceLocalDamping`, `TorqueLocalDamping`	Optional local damping, both default 0.7.
	`ConstitutiveModel`, `YoungModulus`, `PoissonRatio`, `Coefficient`, `Tensile`, `Tensile1`, `Tensile2`	LSMPM soft finite-strain material options. Supported soft models are `NeoHookean`, `HenckyElastic`, `MooneyRivlin`, `Gent`, and `Hydrogel`.
`add_body/create_body`	`GenerateType`, `BodyType`, `RegionName`, `Template`	Required generation mode, DEM body type, region, and body template.
	`Period`, `CheckHistory`, `WriteFile`, `WritePath`, `Visualize`, `SavePath`	Insertion time window, overlap/history behavior, and output controls.
	`TryNumber`, `PoissonSampling`, `Porosity`	Random packing controls.
`DEMBodyTemplate`	`Name`	Template name for clumps, level-set bodies, or generated shapes.
	`BodyNumber`	Number of bodies to generate.
	`BodyPoint`	Explicit center point for a single inserted body.
	`Radius`, `MinRadius`, `MaxRadius`, `RadiusDistribution`, `Fraction`	Sphere radius and grading controls.
	`BoundingRadius`, `MinBoundingRadius`, `MaxBoundingRadius`, `ScaleFactor`	Level-set or template scaling controls.
	`GroupID`, `MaterialID`	Output grouping and material slot.
	`InitialVelocity`, `InitialAngularVelocity`	Initial translational and angular velocity.
	`FixMotion`, `FixVelocity`, `FixAngularVelocity`	Motion constraints.
	`BodyOrientation`, `CheckOverlap`, `TotalParticles`	Orientation, overlap checking, and generated surface/material point count.

DEM dict	Key	Meaning
	`MaterialPointsPerCell`, `ParticlePerCell`	LSMPM soft material-point sampling density inside each level-set grid cell. `MaterialPointsPerCell` is preferred; `ParticlePerCell` is accepted as a compatibility alias.

AffineBody parameter key	Meaning
`assemble_type`	Linear-system backend selector. Accepted values are `COO`, `HashTriplet`, and `MatrixFree`. `COO` assembles directly into `CoordinateSparseMatrix` and solves by the project PCG solver. `HashTriplet` stores unsummed $(i, j, value)$ entries and uses `BuildTriplet` to sort/merge duplicate entries before PCG; it requires Taichi CUDA. `MatrixFree` is currently accepted for compatibility but is routed through the COO-backed PCG path in the AffineBody engine. `affine_assemble_type` in `set_configuration` is a shortcut for this key.
`YoungModulus`, `young_modulus`	Default affine body rigidity modulus when a body template does not override `YoungModulus`; it weights the $\|\boldsymbol{F}^T\boldsymbol{F} - \boldsymbol{I}\|^2$ affine rigidity energy.
`dhat`, `barrier_stiffness`, `contact_damping_stiffness`	IPC-style activation distance, barrier stiffness κ, and contact damping stiffness. `contact_damping`, `k_CD`, and `k_cd` are accepted aliases.
`soft_background_damping`	Background damping coefficient for LSMPM soft grid dofs in the mixed soft-affine IPC operator. This is a soft-body damping term, not a material-pair contact property.
`friction_epsv`	Regularization velocity used by the AffineBody friction model below the slip-speed threshold.
`max_newton_iteration`, `newton_tolerance`	Nonlinear Newton iteration cap and convergence tolerance on the affine control-point update.
`linear_tolerance`, `linear_max_iteration`, `line_search_max_iteration`	PCG tolerance/iteration cap and Armijo line-search iteration cap. The AffineBody path does not fall back to SciPy when PCG fails.
`direct_hessian_dofs`, `finite_difference`, `hessian_shift`, `max_step`	Hessian assembly diagnostics/fallback controls, diagonal regularization, and maximum Newton step length. The normal contact, friction, rigidity, and damping Hessians are assembled analytically in the production path.
`ccd`, `ccd_type`, `ccd_eta`, `accd_tolerance`, `ccd_max_iteration`	CCD/ACCD switches and tolerances. `ccd_type` accepts `none`, `off`, `ccd`, or `accd`; `accd` uses a conservative thickness controlled by `accd_tolerance`. LevelSet AffineBody ACCD bounds source motion plus target SDF-grid-corner motion.

AffineBody parameter key	Meaning
`levelset_auto_initialize`, `levelset_initial_*`	Optional two-body/general-scene level-set overlap removal before time integration. Controls translation anchor stiffness, continued-barrier ratio, positive-gap tolerance, stiffness growth, stage cap, and nonlinear iteration cap. Dense porosity-driven generation instead uses per-body packing-projection controls.

Contact property dict	Required/optional keys	Meaning
`LinearModel`	`StaticFriction` or `Friction`; `DynamicFriction` or `Friction`; optional `NormalStiffness`, `TangentialStiffness`, `EffectiveModulus`, `NormalToShearRatio`, `RollingFriction`, `NormalViscousDamping`, `TangentialViscousDamping`	Linear spring-slider contact. Stiffness may be given directly or inferred from effective modulus and stiffness ratio.
`LinearRollingModel`	Linear model keys plus `RollingStiffness`, `TwistingStiffness`, `TwistingFriction`, `RollingViscousDamping`, `TwistingViscousDamping`	Adds rolling and twisting springs, limits, and dashpots.
`HertzMindlinModel`	`ShearModulus`, `Poisson`, `StaticFriction` or `Friction`, `DynamicFriction` or `Friction`; optional `RollingFriction`, `Restitution`	Nonlinear Hertz-Mindlin normal/tangential contact.
`JiangRollingModel`	`YoungModulus`, `StiffnessRatio`, `Friction`, `ShapeFactor`, `CrushFactor`, `NormalViscousDamping`, `TangentialViscousDamping`	Rolling resistance model with shape/crushing factors.

Contact property dict	Required/optional keys	Meaning
`EnergyConservingModel`	`NormalStiffness`, `TangentialStiffness`, `Friction`; optional `FreeParameter` (default 2.5, required ≥ 2), `NormalViscousDamping`, `TangentialViscousDamping`	Penalty contact for energy-conserving LSDEM/LSMPM interactions.
`BarrierFunctionModel`	`Stiffness`, `NormalCutOff`, `Friction`; optional `StiffnessRatio`, `NormalViscousDamping`, `TangentialViscousDamping`	Barrier-style LSDEM/LSMPM contact.
`FluidParticle`	`NormalStiffness`, `NormalViscousDamping`; optional `NonSlipFraction`	MPDEM fluid-particle contact used by semi-resolved coupling.
`AffineBodycontact`	`Friction`; `dType` may be `particle-particle`, `particle-wall`, or `all`	Friction coefficient used by AffineBody body-body or body-wall IPC-style contact.

Facet and patch wall contacts with spherical particles additionally compute the sphere-plane contact disk area covered by each triangle. The resulting fraction χ_f must be above `Threshold` for the contact to stay active and is used to scale the stored contact force and equivalent stiffness. See Sphere-Triangle Area Fraction for the exact edge-integration formula.

DEM generation/wall dict	Keys	Meaning
`add_region`	`Name`, `Type`, `BoundingBoxPoint`, `BoundingBoxSize`; optional `rotate`, `rotate2D`, `RotateCenter`, `Vertices`, `RegionFunction`, `RegionVolume`, `all_in`	Same region schema as MPM, but DEM uses the region to estimate generated particle volume/number and to place randomly distributed bodies.
`add_template` DEM clump	`Name`, `Pebble`	Defines a multi-sphere clump template. Each pebble entry supplies local center and radius data consumed by the clump generator.
`add_template` LS-DEM/general shape	`Name`, `Object`	Defines a level-set, poly-superellipsoid, or poly-superquadric rigid template. Object data provide mesh, implicit surface, bounding radius, inertia, and equivalent radius information.

DEM generation/wall dict	Keys	Meaning
`add_template` AffineBody	`Name`, `Object`; optional `ContactRepresentation="TriangleMesh"` or `"LevelSet"`	Defines an AffineBody template. Both modes require a non-empty mesh for mass, surface quadrature, and visualization. `LevelSet` additionally requires the object regular signed-distance grid and uses that field, rather than triangle closest features, for body–body narrow phase.
`create_body` direct sphere	`BodyType="Sphere"`, `Template`; template keys `Radius`, `BodyPoint`, `GroupID`, `MaterialID`	Inserts one or more explicit spheres immediately. Optional keys set translational/angular velocity and fixed velocity flags.
`create_body` clump/-rigid body	`BodyType="Clump"` or `RigidBody`; template keys `Name`, `BodyPoint`, `Radius` or `ScaleFactor`, `GroupID`, `MaterialID`	Instantiates a previously defined template. Rigid body templates may also use `BoundingRadius`, `BodyOrientation`, `CheckOverlap`, and `FixMotion`.
`create_body` LSMPM soft body	`BodyType="SoftBody"`, `Template`; template keys `Name`, `BodyPoint`, `Radius` or `ScaleFactor/BoundingRadius`, `GroupID`, `MaterialID`, optional `MaterialPointsPerCell`, `BodyOrientation`, `InitialVelocity`, `InitialAngularVelocity`, and `FixMotion`	Instantiates a deformable level-set MPM body. The body receives a level-set grid, compact soft MPM grid, material points, precomputed TLMPM shape functions, surface material-point ids, and ordinary DEM body id for search and output.
`create_body/add_body` AffineBody	template keys `Name`, `BodyPoint`, `GroupID`, `MaterialID`; optional `ScaleFactor`, `Radius`, `BoundingRadius`, `BodyOrientation`, `InitialVelocity`, `InitialAngularVelocity`, `FixMotion`, `YoungModulus`, `Friction`	Instantiates a previously defined AffineBody template and stores per-body affine material/contact defaults.

DEM generation/wall dict	Keys	Meaning
`add_joint` AffineBody revolute joint	Required `BodyID1`, `WorldAnchor`, and `WorldAxis`; optional `BodyID2=-1` (world), `JointType="Revolute"`, `Stiffness`, `PositionStiffness`, `AxisStiffness`, `MotorStiffness=0`, `TargetAngle=0`, `AngleLimit=[lower,upper]`, `LimitStiffness`, `Damping=0`, and `CollideConnected=False`	Constrains relative translation and axis tilt while leaving rotation about the joint axis free. Public angles and limits are degrees; limits satisfy $-180 \le lower < upper \le 180$. By default only the connected body pair is excluded from IPC self-collision; contact with all other bodies and walls remains enabled.
`add_body` generator dict	`GenerateType`, `BodyType`, `RegionName`, `Template`; optional `Period`, `TryNumber`, `Porosity`, `PoissonSampling`, `CheckHistory`, `WriteFile`, `Visualize`	Creates a time-windowed generator. It can generate, distribute, or lattice-pack particles depending on `GenerateType`.
`add_body` LSMPM soft generator	`BodyType="SoftBody"`, `GenerateType="Generate"` or `"Lattice"`, `RegionName`, `Template`; template keys include `BodyNumber` for generate mode and `Fraction` for lattice mode	Randomly generates or lattice-packs multiple deformable level-set MPM particles. This path is valid only with `scheme="LSMPM"` and consumes `max_soft_body_number`, `max_material_point_number`, `levelset_grid_number`, and `surface_node_number`.
`add_body_from_file`	`FileType`, `Template`, `ParticleFile` or `File`; optional `Restart`, `ParticleNumber`, `Orientation`, `ScaleFactor`, `Offset`, `CheckHistory`	Reads DEM particles or restart state from file.
`Planewall`	`WallType="Plane"`, `MaterialID`, `WallCenter`, `OuterNormal`; optional `WallID`	Adds an infinite plane wall.
`Facetwall:polygon`	`WallType="Facet"`, `WallShape="Polygon"`, `WallID`, `MaterialID`, `WallVertice`, `OuterNormal`; optional `InitialVelocity`, `Visualize`	Triangulates a polygon into facet walls.

DEM generation/wall dict	Keys	Meaning
`Facetwall:cylinder/circular/file`	`WallShape="Cylinder"` uses `WallBase`, `WallRadius`, `WallHeight`, `Resolution`, `Axis`, `Direction`; `Circular` uses `WallCenter`, `WallRadius`, `Resolution`, `OuterNormal`; `File` uses `WallFile`	Creates triangular facet walls from analytic surfaces or meshes. File meshes also accept `ScaleFactor`, `Translation`, `Orientation`, and `Counterclockwise`.
`Patchwall`	`WallType="Patch"`, `WallID`, `MaterialID`, `WallFile`; optional `ShellOffset`, `ScaleFactor`, `Translation`, `Orientation`, `Counterclockwise`	Creates shell/patch wall geometry with stored facet connectivity.
`DigitalElevationwall`	`WallType="DigitalElevation"`, `WallID`, `MaterialID`, `CellSize`, `DigitalElevation`; optional `MainAxis`, `GridNumber`, `NoData`	Builds wall facets from an elevation grid.
`Servowalladditions`	`ControlType`, `TargetStress`, `LimitVelocity`; optional `Alpha`, `Gain`	Activates servo control for generated facet walls when servo wall memory is allocated.
`add_wall_from_file` restart	`WallFile`; optional `ServoFile`, `FileType`, `Visualize`	Rebuilds wall and servo state from saved NPZ/TXT data.
`select_save_data`	`particle`, `sphere`, `clump`, `surface`, `grid`, `bounding`, `wall`, `servo`, `particle_particle_contact`, `particle_wall_contact`, `material_point`, `soft`	Selects DEM output families. In LSMPM, `particle/soft/material_point` writes `LSMPMSoftPoint*.npz`; `GraphicLSDEMSurface*.vtu` excludes soft bodies and PostPlot reconstructs soft surfaces separately as `GraphicLSMPMSoftSurface*.vtu`.
`read_restart`	`file_number`, `file_path`, `particle`, `sphere`, `clump`, `wall`, `servo`, `ppcontact`, `pwcontact`, `is_continue`	Rebuilds DEM bodies, walls, servo state, and contact history from output files.

23 MPDEM and CFDEM User API

23.1 Coupled Public Functions

Function	Purpose
`DEMPM(dem,mpm,title,log=True)`	Builds a coupled facade around an existing DEM and MPM instance.
`set_configuration(**kwargs)`	Synchronizes domain, gravity, visualization, search, coupling scheme, and interaction flags.
`set_solver(dict)`	Sets coupled timestep/time/output and forwards the same solver dictionary to the DEM and MPM sub-solvers.
`memory_allocate(memory,dem_memory=None,mpm_memory=None)`	Optionally allocates sub-solver memory and sets coupled contact-list capacities.
`add_body(mpm_body=None,dem_particle=None,write_file=False,check_overlap=False,adaptive_boundary_radius=False)`	Inserts coupled MPM and DEM bodies.
`choose_contact_model(pp,pw=None)`	Chooses MPM-DEM and MPM-wall contact models unless `coupling_scheme="CFDEM"`.
`add_property(DEMmaterial,MPMmaterial,property,dType="all")`	Adds surface properties for MPM-DEM contact.
`modify_parameters(**kwargs)`	Updates coupled and sub-solver time, gravity, MPM damping, PIC coefficient, and coupling scheme.
`sync_settings()`, `read_restart(...)`, `load_history_contact()`	Synchronizes restart counters and loads coupled contact histories.
`select_save_data(ppcontact=False,pwcontact=False)`	Selects coupled contact output.
`add_essentials(**kwargs)`	Finalizes coupling, builds engines/recorders, installs optional drag model, callback function, and DEM calm operation.
`add_postfunctions(**functions)`, `update_contact_properties(...)`	Postprocessing and runtime contact update hooks.
`is_dem_lsm_pm_soft_rigid_mode()`, `run_dem_lsm_pm_soft_rigid(**kwargs)`	Routing query and dedicated DEM-driven LSMPM runner used when the MPM sub-solver owns recorder services but has no ordinary MPM material scene. The runner accepts `strict_timestep=True` to reject an automatic critical-step correction.
`run(**kwargs)`, `check_critical_timestep()`, `get_critical_timestep()`, `save_data()`, `postprocessing(...)`	Coupled runtime and output functions.

23.2 Coupling Keywords and Compatibility

Keyword	Meaning and current restrictions
`coupling_scheme`	`MPDEM`, `DEMPM`, `MPM`, `DEM`, or `CFDEM`.
`particle_interaction`, `wall_interaction`	Enable MPM-DEM particle and wall contact.
`CFD_coupling_domain`	Two integers defining influence/dependent domains for CFD-style coupling kernels.
`enhanced_coupling`	Enabled for MPDEM/DEMPM contact enhancement; rejected for `CFDEM`.
`search`	DEM neighbor search used by the coupled setup, usually `LinkedCell`.
`body_coordination_number`, `wall_coordination_number`, `compaction_ratio`	Coupled contact-list size controls.
`drag_model`	Dictionary passed to coupled fluid-particle drag. Keys: `DragForceModel`, `DragLaw`, and optional `ParticleReynold`.
`calm`, `function`	DEM particle calm schedule and callback function.

Compatibility rules currently enforced by source validation:

Mode	Rule
`MPDEM`/`DEMPM`	DEM scheme must be `DEM` or `LSDEM`.
`MPDEM`/`DEMPM` with implicit MPM	Supported only for incompressible fluid FDM: `solver_type="Implicit"`, `material_type="Fluid"`, `discretization="FDM"`, dimension 3.
`MPDEM`/`DEMPM` with semi-implicit MPM	Rejected.
`MPDEM`/`DEMPM` with MPM `sparse_grid`	Allowed for contact-based Lagrangian coupling. The sparse grid is used only by the MPM background grid; DEM and coupled contact pairs still use the DEM/coupled neighbor-search buffers.
`CFDEM`	DEM scheme must be `DEM` or `LSDEM`; MPM must be `material_type="Fluid"`.
`CFDEM` with MPM `sparse_grid`	Rejected. The semi-resolved fluid-particle path uses its own fluid grid sampling and does not support BlockScan remapping.
Explicit `CFDEM`	Requires MPM `discretization="FEM"`.
Implicit `CFDEM`	Requires incompressible fluid FDM and dimension 3.
Fluid contact	MPDEM particle-particle or particle-wall contact model must be `FluidParticle`.

23.3 CFDEM Drag Keywords

Keyword	Meaning
`DragForceModel`	`AbrahamModel`, `EmpiricalModel`, `BrownLawlerModel`, or `SchillerNaumannModel`. The selected model is used for the drag coefficient when $1 < Re_p < 1000$.
`DragLaw`	`Linear` or `Quadratic`.
`ParticleReynold`	Optional constant Reynolds number override. If omitted, Re_p is computed from local fluid fraction, density, viscosity, radius, and relative velocity.

23.4 MPDEM and CFDEM Dictionary Schemas

Related theory. MPM-DEM, incompressible DEM-sphere coupling, and contact models.

Coupled dict	Keys	Meaning
`set_configuration`	`domain`, `coupling_scheme`, `particle_interaction`, `wall_interaction`, `gravity`, `visualize`, `track_energy`, `enable_shell`, `CFD_coupling_domain`, `enhanced_coupling`, `search`	Coupling-level configuration. `domain` is required if the MPM and DEM domains differ. `coupling_scheme` selects `MPDEM`, `DEMPM`, `MPM`, `DEM`, or `CFDEM`.
`set_configuration`	`digital_elevation_contact`	Coupled digital-elevation walls currently use `heightfield`; the mode is synchronized to the DEM sub-solver.
`set_solver`	`Timestep`, `SimulationTime`, `SaveInterval`; optional `CFL`, `AdaptiveStep`, `SavePath`	Sets the coupled time controller and forwards the same solver dictionary to both sub-solvers.
`memory_allocate` coupled memory	`body_coordination_number`, `wall_coordination_number`, `compaction_ratio`	Allocates coupled contact-list capacities. DEM-only memory must be passed through `dem_memory`; MPM-only memory through `mpm_memory`.
`memory_allocate` coupled memory	`max_particle_contact_pairs`	Optional non-negative explicit particle-pair contact-list capacity. Zero keeps the coordination/compaction-derived size; an override larger than the potential-pair allocation warns about wasted memory.

Coupled dict	Keys	Meaning
`memory_allocate` coupled memory	`max_wall_contact_pairs`	Optional non-negative explicit wall-pair contact-list capacity with the same override semantics.
`dem_memory` argument	DEM memory schema from DEM Dictionary Schemas	Allocates DEM bodies, contacts, walls, and search buffers before coupled memory is built.
`mpm_memory` argument	MPM memory schema from MPM Dictionary Schemas and Random Field	Allocates MPM particles, constraints, and implicit/fluid buffers before coupling kernels are initialized.
`add_body`	`mpm_body`, `dem_particle`; flags `write_file`, `check_overlap`, `adaptive_boundary_radius`	Inserts the MPM body dict and DEM particle/body dict through their sub-solvers, then prepares coupled particle sets.
`choose_contact_model`	`particle_particle_contact_model`, optional `particle_wall_contact_model`	Selects coupled MPM-particle and MPM-wall contact models. `CFDEM` skips this path.
`add_property`	`DEMmaterial`, `MPMmaterial`, `property`, optional `dType`	Adds material-pair contact properties. For semi-resolved fluid particles the property dict usually follows the `FluidParticle` contact schema.
`drag_model` in `add_essentials`	`DragForceModel`, `DragLaw`, optional `ParticleReynold`	Selects the drag coefficient formula and whether force is linear or quadratic in relative velocity.
`add_essentials` forwarding keys	`mpm_*`, `dem_*`, `reset_function`, `function`, `calm`	Keys with `mpm_` and `dem_` prefixes are split and forwarded to the corresponding sub-solver. `function` is a callback; `calm` controls DEM particle calm scheduling.
`read_restart` and `load_history_contact`	`file_number`, `file_path`, `ppcontact`, `pwcontact`	Synchronizes DEM and MPM restart counters and optionally reloads coupled contact histories.
`select_save_data`	`particle_particle_contact`, `particle_wall_contact`	Selects coupled contact output in addition to sub-solver recorder output.
`run` for DEM-driven LSMPM	`strict_timestep`	Boolean diagnostic gate, default `False`. When true, raise instead of silently accepting an automatic LSMPM critical-timestep reduction.

23.5 MPDEM Soft-Affine IPC Workflow

Related theory. MPM soft-particle backend, Mixed LSMPM soft-affine IPC, Strict contact derivatives.

This workflow couples level-set MPM soft particles with AffineBody rigid particles through one implicit IPC operator. It is exposed through `MPDEM/DEMPM` because examples keep the same high-level preprocessing pattern as MPDEM: initialize the coupled facade, configure DEM/MPM-compatible settings, add templates/regions/bodies, assign material-pair properties, select outputs, and call `run()`.

Entry point or dict	Required use in soft-affine IPC
`gt.MPDEM()` or `gt.DEMPM()`	Constructs the coupled facade. The current soft-affine example accesses the underlying `dem` object for LSMPM/AffineBody setup, while the coupled facade owns the run loop and recorder coordination.
`DEM.set_configuration`	Use `scheme="LSMPM"` and `soft_rigid_contact="IPC"`. `search` may be `LinkedCell`, `HierarchicalLinkedCell`, or `BVH`. `shape_function` controls the soft TLMPM interpolation.
`DEM.set_affine_body_parameters`	Sets the mixed IPC operator controls: `assemble_type`, `dhat`, `barrier_stiffness`, `contact_damping_stiffness`, `friction_epsv`, `soft_background_damping`, Newton/PCG tolerances, line-search limits, `max_step`, and CCD/ACCD options.
`DEM.memory_allocate`	Requires `max_soft_body_number`, `max_material_point_number`, `max_rigid_template_number`, `levelset_grid_number`, `surface_node_number`, and contact coordination capacities. `max_affine_body_number` is the preferred capacity for affine rigid bodies.
`DEM.add_attribute`	Adds soft finite-strain material parameters. Soft PK1 stress and tangent come from the selected finite-strain class. `estress` is output as von-Mises equivalent stress.
`DEM.add_template`	Add one level-set template for soft bodies and one `TemplateType="AffineBody"` template for affine rigid bodies. Set `ContactRepresentation="LevelSet"` when the object also provides a regular signed-distance grid and implicit normal plus lagged-friction contact is required.
`DEM.add_body` with `GenerateType="Generate"`	Generate both `BodyType="SoftBody"` and `BodyType="AffineBody"` from regions. This uses the same generate/overlap path as DEM/LSDEM rather than a separate "create affine body" only route.

Entry point or dict	Required use in soft-affine IPC
`DEM.add_property(...,dType="all")`	Stores pair contact parameters: `Dhat`, `BarrierStiffness`, `ContactDampingStiffness`, and `Friction`. These are material-pair contact properties; global Newton/PCG controls remain in `set_affine_body_parameters`.
`DEM.select_save_data(particle=True,surface=True,bounding=True)`	Writes `LSMPMSoftPoint*.npz`, `AffineBody*.npz`, bounding boxes/spheres, and level-set grids.
`MPDEM.run()` or `DEMPM.run()`	Detects the LSMPM soft-affine IPC mode, installs the MPM soft-particle bridge, builds the mixed IPC engine, and runs the coupled time loop.

Search is body-level first. The chosen DEM search backend creates body-pair candidates; only then are soft surface material points tested against target soft level sets or affine triangles. This applies to linked-cell and BVH routes and avoids allocating a global contact table over all surface points. For implicit IPC, the target soft level set is expanded by the `dhat` activation margin for search so contacts are found while the sampled gap is still positive.

24 FEM User API

The public `FEM` facade follows the same configuration–scene–build–run lifecycle as MPM and IGA. Taichi must be initialized before `build`; there is no NumPy runtime backend. `linear_solver="Scipy"` is only an explicit linear-solve boundary.

Function	Purpose
`set_configuration(dimension=3,solver_type="Explicit",**kwargs)`	Selects embedded dimension and `Explicit` or `Implicit` integration.
`create_mesh(geometry=None,**kwargs)`	Creates box, rectangle, circle, or cylinder meshes, or reads OBJ/Gmsh and other meshio-supported files. `rest_shape` is optional.
`add_mesh(mesh=None,**kwargs)`	Registers an `FEMMesh`, filename, mesh dictionary, or generated primitive; an explicit rest shape overrides the current-coordinate default.
`add_material(model="StVK",material=None,**kwargs)`	Registers a finite-strain volume material, cloth material, or shared incremental elastoplastic material. Cloth models select the triangle $D_s D_m^{-1}$ membrane mapping automatically. Elastoplastic models select the explicit updated-Lagrangian eight-point HEX8 path.
`add_boundary_condition(...)`	Registers nodal Dirichlet data and nodal-force, traction, pressure, or edge traction Neumann data.

Function	Purpose
`set_bending_model(model)`	Selects `Quadratic`, `Dihedral`, or `None` for cloth.
`add_cloth_energy`, `add_stitch`, `add_spring`, `add_sdf`	Adds analytic garment-stitch, target-spring, or frozen-frame one-sided SDF energies. Stitch relations are also excluded from self-contact culling.
`set_solver(solver=None,**kwargs)`	Sets timestep, Newmark/Newton/line-search controls, sparse assembly, linear solver, Hessian projection, damping, output, and load-step settings.
`add_contact(model="IPC",**kwargs)`	Adds ordinary Barrier IPC contact.
`add_contact_property(body_id1,body_id2,property=None,**kwargs)`	Adds or replaces an unordered IPC body-pair parameter set. Once any pair is listed, unlisted surface pairs are inactive.
`build()`, `run()`	Builds the Taichi volume or cloth engine and executes the configured steps.

24.1 FEM Solver and Contact Keywords

Keyword	Accepted values	Effect
`assemble_type`	`COO`, `HashTriplet`	Selects scalar coordinate triplets or device hash block assembly.
`linear_solver`	`PCG`, `BiCGSTAB`, `Scipy`	Selects the device Krylov method or the explicit host linear solve.
`project_pd`	Boolean	Projects local material, bending, auxiliary, and IPC contact Hessians as supported. It is required for PCG.
`rest_shape`	$n \times 3$ array	Sets reference coordinates; omission uses the input mesh coordinates.
`MinimumJacobian`	Positive scalar	Optional Neo-Hookean determinant floor used by the shared finite-strain material implementation.
`model` in `add_contact`	`IPC`	Selects ordinary strict Barrier IPC contact with CCD and minimum clearance d_{min}.
`broad_phase`	`LinkedCell`, `BVH`	Selects only the spatial AABB backend. Both feed the same exact-distance/CCD collision-culling layer.
`dhat`, `dmin`, `kappa`	Positive scalars; `dmin` may be zero	Set IPC activation distance, thickness offset, and barrier stiffness.
`ccd_safety`, `ccd_max_iterations`	$(0, 1]$, positive integer	Control conservative advancement. Swept candidates are rebuilt every query; there is no FEM IPC Verlet multiplier.

Keyword	Accepted values	Effect
`friction_coefficient`, `eps v`	$\mu \geq 0, \epsilon_v > 0$	Set the lagged smooth Coulomb model.
`body_id1`, `body_id2` in `add_contact_property`	Existing non-negative FEM body IDs	Select one unordered IPC pair. The pair may set its own `broad_phase`, `dhat`, `dmin`, `kappa`, friction, CCD, and `project_pd` values.

The collision-culling flow is

selected linked-cell/BVH AABB query → topology and stitch exclusions → exact distance or swept conservative CCD → prefix-sum compact stencils → contact assembly or minimum line-search step.

See FEM collision culling and CCD.

25 FEM–DEM User API

`FEDEM` owns two-way contact between the oriented triangular boundary of an FEM model and ordinary DEM spheres, rigid LSDEM signed-distance bodies, or AffineBody surface meshes. The DEM/LSDEM routes use explicit FEM and a DEM penalty law. The AffineBody route uses elastic implicit FEM and monolithic IPC. Its lifecycle follows `DEMPM`: configure and populate the two child facades, then configure coupling storage and properties before `run`.

Function	Purpose
`FEDEM(dem,fem)`	Creates the coupling facade. The global `geotaichi.FEDEM()` factory may create omitted child facades.
`fem.add_soft_particle(mesh=None,**kwargs)`	Appends one TET4/HEX8 volume as a disconnected deformable body. Ordinary FEM bulk mechanics are retained, and the extracted TRI3 boundary becomes the explicit contact surface.
`fem.add_soft_particle_contact(model="Linear",**kwargs)`	Selects Linear, Hertz–Mindlin, or energy-conserving deformable–deformable PT/EE contact with linked-cell or BVH search, contact thickness, and a Verlet skin.
`fem.add_soft_particle_property(body_id1,body_id2,property=None,**kwargs)`	Adds or replaces the explicit contact law for one unordered pair of deformable soft-particle bodies.
`set_configuration(**kwargs)`	Sets the shared positive 3D domain, gravity, and `search="LinkedCell"` or `"BVH"`.

Function	Purpose
`set_solver(solver=None,**kwargs)`	Synchronizes `Timestep`, `SimulationTime` (alias `simulation_time`), `SaveInterval` (alias `output_interval`), `SavePath`, and `CFL` across the coupling, DEM, and FEM clocks. AffineBody IPC optionally accepts `enable_step_retry`, `step_retry_max_retries`, `step_retry_reduction`, and `step_retry_minimum_timestep`.
`add_surface(facet_sets=None,modifier=None,body_ids=None)`	Selects explicit triangle facets or the triangulated FEM boundary. Orientation may be `Parallel`, `Centripetal`/legacy `Centrioetal`, `Inverse`, or unchanged; `Value` applies a normal offset. `add_patch` is an alias.
`memory_allocate(memory,dem_memory=None)`	Allocates dynamic face-cell memberships, particle candidate prefixes, active/history contact records, and optional DEM memory.
`choose_contact_model(contact_model="Linear",**kwargs)`	Selects exactly one source-equivalent `Linear`/`HertzMindlin` surface law for DEM or LSDEM, or `IPC` with barrier, lagged-friction, and CCD parameters for AffineBody.
`add_property(DEMmaterial,FEMbody,property,dType="all")`	Stores one property for an actual DEM-material/FEM-body pair. `add_contact_prop` is the legacy alias.
`add_ipc_property(AffineBody,FEMbody,property=None,**kwargs)`	Stores pair-local `dhat`, `dmin`, `kappa`, `friction_coefficient`, `epsv`, CCD, and related IPC parameters for an affine-body/FEM-body pair. Unlisted pairs are inactive after pair properties are present.
`select_save_data(contact=True)`	Enables or disables coupled-contact NPZ output independently of child output.
`add_essentials(**kwargs)`	For DEM/LSDEM, validates explicit FEM and no internal FEM IPC/AL, then constructs the selected broad phase and penalty engine. For AffineBody IPC, validates implicit elastic FEM, rejects fully implicit mixed friction, requires lagged affine self-friction with one self-friction iteration, and permits no standalone FEM contact, then constructs the monolithic engine.
`add_postfunctions(**functions)`	Registers Taichi callback kernels after each accepted coupled step.
`modify_parameters(**kwargs)`	Updates end time, timestep, save interval, or gravity for continuation.
`check_critical_timestep()`	For explicit coupling, applies the configured CFL factor to the minimum DEM, FEM-wave, and linear/Hertz coupling estimates. IPC retains its user-selected implicit step.

Function	Purpose
`run(**kwargs)` / `solve(**kwargs)`	The explicit route precomputes search state and advances DEM/LSDEM contact and FEM once per shared step. The AffineBody route advances a monolithic control/node Newton solve with analytic barrier/friction Hessians, lagged friction fixed-point refresh, CCD/ACCD, and Armijo. Legacy `callback` is forwarded to the DEM engine/servo callback, `dem_*` prefixes child setup options, and `function` registers a coupled post-step Taichi callback.
`save_data()`	Forces synchronized DEM, FEM, and optional active-contact output at the current state.

Dictionary	Required keys	Meaning
Coupling memory	none	Optional `max_particle_number`, `max_surface_facet_number` (legacy alias `max_patch_number`), `contact_coordination_number`, `compaction_ratio`, `max_contact_pairs`, `max_facet_cell_pairs`, `max_levelset_cell_pairs`, `verlet_distance`, and `verlet_distance_multiplier`. The level-set capacity bounds current rigid-AABB/cell memberships; capacity overflow is an error.
Deformable soft-particle contact	`ContactThickness`, `search`, `verlet_distance`, `verlet_distance_multiplier`	Sets the explicit PT/EE surface thickness, linked-cell/BVH search, and neighbor-list skin. A positive thickness is required for unsigned edge–edge response; rebuild occurs after a boundary node moves more than half the skin.
Linear property	`NormalStiffness`, `TangentialStiffness`	Optional `Friction`, `NormalViscousDamping`, and `TangentialViscousDamping`.
Hertz–Mindlin property	`ShearModulus`, `Poisson`, `Restitution`	Optional non-negative `Friction`; restitution is converted to the same damping factor as the source implementation.

Dictionary	Required keys	Meaning
IPC model/pair	none	Optional `dhat`, `dmin`, `kappa`, `ccd_safety`, `ccd_max_iterations`, pair-local `friction_coefficient`, and `epsv`. Mixed friction uses `friction_mode="lagged"`; positive `friction_iterations` refresh the frozen PT/EE frame and barrier normal force. Fully implicit mixed friction is not supported. Sparse assembly is `COO` or `HashTriplet`; the linear solver is device `PCG` or the explicitly selected host `Scipy` path.
Step retry defaults off, catches only `NewtonConvergenceError`, and keeps a reduced synchronized timestep only after a successful bounded retry.		

The FEM child is never converted to an SDF. The LSDEM route samples only the rigid child's existing signed-distance field at current FEM boundary nodes, weighted by current lumped nodal area. Affine IPC uses direct point–triangle and edge–edge geometry. Maintained volume examples are `examples/fedem/explicit_levelset_soft_particle.py` for LSDEM `gapn` contact and `examples/fedem/implicit_affine_ipc_soft_particle.py` for frictional AffineBody IPC. Both expose `GT_FEDEM_SEARCH`, time-step/time/output environment overrides, and use the same low-poly sphere asset. The first implicit step includes Taichi JIT of the analytic volume/contact Hessians, so its preparation time is not representative of subsequent Newton steps. See FEM–DEM coupling theory. The owning implementation is `src/fedem/mainFEDEM.py`.

26 FEM–MPM User API

The implicit coupled solver checks residual convergence before matrix assembly. At an unchanged trial, force and tangent share the Direct ULMPM material response and contact query. The HashTriplet path directly scatters immutable classical FEM/IGA material topology into permanent reduced slots; dynamic MPM/contact assembly retains its existing mapping/reduction. No new solver configuration key is needed. See sparse assembly and Krylov solvers for the numerical invariants.

`FEMPM` owns two-way contact between MPM material points and an oriented FEM boundary. The explicit branch uses DEM contact laws and `ParticleCoupling`. The implicit branch uses elastic FEM, Direct implicit ULMPM with Neo-Hookean, Drucker–Prager, or associated von Mises response, and a monolithic IPC Newton system. FEM internal mechanics is Total Lagrangian; MPM is Updated Lagrangian; IPC contact uses the current spatial con-

figuration. Cartesian 2D DP/von-Mises MPM uses a 3D plane-strain material embedding; axisymmetric MPM uses the 3D no-swirl map.

Function	Purpose
`FEMPM(fem,mpm)`	Creates the coupling facade; the global factory may create omitted children. Non-Direct MPM is configured for Lagrangian explicit coupling; Direct MPM retains its implicit backend.
`set_configuration(**kwargs)`	Sets the shared domain, gravity, and `search="LinkedCell"` or `"BVH"`. Implicit IPC accepts Cartesian 2D, 3D, or axisymmetric 2D. Optional `axisymmetric` and `axis_offset` must match both children; omission inherits their already validated mode. Explicit DEM contact remains 3D.
`set_solver(solver=None,**kwargs)`	Synchronizes `Timestep`, `SimulationTime`, `SaveInterval`, `SavePath`, and `CFL`. IPC also accepts `assemble_type="COO"` or `"HashTriplet"`, device `linear_solver="PCG"` or explicit host `"Scipy"`, Newton tolerances, `project_pd`, and line-search parameters. Nonassociated DP automatically replaces device PCG with BiCGSTAB, retains the unprojected physical Jacobian, and uses residual Armijo without a material-consistency outer loop. Associated materials retain projected PCG and their existing material consistency iteration. `enable_step_retry` defaults false; optional `step_retry_max_retries`, `step_retry_reduction`, and `step_retry_minimum_timestep` bound transactional retries of `NewtonConvergenceError` only.
`add_surface(facet_sets=None,modifier=None,body_ids=None)`	Selects explicit facets or the FEM boundary; defaults are edges in 2D and triangles in 3D. `add_patch` is an alias.
`memory_allocate(memory,mpm_memory=None)`	Allocates compact particle/face candidates, active/history contacts, BVH data or dynamic triangle/cell memberships, and optional MPM capacity.
`choose_contact_model(contact_model="Linear",**kwargs)`	Selects the DEM `Linear`/`HertzMindlin` law or implicit `IPC`. IPC accepts `dhat`, `dmin`, `kappa`, `friction_coefficient`, `epsv`, `friction_mode="lagged"`, and `friction_iterations`.
`add_property(MPMmaterial,FEMbody,property,dType="all")`	Stores one MPM-material/FEM-body property; MPM user material ids start at one. On the Direct IPC branch the first id denotes an MPM body; use `add_ipc_property(MPMbody,FEMbody,...)` for unambiguous spelling. `add_contact_prop` remains an alias.
`select_save_data(contact=True)`	Controls coupled-contact NPZ output.

Function	Purpose
`add_essentials(**kwargs)`	Builds either validated 3D explicit particle/-grid MPM plus explicit FEM, or Direct implicit ULMPM plus implicit elastic FEM. The implicit MPM law may be Neo-Hookean, Drucker–Prager, or associated von Mises. Standalone child IPC, Direct TLMPM plasticity, and coupled FEM elastoplasticity are rejected. Cartesian and axisymmetric 2D IPC use analytic point–edge Hessians and CCD/ACCD; the Cartesian plastic MPM map is 3D plane strain with incremental $F_{33} = 1$, while axisymmetric material maps include $F_{\theta\theta} = r/R$ and revolved reference measures. Passing `plane_strain=False` to Cartesian 2D DP or von Mises is rejected.
`add_postfunctions(**functions)`	Registers post-step Taichi callbacks.
`modify_parameters(**kwargs)`	Updates end time, step, save interval, or gravity for continuation.
`check_critical_timestep()`	Applies shared CFL to the minimum MPM, FEM-wave, and contact estimate.
`run(**kwargs)` / `solve(**kwargs)`	Explicit contact rebuilds at half the skin. IPC assembles FEM, MPM, barrier and lagged-friction blocks into one Newton system, rebuilds current/swept candidates for Newton/Armijo/CCD without a Verlet skin, and advances both clocks once. The implicit step is transactional: a failure restores FEM nodal state plus Direct-MPM particle, grid, deformation-gradient, and plastic-history state. After a successful retry the reduced timestep becomes the synchronized coupling/child timestep. Exhaustion restores the entry timestep and exposes attempt, contact, CCD-step, and linear-solver diagnostics through `diagnostics_snapshot()`.
`save_data()`	Forces synchronized child-state output. The explicit path also writes persistent-contact NPZ data. IPC contact diagnostics are returned in run history; an IPC candidate/frozen-friction NPZ schema is not yet exposed.

Coupling memory accepts `max_particle_number`, `max_surface_facet_number` (legacy alias `max_patch_number`), `max_mpm_material_number`, `contact_coordination_number`, `compaction_ratio`, `max_contact_pairs`, `max_facet_cell_pairs` (LinkedCell), `verlet_distance`, and `verlet_distance_multiplier`. Linear and Hertz–Mindlin property keys match FEDEM. See FEM–MPM coupling theory; the owning implementation is `src/fempm/mainFEMPM.py`.

27 IGA and IGAMPM User API

27.1 IGA

Function	Purpose
`IGA(title="IsoGeometricAnalysisEngine",log=True)`	Creates the IGA facade.
`set_configuration(dimension=3,solver_type="Implicit",**kwargs)`	Sets dimension and selects `Explicit` or `Implicit`. For a no-swirl axisymmetric meridian, set `dimension=2`, `axisymmetric=True`, and `axis_offset`; coordinate zero is radius.
`add_primitives(primitives)`	Adds NURBS/geometry primitives.
`add_boundary_condition(dirichlet=None,neumann=None)`	Adds IGA Dirichlet and Neumann boundary objects.
`add_material(**kwargs)`	Adds IGA material properties such as `young_modulus`, `poisson_ratio`, and `density`.
`add_element(degree)`	Sets NURBS element degree, for example $[p, q]$.
`set_solver(**kwargs)`	Stores solver parameters.
`build()`	Creates `ImplicitIGA` or `ExplicitIGA`.
`run(*args,**kwargs)`	Runs the built backend if it implements `run()`.

27.2 IGAMPM

The axisymmetric CPT example accepts `--resume` with a `latest_state.npz` path. It restores accepted physical fields after reference initialization and continues saved frame numbering when resuming in the same output directory. Material, grid resolution and save interval must match; diagnostics newer than the checkpoint are separated from the resumed trajectory. This is an example-specific command-line option.

The implicit coupled solver checks residual convergence before matrix assembly. At an unchanged trial, force and tangent share the Direct ULMPM material response and contact query. The HashTriplet path directly scatters immutable classical FEM/IGA material topology into permanent reduced slots; dynamic MPM/contact assembly retains its existing mapping/reduction. No new solver configuration key is needed. See sparse assembly and Krylov solvers for the numerical invariants.

Function	Purpose
`IGAMPM(iga=None,mpm=None,title="IGA-MPMCouplingEngine",log=True,**kwargs)`	Creates the coupling facade around an IGA object and an MPM object. If either object is omitted, default `gt.IGA()` and `gt.MPM()` facades are created.

Function	Purpose
`set_configuration(dimension=None,coupling_scheme="IGAMPM",contact_model="IPC",activate_friction=False,axisymmetric=False,axis_offset=0.0)`	Sets the coupling dimension and selects implicit `IPC` or explicit `Linear`/`HertzMindlin`. Axisymmetric coupling is IPC-only; the two children must share `axis_offset`.
`choose_contact_model(contact_model="IPC",**kwargs)`	Chooses IPC barrier contact or one explicit DEM law. IPC kwargs include `kappa`, `dhat`, `mu`, `epsv`, capacities, and `assemble_type`.
`add_property(MPMmaterial,IGAbody,property)`	Registers one explicit DEM law for an MPM-material/IGA-patch pair before `build()`.
`set_solver(iga=None,mpm=None)`	Forwards solver dictionaries to the IGA and MPM sub-solvers before the coupling engine is built.
`build()`	Builds compatible children: implicit/implicit for IPC or explicit/explicit for DEM-law contact. Explicit contact requires native Lagrangian-coupling MPM and 3D Cartesian geometry.
`run(grid_disp=None,include_friction=False,steps=None,verbose=True)`	Runs the selected explicit force-exchange step or IPC monolithic solve.
`__getattr__` forwarding	After build, unresolved attributes are forwarded to the coupling engine so low-level contact assembly helpers remain available for tests and diagnostics.

27.2.1 IGAMPM ContactManager lifecycle

When Direct ULMPM uses nonassociated Drucker–Prager, the coupled engine automatically selects full nonsymmetric storage and device BiCGSTAB with block Jacobi preconditioning (scalar Jacobi for COO). The physical Jacobian includes the trial plastic-volume derivative. Residual Armijo retains contact and material CCD, and no material-consistency outer loop runs. Associated DP retains PCG and its existing frozen-volume consistency loop; lagged friction remains a separate outer iteration.

Function	Purpose
`ContactManager.choose_contact_model(contact_model="IPC",**kwargs)`	Validates and stores IPC, Linear, or Hertz–Mindlin contact configuration before the coupling engine is built.
`ContactManager.freeze_configuration()`	Freezes settings captured by the built coupling operator. Configuration-changing calls after freeze are rejected so engine fields and solver assumptions cannot diverge.

27.3 IGA and IGAMPM Dictionary Schemas

Related theory. IGA, IGA-MPM, IGA-MPM CCD/ACCD, IGA-MPM IPC contact model, and optimization methods.

IGA/IGAMPM dict	Keys	Meaning
`IGA.set_configuration`	`dimension`, `solver_type`, `axisymmetric`, `axis_offset`, plus backend keyword overrides	`dimension` selects 2D/3D and `solver_type` selects `Explicit` or `Implicit`. Axisymmetry requires 2D, uses coordinate zero as radius, and evaluates a no-swirl 3D constitutive map with revolved reference weights. Extra keys are retained in the simulation configuration and consumed by the selected backend.
`IGA.add_primitives`	Primitive object or list of primitives	Supplies NURBS patches/curves or geometry containers used to build IGA basis, quadrature, and boundary sets.
`IGA.add_boundary_condition`	`dirichlet`, `neumann`	Adds boundary-condition objects. Dirichlet objects prescribe control-point degrees of freedom. Implicit `DirichletBoundary.append_velocity(dof_id,velocity)` stores constant velocities and refreshes increments with the current timestep, including retries. Ordinary `append` keeps increment semantics; explicit IGA rejects velocity entries. `NeumannBoundary.append` accepts already-integrated generalized control-point forces, not a traction density; axisymmetric callers must include the required $2\pi R$ measure.
`IGA.add_material`	Common keys `young_modulus`, `poisson_ratio`, `density`; model-specific keys forwarded to material manager	Defines the material used by IGA elements. The exact set is backend dependent, so examples should keep key spelling consistent with the selected material class.
`IGA.add_element`	`degree`	NURBS degree list, for example $[p, q]$ in 2D or $[p, q, r]$ in 3D.
`IGA.set_solver`	Solver-specific kwargs such as timestep, nonlinear tolerance, maximum iteration, and output path	Stored and forwarded to `ImplicitIGA` or `ExplicitIGA` during `build()`.

IGA/IGAMPM dict	Keys	Meaning
IGAMPM.set_configuration	dimension, coupling_scheme, contact_model, activate_friction, axisymmetric, axis_offset	Valid coupling schemes are IGAMPM and IGA-MPM. contact_model="IPC" requires implicit children; Linear and HertzMindlin require explicit children, native Lagrangian-coupling MPM, and 3D Cartesian geometry. Axisymmetric mode is IPC-only and requires dimension two with matching child offsets.
IGAMPM.choose_contact_model	contact_model, plus model-specific keywords	Builds the IPC barrier/friction manager or selects one shared DEM Linear/Hertz–Mindlin law. IPC assemble_type selects coupled device COO or HashTriplet.
IGAMPM.add_property	MPMmaterial, IGAbody, property	Assigns one explicit DEM-law property to an MPM-material/IGA-patch pair before build(). Every active pair needs a property.
IGAMPM.set_solver	iga, mpm, and optional coupling solver dictionaries	Forwards child controls and configures bounded transactional retry for implicit IPC.
IGAMPM.build	no user dict	Builds either the monolithic implicit IPC engine or the explicit point–NURBS force-exchange engine under src/igampm/engines.
IGAMPM.run	grid_disp, include_friction, steps, verbose	IPC performs the transactional monolithic solve with device closest-point search, ACCD, assembly, friction, and line search. Linear/Hertz–Mindlin evaluates finite-radius point–NURBS contact, tangential history, and equal/opposite rational-basis force transfer in Taichi before the two explicit advances.
ExplicitLinearproperty	NormalStiffness, TangentialStiffness or EffectiveModulus, NormalToShearRatio; optional Friction, StaticFriction, DynamicFriction, NormalViscousDamping, TangentialViscousDamping	Uses the production DEM linear spring–dashpot/Coulomb law. Fixed and adaptive stiffness parameterizations cannot be mixed in one model.

IGA/IGAMPM dict	Keys	Meaning
`ExplicitHertzMindlin` property	`ShearModulus`, `Poisson`, `Restitution`; optional `Friction`, `StaticFriction`, `DynamicFriction`	Uses the production DEM Hertz–Mindlin/dashpot/Coulomb law. Nonzero `RollingFriction` is rejected for both explicit laws because IGA control points have no rotational DOFs.
`IPCbarrier/frictionkeys`	`kappa`, `dhat`, `friction` or `mu`, `epsv`, `barrier_nnz`, `friction_nnz`, `coordination_number`, `activate_friction`	Control IPC barrier stiffness, activation distance, friction regularization, sparse allocation, and whether friction contacts are assembled. See IGA-MPM IPC contact model and IGA-MPM CCD/ACCD.
`IPCline-searchkeys`	`line_search_work_tol`, `line_search_energy_rtol`, `line_search_energy_atol`, `line_search_max_backtracks`, `line_search_min_alpha`, `line_search_stagnation_alpha`, `line_search_descent_fallback`	Control CCD/backtracking and convergence checks for IPC Newton iterations. See ACCD and Optimization Methods.
`IGAMPMassembly`	`assemble_type="COO"` or `"HashTriplet"`	Selects the device-resident global Hessian representation. Scipy is permitted only as an explicitly selected linear-solve boundary.
`IPCcontactbuffers`	`barrier_set`, `friction_set`, surface/curve contact buffers, monolithic barrier/friction sparse systems	Internal containers exposed by coupling helpers. Users normally size them through the NNZ and coordination-number keys instead of editing kernels.

The nonlinear force criterion uses `monolithic_force_atol` (default 10^{-10}) and `monolithic_force_rtol` (default 5×10^{-4}): $R_k \leq \varepsilon_a + \varepsilon_r \max(R_0, 1)$. For nonassociated DP, R_0 is the first residual of the attempted step, retained across lagged-friction iterations and reset on timestep retry. Both this force criterion and `monolithic_tolerance` must be satisfied. The latter has velocity units and measures IGA control motion, MPM particle motion, grid-spacing-scaled displacement gradients, and, in axisymmetry, the scaled hoop increment. The updated-friction probe uses the same measure with `friction_tolerance` and also verifies force balance for nonassociated DP. See physical convergence criteria. `monolithic_dirichlet_tolerance` defaults to 10^{-12} and bounds the remaining prescribed-displacement mismatch. During prescribed loading, residual Armijo includes the squared mismatch, scaled by the retained force reference divided by its initial displacement mismatch. The weight stays fixed in each Newton solve; exact prescribed corrections reduce this contribution by $(1-\alpha)^2$. This changes only the line-search merit, with no additional configuration key. For nonassociated DP, a CCD/material-feasible line-search trial may also be accepted when both its base and

trial free-force norms satisfy the original force tolerance. Physical correction, prescribed-motion and refreshed-friction criteria still apply; this introduces no additional tolerance setting. `contact_all_mpm_particles` defaults to `False`; setting it true builds contact samples from every MPM particle instead of the surface set. `compact_contact_slots` defaults to `False` and must be boolean. When enabled, prefix-compacted active contacts reserve raw block slots for barrier/friction assembly; `barrier_nnz` and `friction_nnz` remain capacity limits and overflow is an error.

For nonassociated DP, `monolithic_inexact_newton=True` optionally adapts the linear relative tolerance. It defaults to `False`, requires a boolean, and is rejected for other material routes. Its initial relative tolerance is 0.01; subsequent tolerances are $\min(0.01, \max(\varepsilon_{\text{lin}}, 0.9(R_k/R_{k-1})^{1.5}))$, where R_k is the free-force residual norm and `monolithic_linear_solver_relative_tolerance` supplies the floor $0 \leq \varepsilon_{\text{lin}} \leq 0.01$. A new Newton solve resets the sequence. The true linear error bound is $\max(\varepsilon_{\text{lin,a}}, \eta_k R_k)$, where R_k is measured before Dirichlet elimination and η_k is the adaptive forcing term. The eliminated right-hand side does not set this reference. Acceptance requires the configured nonlinear force, physical correction, and Dirichlet tolerances. A terminal inexact correction is rechecked at the configured linear accuracy. Small correction alone cannot terminate the solve. Residual Armijo and contact/material CCD remain active.

The maintained flexible-barrier script supports `--resumelatest_state.npz` in the checkpoint's existing output directory. Physical parameters and output cadence must match; the requested end time may be extended. Finite state, positive deformation determinants, and paired contiguous VTK indices starting at zero are validated before continuing. Existing frames are retained and checkpoint validation errors leave the saved state intact.

The previous standalone IGAMPM MPM facade is not part of the current public API. IGAMPM examples should start from `geotaichi.IGAMPM` and use ordinary `geotaichi.IGA` and `geotaichi.MPM` setup for their two sub-solvers.

28 Detailed Facade API Entries

28.1 MPM

28.1.1 `MPM.__init__`

Signature. `self,title='AHighPerformanceMultiscaleandMultiphysicsSimulator',log=True`

Purpose. constructs the facade object and stores title/logging metadata without allocating simulation fields.

Inputs and call shape. Pass the optional title and logging flag shown in the signature. No simulation field is allocated by the constructor.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as

unsupported by the current code.

Related theory. MPM weak form and transfers.

Source. `src/mpm/mainMPM.py`

28.1.2 `MPM.set_configuration`

Signature. `self,log=True,**kwargs`

Purpose. sets dimension, solver family, coupling mode, boundary policy, and feature switches that select later engine branches.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers.

Source. `src/mpm/mainMPM.py`

28.1.3 `MPM.set_implicit_solver_parameters`

Signature. `self,**implicit_parameters`

Purpose. sets implicit MPM assembly, Krylov, tolerance, tangent, and boundary-reaction controls.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, Implicit solid MPM.

Source. `src/mpm/mainMPM.py`

28.1.4 `MPM.set_semi_implicit_solver_parameters`

Signature. `self,semi_implicit_parameters`

Purpose. sets MGPCG, iteration, tolerance, and pressure-update controls for the documented `solver_type="SemiImplicit"`, `material_type="TwoPhaseDoubleLayer"` route.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, Implicit solid MPM, Semi-implicit two-phase MPM.

Source. `src/mpm/mainMPM.py`

28.1.5 `MPM.set_fbar_parameters`

Signature. `self,**kwargs`

Purpose. sets F-bar or volumetric locking mitigation parameters for solid MPM where supported.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers.

Source. `src/mpm/mainMPM.py`

28.1.6 `MPM.set_solver`

Signature. `self,solver,log=True`

Purpose. sets timestep, final time, save interval, nonlinear or linear tolerances, and integration parameters.

Retry contract. `enable_step_retry` and the three `step_retry_*` controls apply only to AffineBody and LSMPM Soft-Affine IPC. Standard DEM paths are explicit and reject an enabled policy.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, Implicit solid MPM.

Source. `src/mpm/mainMPM.py`

28.1.7 `MPM.memory_allocate`

Signature. `self,memory,log=True`

Purpose. allocates Taichi fields from capacities before kernels compile or bodies are inserted.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers.

Source. `src/mpm/mainMPM.py`

28.1.8 `MPM.print_basic_simulation_info`

Signature. `self`

Purpose. prints top-level module and environment summary.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact

buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers.

Source. `src/mpm/mainMPM.py`

28.1.9 `MPM.print_simulation_info`

Signature. `self`

Purpose. prints configured simulation, material, element, body, and boundary summary.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers.

Source. `src/mpm/mainMPM.py`

28.1.10 `MPM.print_solver_info`

Signature. `self`

Purpose. prints solver parameters and selected numerical branches.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, Implicit solid MPM.

Source. `src/mpm/mainMPM.py`

28.1.11 `MPM.add_contact`

Signature. `self,contact_type,**contact_phys`

Purpose. registers MPM contact or coupling physics where the selected MPM branch supports it.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, IGA-MPM IPC coupling.

Source. `src/mpm/mainMPM.py`

28.1.12 `MPM.add_material`

Signature. `self,model=None,material=None,**kwargs`

Purpose. registers constitutive, fluid, rigid, or contact material parameters referenced by later bodies. With `mpm_backend="Direct"`, implicit ULMPM accepts `NeoHookean`, `DruckerPrager`, or `VonMises`; the latter two are finite-strain associated models rather than the Native small-strain material-manager implementations. They always consume a 3-by-3 material map: Cartesian 2D is automatically embedded as plane strain, axisymmetric 2D uses no-swirl kinematics, and `plane_strain=False` is rejected for Cartesian 2D plasticity.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, Constitutive and contact models.

Source. `src/mpm/mainMPM.py`

28.1.13 `MPM.add_element`

Signature. `self,element`

Purpose. defines MPM background grid, shape function, support, quadrature, and adaptive-grid options.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, Adaptive MPM.

Source. `src/mpm/mainMPM.py`

28.1.14 `MPM.add_region`

Signature. `self,region`

Purpose. creates named geometric selectors reused by body, boundary, delete, update, and coupling calls.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers.

Source. `src/mpm/mainMPM.py`

28.1.15 `MPM.add_body`

Signature. `self,body`

Purpose. generates particles or solver-specific body records and assigns material/body identifiers.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact

buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers.

Source. `src/mpm/mainMPM.py`

28.1.16 `MPM.create_body`

Signature. `self`

Purpose. returns the body container used by the direct MPM backend.

Call order and validation. Set `mpm_backend="Direct"` first, populate the returned `src/mpm/generator/Body.py` object, then pass that object to `MPM.add_body`. Native MPM raises a runtime error.

Related theory. MPM weak form and transfers.

Source. `src/mpm/mainMPM.py`

28.1.17 `MPM.create_ground`

Signature. `self`

Purpose. returns the ground/contact container used by the direct MPM backend.

Call order and validation. Set `mpm_backend="Direct"` first and pass the configured `src/mpm/generator/Ground.py` object to `MPM.add_ground`. Native MPM raises a runtime error.

Related theory. MPM contact.

Source. `src/mpm/mainMPM.py`

28.1.18 `MPM.add_ground`

Signature. `self,ground`

Purpose. registers the ground object consumed while the direct MPM engine is built.

Call order and validation. Available only for `mpm_backend="Direct"`; call it before `build`/`run`. It does not replace native MPM boundary-condition dictionaries.

Related theory. MPM contact.

Source. `src/mpm/mainMPM.py`

28.1.19 `MPM.add_body_from_file`

Signature. `self,body`

Purpose. loads particles, bodies, walls, or geometry from external files.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers.

Source. `src/mpm/mainMPM.py`

28.1.20 `MPM.add_polygons`

Signature. `self,body`

Purpose. adds polygonal geometry as MPM body or boundary input.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers.

Source. `src/mpm/mainMPM.py`

28.1.21 `MPM.read_restart`

Signature. `self,file_number,file_path,is_continue=True`

Purpose. loads saved runtime fields and optionally continues time counters.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers.

Source. `src/mpm/mainMPM.py`

28.1.22 `MPM.add_boundary_condition`

Signature. `self,boundary=None`

Purpose. adds Dirichlet, Neumann, pressure, traction, wall, or module-specific constraints.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers.

Source. `src/mpm/mainMPM.py`

28.1.23 `MPM.add_particle_traction`

Signature. `self,traction`

Purpose. adds particle-based surface traction constraints in MPM.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source. `src/mpm/mainMPM.py`

28.1.24 `MPM.add_virtual_stress_field`

Signature. `self,field,region_name=None,function=None`

Purpose. activates virtual stress boundary loading and stores stress/force evaluators.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source. `src/mpm/mainMPM.py`

28.1.25 `MPM.clean_boundary_condition`

Signature. `self,boundary`

Purpose. removes or rewrites boundary-condition data before solver construction.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers.

Source. `src/mpm/mainMPM.py`

28.1.26 `MPM.write_boundary_condition`

Signature. `self,output_path='OutputData'`

Purpose. writes boundary-condition diagnostics for inspection or restart-style workflows.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary

arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers.

Source. `src/mpm/mainMPM.py`

28.1.27 `MPM.select_save_data`

Signature. `self,particle=True,grid=False,object=True`

Purpose. selects recorder fields and output types.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers.

Source. `src/mpm/mainMPM.py`

28.1.28 `MPM.choose_coupling_region`

Signature. `self,region_name=None,function=None`

Purpose. selects the region where a coupled module exchanges forces or fields.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling.

Source. `src/mpm/mainMPM.py`

28.1.29 `MPM.modify_parameters`

Signature. `self,**kwargs`

Purpose. updates supported runtime scalar parameters after initial setup.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers.

Source. `src/mpm/mainMPM.py`

28.1.30 `MPM.add_spatial_grid`

Signature. `self`

Purpose. constructs spatial grid/search data needed by the configured module.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers.

Source. `src/mpm/mainMPM.py`

28.1.31 `MPM.add_engine`

Signature. `self`

Purpose. constructs the Taichi engine branch from the configured module state.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers.

Source. `src/mpm/mainMPM.py`

28.1.32 `MPM.add_recorder`

Signature. `self`

Purpose. creates recorder objects and output buffers.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers.

Source. `src/mpm/mainMPM.py`

28.1.33 `MPM.add_solver`

Signature. `self,**kwargs`

Purpose. creates the linear, nonlinear, pressure, or contact solver objects.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as

unsupported by the current code.

Related theory. MPM weak form and transfers, Implicit solid MPM.

Source. `src/mpm/mainMPM.py`

28.1.34 `MPM.add_postfunctions`

Signature. `self,**functions`

Purpose. registers callbacks executed during or after time stepping.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers.

Source. `src/mpm/mainMPM.py`

28.1.35 `MPM.set_window`

Signature. `self,window`

Purpose. sets visualization window/camera options.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers.

Source. `src/mpm/mainMPM.py`

28.1.36 `MPM.add_essentials`

Signature. `self,**kwargs`

Purpose. runs the internal build sequence required before stepping.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers.

Source. `src/mpm/mainMPM.py`

28.1.37 `MPM.run`

Signature. `self,visualize=False,**kwargs`

Purpose. builds missing essentials, triggers pre-calculation and Taichi compilation, then advances the time loop.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers.

Source. `src/mpm/mainMPM.py`

28.1.38 `MPM.check_critical_timestep`

Signature. `self`

Purpose. checks timestep against explicit stability estimates.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers.

Source. `src/mpm/mainMPM.py`

28.1.39 `MPM.update_particle_properties`

Signature. `self,property_name,value,override=True,bodyID=None,region_name=None,function=None`

Purpose. updates particle fields selected by body id, region, or user function.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers.

Source. `src/mpm/mainMPM.py`

28.1.40 `MPM.delete_particles`

Signature. `self,bodyID=None,region_name=None,function=None`

Purpose. marks or removes selected particles.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers.

Source. `src/mpm/mainMPM.py`

28.1.41 `MPM.save_data`

Signature. `self`

Purpose. forces recorder output at the current state.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers.

Source. `src/mpm/mainMPM.py`

28.1.42 `MPM.postprocessing`

Signature. `self,start_file=0,end_file=-1,read_path=None,write_path=None,**kwargs`

Purpose. converts saved data to visualization or derived outputs.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers.

Source. `src/mpm/mainMPM.py`

28.2 DEM

28.2.1 `DEM.__init__`

Signature. `self,title='AHighPerformanceMultiscaleandMultiphysicsSimulator',log=True`

Purpose. constructs the facade object and stores title/logging metadata without allocating simulation fields.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact.

Source. `src/dem/mainDEM.py`

28.2.2 `DEM.set_configuration`

Signature. `self,log=True,**kwargs`

Purpose. sets domain, boundary policy, DEM scheme, search mode, AffineBody switches, and feature flags that select later engine branches.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact, Affine body dynamics.

Source. `src/dem/mainDEM.py`

28.2.3 `DEM.set_affine_body_parameters`

Signature. `self,**kwargs`

Purpose. updates AffineBody assembly, barrier, friction, Newton, linear-solver, line-search, and CCD/ACCD parameters.

Inputs and call shape. Call after `DEM.set_configuration(scheme="AffineBody")` or use `affine_body_parameters` inside `set_configuration`. Supported keys are listed in DEM Dictionary Schemas.

Numerical effect. These parameters control the affine nonlinear solve, IPC-style activation distance and barrier stiffness, friction regularization, linear solve tolerance, and CCD/ACCD step capping.

Call order and validation. Set before `add_essentials` or `run` builds the AffineBody engine.

Related theory. Affine body dynamics, Contact Search, ACCD.

Source. `src/dem/mainDEM.py`

28.2.4 `DEM.set_solver`

Signature. `self,solver,log=True`

Purpose. sets timestep, final time, save interval, nonlinear or linear tolerances, and integration parameters.

Retry contract. The four bounded step-retry keys are accepted only for implicit AffineBody and LSMPM Soft-Affine IPC. Standard DEM, LSDEM, and non-IPC LSMPM time integration is explicit.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact, Implicit solid MPM.

Source. `src/dem/mainDEM.py`

28.2.5 `DEM.memory_allocate`

Signature. `self,memory,log=True`

Purpose. allocates Taichi fields from capacities before kernels compile or bodies are inserted.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact.

Source. `src/dem/mainDEM.py`

28.2.6 `DEM.print_basic_simulation_info`

Signature. `self`

Purpose. prints top-level module and environment summary.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact.

Source. `src/dem/mainDEM.py`

28.2.7 `DEM.print_simulation_info`

Signature. `self`

Purpose. prints configured simulation, material, element, body, and boundary summary.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact.

Source. `src/dem/mainDEM.py`

28.2.8 `DEM.print_solver_info`

Signature. `self`

Purpose. prints solver parameters and selected numerical branches.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary

arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact, Implicit solid MPM.

Source. `src/dem/mainDEM.py`

28.2.9 `DEM.add_region`

Signature. `self,region`

Purpose. creates named geometric selectors reused by body, boundary, delete, update, and coupling calls.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact.

Source. `src/dem/mainDEM.py`

28.2.10 `DEM.add_attribute`

Signature. `self,materialID,attribute`

Purpose. adds DEM material/attribute data.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as

unsupported by the current code.

Related theory. DEM dynamics and contact.

Source. `src/dem/mainDEM.py`

28.2.11 `DEM.add_template`

Signature. `self,template`

Purpose. adds DEM clump, rigid, or template geometry.

Inputs and call shape. A template dict may use `TemplateType`; the lower-case `template_type` spelling is accepted as a compatibility alias. Other dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact.

Source. `src/dem/mainDEM.py`

28.2.12 `DEM.preprocess_soft_grid_template`

Signature. `self,name,points_per_cell=1,center_quadrature=False,verlet_distance_multiplier=None,material_points=None,point_volume=None,mechanical_grid_spacing=None,mechanical_grid_refinement=None,reference_volume=None`

Purpose. builds and caches LSMPM material points, mechanical-grid topology, and particle/surface/SDF interpolation support for one registered template, then returns capacity and spacing diagnostics.

Call order and validation. Requires `DEM.set_configuration(scheme="LSMPM")` and a prior `DEM.add_template`. Call it before `memory_allocate` when its returned counts are used to size soft fields. Storage, grid topology, shape function, spacing ratio, and Verlet padding come from the current LSMPM configuration unless explicitly overridden.

Related theory. Level-Set MPM Soft Bodies.

Source. `src/dem/mainDEM.py`, `src/mpm/generator/BodyGenerator.py`

28.2.13 `DEM.create_body`

Signature. `self,body`

Purpose. creates DEM bodies using the configured generator.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact.

Source. `src/dem/mainDEM.py`

28.2.14 `DEM.create_body_batch`

Signature. `self,body`

Purpose. forwards a body dictionary/list to the generator batch path and returns the generator's batch result.

Call order and validation. Uses the same registered regions, templates, material attributes, body schema, and allocation limits as `DEM.create_body`.

Related theory. DEM dynamics and contact.

Source. `src/dem/mainDEM.py`

28.2.15 `DEM.add_body`

Signature. `self,body`

Purpose. generates particles or solver-specific body records and assigns material/body identifiers.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact.

Source. `src/dem/mainDEM.py`

28.2.16 `DEM.add_body_from_file`

Signature. `self,body`

Purpose. loads particles, bodies, walls, or geometry from external files.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact.

Source. `src/dem/mainDEM.py`

28.2.17 `DEM.add_wall`

Signature. `self,body`

Purpose. adds DEM wall geometry.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact.

Source. `src/dem/mainDEM.py`

28.2.18 `DEM.add_wall_from_file`

Signature. `self,body`

Purpose. loads DEM wall geometry from a file.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact.

Source. `src/dem/mainDEM.py`

28.2.19 `DEM.choose_neighbor`

Signature. `self`

Purpose. constructs selected DEM broad-phase neighbor search.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact, DEM BVH/LBVH search.

Source. `src/dem/mainDEM.py`

28.2.20 `DEM.choose_contact_model`

Signature. `self,particle_particle_contact_model=None,particle_wall_contact_model=None`

Purpose. selects DEM or MPDEM contact force/tangent laws before pair properties are added.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact, Constitutive and contact models, IGA-

MPM IPC coupling.

Source. `src/dem/mainDEM.py`

28.2.21 `DEM.add_property`

Signature. `self,materialID1,materialID2,property,dType="all"`

Purpose. fills contact or coupling property tables for material-pair interactions.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact, Constitutive and contact models.

Source. `src/dem/mainDEM.py`

28.2.22 `DEM.inherit_property`

Signature. `self,materialID,property`

Purpose. copies contact/material property values to another material pair.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact, Constitutive and contact models.

Source. `src/dem/mainDEM.py`

28.2.23 `DEM.load_history_contact`

Signature. `self`

Purpose. restores stored tangential/rolling/contact history fields.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact, IGA-MPM IPC coupling.

Source. `src/dem/mainDEM.py`

28.2.24 `DEM.select_save_data`

Signature. `self,particle=True,sphere=False,clump=False,surface=True,grid=False,boundi ng=False,wall=False,particle_particle_contact=False,particle_wall_contact=False`

Purpose. selects recorder fields and output types.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact.

Source. `src/dem/mainDEM.py`

28.2.25 `DEM.read_restart`

Signature. `self,file_number,file_path,particle=True,sphere=True,clump=False,wall=True ,servo=False,ppcontact=True,pwcontact=True,is_continue=True`

Purpose. loads saved runtime fields and optionally continues time counters.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact.

Source. `src/dem/mainDEM.py`

28.2.26 `DEM.modify_parameters`

Signature. `self,**kwargs`

Purpose. updates supported runtime scalar parameters after initial setup.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact.

Source. `src/dem/mainDEM.py`

28.2.27 `DEM.add_engine`

Signature. `self,callback`

Purpose. constructs the Taichi engine branch from the configured module state.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact.

Source. `src/dem/mainDEM.py`

28.2.28 `DEM.add_recorder`

Signature. `self`

Purpose. creates recorder objects and output buffers.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact.

Source. `src/dem/mainDEM.py`

28.2.29 `DEM.add_solver`

Signature. `self,**kwargs`

Purpose. creates the linear, nonlinear, pressure, or contact solver objects.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact, Implicit solid MPM.

Source. `src/dem/mainDEM.py`

28.2.30 `DEM.add_postfunctions`

Signature. `self,**functions`

Purpose. registers callbacks executed during or after time stepping.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact.

Source. `src/dem/mainDEM.py`

28.2.31 `DEM.add_essentials`

Signature. `self,**kwargs`

Purpose. runs the internal build sequence required before stepping.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact.

Source. `src/dem/mainDEM.py`

28.2.32 `DEM.check_verlet_distance_multiplier`

Signature. `self`

Purpose. checks whether the neighbor-search skin is compatible with motion and contact size.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact.

Source. `src/dem/mainDEM.py`

28.2.33 `DEM.set_static_wall`

Signature. `self,static_wall=True`

Purpose. switches DEM walls to static or dynamic handling.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact.

Source. `src/dem/mainDEM.py`

28.2.34 `DEM.servo_switch`

Signature. `self,status="On"`

Purpose. enables or disables servo-wall control.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact.

Source. `src/dem/mainDEM.py`

28.2.35 `DEM.set_window`

Signature. `self,window`

Purpose. sets visualization window/camera options.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary

arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact.

Source. `src/dem/mainDEM.py`

28.2.36 `DEM.run`

Signature. `self,visualize=False,**kwargs`

Purpose. builds missing essentials, triggers pre-calculation and Taichi compilation, then advances the time loop.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact.

Source. `src/dem/mainDEM.py`

28.2.37 `DEM.is_lsmpm_soft_affine_ipc`

Signature. `self`

Purpose. reports whether the scene is routed through mixed LSMPM soft–AffineBody IPC.

Call order and validation. This is an orchestration query used by the facade/coupling run path after configuration; it does not enable IPC by itself.

Related theory. Mixed LSMPM Soft-Affine IPC.

Source. `src/dem/mainDEM.py`

28.2.38 `DEM.check_critical_timestep`

Signature. `self`

Purpose. checks timestep against explicit stability estimates.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact.

Source. `src/dem/mainDEM.py`

28.2.39 `DEM.get_critical_timestep`

Signature. `self`

Purpose. returns estimated DEM critical timestep.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact.

Source. `src/dem/mainDEM.py`

28.2.40 `DEM.update_material_properties`

Signature. `self,materialID,property_name,value,override=True`

Purpose. updates material parameters by material id.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact, Constitutive and contact models.

Source. `src/dem/mainDEM.py`

28.2.41 `DEM.update_particle_properties`

Signature. `self,property_name,value,override=True,bodyID=None,region_name=None,function=None`

Purpose. updates particle fields selected by body id, region, or user function.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact.

Source. `src/dem/mainDEM.py`

28.2.42 `DEM.update_wall_status`

Signature. `self,wallID,property_name,value,override=True`

Purpose. updates wall kinematic/status fields.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact.

Source. `src/dem/mainDEM.py`

28.2.43 `DEM.update_contact_properties`

Signature. `self,materialID1,materialID2,property_name,value,overide=True`

Purpose. updates pair contact properties after contact models are selected.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact, IGA-MPM IPC coupling.

Source. `src/dem/mainDEM.py`

28.2.44 `DEM.delete_particles`

Signature. `self,bodyID=None,region_name=None,function=None`

Purpose. marks or removes selected particles.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact.

Source. `src/dem/mainDEM.py`

28.2.45 `DEM.delete_walls`

Signature. `self,wallID`

Purpose. removes selected DEM walls.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary

arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact.

Source. `src/dem/mainDEM.py`

28.2.46 `DEM.save_data`

Signature. `self`

Purpose. forces recorder output at the current state.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. DEM dynamics and contact.

Source. `src/dem/mainDEM.py`

28.2.47 `DEM.postprocessing`

Signature. `self,start_file=0,end_file=-1,read_path=None,write_path=None,scheme='DEM',**kwargs`

Purpose. converts saved data to visualization or derived outputs.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as

unsupported by the current code.

Related theory. DEM dynamics and contact.

Source. `src/dem/mainDEM.py`

28.3 DEMPM/CFDEM

28.3.1 `DEMPM/CFDEM.__init__`

Signature. `self,dem:DEM,mpm:MPM,title='AHighPerformanceMultiscaleandMultiphysicsSimulatoronGPU',log=True`

Purpose. constructs the facade object and stores title/logging metadata without allocating simulation fields.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM dynamics and contact, DEM-MPM and CFDEM coupling.

Source. `src/mpdem/mainDEMPM.py`

28.3.2 `DEMPM/CFDEM.set_configuration`

Signature. `self,log=True,**kwargs`

Purpose. sets dimension, solver family, coupling mode, boundary policy, and feature switches that select later engine branches.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM dynamics and contact, DEM-MPM

and CFDEM coupling.

Source. `src/mpdem/mainDEMPM.py`

28.3.3 `DEMPM/CFDEM.set_solver`

Signature. `self,solver,log=True`

Purpose. sets timestep, final time, save interval, nonlinear or linear tolerances, and integration parameters.

Retry contract. The four bounded step-retry keys are accepted for LSMPM Soft-Affine IPC and Direct MPM–AffineBody BarrierIPC. The latter rejects enabled retry with Semi-IPC. Other coupling routes retain their own explicit or fluid-projection restrictions.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM dynamics and contact, DEM-MPM and CFDEM coupling, Implicit solid MPM.

Source. `src/mpdem/mainDEMPM.py`

28.3.4 `DEMPM/CFDEM.memory_allocate`

Signature. `self,memory,dem_memory=None,mpm_memory=None`

Purpose. allocates Taichi fields from capacities before kernels compile or bodies are inserted.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM dynamics and contact, DEM-MPM and CFDEM coupling.

Source. `src/mpdem/mainDEMPM.py`

28.3.5 `DEMPM/CFDEM.print_basic_simulation_info`

Signature. `self`

Purpose. prints top-level module and environment summary.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM dynamics and contact, DEM-MPM and CFDEM coupling.

Source. `src/mpdem/mainDEMPM.py`

28.3.6 `DEMPM/CFDEM.print_solver_info`

Signature. `self`

Purpose. prints solver parameters and selected numerical branches.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM dynamics and contact, DEM-MPM and CFDEM coupling, Implicit solid MPM.

Source. `src/mpdem/mainDEMPM.py`

28.3.7 `DEMPM/CFDEM.add_body`

Signature. `self,mpm_body=None,dem_particle=None,write_file=False,check_overlap=False,adaptive_boundary_radius=False`

Purpose. generates particles or solver-specific body records and assigns material/body identifiers.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM dynamics and contact, DEM-MPM and CFDEM coupling.

Source. `src/mpdem/mainDEMPM.py`

28.3.8 `DEMPM/CFDEM.choose_contact_model`

Signature. `self,particle_particle_contact_model,particle_wall_contact_model=None`

Purpose. selects DEM or MPDEM contact force/tangent laws before pair properties are added.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM dynamics and contact, DEM-MPM and CFDEM coupling, Constitutive and contact models, IGA-MPM IPC coupling.

Source. `src/mpdem/mainDEMPM.py`

28.3.9 `DEMPM/CFDEM.add_property`

Signature. `self,DEMmaterial,MPMmaterial,property,dType="all"`

Purpose. fills contact or coupling property tables for material-pair interactions.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM dynamics and contact, DEM-MPM and CFDEM coupling, Constitutive and contact models.

Source. `src/mpdem/mainDEMPM.py`

28.3.10 `DEMPM/CFDEM.modify_parameters`

Signature. `self,**kwargs`

Purpose. updates supported runtime scalar parameters after initial setup.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM dynamics and contact, DEM-MPM and CFDEM coupling.

Source. `src/mpdem/mainDEMPM.py`

28.3.11 `DEMPM/CFDEM.sync_settings`

Signature. `self`

Purpose. synchronizes DEM and MPM settings in coupled workflows.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM dynamics and contact, DEM-MPM and CFDEM coupling.

Source. `src/mpdem/mainDEMPM.py`

28.3.12 `DEMPM/CFDEM.read_restart`

Signature. `self,file_number,file_path,ppcontact=False,pwcontact=False`

Purpose. loads saved runtime fields and optionally continues time counters.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM dynamics and contact, DEM-MPM and CFDEM coupling.

Source. `src/mpdem/mainDEMPM.py`

28.3.13 `DEMPM/CFDEM.load_history_contact`

Signature. `self`

Purpose. restores stored tangential/rolling/contact history fields.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM dynamics and contact, DEM-MPM and CFDEM coupling, IGA-MPM IPC coupling.

Source. `src/mpdem/mainDEMPM.py`

28.3.14 `DEMPM/CFDEM.select_save_data`

Signature. `self,particle_particle_contact=False,particle_wall_contact=False`

Purpose. selects recorder fields and output types.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM dynamics and contact, DEM-MPM and CFDEM coupling.

Source. `src/mpdem/mainDEMPM.py`

28.3.15 `DEMPM/CFDEM.add_essentials`

Signature. `self,**kwargs:dict`

Purpose. runs the internal build sequence required before stepping.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM dynamics and contact, DEM-MPM and CFDEM coupling.

Source. `src/mpdem/mainDEMPM.py`

28.3.16 `DEMPM/CFDEM.add_postfunctions`

Signature. `self,**functions`

Purpose. registers callbacks executed during or after time stepping.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set

before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM dynamics and contact, DEM-MPM and CFDEM coupling.

Source. `src/mpdem/mainDEMPM.py`

28.3.17 `DEMPM/CFDEM.update_contact_properties`

Signature. `self,materialID1,materialID2,property_name,value,overide=True`

Purpose. updates pair contact properties after contact models are selected.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM dynamics and contact, DEM-MPM and CFDEM coupling, IGA-MPM IPC coupling.

Source. `src/mpdem/mainDEMPM.py`

28.3.18 `DEMPM/CFDEM.is_dem_lsm_pm_soft_rigid_mode`

Signature. `self`

Purpose. reports whether the coupled facade should use the DEM-driven LSMPM soft/-rigid run path.

Call order and validation. Returns true when `dem.sims.scheme="LSMPM"` and the MPM sub-solver has no ordinary material allocation. It is an orchestration query and does not configure the scene.

Related theory. MPM Soft-Particle Backend and MPDEM Dispatch.

Source. `src/mpdem/mainDEMPM.py`

28.3.19 DEMPM/CFDEM.run_dem_lsm_pm_soft_rigid

Signature. `self,**kwargs`

Purpose. builds the DEM LSMPM scene, installs the MPM recorder proxy, checks the explicit critical timestep, and runs the DEM-owned solver.

Call order and validation. Used by `run` only for the routing condition above. `visualize` selects solver versus viewer, and `strict_timestep=True` raises if the critical-timestep check changes the requested value.

Related theory. Critical Timestep, MPM Soft-Particle Backend and MPDEM Dispatch.

Source. `src/mpdem/mainDEMPM.py`

28.3.20 DEMPM/CFDEM.run

Signature. `self,**kwargs`

Purpose. builds missing essentials, triggers pre-calculation and Taichi compilation, then advances the time loop.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM dynamics and contact, DEM-MPM and CFDEM coupling.

Source. `src/mpdem/mainDEMPM.py`

28.3.21 DEMPM/CFDEM.check_critical_timestep

Signature. `self`

Purpose. checks timestep against explicit stability estimates.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before

kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM dynamics and contact, DEM-MPM and CFDEM coupling.

Source. `src/mpdem/mainDEMPM.py`

28.3.22 `DEMPM/CFDEM.get_critical_timestep`

Signature. `self`

Purpose. returns estimated DEM critical timestep.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM dynamics and contact, DEM-MPM and CFDEM coupling.

Source. `src/mpdem/mainDEMPM.py`

28.3.23 `DEMPM/CFDEM.save_data`

Signature. `self`

Purpose. forces recorder output at the current state.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM dynamics and contact, DEM-MPM and CFDEM coupling.

Source. `src/mpdem/mainDEMPM.py`

28.3.24 `DEMPM/CFDEM.postprocessing`

Signature. `self,start_file=0,end_file=-1,read_path=None,write_path=None,scheme=None,**kwargs`

Purpose. converts saved data to visualization or derived outputs.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM dynamics and contact, DEM-MPM and CFDEM coupling.

Source. `src/mpdem/mainDEMPM.py`

28.4 FEM

28.4.1 `FEM.set_configuration`

Signature. `self,dimension=3,solver_type="Explicit",**kwargs`

Purpose. Selects the FEM dimension and explicit or implicit solver family; additional keywords are forwarded to solver construction.

Related theory. Finite Element Method.

Source. `src/fem/mainFEM.py`

28.4.2 `FEM.create_mesh`

Signature. `self,geometry=None,**kwargs`

Purpose. Generates `box`, `rectangle`, `circle`, or `cylinder` meshes, or reads a mesh file. Geometry sizes/divisions and `rest_shape` are normalized before dispatch; rectangles/circles require TRI3 and cylinders currently require TET4.

Related theory. Finite Element Method.

Source. `src/fem/mainFEM.py`

28.4.3 `FEM.add_mesh`

Signature. `self,mesh=None,**kwargs`

Purpose. Registers an `FEMMesh`, filename, generation dictionary, or primitive parameters. If `rest_shape` is omitted, current coordinates are used as the reference configuration.

Related theory. Finite Element Method.

Source. `src/fem/mainFEM.py`

28.4.4 `FEM.add_material`

Signature. `self,model="StVK",material=None,**kwargs`

Purpose. Registers a supported finite-strain or cloth constitutive model, or one of the shared incremental elastoplastic models `ElasticPerfectlyPlastic`, `MohrCoulomb`, `DruckerPrager`, `StateDependentMohrCoulomb`, `ModifiedCamClay`, `GranularMaterial`, `SanisandMS`, and `NorSand`. A cloth model triggers the TRI3 cloth assembly path; an elastoplastic model requires explicit HEX8 and cannot be combined with FEM IPC/AL. Modified Cam Clay has no single preprocessing modulus and therefore requires a finite explicit `dt` instead of `"auto"`.

Related theory. Finite Element Method.

Source. `src/fem/mainFEM.py`

28.4.5 `FEM.add_boundary_condition`

Signature. `self,boundary=None,dirichlet=None,neumann=None`

Purpose. Registers object-based or dictionary-based displacement, nodal-force, traction, pressure, and edge-traction conditions.

Related theory. Finite Element Method.

Source. `src/fem/mainFEM.py`

28.4.6 `FEM.set_bending_model` and cloth energies

Signatures. `set_bending_model(model)`, `add_stitch(stitches,stiffness,ratios=None)`, `add_spring(nodes,stiffness,targets=None)`, and `add_sdf(...)`.

Purpose. Selects quadratic/dihedral/no bending and adds analytic garment stitch, target spring, or frozen-frame SDF energy. Garment stitch point–edge relations are passed into FEM collision culling as exclusions.

Related theory. Finite Element Method and FEM collision culling.

Source. `src/fem/mainFEM.py`

28.4.7 `FEM.set_solver`

Signature. `self,solver=None,**kwargs`

Purpose. Stores time integration, nonlinear iteration, Armijo line search, output, `assemble_type`, `linear_solver`, and `project_pd` settings. PCG requires projected local Hessians.

Retry contract. Implicit FEM accepts `enable_step_retry`, `step_retry_max_retries`, `step_retry_reduction`, and `step_retry_minimum_timestep`. Explicit FEM and implicit Barrier IPC/contact-free FEM restore nodal state before retry.

Related theory. Finite Element Method.

Source. `src/fem/mainFEM.py`

28.4.8 `FEM.add_contact`

Signature. `self,model="IPC",**kwargs`

Purpose. Registers ordinary Barrier `IPC` contact. `broad_phase` accepts `LinkedCell` or `BVH`; this selects the spatial backend before common distance/CCD pruning and compact assembly. Contact currently requires the implicit solver.

Related theory. FEM collision culling and CCD.

Source. `src/fem/mainFEM.py` and `src/fem/contact/ContactModel.py`

28.4.9 `FEM.add_contact_property`

Signature. `self,body_id1,body_id2,property=None,**kwargs`

Purpose. Adds or replaces the IPC parameters for one unordered pair of disconnected FEM bodies. Body IDs come from `cell_body_ids`, or are inferred from mesh connected components. Pair-specific broad phase, activation/minimum distance, stiffness, friction, CCD, and PSD projection are used for PT/EE culling and assembly.

Related theory. FEM collision culling and CCD.

Source. `src/fem/mainFEM.py` and `src/fem/contact/ContactModel.py`

28.4.10 `FEM.build` and `FEM.run`

Signatures. `build()` and `run(*args,**kwargs)`.

Purpose. Validates mesh/material/contact compatibility, constructs the Taichi volume or cloth engine, and runs it. Requesting a NumPy/Scipy assembly backend is rejected; Scipy is available only as a linear solver.

Related theory. Finite Element Method.

Source. `src/fem/mainFEM.py`

28.5 IGA

28.5.1 `IGA.__init__`

Signature. `self,title="IsoGeometricAnalysisEngine",log=True`

Purpose. constructs the facade object and stores title/logging metadata without allocating simulation fields.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. IGA and IGAMPM.

Source. `src/iga/mainIGA.py`

28.5.2 `IGA.set_configuration`

Signature. `self,dimension=3,solver_type="Implicit",**kwargs`

Purpose. sets dimension, solver family, coupling mode, boundary policy, and feature switches that select later engine branches.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. IGA and IGAMPM.

Source. `src/iga/mainIGA.py`

28.5.3 `IGA.add_primitives`

Signature. `self,primitives`

Purpose. adds IGA geometric primitives, control points, knots, and weights.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. IGA and IGAMPM.

Source. `src/iga/mainIGA.py`

28.5.4 `IGA.add_boundary_condition`

Signature. `self,dirichlet=None,neumann=None`

Purpose. adds Dirichlet, Neumann, pressure, traction, wall, or module-specific constraints.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. IGA and IGAMPM.

Source. `src/iga/mainIGA.py`

28.5.5 `IGA.add_material`

Signature. `self,**kwargs`

Purpose. registers constitutive, fluid, rigid, or contact material parameters referenced by later bodies.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before

kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. IGA and IGAMPM, Constitutive and contact models.

Source. `src/iga/mainIGA.py`

28.5.6 `IGA.add_element`

Signature. `self,degree`

Purpose. defines MPM background grid, shape function, support, quadrature, and adaptive-grid options.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. IGA and IGAMPM, Adaptive MPM.

Source. `src/iga/mainIGA.py`

28.5.7 `IGA.set_solver`

Signature. `self,**kwargs`

Purpose. sets timestep, final time, save interval, nonlinear or linear tolerances, and integration parameters.

Retry contract. The four bounded step-retry keys apply only to implicit IGA. Non-convergence is raised before control-point state advance; explicit IGA rejects an enabled policy.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. IGA and IGAMPM, Implicit solid MPM.

Source. `src/iga/mainIGA.py`

28.5.8 `IGA.build`

Signature. `self`

Purpose. constructs lower-level IGA, MPM, or coupling solver objects after all primitives and options are registered.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. IGA and IGAMPM.

Source. `src/iga/mainIGA.py`

28.5.9 `IGA.run`

Signature. `self,*args,**kwargs`

Purpose. builds missing essentials, triggers pre-calculation and Taichi compilation, then advances the time loop.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. IGA and IGAMPM.

Source. `src/iga/mainIGA.py`

28.6 IGAMPM

28.6.1 `IGAMPM.__init__`

Signature. `self,iga=None,mpm=None,title="IGA-MPMCouplingEngine",log=True,**kwargs`

Purpose. constructs the current IGA-MPM coupling facade around existing or newly created IGA and MPM objects.

Inputs and call shape. Pass optional `iga` and `mpm` facade/engine objects. Extra keyword arguments are stored as contact-model defaults and forwarded to `ContactManager`.

Numerical effect. No kernels are built at construction time; the call stores the sub-solvers, simulation configuration, and contact manager.

Call order and validation. Call before configuring contact or building the coupled engine.

Related theory. IGA-MPM.

Source. `src/igampm/mainIGAMPM.py`

28.6.2 `IGAMPM.set_configuration`

Signature. `self,dimension=None,coupling_scheme="IGAMPM",contact_model="IPC",activate_friction=False,axisymmetric=False,axis_offset=0.0`

Purpose. sets coupling dimension and coordinate mode, validates `IGAMPM`/`IGA-MPM`, and selects implicit IPC or explicit DEM-law contact.

Inputs and call shape. `contact_model` accepts `IPC`, `Linear`, or `HertzMindlin`. Axisymmetric contact accepts IPC only.

Numerical effect. Updates the coupling simulation object and rebuilds the contact manager's model selection.

Call order and validation. Call before `build()`; IPC requires implicit/implicit children, while a DEM law requires explicit/explicit children and native Lagrangian-coupling MPM. Construct an explicit `IGAMPM` before MPM particle allocation so the facade can select the coupling particle field.

Related theory. Current IGA-MPM facade.

Source. `src/igampm/mainIGAMPM.py`

28.6.3 `IGAMPM.choose_contact_model`

Signature. `self,contact_model="IPC",**kwargs`

Purpose. configures the coupling contact manager and updates a built engine if present.

Inputs and call shape. Select `IPC`, `Linear`, or `HertzMindlin`. IPC kwargs include `kappa`, `dhat`, `mu`, `epsv`, `activate_friction`, `barrier_nnz`, `friction_nnz`, and coupled `assemble_type` equal to `COO` or `HashTriplet`. Explicit material-pair values are supplied separately through `add_property`.

Numerical effect. Creates IPC barrier/friction objects or one host-side DEM property owner whose force evaluation remains in Taichi; storage choices freeze at build time.

Call order and validation. Call after construction and before `build()` unless deliberately updating an already built engine.

Related theory. IGA-MPM IPC contact model.

Source. `src/igampm/mainIGAMPM.py`

28.6.4 `IGAMPM.add_property`

Signature. `self,MPMmaterial,IGAbody,property,dType="all"`

Purpose. registers an explicit contact property for one MPM-material/IGA-patch pair.

Inputs and call shape. `property` contains either the Linear stiffness/damping/friction keys or the Hertz–Mindlin modulus/Poisson/restitution/friction keys. `dType` is retained only for facade spelling compatibility.

Numerical effect. Stores the pair values until build allocates the device property table. The same Taichi DEM force functions used by DEM coupling evaluate the pair during contact.

Call order and validation. Select an explicit law first and call before `build()`. Every active material/patch pair must be registered. Nonzero rolling friction is rejected.

Related theory. Explicit DEM-law IGA–MPM contact.

Source. `src/igampm/mainIGAMPM.py`

28.6.5 `IGAMPM.set_solver`

Signature. `self,iga=None,mpm=None,coupling=None,**kwargs`

Purpose. forwards solver dictionaries to the IGA and MPM sub-solvers and stores coupling retry controls before building the coupling engine.

Inputs and call shape. Pass `iga=\{...\}` and/or `mpm=\{...\}` dictionaries using the corresponding sub-solver API. The optional `coupling` dictionary accepts `enable_step_retry`, `step_retry_max_retries`, `step_retry_reduction`, and `step_retry_minimum_timestep`.

Numerical effect. Implicit IPC uses the existing full IGA/MPM/contact transaction to retry recognized nonconvergence at a bounded reduced timestep. Explicit DEM-law contact rejects an enabled retry policy.

Call order and validation. Call before `build()`.

Related theory. IGA-MPM.

Source. `src/igampm/mainIGAMPM.py`

28.6.6 `IGAMPM.build`

Signature. `self`

Purpose. builds or reuses the IGA engine and MPM engine, checks solver-family compatibility, prepares MPM active dofs, and constructs the IGA-MPM coupling engine.

Inputs and call shape. No user dictionary is required; all sub-solver state must already be registered.

Numerical effect. Creates the selected engine under `src/igampm/engines`. IPC installs barrier/friction buffers and COO/HashTriplet storage. Explicit contact allocates a stable particle/surface history table and NURBS-surface bounds.

Call order and validation. IPC requires two implicit children. Linear/Hertz–Mindlin requires two explicit children, native Lagrangian MPM, matching positive timesteps, 3D Cartesian geometry, coupled points, and all active pair properties. Mixed pairs are rejected.

Related theory. Current IGA-MPM facade.

Source. `src/igampm/mainIGAMPM.py`

28.6.7 `IGAMPM.run`

Signature. `self,grid_disp=None,include_friction=False,steps=None,verbose=True`

Purpose. runs the selected IGA-MPM contact solve.

Inputs and call shape. May receive an MPM grid displacement, an `include_friction` flag, a step count, and verbosity.

Numerical effect. IPC performs per-pair Taichi point–NURBS ACCD with virtual moving control points, device TOC reduction, sparse assembly, and the monolithic solve. Explicit contact rebuilds current NURBS bounds, projects finite-radius points, applies the DEM force law and equal/opposite rational-basis forces, and advances MPM and IGA. Its CFL gate uses the MPM material and contact estimates; IGA currently supplies no independent spectral stability estimator.

Call order and validation. Calls `build()` lazily if needed.

Related theory. IGA-MPM IPC contact model, Explicit DEM-law IGA–MPM contact.

Source. `src/igampm/mainIGAMPM.py`

28.7 IGA-MPM IPC Coupling

28.7.1 `IGA-MPMIPCCoupling.__init__`

Signature. `self,iga:ImplicitIGA`

Purpose. constructs the facade object and stores title/logging metadata without allocating simulation fields.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

28.7.2 `IGA-MPMIPCCoupling.update_from_patch`

Signature. `self,control_points:ti.template(`

Purpose. provides a user-visible or coupling-visible operation in this facade or helper object.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

28.7.3 `IGA-MPMIPCCoupling.__init__`

Signature. `self,iga:ImplicitIGA,mpm:ImplicitMPM,**kwargs`

Purpose. constructs the facade object and stores title/logging metadata without allocating simulation fields.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact

buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

28.7.4 `IGA-MPMIPCCoupling.total_degree_of_freedom`

Signature. `self`

Purpose. returns total coupled IGA-MPM degree of freedom count.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

28.7.5 `IGA-MPMIPCCoupling.reset_contacts`

Signature. `self`

Purpose. clears stored IPC barrier contacts.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA

and IGAMPM, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

28.7.6 `IGA-MPMIPCCoupling.reset_friction_contacts`

Signature. `self`

Purpose. clears stored IPC friction contacts.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

28.7.7 `IGA-MPMIPCCoupling.load_contacts_from_buffer`

Signature. `self`

Purpose. loads IPC contact buffers into active contact fields.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

28.7.8 `IGA-MPMIPCCoupling.count_contacts`

Signature. `self`

Purpose. returns active IPC barrier contact count.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

28.7.9 `IGA-MPMIPCCoupling.count_friction_contacts`

Signature. `self`

Purpose. returns active IPC friction contact count.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

28.7.10 `IGA-MPMIPCCoupling.update_particle_pos`

Signature. `self,grid_disp:ti.template(`

Purpose. updates MPM particle positions from grid displacement for coupling search.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending

on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

28.7.11 `IGA-MPMIPCCoupling.find_closest_surface_contacts`

Signature. `self,surface_id:ti.i32,prefix_num_knot_u:ti.i32,prefix_num_knot_v:ti.i32,prefix_num_ctrlpts:ti.i32,num_knot_u:ti.i32,num_knot_v:ti.i32,surface:ti.template(`

Purpose. searches closest NURBS surface contact candidates for MPM particles.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. Rebuilds current positive-weight NURBS control-hull bounds. Near pairs and positive SemiIPC multipliers use full closest-point projection. Far inactive pairs retain a conservative distance lower bound for ACCD; their reported distance or global minimum need not be the exact closest distance. Active barrier geometry and derivatives are unchanged. All pairs remain represented for conservative stepping.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Performance. Three-dimensional queries share refreshed knot-span hulls and precomputed Greville coordinates. A fixed-topology span BVH is refitted after deformation and prunes span subtrees while also locating the exact nearest control-point seed. Before projection, span bounds can certify that an entire pair lies outside the activation and strict-feasibility thresholds. Such inactive pairs retain conservative distance bounds for ACCD. Previous closest parameters provide extra seeds without removing potentially improving spans or boundary minima. Built-in Newton and Armijo reuse queries at identical geometry; standalone calls still prepare their own query. CCD first screens whole segments with a per-surface control-direction box, retaining the original tight motion bound for unresolved pairs. Moving ACCD queries retain moving hull evaluation.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, closest-point implicit differentiation, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

28.7.12 `IGA-MPMIPCCoupling.find_closest_curve_contacts`

Signature. `self,surface_id:ti.i32,prefix_num_knot_u:ti.i32,prefix_num_ctrlpts:ti.i32,num_knot_u:ti.i32,surface:ti.template(`

Purpose. searches closest NURBS curve contact candidates for MPM particles.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. Rebuilds current positive-weight NURBS control-hull bounds. Near pairs and positive SemiIPC multipliers use full closest-point projection. Far inactive pairs retain a conservative distance lower bound for ACCD; their reported distance or global minimum need not be the exact closest distance. Active barrier geometry and derivatives are unchanged. All pairs remain represented for conservative stepping.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, closest-point implicit differentiation, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

28.7.13 `IGA-MPMIPCCoupling.clear_barrier_system`

Signature. `self`

Purpose. zeros IPC barrier matrix and right-hand-side buffers.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

28.7.14 `IGA-MPMIPCCoupling.clear_friction_system`

Signature. `self`

Purpose. zeros IPC friction matrix and right-hand-side buffers.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

28.7.15 `IGA-MPMIPCCoupling.add_barrier_triplet`

Signature. `self,row,col,value`

Purpose. inserts one barrier matrix triplet.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, Implicit solid MPM, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

28.7.16 `IGA-MPMIPCCoupling.add_friction_triplet`

Signature. `self,row,col,value`

Purpose. inserts one friction matrix triplet.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, Implicit solid MPM, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

28.7.17 `IGA-MPMIPCCoupling.scatter_barrier_rhs`

Signature. `self,active_mpm_dof:ti.i32`

Purpose. scatters IPC barrier residual into subsystem right-hand sides.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

28.7.18 `IGA-MPMIPCCoupling.scatter_friction_rhs`

Signature. `self,active_mpm_dof:ti.i32`

Purpose. scatters IPC friction residual into subsystem right-hand sides.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

28.7.19 `IGA-MPMIPCCoupling.assemble_barrier_matrix_for_surface`

Signature. `self,surface_id:ti.i32,prefix_num_knot_u:ti.i32,prefix_num_knot_v:ti.i32,prefix_num_ctrlpts:ti.i32,num_knot_u:ti.i32,num_knot_v:ti.i32,num_ctrlpts_u:ti.i32,surface:ti.template(`

Purpose. assembles IPC barrier Hessian blocks for NURBS surface contacts.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, Implicit solid MPM, IPC gradient/Hessian derivation, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

28.7.20 `IGA-MPMIPCCoupling.assemble_barrier_matrix_for_curve`

Signature. `self,surface_id:ti.i32,prefix_num_knot_u:ti.i32,prefix_num_ctrlpts:ti.i32,num_knot_u:ti.i32,surface:ti.template(`

Purpose. assembles IPC barrier Hessian blocks for NURBS curve contacts.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, Implicit solid MPM, IPC gradient/Hessian derivation, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

28.7.21 `IGA-MPMIPCCoupling.initialize_surface_friction_contacts`

Signature. `self,surface_id:ti.i32,prefix_num_knot_u:ti.i32,prefix_num_knot_v:ti.i32,prefix_num_ctrlpts:ti.i32,num_knot_u:ti.i32,num_knot_v:ti.i32,surface:ti.template(`

Purpose. initializes friction data for surface contacts.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

28.7.22 `IGA-MPMIPCCoupling.initialize_curve_friction_contacts`

Signature. `self,surface_id:ti.i32,prefix_num_knot_u:ti.i32,prefix_num_ctrlpts:ti.i32,num_knot_u:ti.i32,surface:ti.template(`

Purpose. initializes friction data for curve contacts.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

28.7.23 `IGA-MPMIPCCoupling.assemble_friction_matrix_for_surface`

Signature. `self,surface_id:ti.i32,prefix_num_knot_u:ti.i32,prefix_num_knot_v:ti.i32,prefix_num_ctrlpts:ti.i32,num_knot_u:ti.i32,num_knot_v:ti.i32,num_ctrlpts_u:ti.i32,surface:ti.template(`

Purpose. assembles IPC friction tangent blocks for surface contacts.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, Implicit solid MPM, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

28.7.24 `IGA-MPMIPCCoupling.assemble_friction_matrix_for_curve`

Signature. `self,surface_id:ti.i32,prefix_num_knot_u:ti.i32,prefix_num_ctrlpts:ti.i32,num_knot_u:ti.i32,surface:ti.template(`

Purpose. assembles IPC friction tangent blocks for curve contacts.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, Implicit solid MPM, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

28.7.25 `IGA-MPMIPCCoupling.initialize_curve_barrier_python`

Signature. `self`

Purpose. initializes curve barrier contacts using the Python closest-point helper.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact

buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

28.7.26 `IGA-MPMIPCCoupling.initialize_barrier`

Signature. `self,grid_disp=None`

Purpose. finds active barrier contacts and prepares barrier assembly.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. Rebuilds current positive-weight NURBS control-hull bounds. Near pairs and positive SemiIPC multipliers use full closest-point projection. Far inactive pairs retain a conservative distance lower bound for ACCD; their reported distance or global minimum need not be the exact closest distance. Active barrier geometry and derivatives are unchanged. All pairs remain represented for conservative stepping.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

28.7.27 `IGA-MPMIPCCoupling.assemble_barrier_system`

Signature. `self`

Purpose. assembles the active IPC barrier matrix and residual.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as

unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

28.7.28 `IGA-MPMIPCCoupling.add_barrier_rhs_to_subsystems`

Signature. `self`

Purpose. adds barrier residuals to separate IGA and MPM subsystem vectors.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

28.7.29 `IGA-MPMIPCCoupling.add_friction_rhs_to_subsystems`

Signature. `self`

Purpose. adds friction residuals to separate IGA and MPM subsystem vectors.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

28.7.30 `IGA-MPMIPCCoupling.barrier_matrix`

Signature. `self`

Purpose. returns the assembled barrier matrix.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, Implicit solid MPM, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

28.7.31 `IGA-MPMIPCCoupling.friction_matrix`

Signature. `self`

Purpose. returns the assembled friction matrix.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, Implicit solid MPM, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

28.7.32 `IGA-MPMIPCCoupling.barrier_blocks`

Signature. `self,active_mpm_dof=None`

Purpose. returns block partitions of the barrier matrix.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set

before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

28.7.33 `IGA-MPMIPCCoupling.friction_blocks`

Signature. `self,active_mpm_dof=None`

Purpose. returns block partitions of the friction matrix.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

28.7.34 `IGA-MPMIPCCoupling.barrier_contact_forces`

Signature. `self,active_mpm_dof=None`

Purpose. returns barrier contact force vectors.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as

unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

28.7.35 `IGA-MPMIPCCoupling.friction_contact_forces`

Signature. `self,active_mpm_dof=None`

Purpose. returns friction contact force vectors.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

28.7.36 `IGA-MPMIPCCoupling.monolithic_barrier_system`

Signature. `self,K_iga=None,K_mpm=None,rhs_iga=None,rhs_mpm=None,active_mpm_dof=None,include_friction=False`

Purpose. assembles and returns the selected coupled IGA–MPM Taichi COO or HashTriplet matrix and device right-hand-side fields, including IPC terms.

Inputs and call shape. Set `include_friction` to include the configured lagged or fully implicit friction law. The legacy external block arguments must remain `None`; injecting SciPy or NumPy subsystem blocks is rejected. The returned dictionary contains `matrix`, `rhs`, `unconstrained_rhs`, `correction`, `active_nodes`, and active-DOF metadata.

Numerical effect. Body, barrier, friction, constraint-elimination, and sparse-reduction work executes in Taichi. Convert the selected matrix to Scipy only at an explicitly selected external linear-solver or output boundary.

Call order and validation. Build an implicit IPC coupling first and select coupled `assemble_type="COO"` or `"HashTriplet"`. Subsystem block sources and global scatter remain in Taichi; external matrix injection does not enable a host assembly fallback.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, IGA-MPM IPC coupling, Implicit Dirichlet constraints.

Source. `src/igampm/engines/ImplicitEngine.py`

28.7.37 `IGA-MPMIPCCoupling.initialize_friction`

Signature. `self,grid_disp=None`

Purpose. finds active friction contacts and prepares friction assembly.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

28.7.38 `IGA-MPMIPCCoupling.assemble_friction_system`

Signature. `self`

Purpose. assembles the active IPC friction matrix and residual.

Inputs and call shape. Pass the parameters shown in the signature fragment. Dictionary arguments are expanded in the keyword reference below; capacity dictionaries must be set before field allocation and solver dictionaries before solver construction.

Numerical effect. This call changes configuration, allocated fields, body state, contact buffers, boundary residuals, solver matrices, recorder output, or coupling forces depending on the owning module.

Call order and validation. Use it after referenced data have been registered and before kernels that consume those fields compile. Rejected combinations should be treated as unsupported by the current code.

Related theory. MPM weak form and transfers, DEM-MPM and CFDEM coupling, IGA and IGAMPM, IGA-MPM IPC coupling.

Source. `src/igampm/Coupling/iga_mpm.py`

29 Detailed Keyword Reference

29.1 Common Literals and Solver Aliases

29.1.1 `"BVH"`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. controls spatial search or broad-phase neighbor construction. In DEM it selects the LBVH path for particles and mesh-like walls, while DigitalElevation walls keep their elevation-grid search.

Numerical or workflow effect. trades neighbor build cost against candidate count and contact-list stability. DEM BVH builds from current active AABB batch prefixes; static patch/facet walls can keep their wall batch while the particle batch is rebuilt or refit.

Related theory. DEM BVH/LBVH search, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.1.2 `"MGPCG"`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2 A–C Keywords and Calls

29.2.1 `AbrahamModel`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.2 `AbsorbingConstraint`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.3 `acceleration`

Scope. DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/generator/LoadFromFile.py`

29.2.4 `active`

Scope. MPM, DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/generator/LoadFromFile.py;src/dem/generator/WallGenerator.py;src/mpm/generator/LoadFromFile.py`

29.2.5 `AdaptiveGrid`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. sets geometry, interpolation, grid spacing, or boundary classification

Numerical or workflow effect. changes support size, active node count, pressure-cell classification, and transfer accuracy

Related theory. MPM weak form and transfers.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpm/SceneManager.py`

29.2.6 `AdaptiveStep`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py;src/mpdem/mainDEMPM.py;src/mpm/mainMPM.py`

29.2.7 `add_attribute(materialID,attribute)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.8 `add_body`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.9 `add_body(bodies)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.10 `add_body(body)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.11 `add_body/create_body`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.12 `add_body_from_file(body)`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. selects input or output files

Numerical or workflow effect. controls restart, geometry loading, recorder writes, and postprocessing data flow

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.13 `add_boundary_condition`

Scope. shared

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. sets geometry, interpolation, grid spacing, or boundary classification

Numerical or workflow effect. changes support size, active node count, pressure-cell classification, and transfer accuracy

Related theory. MPM weak form and transfers.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.14 `add_boundary_condition(boundary)`

Scope. shared

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. sets geometry, interpolation, grid spacing, or boundary classification

Numerical or workflow effect. changes support size, active node count, pressure-cell classification, and transfer accuracy

Related theory. MPM weak form and transfers.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.15 `add_boundary_condition(dirichlet=None,neumann=None)`

Scope. shared

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. sets geometry, interpolation, grid spacing, or boundary classification

Numerical or workflow effect. changes support size, active node count, pressure-cell classification, and transfer accuracy

Related theory. MPM weak form and transfers.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.16 `add_circle`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.17 `add_contact(contact_type,**contact_phys)`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.18 `add_cube`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.19 `add_cylinder`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.20 `add_element`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. sets geometry, interpolation, grid spacing, or boundary classification

Numerical or workflow effect. changes support size, active node count, pressure-cell classification, and transfer accuracy

Related theory. MPM weak form and transfers.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.21 `add_element(degree)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. sets geometry, interpolation, grid spacing, or boundary classification

Numerical or workflow effect. changes support size, active node count, pressure-cell classification, and transfer accuracy

Related theory. MPM weak form and transfers.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.22 `add_element(element)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. sets geometry, interpolation, grid spacing, or boundary classification

Numerical or workflow effect. changes support size, active node count, pressure-cell classification, and transfer accuracy

Related theory. MPM weak form and transfers.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.23 `add_engine()`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.24 `add_engine(callback)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.25 `add_essentials`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.26 `add_essentials(**kwargs)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.27 `add_from_file`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. selects input or output files

Numerical or workflow effect. controls restart, geometry loading, recorder writes, and postprocessing data flow

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.28 `add_ground(ground)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.29 `add_material(**kwargs)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.30 `add_material(model,material)`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.31 `add_particle_traction(traction)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.32 `add_particles`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.33 `add_polygons(body)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.34 `add_postfunctions(**functions)`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.35 `add_primitives(primitives)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.36 `add_property(DEMmaterial,MPMmaterial,property,dType="all")`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.37 `add_property(materialID1,materialID2,property,dType="all")`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.38 `add_recorder()`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.39 `add_rectangle`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.40 `add_region`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.41 `add_region(region)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.42 `add_ring`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.43 `add_semi_circle`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.44 `add_solver(**kwargs)`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. chooses a solver, assembly format, or preconditioner parameter

Numerical or workflow effect. affects compilation path, memory layout, convergence, and boundary-condition application

Related theory. Implicit solid MPM, MGPCG multilevel solver.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.45 `add_spatial_grid()`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. sets geometry, interpolation, grid spacing, or boundary classification

Numerical or workflow effect. changes support size, active node count, pressure-cell classification, and transfer accuracy

Related theory. MPM weak form and transfers.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.46 `add_sphere`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.47 `add_template(template)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.48 `add_virtual_stress_field(field,region_name=None,function=None)`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.49 `add_wall`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.50 `add_wall(body)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.51 `add_wall_from_file(body)`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. selects input or output files

Numerical or workflow effect. controls restart, geometry loading, recorder writes, and postprocessing data flow

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.52 `Affine`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference`

29.2.53 `AffineBody`

Scope. DEM

Accepted value. `scheme="AffineBody"` or `affine_body=True`

Meaning. selects the DEM AffineBody branch for mesh-surface affine bodies and IPC-style body-body/body-wall contact.

Numerical or workflow effect. changes body storage, template loading, contact property routing, neighbor search ownership, engine construction, solver construction, and recorder output to the AffineBody path.

Related theory. Affine body dynamics, Contact Search.

Source hint. `src/dem/mainDEM.py;src/dem/Simulation.py;src/dem/engines/AffineBodyEngine.py`

29.2.54 `affine_assemble_type`

Scope. DEM

Accepted value. string backend selector, default `MatrixFree`

Meaning. shortcut passed to AffineBody `assemble_type`.

Numerical or workflow effect. selects the affine linear-system assembly/solve backend before the AffineBody engine is built.

Related theory. Affine body dynamics, Linear Solver.

Source hint. `src/dem/mainDEM.py;src/dem/Simulation.py;src/dem/engines/AffineBodyEngine.py`

29.2.55 `affine_body`

Scope. DEM

Accepted value. boolean

Meaning. convenience switch equivalent to selecting `scheme="AffineBody"`.

Numerical or workflow effect. activates AffineBody validation, memory rules, template/body insertion, engine, solver, contact property routing, and recorder behavior.

Related theory. Affine body dynamics.

Source hint. `src/dem/mainDEM.py`

29.2.56 `affine_body_parameters`

Scope. DEM

Accepted value. dict

Meaning. nested parameter dictionary forwarded to `DEM.set_affine_body_parameters`.

Numerical or workflow effect. controls AffineBody barrier/contact parameters, nonlinear and linear solver tolerances, line search, and CCD/ACCD behavior.

Related theory. Affine body dynamics, ACCD.

Source hint. `src/dem/mainDEM.py;src/dem/Simulation.py`

29.2.57 `all_in`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/utils/RegionFunction.py`

29.2.58 `Alpha`

Scope. DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/dem/generator/WallGenerator.py`

29.2.59 `alpha`

Scope. MPM, DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `manual/reference;src/dem/generator/WallGenerator.py;src/physics_model/consititutive_model/infinitesimal_strain/ElasPlasticity.py`

29.2.60 `alphaPIC`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpdem/mainDEMPM.py;src/mpm/mainMPM.py`

29.2.61 `angular_moment`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/generator/LoadFromFile.py`

29.2.62 `AngularVelocity`

Scope. DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `src/dem/SceneManager.py;src/dem/structs/Geometry.py`

29.2.63 `AOSOA`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.2.64 `AppliedPoint`

Scope. DEM

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/structs/Geometry.py`

29.2.65 `arch`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.66 `assemble_type`

Scope. MPM

Accepted value. string, callable, path, or enumerated choice

Meaning. selects MatrixFree, COO, HashTriplet, or legacy CSR alias for implicit MPM tangent handling

Numerical or workflow effect. changes memory footprint, Dirichlet enforcement, reaction-force reconstruction, and solver compatibility

Related theory. Implicit solid MPM, Implicit Dirichlet constraints.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpm/mainMPM.py`

29.2.67 `assemble_type="HashTriplet"`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. chooses a solver, assembly format, or preconditioner parameter

Numerical or workflow effect. affects compilation path, memory layout, convergence, and boundary-condition application

Related theory. Implicit solid MPM, MGPCG multilevel solver.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.68 `assemble_type="MatrixFree"`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. chooses a solver, assembly format, or preconditioner parameter

Numerical or workflow effect. affects compilation path, memory layout, convergence, and boundary-condition application

Related theory. Implicit solid MPM, MGPCG multilevel solver.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.69 `atmospheric_pressure`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets fluid, pore-pressure, porous-media, or viscous parameter

Numerical or workflow effect. enters pressure projection, Darcy or Ergun drag, pressure wave speed, and mixture mass balance

Related theory. Incompressible fluid MPM, Semi-implicit two-phase MPM.

Source hint. `manual/reference;src/physics_model/consititutive_model/strain_rate/Bingham.py;src/physics_model/consititutive_model/strain_rate/Newtonian.py`

29.2.70 `Auto`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.71 `AverageDiameter`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/physics_model/consititutive_model/infinitesimal_strain/RateDependent.py`

29.2.72 `axis`

Scope. MPM

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/mpm/PostPlot.py`

29.2.73 `Axis`

Scope. MPM, DEM

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/WallGenerator.py`

29.2.74 `B-BarMethod`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.75 `background_damping`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpdem/mainDEMPM.py;src/mpm/mainMPM.py`

29.2.76 `BackgroundColor`

Scope. MPM, DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/Si`

mulation.py;src/mpm/Simulation.py

29.2.77 `BarrierFunctionModel`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.78 `BarrierModel`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.79 `barrier_nnz`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. allocates barrier matrix triplet capacity for IPC coupling

Numerical or workflow effect. must scale with contact count times squared active contact degrees of freedom

Related theory. Contact Model, IGA-MPM IPC contact model, MPM self-collision IPC.

Source hint. `manual/reference`

29.2.80 `beta`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/physics_model/consititutive_model/infinitesimal_strain/ElasPlasticity.py;src/physics_model/consititutive_model/infinitesimal_strain/ModifiedCamClay.py`

29.2.81 `BiCG`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.82 `BiCGSTAB`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `manual/reference`

29.2.83 `Bingham`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.84 `BodyTemplate`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.85 `body_coordination_number`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. integer count, index, level, or interval

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py;src/mpdem/mainDEMPM.py`

29.2.86 `body_num`

Scope. MPM, DEM

Accepted value. value type defined by the owning call site

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `src/dem/generator/LoadFromFile.py;src/dem/generator/WallGenerator.py;src/mpm/generator/LoadFromFile.py`

29.2.87 `bodyID`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `src/mpm/PostPlot.py;src/mpm/generator/LoadFromFile.py`

29.2.88 `BodyID`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpm/generator/BodyGenerator.py;src/mpm/generator/LoadFromFile.py`

29.2.89 `BodyNumber`

Scope. MPM, DEM

Accepted value. integer count, index, level, or interval

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/BodyGenerator.py;src/dem/generator/LoadFromFile.py`

29.2.90 `BodyOrientation`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/BodyGenerator.py;src/mpdem/generator/LSBodyGenerate.py`

29.2.91 `BodyPoint`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/BodyGenerator.py;src/mpdem/generator/LSBodyGenerate.py`

29.2.92 `BodyType`

Scope. MPM, DEM

Accepted value. string, callable, path, or enumerated choice

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/BodyGenerator.py;src/dem/generator/LoadFromFile.py`

29.2.93 `bottom_smoothing`

Scope. MPM

Accepted value. integer count, index, level, or interval

Meaning. chooses a solver, assembly format, or preconditioner parameter

Numerical or workflow effect. affects compilation path, memory layout, convergence, and boundary-condition application

Related theory. Implicit solid MPM, MGPCG multilevel solver.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpm/mainMPM.py`

29.2.94 `boundary`

Scope. MPM, DEM

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. sets geometry, interpolation, grid spacing, or boundary classification

Numerical or workflow effect. changes support size, active node count, pressure-cell classification, and transfer accuracy

Related theory. MPM weak form and transfers.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py;src/mpm/mainMPM.py`

29.2.95 `boundary_direction_detection`

Scope. MPM

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. sets geometry, interpolation, grid spacing, or boundary classification

Numerical or workflow effect. changes support size, active node count, pressure-cell classification, and transfer accuracy

Related theory. MPM weak form and transfers.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.2.96 `BoundaryType`

Scope. MPM

Accepted value. string, callable, path, or enumerated choice

Meaning. sets geometry, interpolation, grid spacing, or boundary classification

Numerical or workflow effect. changes support size, active node count, pressure-cell classification, and transfer accuracy

Related theory. MPM weak form and transfers.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/boundaries/BoundaryConstraint.py`

29.2.97 `BoundingBoxFile`

Scope. MPM, DEM

Accepted value. string, callable, path, or enumerated choice

Meaning. selects input or output files

Numerical or workflow effect. controls restart, geometry loading, recorder writes, and postprocessing data flow

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/LoadFromFile.py`

29.2.98 `BoundingBoxPoint`

Scope. shared

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/utils/RegionFunction.py`

29.2.99 `BoundingBoxSize`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/utils/RegionFunction.py`

29.2.100 `BoundingRadius`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/BodyGenerator.py;src/mpdem/generator/LSBodyGenerate.py`

29.2.101 `BoundingSphereFile`

Scope. MPM, DEM

Accepted value. string, callable, path, or enumerated choice

Meaning. selects input or output files

Numerical or workflow effect. controls restart, geometry loading, recorder writes, and postprocessing data flow

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/LoadFromFile.py`

29.2.102 `BridgingCells`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. controls adaptive MPM refinement, particle splitting, or coarse-fine interface treatment

Numerical or workflow effect. changes grid topology, support, transfer consistency, and implicit tangent structure

Related theory. Adaptive MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/elements/AdaptiveHexahedronElement.py;src/mpm/elements/AdaptiveQuadrilateralElement.py`

29.2.103 `BridgingDomain`

Scope. MPM

Accepted value. boolean switch or small enumerated mode

Meaning. controls adaptive MPM refinement, particle splitting, or coarse-fine interface treatment

Numerical or workflow effect. changes grid topology, support, transfer consistency, and implicit tangent structure

Related theory. Adaptive MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpm/SceneManager.py;src/mpm/elements/AdaptiveHexahedronElement.py;src/mpm/elements/AdaptiveQuadrilateralElement.py`

29.2.104 `BrownLawlerModel`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.105 `Brust`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.106 `BufferCells`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. controls spatial search or broad-phase neighbor construction

Numerical or workflow effect. trades neighbor build cost against candidate count and contact-list stability

Related theory. DEM BVH/LBVH search, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/elements/AdaptiveHexahedronElement.py;src/mpm/elements/AdaptiveQuadrilateralElement.py`

29.2.107 `build()`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.108 `BVH`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. selects bounding-volume hierarchy neighbor search

Numerical or workflow effect. reduces broad-phase candidate generation cost for DEM and wall interactions

Related theory. DEM BVH/LBVH search.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.109 `bvh_rebuild_interval`

Scope. MPM, DEM

Accepted value. integer count, index, level, or interval

Meaning. sets steps between full LBVH rebuilds

Numerical or workflow effect. balances radix-sort build cost against broad-phase query quality

Related theory. DEM BVH/LBVH search.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py`

29.2.110 `calculate_reaction_force`

Scope. MPM

Accepted value. boolean switch or small enumerated mode

Meaning. requests reaction force output for implicit boundaries

Numerical or workflow effect. strong constraints recover eliminated residuals while MatrixFree penalty recovers an approximate penalty reaction

Related theory. Implicit Dirichlet constraints, Implicit solid MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpm/mainMPM.py`

29.2.111 `callback`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/mainDEM.py`

29.2.112 `calm`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py;src/mpdem/mainDEMPM.py`

29.2.113 `CameraUp`

Scope. MPM, DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/dem/Simulation.py;src/mpm/Simulation.py`

29.2.114 `cell_size`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. controls spatial search or broad-phase neighbor construction

Numerical or workflow effect. trades neighbor build cost against candidate count and contact-list stability

Related theory. DEM BVH/LBVH search, DEM dynamics and contact.

Source hint. `manual/reference`

29.2.115 `CellSize`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. controls spatial search or broad-phase neighbor construction

Numerical or workflow effect. trades neighbor build cost against candidate count and contact-list stability

Related theory. DEM BVH/LBVH search, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/dem/generator/WallGenerator.py`

29.2.116 `CellThickness`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. controls spatial search or broad-phase neighbor construction

Numerical or workflow effect. trades neighbor build cost against candidate count and contact-list stability

Related theory. DEM BVH/LBVH search, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/boundaries/BoundaryConstraint.py`

29.2.117 `center`

Scope. MPM, DEM

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/generator/LoadFromFile.py;src/mpm/PostPlot.py`

29.2.118 `centerOfMass`

Scope. DEM

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/PostPlot.py;src/dem/generator/LoadFromFile.py`

29.2.119 `CFD_coupling_domain`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. sets geometry, interpolation, grid spacing, or boundary classification

Numerical or workflow effect. changes support size, active node count, pressure-cell classification, and transfer accuracy

Related theory. MPM weak form and transfers.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpdem/mainDEMPM.py`

29.2.120 `CFDEM`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.121 `CFL`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py;src/mpdem/mainDEMPM.py;src/mpm/mainMPM.py`

29.2.122 `CG`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.123 `characteristic_length`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `manual/reference`

29.2.124 `check_critical_timestep()`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.125 `CheckHistory`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. boolean switch or small enumerated mode

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/BodyGenerator.py;src/mpdem/generator/LSBodyGenerate.py;src/mpm/generator/BodyGenerator.py;src/mpm/generator/LoadFromFile.py`

29.2.126 `CheckOverlap`

Scope. MPM, DEM

Accepted value. boolean switch or small enumerated mode

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/BodyGenerator.py`

29.2.127 `choose_contact_model(pp,pw)`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.128 `choose_contact_model(pp,pw=None)`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.129 `choose_coupling_region(region_name=None,function=None)`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.130 `choose_neighbor()`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.131 `cL`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `manual/reference;src/physics_model/consititutive_model/strain_rate/Newtonian.py`

29.2.132 `cl`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.133 `clean_boundary_condition(boundary)`

Scope. shared

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. sets geometry, interpolation, grid spacing, or boundary classification

Numerical or workflow effect. changes support size, active node count, pressure-cell classification, and transfer accuracy

Related theory. MPM weak form and transfers.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.134 `ClumpFile`

Scope. MPM, DEM

Accepted value. string, callable, path, or enumerated choice

Meaning. selects input or output files

Numerical or workflow effect. controls restart, geometry loading, recorder writes, and postprocessing data flow

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/LoadFromFile.py`

29.2.135 `ClumpProperty`

Scope. MPM, DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/ClumpTemplate.py`

29.2.136 `coeff`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.137 `Coefficient`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `manual/reference;src/physics_model/consititutive_model/finite_strain/MooneyRivlin.py`

29.2.138 `Cohesion`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `manual/reference;src/physics_model/consititutive_model/infinitesimal_strain/DruckerPrager.py;src/physics_model/consititutive_model/infinitesimal_strain/MohrCoulomb.py;src/physics_model/consititutive_model/infinitesimal_strain/StateDependentMohrCoulomb.py`

29.2.139 `compaction_ratio`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py;src/mpdem/mainDEMPM.py`

29.2.140 `compressible`

Scope. MPM

Accepted value. boolean switch or small enumerated mode

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `manual/reference`

29.2.141 `CompressionType`

Scope. DEM

Accepted value. string, callable, path, or enumerated choice

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/engines/ElementTest.py`

29.2.142 `Cone2D`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.143 `configuration`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpm/mainMPM.py`

29.2.144 `ConfiningPressure`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets fluid, pore-pressure, porous-media, or viscous parameter

Numerical or workflow effect. enters pressure projection, Darcy or Ergun drag, pressure wave speed, and mixture mass balance

Related theory. Incompressible fluid MPM, Semi-implicit two-phase MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpm/boundaries/BoundaryConstraint.py`

29.2.145 `connectivity`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/PostPlot.py;src/dem/generator/LoadFromFile.py`

29.2.146 `contact_detection`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `manual/reference`

29.2.147 `contact_force`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `src/mpm/PostPlot.py`

29.2.148 `contact_num`

Scope. DEM, MPDEM/CFDEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/contact/ContactModelBase.py;src/dem/contact/RollingModelBase.py; src/mpdem/contact/ContactModelBase.py`

29.2.149 `Container`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/engines/ElementTest.py`

29.2.150 `ControlType`

Scope. MPM, DEM

Accepted value. string, callable, path, or enumerated choice

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/WallGenerator.py`

29.2.151 `COO`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. selects coordinate triplet assembly

Numerical or workflow effect. supports direct strong Dirichlet row/column treatment at higher memory cost

Related theory. Implicit solid MPM, Implicit Dirichlet constraints.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference`

29.2.152 `coordination_number`

Scope. MPM

Accepted value. integer count, index, level, or interval

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `manual/reference`

29.2.153 `coords`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/mpm/PostPlot.py`

29.2.154 `Counterclockwise`

Scope. MPM, DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/WallGenerator.py`

29.2.155 `coupling`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/generator/LoadFromFile.py`

29.2.156 `coupling_scheme`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. string, callable, path, or enumerated choice

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpdem/mainDEMPM.py`

29.2.157 `coupling_scheme="CFDEM"`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.158 `cpu`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.159 `cpu_max_num_threads`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.160 `cQ`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `manual/reference;src/physics_model/consititutive_model/strain_rate/Newtonian.py`

29.2.161 `cq`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.162 `create_body(body)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.163 `CriticalStrainRate`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/physics_model/consititutive_model/strain_rate/Bingham.py`

29.2.164 `CrushFactor`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/contact/JiangRolling.py`

29.2.165 `CSR`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference`

29.2.166 `csr`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.2.167 `CubicBSpline`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference`

29.2.168 `cut_off`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/mpm/Contact.py`

29.2.169 `Cylinder`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3 D–F Keywords and Calls

29.3.1 `Damping`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `manual/reference`

29.3.2 `debug`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3.3 `default_fp`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference`

29.3.4 `default_ip`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3.5 `delayed_fluid_advection`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.3.6 `delete_particles`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3.7 `delete_particles(...)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3.8 `delete_walls`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3.9 `DEM`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3.10 `DEM(title,log=True)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3.11 `DEMContact`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3.12 `DEMPM`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3.13 `DEMPM(dem,mpm,title,log=True)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3.14 `Density`

Scope. MPM, DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets mass density or phase density

Numerical or workflow effect. scales particle mass, body force, wave speed, pressure response, drag, and virtual-mass forces

Related theory. Constitutive and contact models.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/structs/BaseStruct.py;src/dem/structs/Geometry.py;src/mpm/generator/BodyGenerator.py;src/mpm/generator/LoadFromFile.py;...`

29.3.15 `density`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets mass density or phase density

Numerical or workflow effect. scales particle mass, body force, wave speed, pressure response, drag, and virtual-mass forces

Related theory. Constitutive and contact models.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3.16 `density_projection`

Scope. MPM

Accepted value. boolean switch or small enumerated mode

Meaning. activates density-error pressure correction in addition to divergence projection

Numerical or workflow effect. should reuse valid fluid classification rather than forcing unnecessary full-domain rebuilds

Related theory. Incompressible fluid MPM, Fluid level set and density projection.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpm/mainMPM.py`

29.3.17 `density_projection=True`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets mass density or phase density

Numerical or workflow effect. scales particle mass, body force, wave speed, pressure response, drag, and virtual-mass forces

Related theory. Constitutive and contact models.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3.18 `density_projection_error_clamp`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets mass density or phase density

Numerical or workflow effect. scales particle mass, body force, wave speed, pressure response, drag, and virtual-mass forces

Related theory. Constitutive and contact models.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.3.19 `density_projection_interior_only`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets mass density or phase density

Numerical or workflow effect. scales particle mass, body force, wave speed, pressure response, drag, and virtual-mass forces

Related theory. Constitutive and contact models.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.3.20 `density_projection_interior_only=True`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets mass density or phase density

Numerical or workflow effect. scales particle mass, body force, wave speed, pressure response, drag, and virtual-mass forces

Related theory. Constitutive and contact models.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3.21 `density_projection_max_shift_ratio`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets mass density or phase density

Numerical or workflow effect. scales particle mass, body force, wave speed, pressure response, drag, and virtual-mass forces

Related theory. Constitutive and contact models.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.3.22 `density_projection_tolerance`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets mass density or phase density

Numerical or workflow effect. scales particle mass, body force, wave speed, pressure response, drag, and virtual-mass forces

Related theory. Constitutive and contact models.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.3.23 `DensityProjection`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets mass density or phase density

Numerical or workflow effect. scales particle mass, body force, wave speed, pressure response, drag, and virtual-mass forces

Related theory. Constitutive and contact models.

Source hint. `manual/reference`

29.3.24 `Destroy`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3.25 `device_memory_fraction`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference`

29.3.26 `device_memory_GB`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference`

29.3.27 `dhat`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets IPC barrier activation distance

Numerical or workflow effect. larger values create more active contacts and larger matrices

Related theory. Contact Model, IGA-MPM IPC contact model, MPM self-collision IPC.

Source hint. `manual/reference`

29.3.28 `DigitalElevation`

Scope. MPM, DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/WallGenerator.py`

29.3.29 `Dilation`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `manual/reference;src/physics_model/consititutive_model/infinitesimal_strain/DruckerPrager.py;src/physics_model/consititutive_model/infinitesimal_strain/MohrCoulomb.py`

29.3.30 `dim`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3.31 `Direction`

Scope. MPM, DEM

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/WallGenerator.py`

29.3.32 `DirichletBoundary.append`

Scope. shared

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. sets geometry, interpolation, grid spacing, or boundary classification

Numerical or workflow effect. changes support size, active node count, pressure-cell classification, and transfer accuracy

Related theory. MPM weak form and transfers.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3.33 `discretization`

Scope. MPM

Accepted value. `"FEM"` or `"FDM"` in `MPM.set_configuration`

Meaning. selects the top-level MPM element/discretization path, not every internal linear-system stencil.

Numerical or workflow effect. `"FEM"` is required by explicit MPM, implicit solid MPM, and the documented double-layer semi-implicit two-phase route because those paths need particle-grid shape functions and nodal transfers. `"FDM"` is the implicit incompressible-fluid staggered-grid path. The double-layer pressure projection uses auxiliary cell-centered/MAC finite-difference stencils internally while its public `discretization` setting remains `"FEM"`.

Related theory. Semi-implicit two-phase MPM, Incompressible fluid MPM, MPM weak form and transfers.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.3.34 `discretization="FDM"`

Scope. MPM implicit incompressible fluid

Accepted value. literal value for `MPM.set_configuration(discretization="FDM")`

Meaning. selects the staggered-grid incompressible fluid branch.

Numerical or workflow effect. valid with `solver_type="Implicit"`, `material_type="Fluid"`, and `add_element(ElementType="Staggered")`. It is not the public setting for double-layer `SemiImplicit` two-phase MPM. That pressure projection uses internal FDM/MAC stencils, but the top-level solver still runs through the FEM MPM transfer path.

Related theory. Incompressible fluid MPM, Fluid level set pressure projection and density projection.

Source hint. `src/mpm/Simulation.py;src/mpm/engines/IncompressibleEngine.py`

29.3.35 `discretization="FEM"`

Scope. MPM solid and two-phase paths

Accepted value. literal value for `MPM.set_configuration(discretization="FEM")`

Meaning. selects particle-grid interpolation through FEM-style element nodes and shape functions.

Numerical or workflow effect. required by explicit MPM, implicit solid MPM, and the documented double-layer `SemiImplicit` two-phase route. For that route the particle transfer and solid skeleton use FEM-style shape functions, while pressure is solved on the cell-centered/MAC MGPCG hierarchy.

Related theory. MPM weak form and transfers, Semi-implicit two-phase MPM.

Source hint. `src/mpm/Simulation.py;src/mpm/SceneManager.py`

29.3.36 `DiscretizeDomain`

Scope. MPM, DEM

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. sets geometry, interpolation, grid spacing, or boundary classification

Numerical or workflow effect. changes support size, active node count, pressure-cell classification, and transfer accuracy

Related theory. MPM weak form and transfers.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/LoadFromFile.py`

29.3.37 `Displacement`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/boundaries/BoundaryConstraint.py`

29.3.38 `DisplacementF-BarMethod`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3.39 `displacement_tolerance`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.3.40 `DisplacementConstraint`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3.41 `DisplacementX`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/boundaries/BoundaryConstraint.py`

29.3.42 `DisplacementY`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/boundaries/BoundaryConstraint.py`

29.3.43 `DisplacementZ`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/boundaries/BoundaryConstraint.py`

29.3.44 `distance_field`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/generator/LoadFromFile.py`

29.3.45 `dof_multiplier`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.3.46 `domain`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. sets geometry, interpolation, grid spacing, or boundary classification

Numerical or workflow effect. changes support size, active node count, pressure-cell classification, and transfer accuracy

Related theory. MPM weak form and transfers.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py;src/mpdem/mainDEMPM.py;src/mpm/mainMPM.py`

29.3.47 `Down`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3.48 `dpType`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/physics_model/consititutive_model/infinitesimal_strain/DruckerPrager.py`

29.3.49 `drag_model`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. string, callable, path, or enumerated choice

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpdem/mainDEMPM.py;src/physics_model/consititutive_model/MaterialModel.py`

29.3.50 `DragForceModel`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpdem/fluid_dynamics/DragForceModel.py`

29.3.51 `DragLaw`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. string, callable, path, or enumerated choice

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpdem/fluid_dynamics/DragForceModel.py`

29.3.52 `DragModel`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/physics_model/consititutive_model/MaterialModel.py`

29.3.53 `drift_correct`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.3.54 `DruckerPrager`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3.55 `druckerPrager`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3.56 `DynamicFriction`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets friction coefficient or tangential resistance parameter

Numerical or workflow effect. limits tangential contact forces, changes plastic strength, or activates IPC friction

Related theory. Contact Model, DEM, IGA-MPM IPC contact model, DP/MC rate dependence.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/contact/HertzMindlin.py;src/dem/contact/Linear.py;src/mpdem/contact/HertzMindlin.py;src/mpdem/contact/Linear.py;...`

29.3.57 `e0`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/physics_model/consititutive_model/infinitesimal_strain/StateDependentMohrCoulomb.py`

29.3.58 `e_ref`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3.59 `e_Tao`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/physics_model/consititutive_model/infinitesimal_strain/StateDependentMohrCoulomb.py`

29.3.60 `EffectiveModulus`

Scope. DEM, MPDEM/CFDEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets elastic constants used by stress increments and tangent stiffness

Numerical or workflow effect. controls wave speed, timestep restrictions, incompressibility, and implicit matrix conditioning

Related theory. Constitutive Equations, DP/MC softening and rate dependence, Return mapping and substepping, Implicit solid MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/dem/contact/Linear.py;src/dem/contact/LinearRolling.py;src/mpdem/contact/Linear.py`

29.3.61 `ElasticPerfectlyPlastic`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3.62 `ElementLength`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. sets geometry, interpolation, grid spacing, or boundary classification

Numerical or workflow effect. changes support size, active node count, pressure-cell classification, and transfer accuracy

Related theory. MPM weak form and transfers.

Source hint. `manual/reference;src/physics_model/consititutive_model/strain_rate/Newtonian.py`

29.3.63 `ElementSize`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. sets geometry, interpolation, grid spacing, or boundary classification

Numerical or workflow effect. changes support size, active node count, pressure-cell classification, and transfer accuracy

Related theory. MPM weak form and transfers.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/SceneManager.py`

29.3.64 `ElementType`

Scope. MPM

Accepted value. string, callable, path, or enumerated choice

Meaning. sets geometry, interpolation, grid spacing, or boundary classification

Numerical or workflow effect. changes support size, active node count, pressure-cell classification, and transfer accuracy

Related theory. MPM weak form and transfers.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/SceneManager.py`

29.3.65 `ElementType="Staggered"`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. sets geometry, interpolation, grid spacing, or boundary classification

Numerical or workflow effect. changes support size, active node count, pressure-cell classification, and transfer accuracy

Related theory. MPM weak form and transfers.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3.66 `EmpiricalModel`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3.67 `enable_shell`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. boolean switch or small enumerated mode

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/dem/mainDEM.py;src/mpdem/mainDEMPM.py`

29.3.68 `end1`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/PostPlot.py`

29.3.69 `end2`

Scope. DEM, MPDEM/CFDEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/PostPlot.py;src/dem/contact/ContactModelBase.py;src/dem/contact/RollingModelBase.py;src/mpdem/contact/ContactModelBase.py`

29.3.70 `endIndex`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/PostPlot.py;src/dem/generator/LoadFromFile.py;src/dem/generator/WallGenerator.py`

29.3.71 `endNode`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/PostPlot.py;src/dem/generator/LoadFromFile.py`

29.3.72 `EndPoint`

Scope. MPM

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/boundaries/BoundaryConstraint.py`

29.3.73 `EndTime`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/utils/functions/SinFunction.py`

29.3.74 `EnergyConservingModel`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3.75 `engine`

Scope. MPM, DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py`

29.3.76 `enhanced_coupling`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpdem/mainDEMPM.py`

29.3.77 `eps`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `manual/reference;src/physics_model/consititutive_model/infinitesimal_strain/RateDependent.py`

29.3.78 `epsv`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets IPC friction velocity regularization scale

Numerical or workflow effect. keeps the friction Hessian finite near zero slip

Related theory. Contact Model, IGA-MPM IPC contact model, MPM self-collision IPC.

Source hint. `manual/reference`

29.3.79 `equivalentRadius`

Scope. DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/PostPlot.py;src/dem/generator/LoadFromFile.py`

29.3.80 `Explicit`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3.81 `ExplicitIGA`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3.82 `ExplicitTLMPM`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3.83 `ExplicitULMPM`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3.84 `extent`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/generator/LoadFromFile.py;src/sdf/LevelSetGrid.py`

29.3.85 `external_force`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `src/mpm/PostPlot.py`

29.3.86 `ExternalForce`

Scope. MPM, DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/dem/structs/Geometry.py;src/mpm/boundaries/BoundaryConstraint.py`

29.3.87 `ExternalForceX`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `src/mpm/boundaries/BoundaryConstraint.py`

29.3.88 `ExternalForceY`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `src/mpm/boundaries/BoundaryConstraint.py`

29.3.89 `ExternalForceZ`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `src/mpm/boundaries/BoundaryConstraint.py`

29.3.90 `ExternalTorque`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/structs/Geometry.py`

29.3.91 `F-BarMethod`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3.92 `face_index`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/generator/LoadFromFile.py`

29.3.93 `fai_c`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/physics_model/consititutive_model/infinitesimal_strain/StateDependentMohrCoulomb.py`

29.3.94 `fbar_fraction`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.3.95 `FDM`

Scope. MPM implicit fluid and internal pressure projections

Accepted value. finite-difference discretization concept; as a public `discretization` value it is `"FDM"`.

Meaning. pressure unknowns live at cell centers and velocity components on staggered/-MAC faces for incompressible-fluid pressure projection.

Numerical or workflow effect. as a user-facing `discretization` value, `FDM` selects the implicit incompressible-fluid staggered element path. The double-layer semi-implicit pressure solve uses FDM/MAC stencils internally, but still requires `discretization="FEM"` at configuration time.

Related theory. Incompressible fluid MPM, Semi-implicit two-phase MPM.

Source hint. `src/mpm/Simulation.py;src/mpm/engines/IncompressibleEngine.py;src/mpm/engines/ULSemiImplicitTwoPhaseDoubleLayerEngine.py`

29.3.96 `FEM`

Scope. MPM particle-grid transfer and solid/two-phase solves

Accepted value. finite-element-style particle-grid discretization concept; as a public `discretization` value it is `"FEM"`.

Meaning. uses element nodes, shape functions, gradients, and cached particle support for P2G/G2P transfer and FEM-style weak-form assembly.

Numerical or workflow effect. required by the double-layer semi-implicit two-phase engine because its solid skeleton and particle transfers are not the staggered-grid implicit-fluid branch. Its pressure subproblem is solved on the internal cell-centered/MAC MGPCG hierarchy.

Related theory. MPM weak form and transfers, Semi-implicit two-phase MPM.

Source hint. `src/mpm/Simulation.py;src/mpm/SceneManager.py`

29.3.97 `File`

Scope. MPM, DEM

Accepted value. string, callable, path, or enumerated choice

Meaning. selects input or output files

Numerical or workflow effect. controls restart, geometry loading, recorder writes, and postprocessing data flow

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/LoadFromFile.py;src/mpm/generator/Generator.py;src/mpm/generator/LoadFromFile.py`

29.3.98 `file_number`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. integer count, index, level, or interval

Meaning. selects input or output files

Numerical or workflow effect. controls restart, geometry loading, recorder writes, and postprocessing data flow

Related theory. Theory overview.

Source hint. `src/dem/mainDEM.py;src/mpdem/mainDEMPM.py`

29.3.99 `FileType`

Scope. MPM, DEM

Accepted value. string, callable, path, or enumerated choice

Meaning. selects input or output files

Numerical or workflow effect. controls restart, geometry loading, recorder writes, and postprocessing data flow

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/LoadFromFile.py;src/dem/generator/WallGenerator.py;src/mpm/generator/LoadFromFile.py`

29.3.100 `fix_v`

Scope. MPM, DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/generator/LoadFromFile.py;src/mpm/generator/LoadFromFile.py`

29.3.101 `fix_w`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/generator/LoadFromFile.py`

29.3.102 `FixAngularVelocity`

Scope. MPM, DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/BodyGenerator.py;src/dem/generator/LoadFromFile.py`

29.3.103 `FixMotion`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/BodyGenerator.py;src/dem/generator/LoadFromFile.py;src/mpdem/generator/LSBodyGenerate.py`

29.3.104 `FixVelocity`

Scope. MPM, DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/BodyGenerator.py;src/dem/generator/LoadFromFile.py;src/mpm/generator/BodyGenerator.py;src/mpm/generator/LoadFromFile.py`

29.3.105 `FLIP`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3.106 `float32`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3.107 `float64`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3.108 `Fluid`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3.109 `FluidParticle`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.3.110 `fluid_level_set`

Scope. MPM

Accepted value. boolean

Meaning. enables the free-surface ghost-fluid level-set path for implicit incompressible fluid MPM with FDM staggered elements.

Numerical or workflow effect. builds or updates a cell-centered ϕ field with $\phi < 0$ in fluid and $\phi > 0$ in air, marks fluid/air/solid pressure cells, inserts the free-surface fraction θ into the pressure matrix and right-hand side, and uses the same θ during post-solve face-velocity correction. It is a pressure-projection option, not only an output flag. With `solid_sdf_cut_cell`, open face fractions additionally scale the pressure coefficients.

Related theory. Incompressible fluid MPM, Fluid level set pressure projection and density projection, SDF and level set geometry, compatibility matrix.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpm/mainMPM.py`

29.3.111 `fluid_level_set=True`

Scope. MPM

Accepted value. literal configuration value in `MPM.set_configuration`

Meaning. selects the free-surface pressure-projection branch described by fluid_level_set.

Numerical or workflow effect. requires `solver_type="Implicit"`, `material_type="Fluid"`, `discretization="FDM"`, and a staggered grid. Unsupported solver/material combinations are rejected during compatibility checks.

Related theory. Fluid level set pressure projection and density projection.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/Simulation.py`

29.3.112 `fluid_level_set_function`

Scope. MPM

Accepted value. callable SDF/level-set update function accepted by the selected fluid engine

Meaning. overrides or augments the engine's internal particle-based fluid SDF update when the engine exposes `update_external_fluid_level_set`.

Numerical or workflow effect. the callable must populate distances using the same sign convention as the internal field: negative in fluid and positive in air. Wrong signs invert the ghost-fluid boundary and can create pressure spikes at the free surface.

Related theory. Fluid level set pressure projection and density projection, SDF and level set geometry, Level-set redistance and advection.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.3.113 `solid_sdf_functionandibm_source_function`

Scope. MPM incompressible FDM

Accepted value. Callables supplied to `MPM.run` or `MPM.add_essentials`. `ibm_function` is a compatibility alias for `ibm_source_function`.

Meaning. `solid_sdf_function` replaces the selected engine's external solid-cell/SDF update, while `ibm_source_function` replaces its external immersed-boundary momentum-source update when those hooks exist.

Numerical or workflow effect. The callbacks execute through engine hooks during the build/time-stepping workflow and therefore must use the engine's grid layout, sign convention, units, and active-domain assumptions. On an engine without the matching hook the supplied callback has no effect.

Related theory. Incompressible fluid MPM, SDF and level set geometry.

Source hint. `src/mpm/mainMPM.py;src/mpm/engines/IncompressibleEngine.py`

29.3.114 `FluidBulkModulus`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets elastic constants used by stress increments and tangent stiffness

Numerical or workflow effect. controls wave speed, timestep restrictions, incompressibility, and implicit matrix conditioning

Related theory. Constitutive Equations, DP/MC softening and rate dependence, Return mapping and substepping, Implicit solid MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/physics_model/consititutive_model/MaterialModel.py`

29.3.115 `FluidDensity`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets mass density or phase density

Numerical or workflow effect. scales particle mass, body force, wave speed, pressure response, drag, and virtual-mass forces

Related theory. Constitutive and contact models.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/physics_model/consititutive_model/MaterialModel.py`

29.3.116 `FluidPressure`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets fluid, pore-pressure, porous-media, or viscous parameter

Numerical or workflow effect. enters pressure projection, Darcy or Ergun drag, pressure wave speed, and mixture mass balance

Related theory. Incompressible fluid MPM, Semi-implicit two-phase MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/boundaries/BoundaryConstraint.py`

29.3.117 `FluidUnitWeight`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/physics_model/consititutive_model/MaterialModel.py`

29.3.118 `FluidViscosity`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets fluid, pore-pressure, porous-media, or viscous parameter

Numerical or workflow effect. enters pressure projection, Darcy or Ergun drag, pressure wave speed, and mixture mass balance

Related theory. Incompressible fluid MPM, Semi-implicit two-phase MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/physics_model/consititutive_model/MaterialModel.py`

29.3.119 `ForceLocalDamping`

Scope. DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `src/dem/structs/BaseStruct.py`

29.3.120 `Fraction`

Scope. MPM, DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/BodyGenerator.py`

29.3.121 `free_surface`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/mpm/PostPlot.py;src/mpm/generator/LoadFromFile.py`

29.3.122 `free_surface_detection`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.3.123 `FreeParameter`

Scope. `EnergyConservingModel` contact property in DEM/LSDEM/LSMPM.

Accepted value. Real exponent $\theta \geq 2$; default 2.5.

Meaning. Sets the power in the energy-conserving normal penalty law.

Numerical or workflow effect. The LSMPM critical-timestep estimate uses its tangent at a conservative penetration bound. Values below two are rejected because the explicit-contact tangent is singular at zero gap.

Related theory. Critical timestep.

Source hint. `src/dem/contact/EnergyConservation.py`

29.3.124 `friction`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets friction coefficient or tangential resistance parameter

Numerical or workflow effect. limits tangential contact forces, changes plastic strength, or activates IPC friction

Related theory. Contact Model, DEM, IGA-MPM IPC contact model, DP/MC rate dependence.

Source hint. `src/mpm/Contact.py`

29.3.125 `Friction`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets friction coefficient or tangential resistance parameter

Numerical or workflow effect. limits tangential contact forces, changes plastic strength, or activates IPC friction

Related theory. DEM dynamics and contact, IGA-MPM IPC coupling, Constitutive and contact models.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/contact/EnergyConservation.py;src/dem/contact/JiangRolling.py;src/dem/contact/LinearRolling.py;src/mpdem/contact/EnergyConservation.py;...`

29.3.126 `friction_nnz`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. allocates friction matrix triplet capacity for IPC coupling

Numerical or workflow effect. same dof-squared scaling as barrier capacity when friction is active

Related theory. Contact Model, IGA-MPM IPC contact model, MPM self-collision IPC.

Source hint. `manual/reference`

29.3.127 `FrictionConstraint`

Scope. Native explicit MPM, stationary axis-aligned wall.

Accepted values. `boundary.BoundaryType="FrictionConstraint"`; `boundary.Friction` is the dimensionless Coulomb coefficient; `boundary.Norm` is an outward axis-aligned normal (e.g. $[0, -1]$ for a 2D floor). `boundary.StartPoint` and `boundary.EndPoint` select grid nodes.

Numerical effect. After the free grid update, an incoming normal velocity is projected to zero; the tangential velocity is reduced by at most μ times the normal impulse per unit mass. The acceleration is corrected by the same velocity increment divided by the time step, preserving the FLIP increment. Separating nodes are unchanged. A later velocity constraint overrides its prescribed components. Ordinary/contact nodes share this law.

Related theory. MPM wall friction.

Source. `src/mpm/structs/GridNode.py`; tests in `tests/unit/mpm/test_grid_wall_friction.py`.

29.3.128 `function`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. string, callable, path, or enumerated choice

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py;src/mpdem/mainDEMPM.py;src/mpm/mainMPM.py`

29.3.129 `preintegration_function`

Scope. DEM

Accepted value. one Python callable, or a list/dictionary of callables that Taichi can decorate as kernels; default `None`.

Meaning. registers device diagnostics or state adjustments that run after contact/neighbor updates and immediately before the integration kernel in each DEM step.

Numerical or workflow effect. callbacks see the completed contact state for the current step but execute before positions and velocities advance. With `reset_function=True`, previously registered pre-integration and postprocess callbacks are cleared when the solver is rebuilt.

Related theory. DEM dynamics and contact.

Source hint. `src/dem/mainDEM.py;src/dem/DEMBase.py`

29.4 G–I Keywords and Calls

29.4.1 `G2P2G`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.2 `GEOTAICHI_INCOMPRESSIBLE_P2G`

Scope. Incompressible fluid MPM with `solver_type="Implicit"`, `material_type="Fluid"`, and `discretization="FDM"`.

Accepted value. `auto`, `atomic`, `shared`, `shared_memory`, or `cell_reduced`; default `auto`.

Meaning. Selects the particle-to-MAC-grid implementation for diagnosis and performance comparison. Auto chooses CUDA tile-local shared-memory reduction on CUDA and direct atomic scatter elsewhere. `shared_memory` and `cell_reduced` normalize to the shared request, which uses the cell-reduced fallback off CUDA.

Numerical or workflow effect. The routes compute the same mass and momentum sums. Reduced modes add cell-bin and prefix-scan workspace but reduce same-face atomic contention when many particles occupy each cell.

Related theory. Incompressible Fluid MPM.

Source hint. `src/mpm/engines/IncompressibleEngine.py`

29.4.3 `Gain`

Scope. MPM, DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/WallGenerator.py`

29.4.4 `gain`

Scope. DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/generator/WallGenerator.py`

29.4.5 `gamma`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/physics_model/consititutive_model/strain_rate/Bingham.py`

29.4.6 `gauss_number`

Scope. MPM

Accepted value. integer count, index, level, or interval

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.4.7 `GenerateType`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. string, callable, path, or enumerated choice

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/BodyGenerator.py;src/mpdem/generator/LSBodyGenerate.py`

29.4.8 `Gent`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.9 `GeoContact`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.10 `GeometryFile`

Scope. MPM, DEM

Accepted value. string, callable, path, or enumerated choice

Meaning. selects input or output files

Numerical or workflow effect. controls restart, geometry loading, recorder writes, and postprocessing data flow

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/LoadFromFile.py`

29.4.11 `geotaichi`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.12 `get_critical_timestep()`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.13 `GhostCell`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. controls spatial search or broad-phase neighbor construction

Numerical or workflow effect. trades neighbor build cost against candidate count and contact-list stability

Related theory. DEM BVH/LBVH search, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/SceneManager.py`

29.4.14 `GIMP`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference`

29.4.15 `GJK`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.16 `gpu`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.17 `GrainDensity`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets mass density or phase density

Numerical or workflow effect. scales particle mass, body force, wave speed, pressure response, drag, and virtual-mass forces

Related theory. Constitutive and contact models.

Source hint. `src/physics_model/consititutive_model/infinitesimal_strain/RateDependent.py`

29.4.18 `GrainDiameter`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/physics_model/consititutive_model/MaterialModel.py`

29.4.19 `grainIndex`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/PostPlot.py;src/dem/generator/LoadFromFile.py`

29.4.20 `GranularMaterial`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.21 `gravity`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py;src/mpdem/mainDEMPM.py;src/mpm/mainMPM.py`

29.4.22 `Gravity`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `manual/reference;src/physics_model/consititutive_model/MaterialModel.py`

29.4.23 `gravity_field`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.4.24 `grid_num`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. sets geometry, interpolation, grid spacing, or boundary classification

Numerical or workflow effect. changes support size, active node count, pressure-cell classification, and transfer accuracy

Related theory. MPM weak form and transfers.

Source hint. `src/dem/generator/LoadFromFile.py`

29.4.25 `grid_space`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. sets geometry, interpolation, grid spacing, or boundary classification

Numerical or workflow effect. changes support size, active node count, pressure-cell classification, and transfer accuracy

Related theory. MPM weak form and transfers.

Source hint. `src/dem/generator/LoadFromFile.py`

29.4.26 `GridFile`

Scope. MPM, DEM

Accepted value. string, callable, path, or enumerated choice

Meaning. selects input or output files

Numerical or workflow effect. controls restart, geometry loading, recorder writes, and postprocessing data flow

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/LoadFromFile.py`

29.4.27 `GridNumber`

Scope. MPM, DEM

Accepted value. integer count, index, level, or interval

Meaning. sets geometry, interpolation, grid spacing, or boundary classification

Numerical or workflow effect. changes support size, active node count, pressure-cell classification, and transfer accuracy

Related theory. MPM weak form and transfers.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/WallGenerator.py`

29.4.28 `Ground.append`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.29 `groupID`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `src/dem/PostPlot.py;src/dem/generator/LoadFromFile.py`

29.4.30 `GroupID`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. value type defined by the owning call site

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/BodyGenerator.py;src/dem/generator/LoadFromFile.py;src/mpdem/generator/LSBodyGenerate.py`

29.4.31 `gt.box`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.32 `gt.circle`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.33 `gt.cylinder`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.34 `gt.DEM`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.35 `gt.DEMPM`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.36 `gt.IGA`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.37 `gt.init`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.38 `gt.init(...)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.39 `gt.make_print_to_file(path)`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. selects input or output files

Numerical or workflow effect. controls restart, geometry loading, recorder writes, and postprocessing data flow

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.40 `gt.IGAMPM`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. constructs the current IGA-MPM coupling facade.

Numerical or workflow effect. wraps implicit IGA and MPM sub-solvers and builds the monolithic IPC coupling engine on demand.

Related theory. IGA-MPM.

Source hint. `src/igampm/mainIGAMPM.py`

29.4.41 `gt.MPM`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.42 `gt.rectangle`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.43 `gt.sphere`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.44 `HangingConstraintMode`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. chooses penalty, shape, or shape-function treatment for coarse-fine interface nodes

Numerical or workflow effect. implicit adaptive solid MPM requires the strong shape-function path in the current implementation

Related theory. Adaptive MPM, Implicit solid MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpm/SceneManager.py;src/mpm/elements/AdaptiveHexahedronElement.py;src/mpm/elements/AdaptiveQuadrilateralElement.py`

29.4.45 `HangingShapeCapacityFactor`

Scope. MPM

Accepted value. string, callable, path, or enumerated choice

Meaning. controls adaptive MPM refinement, particle splitting, or coarse-fine interface treatment

Numerical or workflow effect. changes grid topology, support, transfer consistency, and implicit tangent structure

Related theory. Adaptive MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpm/SceneManager.py;src/mpm/elements/AdaptiveHexahedronElement.py;src/mpm/elements/AdaptiveQuadrilateralElement.py`

29.4.46 `HashTriplet`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. selects hash-reduced triplet assembly

Numerical or workflow effect. reduces duplicate matrix-entry writes but requires careful capacity and symmetry handling

Related theory. Implicit solid MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference`

29.4.47 `HenckyElastic`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.48 `HertzMindlinModel`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.49 `hierarchical_level`

Scope. MPM, DEM

Accepted value. integer count, index, level, or interval

Meaning. controls spatial search or broad-phase neighbor construction

Numerical or workflow effect. trades neighbor build cost against candidate count and contact-list stability

Related theory. DEM BVH/LBVH search, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py`

29.4.50 `hierarchical_size`

Scope. MPM, DEM

Accepted value. value type defined by the owning call site

Meaning. controls spatial search or broad-phase neighbor construction

Numerical or workflow effect. trades neighbor build cost against candidate count and contact-list stability

Related theory. DEM BVH/LBVH search, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py`

29.4.51 `HierarchicalLinkedCell`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. controls spatial search or broad-phase neighbor construction

Numerical or workflow effect. trades neighbor build cost against candidate count and contact-list stability

Related theory. DEM BVH/LBVH search, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.52 `Hydrogel`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.53 `IGA`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.54 `IGA(title="IsoGeometricAnalysisEngine",log=True)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.55 `IGAMPM`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.56 `Implicit`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.57 `implicit_solver`

Scope. MPM

Accepted value. string, callable, path, or enumerated choice

Meaning. chooses a solver, assembly format, or preconditioner parameter

Numerical or workflow effect. affects compilation path, memory layout, convergence, and boundary-condition application

Related theory. Implicit solid MPM, MGPCG multilevel solver.

Source hint. `manual/reference`

29.4.58 `ImplicitIGA`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.59 `ImplicitTLMPM`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.60 `ImplicitTwoPhasePenaltyContactAssembler`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. controls adaptive MPM refinement, particle splitting, or coarse-fine interface treatment

Numerical or workflow effect. changes grid topology, support, transfer consistency, and implicit tangent structure

Related theory. Adaptive MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.61 `ImplicitTwoPhasePenaltyContactSolver`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. controls adaptive MPM refinement, particle splitting, or coarse-fine interface treatment

Numerical or workflow effect. changes grid topology, support, transfer consistency, and implicit tangent structure

Related theory. Adaptive MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.62 `ImplicitTwoPhaseRigidMonolithicAssembler`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. chooses a solver, assembly format, or preconditioner parameter

Numerical or workflow effect. affects compilation path, memory layout, convergence, and boundary-condition application

Related theory. Implicit solid MPM, MGPCG multilevel solver.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.63 `ImplicitTwoPhaseRigidMonolithicSolver`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. chooses a solver, assembly format, or preconditioner parameter

Numerical or workflow effect. affects compilation path, memory layout, convergence,

and boundary-condition application

Related theory. Implicit solid MPM, MGPCG multilevel solver.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.64 `ImplicitULMPM`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.65 `Index`

Scope. MPM, DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/PostPlot.py;src/dem/generator/LoadFromFile.py`

29.4.66 `InertialNumber`

Scope. shared

Accepted value. integer count, index, level, or interval

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/physics_model/consititutive_model/infinitesimal_strain/RateDependent.py`

29.4.67 `InertiaMoment`

Scope. MPM, DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/ClumpTemplate.py`

29.4.68 `initial_pressure`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets fluid, pore-pressure, porous-media, or viscous parameter

Numerical or workflow effect. enters pressure projection, Darcy or Ergun drag, pressure wave speed, and mixture mass balance

Related theory. Incompressible fluid MPM, Semi-implicit two-phase MPM.

Source hint. `manual/reference`

29.4.69 `InitialAngularVelocity`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/BodyGenerator.py;src/dem/generator/LoadFromFile.py;src/mpdem/generator/LSBodyGenerate.py`

29.4.70 `InitialVelocity`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/BodyGenerator.py;src/dem/generator/LoadFromFile.py;src/dem/generator/WallGenerator.py;src/mpdem/generator/LSBodyGenerate.py;...`

29.4.71 `insert.py`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.72 `int32`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.73 `int64`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.74 `integration_scheme`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.4.75 `InternalStress`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/generator/Generator.py`

29.4.76 `inverse_inertia`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/generator/LoadFromFile.py`

29.4.77 `inverseInertia`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/generator/LoadFromFile.py`

29.4.78 `IPCTLMPM`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.79 `IPCULMPM`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.4.80 `is_2DAxisy`

Scope. MPM

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.4.81 `is_fix`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/generator/LoadFromFile.py`

29.4.82 `iterative_model`

Scope. MPM, DEM

Accepted value. string, callable, path, or enumerated choice

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py`

29.5 J–M Keywords and Calls

29.5.1 `jacobian_clamp`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.5.2 `JiangRollingModel`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.5.3 `Jm`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.5.4 `Jm1`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.5.5 `Jm2`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.5.6 `k0`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/physics_model/consititutive_model/MaterialModel.py`

29.5.7 `kappa`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets IPC barrier stiffness coefficient

Numerical or workflow effect. larger values enforce separation more strongly but worsen conditioning

Related theory. Contact Model, IGA-MPM IPC contact model, MPM self-collision IPC, MCC return mapping.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/physics_model/consititutive_model/infinitesimal_strain/ModifiedCamClay.py`

29.5.8 `kernel_assemble_bc_map`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. chooses a solver, assembly format, or preconditioner parameter

Numerical or workflow effect. affects compilation path, memory layout, convergence, and boundary-condition application

Related theory. Implicit solid MPM, MGPCG multilevel solver.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.5.9 `kernel_profiler`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. selects input or output files

Numerical or workflow effect. controls restart, geometry loading, recorder writes, and postprocessing data flow

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.5.10 `ksi`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/physics_model/consititutive_model/infinitesimal_strain/StateDependentMohrCoulomb.py`

29.5.11 `LagrangianMultiplier`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.5.12 `lambda`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `manual/reference;src/physics_model/consititutive_model/infinitesimal_strain/ModifiedCamClay.py`

29.5.13 `lambda_`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.5.14 `lambda_c`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `manual/reference;src/physics_model/consititutive_model/infinitesimal_strain/StateDependentMohrCoulomb.py`

29.5.15 `LengthSize`

Scope. MPM, DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/GeneralShapeTemplate.py`

29.5.16 `levelset_grid_number`

Scope. MPM, DEM

Accepted value. integer vector or list giving level-set grid resolution in each direction

Meaning. sets the sampling resolution for level-set/SDF geometry used by DEM, LSDEM, LSMPM, or MPM geometry loaders

Numerical or workflow effect. controls distance interpolation accuracy, surface extraction quality, contact normals, mass-property estimates, and memory cost

Related theory. SDF and level set geometry, Contact-search narrow phase.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py`

29.5.17 `Lightweight`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.5.18 `LimitVelocity`

Scope. MPM, DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/WallGenerator.py`

29.5.19 `line_search_descent_fallback`

Scope. MPM

Accepted value. string, callable, path, or enumerated choice

Meaning. controls spatial search or broad-phase neighbor construction

Numerical or workflow effect. trades neighbor build cost against candidate count and contact-list stability

Related theory. DEM BVH/LBVH search, DEM dynamics and contact.

Source hint. `manual/reference`

29.5.20 `line_search_energy_atol`

Scope. MPM

Accepted value. string, callable, path, or enumerated choice

Meaning. controls spatial search or broad-phase neighbor construction

Numerical or workflow effect. trades neighbor build cost against candidate count and contact-list stability

Related theory. DEM BVH/LBVH search, DEM dynamics and contact.

Source hint. `manual/reference`

29.5.21 `line_search_energy_rtol`

Scope. MPM

Accepted value. string, callable, path, or enumerated choice

Meaning. controls spatial search or broad-phase neighbor construction

Numerical or workflow effect. trades neighbor build cost against candidate count and contact-list stability

Related theory. DEM BVH/LBVH search, DEM dynamics and contact.

Source hint. `manual/reference`

29.5.22 `line_search_max_backtracks`

Scope. MPM

Accepted value. string, callable, path, or enumerated choice

Meaning. controls spatial search or broad-phase neighbor construction

Numerical or workflow effect. trades neighbor build cost against candidate count and contact-list stability

Related theory. DEM BVH/LBVH search, DEM dynamics and contact.

Source hint. `manual/reference`

29.5.23 `line_search_min_alpha`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. controls spatial search or broad-phase neighbor construction

Numerical or workflow effect. trades neighbor build cost against candidate count and contact-list stability

Related theory. DEM BVH/LBVH search, DEM dynamics and contact.

Source hint. `manual/reference`

29.5.24 `line_search_stagnation_alpha`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. controls spatial search or broad-phase neighbor construction

Numerical or workflow effect. trades neighbor build cost against candidate count and contact-list stability

Related theory. DEM BVH/LBVH search, DEM dynamics and contact.

Source hint. `manual/reference`

29.5.25 `line_search_work_tol`

Scope. MPM

Accepted value. string, callable, path, or enumerated choice

Meaning. controls spatial search or broad-phase neighbor construction

Numerical or workflow effect. trades neighbor build cost against candidate count and contact-list stability

Related theory. DEM BVH/LBVH search, DEM dynamics and contact.

Source hint. `manual/reference`

29.5.26 `Linear`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference`

29.5.27 `LinearModel`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.5.28 `LinearRollingModel`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.5.29 `linear_solver`

Scope. MPM

Accepted value. string, callable, path, or enumerated choice

Meaning. chooses a solver, assembly format, or preconditioner parameter

Numerical or workflow effect. affects compilation path, memory layout, convergence, and boundary-condition application

Related theory. Implicit solid MPM, MGPCG multilevel solver.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.5.30 `linear_solver="PCG"`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. chooses a solver, assembly format, or preconditioner parameter

Numerical or workflow effect. affects compilation path, memory layout, convergence, and boundary-condition application

Related theory. Implicit solid MPM, MGPCG multilevel solver.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.5.31 `LinearElastic`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.5.32 `linearElastic`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.5.33 `LinearSolver`

Scope. MPM

Accepted value. string, callable, path, or enumerated choice

Meaning. chooses a solver, assembly format, or preconditioner parameter

Numerical or workflow effect. affects compilation path, memory layout, convergence, and boundary-condition application

Related theory. Implicit solid MPM, MGPCG multilevel solver.

Source hint. `manual/reference`

29.5.34 `LinkedCell`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. controls spatial search or broad-phase neighbor construction

Numerical or workflow effect. trades neighbor build cost against candidate count and contact-list stability

Related theory. DEM BVH/LBVH search, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.5.35 `load_history_contact()`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.5.36 `localNode`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/PostPlot.py;src/dem/generator/LoadFromFile.py`

29.5.37 `LocalStiffness`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `manual/reference`

29.5.38 `log`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.5.39 `LookAt`

Scope. MPM, DEM

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/Simulation.py;src/mpm/Simulation.py`

29.5.40 `LookFrom`

Scope. MPM, DEM

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/Simulation.py;src/mpm/Simulation.py`

29.5.41 `LSDEM`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.5.42 `LSMPM`

Scope. DEM

Accepted value. `DEM.set_configuration(scheme="LSMPM")`

Meaning. selects the level-set MPM soft-body branch inside the DEM facade.

Numerical or workflow effect. allocates level-set rigid-body storage plus soft material points, compact soft MPM grids, soft surface point lists, TLMPM shape functions, LSMPM contact-node tables, deforming local SDF storage, monitor-triggered SDF redistance, and soft-point output. Rigid bodies can coexist with soft bodies; soft-rigid and soft-soft contact use the LSDEM contact laws with LSMPM soft force transfer.

Related theory. Level-set MPM soft bodies, contact search implementation.

Source hint. `src/dem/Simulation.py;src/dem/generator/BodyGenerator.py;src/dem/engines/SoftBodyKernel.py;src/dem/contact/ContactKernel.py`

29.5.43 `soft_levelset_reinitialization`

Scope. DEM LSMPM

Accepted value. Boolean or dict in `DEM.set_configuration`. A dict accepts the redistance keys `enabled`, `grad_threshold`/`threshold`, `check_interval`, `monitor_band`/`monitor_band_cells`, `reinit_band`/`reinit_band_cells`, `cfl`/`pseudo_time_cfl`, and `iterations`; transport keys `advection_scheme`, `advection_cfl`, and `advection_interval`; volume-correction keys `volume_correction`/`preserve_volume`, `volume_epsilon_cells`, `volume_tolerance`, `volume_max_shift_cells`, `volume_iterations`, and `volume_interval`; and `domain_check` plus `domain_tolerance_cells`. The compatibility spelling `lsmpm_levelset_reinitialization` is also accepted.

Meaning. controls transport, monitor-triggered redistance, optional volume preservation, and compact-domain auditing for deforming LSMPM soft-body level sets.

Numerical or workflow effect. `ReferenceMap` is the default update: it reconstructs total inverse motion from material-point positions and $\boldsymbol{F}$, samples an immutable initial SDF, and applies a local inverse-normal-stretch distance correction. `MacCormack` and `WENO5` are incremental transport alternatives; WENO5 uses Jiang–Shu reconstruction, SSPRK3, and CFL subcycling. Redistance defaults to `False` for `ReferenceMap`; selecting MacCormack or WENO5 without an explicit `enabled` value sets it to `True`. Explicit flags take precedence. When redistance is enabled, the engine checks the monitoring band every `check_interval` steps. If $\max |\|\nabla\phi\| - 1|$ exceeds `grad_threshold`, the current ϕ is stored as ϕ_0 and a Godunov pseudo-time solve restores signed-distance scaling in the redistance band. This pseudo-time source does not overwrite the immutable initial reference-map field. Maintenance can move the zero contour. Redistance parameters are threshold 0.2, check interval 5, monitoring band 4 cells, redistance band 8 cells, pseudo-time CFL 0.3, and automatic iteration count.

Related theory. Level-set MPM soft bodies, Level-set redistance and advection.

Source hint. `src/mpm/Simulation.py;src/dem/mainDEM.py;src/mpm/soft_particle/ReferenceMap.py;src/mpm/engines/SoftParticleEngine.py`

29.5.44 `soft_levelset_advection_*`

Scope. DEM LSMPM

Accepted value. Flat configuration keys are `soft_levelset_advection_scheme`, `soft_levelset_advection_cfl`, and `soft_levelset_advection_interval`. Scheme values are `ReferenceMap` (default), its aliases `InverseMapping` and `SemiLagrangian`, the explicit legacy selector `MacCormack`, or `WENO5`/`WENOFive`; CFL and interval must be positive.

Numerical or workflow effect. ReferenceMap uses current positions, initial positions, and total $\boldsymbol{F}$ to reconstruct affine inverse patches on the SDF grid. Its metric correction is approximate for curved/nonuniform motion; unsupported exterior values are positive caps, not exact far-field distances. It rejects invalid particle determinants, invalid near-interface inverse Jacobians, and missing support cutting the material interior. WENO5 defaults to CFL 0.20, uses SSPRK3 and a three-cell reconstruction halo, and falls back to first-order upwinding near compact-domain boundaries. Other schemes default to CFL 1.0; ReferenceMap does not subcycle on it. The interval defaults to one and accumulates that many mechanical timesteps before an update. ReferenceMap reconstructs total motion at the scheduled step rather than advecting the previous field over the accumulated interval.

Related theory. Level-set MPM soft bodies, Level-set redistance and advection.

Source hint. `src/mpm/Simulation.py;src/mpm/soft_particle/ReferenceMap.py;src/mpm/soft_particle/SoftBodyKernel.py;src/mpm/soft_particle/DEMPMBridge.py`

29.5.45 `soft_levelset_reinit_*`

Scope. DEM LSMPM

Accepted value. flat `DEM.set_configuration` keys `soft_levelset_reinit_grad_threshold`,

soft_levelset_reinit_check_interval, soft_levelset_reinit_monitor_band, soft_levelset_reinit_band, soft_levelset_reinit_cfl, and soft_levelset_reinit_iterations.

Meaning. convenience aliases for the nested soft_levelset_reinitialization dictionary.

Related theory. Level-set MPM soft bodies.

Source hint. src/dem/mainDEM.py;src/dem/Simulation.py

29.5.46 soft_grid_*andsoft_pic_fraction

Scope. DEM LSMPM

Accepted value. soft_grid_storage is Dense or Compact; legacy Sparse, BlockSparse, and FixedSparse values normalize to compact contiguous storage. soft_grid_type is hexahedral or tetrahedral, soft_mechanical_grid_spacing_ratio is positive, and soft_pic_fraction lies in $[0, 1]$.

Numerical or workflow effect. These keys select mechanical-grid topology and storage, grid spacing relative to template equivalent radius, and the PIC/FLIP blend. Tetrahedral support forces the linear basis and four support nodes. Compact storage is precomputed support storage, not a pointer SNode sparse grid.

Related theory. Level-set MPM soft bodies.

Source hint. src/dem/Simulation.py;src/dem/SceneManager.py;src/mpm/soft_particle/DEMPMBridge.py

29.5.47 m_theta

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. docs/helper/geotaichi_user_theory_manual.tex

29.5.48 Magnitude

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/utils/functions/SinFunction.py`

29.5.49 `MainAxis`

Scope. MPM, DEM

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/WallGenerator.py`

29.5.50 `mapping`

Scope. MPM

Accepted value. string, callable, path, or enumerated choice

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpm/mainMPM.py`

29.5.51 `mass`

Scope. MPM, DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/generator/LoadFromFile.py;src/mpm/PostPlot.py;src/mpm/generator/LoadFromFile.py`

29.5.52 `mass_center`

Scope. DEM

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/PostPlot.py;src/dem/generator/LoadFromFile.py`

29.5.53 `mass_density`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets mass density or phase density

Numerical or workflow effect. scales particle mass, body force, wave speed, pressure response, drag, and virtual-mass forces

Related theory. Constitutive and contact models.

Source hint. `src/mpm/generator/LoadFromFile.py`

29.5.54 `master`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/PostPlot.py;src/dem/generator/LoadFromFile.py`

29.5.55 `material`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.5.56 `material_type`

Scope. MPM

Accepted value. string, callable, path, or enumerated choice

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpm/mainMPM.py`

29.5.57 `material_type="Fluid"`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.5.58 `material_type="Solid"`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.5.59 `material_type="TwoPhaseDoubleLayer"`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.5.60 `material_type="TwoPhaseSingleLayer"`

Scope. MPM two-phase material selector

Accepted value. literal `"TwoPhaseSingleLayer"` in `MPM.set_configuration(material_type=...)`

Meaning. stores solid and pore-fluid state on the same material point.

Numerical or workflow effect. with `solver_type="Explicit"`, this selects the Cartesian ULMPM single-layer mixture path in two or three dimensions with PIC/FLIP transfer.

Related theory. MPM weak form and transfers.

Source hint. `src/mpm/Simulation.py;src/mpm/SceneManager.py;src/mpm/engines/ULExplicitTwoPhaseEngine.py`

29.5.61 `MaterialFile`

Scope. MPM

Accepted value. text file path in a material dictionary when `random_field=True`

Meaning. supplies per-particle material parameters for supported random-field material readers.

Numerical or workflow effect. the file is read with `np.loadtxt`; hash-prefixed lines are comments, rows are consumed in inserted-particle order, and each supported material expects its own column layout. See the random-field schema before using this key.

Related theory. Random-field material parameters; schema: random-field material files.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/physics_model/consititutive_model/MaterialModel.py`

29.5.62 `materialID`

Scope. MPM, DEM

Accepted value. value type defined by the owning call site

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `src/dem/generator/LoadFromFile.py;src/dem/generator/WallGenerator.py;src/mpm/Contact.py;src/mpm/MaterialManager.py;src/mpm/PostPlot.py;src/mpm/generator/LoadFromFile.py`

29.5.63 `MaterialID`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. value type defined by the owning call site

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/BodyGenerator.py;src/dem/generator/LoadFromFile.py;src/dem/generator/WallGenerator.py;src/mpdem/generator/LSBodyGenerate.py;...`

29.5.64 `MaterialPointsPerCell`

Scope. DEM LSMPM

Accepted value. positive integer in a `BodyType="SoftBody"` template; `ParticlePerCell` is accepted as an alias.

Meaning. sets how many LSMPM soft material points are sampled per local level-set grid cell when a soft body is created, generated, or lattice-packed.

Numerical or workflow effect. controls soft-body mass/stress resolution, number of surface material points, contact smoothness, memory use, and TLMPM/level-set update cost. It must be chosen consistently with `max_material_point_number` and `levelset_grid_number`.

Related theory. Level-set MPM soft bodies.

Source hint. `src/dem/generator/BodyGenerator.py;src/dem/generator/InsertionKernel.py`

29.5.65 `ParticlePerCell`

Scope. DEM LSMPM

Accepted value. positive integer alias for `MaterialPointsPerCell`.

Meaning. compatibility spelling for LSMPM soft material-point sampling density.

Numerical or workflow effect. same as `MaterialPointsPerCell`; if both keys are supplied, `MaterialPointsPerCell` is the preferred explicit spelling.

Related theory. Level-set MPM soft bodies.

Source hint. `src/dem/generator/BodyGenerator.py`

29.5.66 `matrix_free`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. chooses a solver, assembly format, or preconditioner parameter

Numerical or workflow effect. affects compilation path, memory layout, convergence, and boundary-condition application

Related theory. Implicit solid MPM, MGPCG multilevel solver.

Source hint. `manual/reference`

29.5.67 `matrix_symmetric`

Scope. MPM

Accepted value. boolean switch or small enumerated mode

Meaning. declares whether the tangent may be treated as symmetric

Numerical or workflow effect. must be false for nonassociated plasticity, nonsymmetric stabilization, or friction blocks unless projected

Related theory. Implicit solid MPM, Constitutive and contact models.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpm/mainMPM.py`

29.5.68 `MatrixFree`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. selects operator-application without storing full matrix entries

Numerical or workflow effect. reduces memory but keeps current solid Dirichlet handling on penalty path

Related theory. Implicit solid MPM, Implicit Dirichlet constraints.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference`

29.5.69 `MatrixFreeMGPCG`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. chooses a solver, assembly format, or preconditioner parameter

Numerical or workflow effect. affects compilation path, memory layout, convergence, and boundary-condition application

Related theory. Implicit solid MPM, MGPCG multilevel solver.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.5.70 `max_absorbing_constraint`

Scope. MPM

Accepted value. integer count, index, level, or interval

Meaning. declares memory capacity before Taichi fields are allocated

Numerical or workflow effect. undersizing causes insertion, refinement, contact, split, or recorder failure; oversizing increases GPU memory use

Related theory. MPM weak form and transfers, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/Simulation.py`

29.5.71 `max_affine_body_number`

Scope. DEM

Accepted value. positive integer capacity

Meaning. declares AffineBody body capacity before Taichi fields are allocated.

Numerical or workflow effect. undersizing prevents affine bodies from being inserted; oversizing increases memory use. If this key is omitted in `AffineBody` mode, `max_rigid_body_number` is accepted as a compatibility fallback.

Related theory. Affine body dynamics.

Source hint. `src/dem/mainDEM.py;src/dem/SceneManager.py`

29.5.72 `max_bounding_radius`

Scope. DEM

Accepted value. integer count, index, level, or interval

Meaning. declares memory capacity before Taichi fields are allocated

Numerical or workflow effect. undersizing causes insertion, refinement, contact, split, or recorder failure; oversizing increases GPU memory use

Related theory. MPM weak form and transfers, DEM dynamics and contact.

Source hint. `src/dem/ContactManager.py`

29.5.73 `max_box`

Scope. DEM

Accepted value. integer count, index, level, or interval

Meaning. declares memory capacity before Taichi fields are allocated

Numerical or workflow effect. undersizing causes insertion, refinement, contact, split, or recorder failure; oversizing increases GPU memory use

Related theory. MPM weak form and transfers, DEM dynamics and contact.

Source hint. `src/dem/generator/LoadFromFile.py`

29.5.74 `max_clump_number`

Scope. MPM, DEM

Accepted value. integer count, index, level, or interval

Meaning. declares memory capacity before Taichi fields are allocated

Numerical or workflow effect. undersizing causes insertion, refinement, contact, split, or recorder failure; oversizing increases GPU memory use

Related theory. MPM weak form and transfers, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py`

29.5.75 `max_constraint_number`

Scope. MPM

Accepted value. integer count, index, level, or interval

Meaning. declares memory capacity before Taichi fields are allocated

Numerical or workflow effect. undersizing causes insertion, refinement, contact, split, or recorder failure; oversizing increases GPU memory use

Related theory. MPM weak form and transfers, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.5.76 `max_digital_elevation_facet_number`

Scope. MPM, DEM

Accepted value. integer count, index, level, or interval

Meaning. declares memory capacity before Taichi fields are allocated

Numerical or workflow effect. undersizing causes insertion, refinement, contact, split, or recorder failure; oversizing increases GPU memory use

Related theory. MPM weak form and transfers, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py`

29.5.77 `max_displacement_constraint`

Scope. MPM

Accepted value. integer count, index, level, or interval

Meaning. declares memory capacity before Taichi fields are allocated

Numerical or workflow effect. undersizing causes insertion, refinement, contact, split, or recorder failure; oversizing increases GPU memory use

Related theory. MPM weak form and transfers, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/Simulation.py`

29.5.78 `max_facet_number`

Scope. MPM, DEM

Accepted value. integer count, index, level, or interval

Meaning. declares memory capacity before Taichi fields are allocated

Numerical or workflow effect. undersizing causes insertion, refinement, contact, split, or recorder failure; oversizing increases GPU memory use

Related theory. MPM weak form and transfers, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py`

29.5.79 `max_friction_constraint`

Scope. MPM

Accepted value. integer count, index, level, or interval

Meaning. declares memory capacity before Taichi fields are allocated

Numerical or workflow effect. undersizing causes insertion, refinement, contact, split, or recorder failure; oversizing increases GPU memory use

Related theory. MPM weak form and transfers, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/Simulation.py`

29.5.80 `max_iteration_number`

Scope. MPM

Accepted value. integer count, index, level, or interval

Meaning. declares memory capacity before Taichi fields are allocated

Numerical or workflow effect. undersizing causes insertion, refinement, contact, split, or recorder failure; oversizing increases GPU memory use

Related theory. MPM weak form and transfers, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.5.81 `max_material_number`

Scope. MPM, DEM

Accepted value. integer count, index, level, or interval

Meaning. declares memory capacity before Taichi fields are allocated

Numerical or workflow effect. undersizing causes insertion, refinement, contact, split, or recorder failure; oversizing increases GPU memory use

Related theory. MPM weak form and transfers, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py;src/mpm/mainMPM.py`

29.5.82 `max_material_point_number`

Scope. MPM, DEM

Accepted value. integer count, index, level, or interval

Meaning. declares memory capacity before Taichi fields are allocated

Numerical or workflow effect. undersizing causes insertion, refinement, contact, split, or recorder failure; oversizing increases GPU memory use

Related theory. MPM weak form and transfers, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py`

29.5.83 `max_node_pressure_constraint`

Scope. MPM

Accepted value. integer count, index, level, or interval

Meaning. declares memory capacity before Taichi fields are allocated

Numerical or workflow effect. undersizing causes insertion, refinement, contact, split, or recorder failure; oversizing increases GPU memory use

Related theory. MPM weak form and transfers, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/Simulation.py`

29.5.84 `max_particle_number`

Scope. MPM, DEM

Accepted value. integer count, index, level, or interval

Meaning. declares memory capacity before Taichi fields are allocated

Numerical or workflow effect. undersizing causes insertion, refinement, contact, split, or recorder failure; oversizing increases GPU memory use

Related theory. MPM weak form and transfers, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py;src/mpm/mainMPM.py`

29.5.85 `max_particle_traction_constraint`

Scope. MPM

Accepted value. integer count, index, level, or interval

Meaning. declares memory capacity before Taichi fields are allocated

Numerical or workflow effect. undersizing causes insertion, refinement, contact, split, or recorder failure; oversizing increases GPU memory use

Related theory. MPM weak form and transfers, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/Simulation.py`

29.5.86 `max_patch_number`

Scope. MPM, DEM

Accepted value. integer count, index, level, or interval

Meaning. declares memory capacity before Taichi fields are allocated

Numerical or workflow effect. undersizing causes insertion, refinement, contact, split, or recorder failure; oversizing increases GPU memory use

Related theory. MPM weak form and transfers, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py`

29.5.87 `max_plane_number`

Scope. MPM, DEM

Accepted value. integer count, index, level, or interval

Meaning. declares memory capacity before Taichi fields are allocated

Numerical or workflow effect. undersizing causes insertion, refinement, contact, split, or recorder failure; oversizing increases GPU memory use

Related theory. MPM weak form and transfers, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py`

29.5.88 `max_reflection_constraint`

Scope. MPM

Accepted value. integer count, index, level, or interval

Meaning. declares memory capacity before Taichi fields are allocated

Numerical or workflow effect. undersizing causes insertion, refinement, contact, split, or recorder failure; oversizing increases GPU memory use

Related theory. MPM weak form and transfers, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/Simulation.py`

29.5.89 `max_rigid_body_number`

Scope. MPM, DEM

Accepted value. integer count, index, level, or interval

Meaning. declares memory capacity before Taichi fields are allocated

Numerical or workflow effect. undersizing causes insertion, refinement, contact, split, or recorder failure; oversizing increases GPU memory use

Related theory. MPM weak form and transfers, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py`

29.5.90 `max_rigid_template_number`

Scope. MPM, DEM

Accepted value. integer count, index, level, or interval

Meaning. declares memory capacity before Taichi fields are allocated

Numerical or workflow effect. undersizing causes insertion, refinement, contact, split, or recorder failure; oversizing increases GPU memory use

Related theory. MPM weak form and transfers, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py`

29.5.91 `max_servo_wall_number`

Scope. MPM, DEM

Accepted value. integer count, index, level, or interval

Meaning. declares memory capacity before Taichi fields are allocated

Numerical or workflow effect. undersizing causes insertion, refinement, contact, split, or recorder failure; oversizing increases GPU memory use

Related theory. MPM weak form and transfers, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py`

29.5.92 `max_soft_body_number`

Scope. MPM, DEM

Accepted value. integer count, index, level, or interval

Meaning. declares memory capacity before Taichi fields are allocated

Numerical or workflow effect. undersizing causes insertion, refinement, contact, split, or recorder failure; oversizing increases GPU memory use

Related theory. MPM weak form and transfers, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py`

29.5.93 `max_sphere_number`

Scope. MPM, DEM

Accepted value. integer count, index, level, or interval

Meaning. declares memory capacity before Taichi fields are allocated

Numerical or workflow effect. undersizing causes insertion, refinement, contact, split, or recorder failure; oversizing increases GPU memory use

Related theory. MPM weak form and transfers, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py`

29.5.94 `max_traction_constraint`

Scope. MPM

Accepted value. integer count, index, level, or interval

Meaning. declares memory capacity before Taichi fields are allocated

Numerical or workflow effect. undersizing causes insertion, refinement, contact, split, or recorder failure; oversizing increases GPU memory use

Related theory. MPM weak form and transfers, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/Simulation.py`

29.5.95 `max_velocity`

Scope. DEM

Accepted value. integer count, index, level, or interval

Meaning. declares memory capacity before Taichi fields are allocated

Numerical or workflow effect. undersizing causes insertion, refinement, contact, split, or recorder failure; oversizing increases GPU memory use

Related theory. MPM weak form and transfers, DEM dynamics and contact.

Source hint. `src/dem/generator/WallGenerator.py`

29.5.96 `max_velocity_constraint`

Scope. MPM

Accepted value. integer count, index, level, or interval

Meaning. declares memory capacity before Taichi fields are allocated

Numerical or workflow effect. undersizing causes insertion, refinement, contact, split, or recorder failure; oversizing increases GPU memory use

Related theory. MPM weak form and transfers, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/Simulation.py`

29.5.97 `MaxBoundingRadius`

Scope. MPM, DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/BodyGenerator.py`

29.5.98 `MaxRadius`

Scope. MPM, DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/BodyGenerator.py;src/dem/generator/LoadFromFile.py`

29.5.99 `memory_allocate`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.5.100 `memory_allocate(dict)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.5.101 `memory_allocate(memory,dem_memory=None,mpm_memory=None)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.5.102 `MGPCG`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. selects multigrid-preconditioned conjugate gradient pressure or linear solver branch

Numerical or workflow effect. uses restriction, smoothing, and prolongation to precondition low-frequency residuals

Related theory. MGPCG multilevel solver.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference`

29.5.103 `MiddleCircumscribed`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.5.104 `min_bounding_radius`

Scope. DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/ContactManager.py`

29.5.105 `min_box`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/generator/LoadFromFile.py`

29.5.106 `MinBoundingRadius`

Scope. MPM, DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/BodyGenerator.py`

29.5.107 `MinRadius`

Scope. MPM, DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/BodyGenerator.py;src/dem/generator/LoadFromFile.py`

29.5.108 `mode`

Scope. MPM

Accepted value. string, callable, path, or enumerated choice

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.5.109 `ModifiedCamClay`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.5.110 `modify_parameters(**kwargs)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.5.111 `Modulus`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets elastic constants used by stress increments and tangent stiffness

Numerical or workflow effect. controls wave speed, timestep restrictions, incompressibility, and implicit matrix conditioning

Related theory. Constitutive and contact models, Implicit solid MPM.

Source hint. `manual/reference;src/physics_model/consititutive_model/strain_rate/Bingham.py;src/physics_model/consititutive_model/strain_rate/Newtonian.py`

29.5.112 `MohrCoulomb`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.5.113 `MooneyRivlin`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.5.114 `MoveVelocity`

Scope. MPM, DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/Simulation.py;src/mpm/Simulation.py`

29.5.115 `moving_least_square`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.5.116 `MPDEM`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.5.117 `MPM`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.5.118 `MPM(title,log=True)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.5.119 `MPMContact`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.5.120 `multilevel`

Scope. MPM

Accepted value. integer count, index, level, or interval

Meaning. requests the number of MGPCG levels

Numerical or workflow effect. actual valid levels are capped by grid dimensions

Related theory. MGPCG multilevel solver.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpm/mainMPM.py`

29.5.121 `MUSL`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference`

29.6 N–R Keywords and Calls

29.6.1 `Name`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. string, callable, path, or enumerated choice

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/GenerateManager.py;src/dem/generator/BodyGenerator.py;src/dem/generator/ClumpTemplate.py;src/dem/generator/GeneralShapeTemplate.py;...`

29.6.2 `nd`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/physics_model/consititutive_model/infinitesimal_strain/StateDependentMohrCoulomb.py`

29.6.3 `NeoHookean`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.4 `neoHookean`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.5 `NeumannBoundary.append`

Scope. IGA and Direct MPM control-point/grid degrees of freedom

Accepted value. degree-of-freedom identifiers and already-integrated generalized force values

Meaning. adds resultant Neumann loads; it does not perform boundary quadrature of a traction density

Numerical or workflow effect. contributes $-\boldsymbol{f}_{\mathrm{N}} \cdot \boldsymbol{u}$ to implicit energy and $\boldsymbol{f}_{\mathrm{N}}$ to the residual; axisymmetric input must already include its revolved $2\pi R$ measure

Related theory. IGA.

Source hint. `src/iga/boundaries/BoundaryCondition.py`

29.6.6 `Newmark`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.7 `newmark_parameter`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpm/mainMPM.py`

29.6.8 `Newtonian`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.9 `nf`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/physics_model/consititutive_model/infinitesimal_strain/StateDependentMohrCoulomb.py`

29.6.10 `NLevel`

Scope. MPM

Accepted value. integer count, index, level, or interval

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/boundaries/BoundaryConstraint.py`

29.6.11 `NoData`

Scope. MPM, DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/WallGenerator.py`

29.6.12 `None`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.13 `NonSlipFraction`

Scope. DEM, MPDEM/CFDEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpdem/contact/ParticleFluid.py`

29.6.14 `norm`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/PostPlot.py;src/dem/generator/WallGenerator.py`

29.6.15 `Norm`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/boundaries/BoundaryConstraint.py`

29.6.16 `normal`

Scope. MPM

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/mpm/PostPlot.py;src/mpm/generator/LoadFromFile.py`

29.6.17 `Normal`

Scope. shared

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.18 `normal_force`

Scope. DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `src/dem/PostPlot.py;src/dem/contact/ContactModelBase.py;src/dem/contact/RollingModelBase.py`

29.6.19 `NormalCutOff`

Scope. DEM

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/contact/EnergyConservation.py`

29.6.20 `NormalStiffness`

Scope. DEM, MPDEM/CFDEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/dem/contact/EnergyConservation.py;src/dem/contact/Linear.py;src/dem/contact/LinearRolling.py;src/mpdem/contact/EnergyConservation.py;src/mpdem/contact/Linear.py;...`

29.6.21 `NormalToShearRatio`

Scope. DEM, MPDEM/CFDEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/dem/contact/Linear.py;src/dem/contact/LinearRolling.py;src/mpdem/contact/Linear.py`

29.6.22 `NormalViscousDamping`

Scope. DEM, MPDEM/CFDEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/dem/contact/EnergyConservation.py;src/dem/contact/JiangRolling.py;src/dem/contact/Linear.py;src/dem/contact/LinearRolling.py;src/mpdem/contact/EnergyConservation.py;...`

29.6.23 `nParticlesPerCell`

Scope. `MPM.add_body.body.Template` and coupled MPM body generation.

Accepted value. Positive integer number of particles per cell axis.

Meaning. A value q creates q^d samples in a fully filled d-dimensional cell: q^2 in 2D and q^3 in 3D. It is not the total particle count per cell.

Numerical or workflow effect. Controls particle quadrature, cell-occupancy continuity, memory, and P2G contention. In incompressible free-surface MPM, an under-resolved moving fluid may leave false empty cells, fragment the pressure domain, and create spurious gas pockets. Size `max_particle_number` from the per-axis interpretation and validate occupancy, volume, pressure, and free surface under refinement.

Related theory. Incompressible Fluid MPM, MPM particle quadrature.

Source hint. `src/mpm/generator/BodyGenerator.py`, `src/mpm/engines/IncompressibleEngine.py`

29.6.24 `NSphere`

Scope. MPM, DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/ClumpTemplate.py`

29.6.25 `Object`

Scope. MPM, DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/GeneralShapeTemplate.py`

29.6.26 `ocr`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.27 `offline_cache`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.28 `Offset`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/generator/LoadFromFile.py`

29.6.29 `oldRollAngle`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/contact/RollingModelBase.py`

29.6.30 `oldTangentialOverlap`

Scope. DEM, MPDEM/CFDEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/contact/ContactModelBase.py;src/dem/contact/RollingModelBase.py;src/mpdem/contact/ContactModelBase.py`

29.6.31 `oldTwistAngle`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/contact/RollingModelBase.py`

29.6.32 `omega`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/PostPlot.py;src/dem/generator/LoadFromFile.py`

29.6.33 `Orientation`

Scope. MPM, DEM

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/SceneManager.py;src/dem/generator/LoadFromFile.py;src/dem/generator/WallGenerator.py;src/mpm/generator/LoadFromFile.py`

29.6.34 `OuterNormal`

Scope. MPM, DEM

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/WallGenerator.py;src/mpm/boundaries/BoundaryConstraint.py`

29.6.35 `OverConsolidationRatio`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/physics_model/consititutive_model/infinitesimal_strain/ModifiedCamClay.py`

29.6.36 `p_ref`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.37 `parameters`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/generator/LoadFromFile.py`

29.6.38 `particle_interaction`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpdem/mainDEMPM.py`

29.6.39 `particle_shifting`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.6.40 `particle_traction_method`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpm/mainMPM.py`

29.6.41 `ParticleColor`

Scope. MPM, DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/Simulation.py;src/mpm/Simulation.py`

29.6.42 `ParticleFile`

Scope. MPM, DEM

Accepted value. string, callable, path, or enumerated choice

Meaning. selects input or output files

Numerical or workflow effect. controls restart, geometry loading, recorder writes, and postprocessing data flow

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/LoadFromFile.py;src/mpm/generator/LoadFromFile.py`

29.6.43 `particleID`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/mpm/PostPlot.py;src/mpm/generator/LoadFromFile.py`

29.6.44 `ParticleNumber`

Scope. MPM, DEM

Accepted value. integer count, index, level, or interval

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/LoadFromFile.py;src/mpm/generator/LoadFromFile.py`

29.6.45 `ParticleReynold`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpdem/fluid_dynamics/DragForceModel.py`

29.6.46 `ParticleSplitBatch`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/elements/AdaptiveHexahedronElement.py`

29.6.47 `ParticleSplitInterval`

Scope. MPM

Accepted value. integer count, index, level, or interval

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/elements/AdaptiveHexahedronElement.py;src/mpm/elements/AdaptiveQuadrilateralElement.py`

29.6.48 `SplitInteriorOnly`

Scope. MPM adaptive 3D

Accepted value. boolean switch

Meaning. restricts 1-to-8 particle splitting to refined coarse cells whose particle volume fraction is high enough to be treated as interior material

Numerical or workflow effect. prevents symmetric child placement from creating par-

ticles outside free surfaces or slope boundaries; grid cells can still refine without splitting their boundary particles

Related theory. Adaptive MPM.

Source hint. `src/mpm/elements/AdaptiveHexahedronElement.py;src/mpm/elements/AdaptiveHexahedronKernel.py`

29.6.49 `SplitInteriorVolumeFraction`

Scope. MPM adaptive 3D

Accepted value. real scalar in $(0, 1]$

Meaning. minimum coarse-cell particle-volume fraction used by `SplitInteriorOnly`

Numerical or workflow effect. larger values split fewer boundary/partially filled cells; smaller values split more cells but can reintroduce boundary child particles outside the original material domain

Related theory. Adaptive MPM.

Source hint. `src/mpm/elements/AdaptiveHexahedronElement.py;src/mpm/elements/AdaptiveHexahedronKernel.py`

29.6.50 `ParticleStress`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/generator/BodyGenerator.py;src/mpm/generator/LoadFromFile.py`

29.6.51 `ParticleTractionConstraint`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.52 `path`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. selects input or output files

Numerical or workflow effect. controls restart, geometry loading, recorder writes, and postprocessing data flow

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.53 `pc`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/physics_model/consititutive_model/infinitesimal_strain/ModifiedCamClay.py`

29.6.54 `PCG`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference`

29.6.55 `PCN`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.56 `Pebble`

Scope. MPM, DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/ClumpTemplate.py`

29.6.57 `PebbleFile`

Scope. MPM, DEM

Accepted value. string, callable, path, or enumerated choice

Meaning. selects input or output files

Numerical or workflow effect. controls restart, geometry loading, recorder writes, and postprocessing data flow

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/LoadFromFile.py`

29.6.58 `penalty`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. controls adaptive MPM refinement, particle splitting, or coarse-fine interface treatment

Numerical or workflow effect. changes grid topology, support, transfer consistency, and

implicit tangent structure

Related theory. Adaptive MPM.

Source hint. `src/mpm/Contact.py`

29.6.59 `Penalty`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. controls adaptive MPM refinement, particle splitting, or coarse-fine interface treatment

Numerical or workflow effect. changes grid topology, support, transfer consistency, and implicit tangent structure

Related theory. Adaptive MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpm/elements/AdaptiveHexahedronElement.py;src/mpm/elements/AdaptiveQuadrilateralElement.py`

29.6.60 `PenaltyBeta`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. controls adaptive MPM refinement, particle splitting, or coarse-fine interface treatment

Numerical or workflow effect. changes grid topology, support, transfer consistency, and implicit tangent structure

Related theory. Adaptive MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpm/elements/AdaptiveHexahedronElement.py;src/mpm/elements/AdaptiveQuadrilateralElement.py`

29.6.61 `PenaltyIterations`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. controls adaptive MPM refinement, particle splitting, or coarse-fine interface treatment

Numerical or workflow effect. changes grid topology, support, transfer consistency, and

implicit tangent structure

Related theory. Adaptive MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpm/elements/AdaptiveHexahedronElement.py;src/mpm/elements/AdaptiveQuadrilateralElement.py`

29.6.62 `Period`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/BodyGenerator.py;src/dem/generator/LoadFromFile.py;src/mpdem/generator/LSBodyGenerate.py;src/mpm/generator/BodyGenerator.py;...`

29.6.63 `Permeability`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets fluid, pore-pressure, porous-media, or viscous parameter

Numerical or workflow effect. enters pressure projection, Darcy or Ergun drag, pressure wave speed, and mixture mass balance

Related theory. Incompressible fluid MPM, Semi-implicit two-phase MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/physics_model/consititutive_model/MaterialModel.py`

29.6.64 `Phase`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/generator/BodyGenerator.py;src/utils/functions/SinFunction.py`

29.6.65 `PIC`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.66 `PIC/FLIP`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.67 `PlasticDevStrain`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `manual/reference;src/physics_model/consititutive_model/infinitesimal_strain/DruckerPrager.py;src/physics_model/consititutive_model/infinitesimal_strain/ElasticPerfectlyPlastic.py;src/physics_model/consititutive_model/infinitesimal_strain/MohrCoulomb.py`

29.6.68 `point`

Scope. DEM

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/PostPlot.py;src/dem/generator/WallGenerator.py`

29.6.69 `point1`

Scope. DEM

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/PostPlot.py;src/dem/generator/WallGenerator.py`

29.6.70 `point2`

Scope. DEM

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/PostPlot.py;src/dem/generator/WallGenerator.py`

29.6.71 `point3`

Scope. DEM

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/PostPlot.py;src/dem/generator/WallGenerator.py`

29.6.72 `point_coordination_number`

Scope. MPM, DEM

Accepted value. integer count, index, level, or interval

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py`

29.6.73 `PointLight`

Scope. MPM, DEM

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/Simulation.py;src/mpm/Simulation.py`

29.6.74 `Poisson`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. value type defined by the owning call site

Meaning. sets elastic constants used by stress increments and tangent stiffness

Numerical or workflow effect. controls wave speed, timestep restrictions, incompressibility, and implicit matrix conditioning

Related theory. Constitutive and contact models, Implicit solid MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/contact/HertzMindlin.py;src/mpdem/contact/HertzMindlin.py`

29.6.75 `poisson_ratio`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets elastic constants used by stress increments and tangent stiffness

Numerical or workflow effect. controls wave speed, timestep restrictions, incompressibility, and implicit matrix conditioning

Related theory. Constitutive and contact models, Implicit solid MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.76 `PoissonRatio`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets elastic constants used by stress increments and tangent stiffness

Numerical or workflow effect. controls wave speed, timestep restrictions, incompressibility, and implicit matrix conditioning

Related theory. Constitutive and contact models, Implicit solid MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/physics_model/consititutive_model/finite_strain/Gent.py;src/physics_model/consititutive_model/finite_strain/HenckyElastic.py;src/physics_model/consititutive_model/finite_strain/Hydrogel.py;src/physics_model/consititutive_model/finite_strain/MooneyRivlin.py;...`

29.6.77 `PoissonSampling`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. value type defined by the owning call site

Meaning. sets elastic constants used by stress increments and tangent stiffness

Numerical or workflow effect. controls wave speed, timestep restrictions, incompressibility, and implicit matrix conditioning

Related theory. Constitutive and contact models, Implicit solid MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/BodyGenerator.py;src/mpdem/generator/LSBodyGenerate.py`

29.6.78 `Polygon2D`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.79 `PolySuperEllipsoid`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.80 `PolySuperQuadrics`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.81 `PorePressure`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets fluid, pore-pressure, porous-media, or viscous parameter

Numerical or workflow effect. enters pressure projection, Darcy or Ergun drag, pressure wave speed, and mixture mass balance

Related theory. Incompressible fluid MPM, Semi-implicit two-phase MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/generator/Generator.py`

29.6.82 `Porosity`

Scope. MPM, DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets fluid, pore-pressure, porous-media, or viscous parameter

Numerical or workflow effect. enters pressure projection, Darcy or Ergun drag, pressure wave speed, and mixture mass balance

Related theory. Incompressible fluid MPM, Semi-implicit two-phase MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/BodyGenerator.py;src/physics_model/consititutive_model/MaterialModel.py`

29.6.83 `position`

Scope. MPM, DEM

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/PostPlot.py;src/dem/generator/LoadFromFile.py;src/mpm/PostPlot.py;src/mpm/generator/LoadFromFile.py`

29.6.84 `Position`

Scope. MPM, DEM

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/ClumpTemplate.py`

29.6.85 `postprocessing`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.86 `postprocessing(...)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.87 `ppcontact`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py;src/mpdem/mainDEMPM.py`

29.6.88 `pre_and_post_smoothing`

Scope. MPM

Accepted value. integer count, index, level, or interval

Meaning. chooses a solver, assembly format, or preconditioner parameter

Numerical or workflow effect. affects compilation path, memory layout, convergence, and boundary-condition application

Related theory. Implicit solid MPM, MGPCG multilevel solver.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpm/mainMPM.py`

29.6.89 `PredictCorrector`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.90 `pressure`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets fluid, pore-pressure, porous-media, or viscous parameter

Numerical or workflow effect. enters pressure projection, Darcy or Ergun drag, pressure wave speed, and mixture mass balance

Related theory. Incompressible fluid MPM, Semi-implicit two-phase MPM.

Source hint. `src/mpm/PostPlot.py;src/mpm/generator/LoadFromFile.py`

29.6.91 `Pressure`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets fluid, pore-pressure, porous-media, or viscous parameter

Numerical or workflow effect. enters pressure projection, Darcy or Ergun drag, pressure wave speed, and mixture mass balance

Related theory. Incompressible fluid MPM, Semi-implicit two-phase MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpm/boundaries/BoundaryConstraint.py`

29.6.92 `pressure_bate`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets fluid, pore-pressure, porous-media, or viscous parameter

Numerical or workflow effect. enters pressure projection, Darcy or Ergun drag, pressure wave speed, and mixture mass balance

Related theory. Incompressible fluid MPM, Semi-implicit two-phase MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.6.93 `pressure_beta`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets fluid, pore-pressure, porous-media, or viscous parameter

Numerical or workflow effect. enters pressure projection, Darcy or Ergun drag, pressure wave speed, and mixture mass balance

Related theory. Incompressible fluid MPM, Semi-implicit two-phase MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.6.94 `pressure_ref`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets fluid, pore-pressure, porous-media, or viscous parameter

Numerical or workflow effect. enters pressure projection, Darcy or Ergun drag, pressure wave speed, and mixture mass balance

Related theory. Incompressible fluid MPM, Semi-implicit two-phase MPM.

Source hint. `src/physics_model/consititutive_model/infinitesimal_strain/ModifiedCamClay.py`

29.6.95 `pressure_smoothing`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets fluid, pore-pressure, porous-media, or viscous parameter

Numerical or workflow effect. enters pressure projection, Darcy or Ergun drag, pressure wave speed, and mixture mass balance

Related theory. Incompressible fluid MPM, Semi-implicit two-phase MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.6.96 `pressure_solver`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets fluid, pore-pressure, porous-media, or viscous parameter

Numerical or workflow effect. enters pressure projection, Darcy or Ergun drag, pressure wave speed, and mixture mass balance

Related theory. Incompressible fluid MPM, Semi-implicit two-phase MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.6.97 `pressure_solver="MGPCG"`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets fluid, pore-pressure, porous-media, or viscous parameter

Numerical or workflow effect. enters pressure projection, Darcy or Ergun drag, pressure wave speed, and mixture mass balance

Related theory. Incompressible fluid MPM, Semi-implicit two-phase MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.98 `pressure_solver="PCG"`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets fluid, pore-pressure, porous-media, or viscous parameter

Numerical or workflow effect. enters pressure projection, Darcy or Ergun drag, pressure wave speed, and mixture mass balance

Related theory. Incompressible fluid MPM, Semi-implicit two-phase MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.99 `pressure_solver=MGPCG`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets fluid, pore-pressure, porous-media, or viscous parameter

Numerical or workflow effect. enters pressure projection, Darcy or Ergun drag, pressure wave speed, and mixture mass balance

Related theory. Incompressible fluid MPM, Semi-implicit two-phase MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.100 `pressure_stabilize`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets fluid, pore-pressure, porous-media, or viscous parameter

Numerical or workflow effect. enters pressure projection, Darcy or Ergun drag, pressure wave speed, and mixture mass balance

Related theory. Incompressible fluid MPM, Semi-implicit two-phase MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.101 `pressure_stabilize="FIC"`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets fluid, pore-pressure, porous-media, or viscous parameter

Numerical or workflow effect. enters pressure projection, Darcy or Ergun drag, pressure wave speed, and mixture mass balance

Related theory. Incompressible fluid MPM, Semi-implicit two-phase MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.102 `print_ir`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.103 `psize`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/mpm/generator/LoadFromFile.py`

29.6.104 `pwcontact`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py;src/mpdem/mainDEMPM.py`

29.6.105 `Q4N2D`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.106 `QuadBSpline`

Scope. MPM

Accepted value. `MPM.add_element.element.ShapeFunction="QuadBSpline"` on Direct MPM; Native MPM also accepts it through its element/shape-function configuration.

Meaning. Selects quadratic B-spline interpolation, with 3^d support nodes in dimension d.

Numerical or workflow effect. Direct MPM requires at least four nodes per grid axis. Near grid boundaries, the first/second/penultimate/last-node polynomials reproduce constants and affine fields; values, gradients and IPC position Hessians use the same boundary type. Outside-grid particles are rejected. The implicit IGA CPT and flexible-barrier examples use this choice. Linear CPT checkpoints cannot be resumed into a QuadBSpline run; begin at frame zero.

Related theory. MPM shape functions.

Source hint. `src/mpm/engines/direct/MPMSolver.py;src/utils/ShapeFunctions.py;examples/igampm/cpt_dp/cpt_dp.py;examples/igampm/flexible_barrier/flexible_barrier.py`

29.6.107 `Quadratic`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.108 `quanternion`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/PostPlot.py;src/dem/generator/LoadFromFile.py`

29.6.109 `QuarterCylinder`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.110 `quasi_static`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.6.111 `R8N3D`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.112 `radius`

Scope. MPM, DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/PostPlot.py;src/dem/generator/LoadFromFile.py;src/mpm/generator/LoadFromFile.py`

29.6.113 `Radius`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/BodyGenerator.py;src/dem/generator/ClumpTemplate.py;src/mpdem/generator/LSBodyGenerate.py`

29.6.114 `RadiusDistribution`

Scope. MPM, DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/BodyGenerator.py`

29.6.115 `random_field`

Scope. MPM

Accepted value. boolean in `MPM.set_configuration`

Meaning. enables per-particle material parameters read from each material's `MaterialFile` before body insertion.

Numerical or workflow effect. supported readers consume rows in particle order and store density plus converted elastic/strength parameters on the particle state. File-supported models are currently `GranularMaterial`, `ElasticPerfectlyPlastic`, `MohrCoulomb`, and `DruckerPrager`; `LinearElastic` and `ModifiedCamClay` do not provide active file readers here, while `NorSand` and `SanisandMS` reject the mode.

Related theory. Random-field material parameters; schema: random-field material files.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.6.116 `random_seed`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.117 `RateDependent`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `manual/reference;src/physics_model/consititutive_model/infinitesimal_strain/ElasPlasticity.py`

29.6.118 `rayleigh_damping`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/mpm/mainMPM.py`

29.6.119 `RayPath`

Scope. MPM, DEM

Accepted value. string, callable, path, or enumerated choice

Meaning. selects input or output files

Numerical or workflow effect. controls restart, geometry loading, recorder writes, and postprocessing data flow

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/GeneralShapeTemplate.py`

29.6.120 `read_restart(...)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.121 `read_restart(file_number,file_path,is_continue=True)`

Scope. shared

Accepted value. integer count, index, level, or interval

Meaning. selects input or output files

Numerical or workflow effect. controls restart, geometry loading, recorder writes, and postprocessing data flow

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.122 `Rectangle`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.123 `Rectangle2D`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.124 `RefineCriterion`

Scope. MPM

Accepted value. string, callable, path, or enumerated choice

Meaning. selects the particle state variable used to seed adaptive refinement

Numerical or workflow effect. connects material history variables to cell marking and particle splitting

Related theory. Adaptive MPM, Constitutive and contact models.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpm/elements/AdaptiveHexahedronElement.py;src/mpm/elements/AdaptiveQuadrilateralElement.py`

29.6.125 `RefineInterval`

Scope. MPM

Accepted value. integer count, index, level, or interval

Meaning. controls adaptive MPM refinement, particle splitting, or coarse-fine interface treatment

Numerical or workflow effect. changes grid topology, support, transfer consistency, and implicit tangent structure

Related theory. Adaptive MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpm/elements/AdaptiveHexahedronElement.py;src/mpm/elements/AdaptiveQuadrilateralElement.py`

29.6.126 `RefineParticles`

Scope. MPM

Accepted value. boolean switch or small enumerated mode

Meaning. controls adaptive MPM refinement, particle splitting, or coarse-fine interface treatment

Numerical or workflow effect. changes grid topology, support, transfer consistency, and implicit tangent structure

Related theory. Adaptive MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpm/elements/AdaptiveHexahedronElement.py;src/mpm/elements/AdaptiveQuadrilateralElement.py`

29.6.127 `RefineRatio`

Scope. MPM

Accepted value. real scalar in $(0, 1]$

Meaning. limits adaptive refinement by the fraction of original coarse cells that may be refined

Numerical or workflow effect. if n_r coarse cells are refined out of N_c, refinement is rejected when n_r/N_c exceeds `RefineRatio`; internally the corresponding added fine-cell budget is $(2^d - 1)n_r/N_c$, so the same user ratio reserves three added cells per refined coarse cell in 2D and seven in 3D

Related theory. Adaptive MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpm/elements/AdaptiveHexahedronElement.py;src/mpm/elements/AdaptiveQuadrilateralElement.py`

29.6.128 `RefineThreshold`

Scope. MPM

Accepted value. positive real scalar, with units determined by the selected refinement indicator

Meaning. sets the minimum indicator that marks a coarse cell for adaptive refinement

Numerical or workflow effect. lower values refine more cells and can exhaust refined-cell, node-map, or particle-split buffers. The same threshold applies at both supported levels. Each scheduled qualifying check advances the original coarse cell by one level, capped by `MaxLevel`; a second qualifying check promotes level one to level two. There is no separate level-two threshold or coarsening criterion. Buffer dilation and neighbor balancing can also promote cells.

Related theory. Adaptive MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpm/elements/AdaptiveHexahedronElement.py;src/mpm/elements/AdaptiveQuadrilateralElement.py`

29.6.129 `Reflect`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.130 `ReflectionConstraint`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.131 `RegionFunction`

Scope. MPM

Accepted value. string, callable, path, or enumerated choice

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/boundaries/BoundaryConstraint.py;src/utils/RegionFunction.py`

29.6.132 `RegionName`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. string, callable, path, or enumerated choice

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/BodyGenerator.py;src/mpdem/generator/LSBodyGenerate.py;src/mpm/boundaries/BoundaryConstraint.py;src/mpm/generator/BodyGenerator.py`

29.6.133 `RegionVolume`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/utils/RegionFunction.py`

29.6.134 `reset_function`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. string, callable, path, or enumerated choice

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/mainDEM.py;src/mpdem/mainDEMPM.py;src/mpm/mainMPM.py`

29.6.135 `residual_tolerance`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.6.136 `ResidualCohesion`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/physics_model/consititutive_model/infinitesimal_strain/DruckerPrager.py;src/physics_model/consititutive_model/infinitesimal_strain/MohrCoulomb.py`

29.6.137 `ResidualDilation`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/physics_model/consititutive_model/infinitesimal_strain/DruckerPrager.py;src/physics_model/consititutive_model/infinitesimal_strain/MohrCoulomb.py`

29.6.138 `ResidualFriction`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets friction coefficient or tangential resistance parameter

Numerical or workflow effect. limits tangential contact forces, changes plastic strength, or activates IPC friction

Related theory. DEM dynamics and contact, IGA-MPM IPC coupling, Constitutive and contact models.

Source hint. `src/physics_model/consititutive_model/infinitesimal_strain/DruckerPrager.py;src/physics_model/consititutive_model/infinitesimal_strain/MohrCoulomb.py`

29.6.139 `ResidualPlasticDevStrain`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/physics_model/consititutive_model/infinitesimal_strain/DruckerPrager.py;src/physics_model/consititutive_model/infinitesimal_strain/ElasticPerfectlyPlastic.py;src/physics_model/consititutive_model/infinitesimal_strain/MohrCoulomb.py`

29.6.140 `ResidualYieldStress`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `src/physics_model/consititutive_model/infinitesimal_strain/ElasticPerfectlyPlastic.py`

29.6.141 `Resolution`

Scope. MPM, DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/ClumpTemplate.py;src/dem/generator/WallGenerator.py`

29.6.142 `Restart`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/generator/LoadFromFile.py`

29.6.143 `Restitution`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/contact/HertzMindlin.py;src/mpdem/contact/HertzMindlin.py`

29.6.144 `ReturnMapping`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion,

solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.145 `RigidBody`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/generator/BodyGenerator.py;src/mpm/generator/LoadFromFile.py`

29.6.146 `RigidFile`

Scope. MPM, DEM

Accepted value. string, callable, path, or enumerated choice

Meaning. selects input or output files

Numerical or workflow effect. controls restart, geometry loading, recorder writes, and postprocessing data flow

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/LoadFromFile.py`

29.6.147 `RigidTwoPhaseBondedCoupler`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.148 `RollingFriction`

Scope. DEM, MPDEM/CFDEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets friction coefficient or tangential resistance parameter

Numerical or workflow effect. limits tangential contact forces, changes plastic strength, or activates IPC friction

Related theory. DEM dynamics and contact, IGA-MPM IPC coupling, Constitutive and contact models.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/dem/contact/HertzMindlin.py;src/dem/contact/Linear.py;src/dem/contact/LinearRolling.py;src/mpdem/contact/HertzMindlin.py;src/mpdem/contact/Linear.py`

29.6.149 `RollingStiffness`

Scope. DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/dem/contact/LinearRolling.py`

29.6.150 `RollingViscousDamping`

Scope. DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/dem/contact/LinearRolling.py`

29.6.151 `rotate`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/utils/RegionFunction.py`

29.6.152 `rotate2D`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/utils/RegionFunction.py`

29.6.153 `RotateCenter`

Scope. DEM

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/dem/SceneManager.py;src/dem/structs/Geometry.py;src/utils/RegionFunction.py`

29.6.154 `run`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.155 `run()`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.156 `run(**kwargs)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.157 `run(*args,**kwargs)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.6.158 `run(visualize=False,**kwargs)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7 S–U Keywords and Calls

29.7.1 `save_data`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.2 `save_data()`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.3 `SaveInterval`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. integer count, index, level, or interval

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py;src/mpdem/mainDEMPM.py;src/mpm/mainMPM.py`

29.7.4 `SavePath`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. string, callable, path, or enumerated choice

Meaning. selects input or output files

Numerical or workflow effect. controls restart, geometry loading, recorder writes, and postprocessing data flow

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/BodyGenerator.py;src/dem/generator/ClumpTemplate.py;src/dem/generator/GeneralShapeTemplate.py;src/dem/mainDEM.py;...`

29.7.5 `scale`

Scope. DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/PostPlot.py;src/dem/generator/LoadFromFile.py`

29.7.6 `ScaleFactor`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/BodyGenerator.py;src/dem/generator/LoadFromFile.py;src/dem/generator/WallGenerator.py;src/mpdem/generator/LSBodyGenerate.py;...`

29.7.7 `scheme`

Scope. MPM, DEM

Accepted value. string, callable, path, or enumerated choice

Meaning. selects the DEM body/contact scheme; DEM accepts `DEM`, `LSDEM`, `LSMPM`, `PolySuperEllipsoid`, `PolySuperQuadrics`, and `AffineBody`.

Numerical or workflow effect. controls body storage, template format, contact geometry kernels, engine construction, solver construction, and recorder output.

Related theory. DEM dynamics and contact, Affine body dynamics.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py`

29.7.8 `SchillerNaumannModel`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.9 `search`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. string, callable, path, or enumerated choice

Meaning. controls spatial search or broad-phase neighbor construction

Numerical or workflow effect. trades neighbor build cost against candidate count and contact-list stability

Related theory. DEM BVH/LBVH search, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py;src/mpdem/mainDEMPM.py`

29.7.10 `search="LinkedCell"`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. controls spatial search or broad-phase neighbor construction

Numerical or workflow effect. trades neighbor build cost against candidate count and contact-list stability

Related theory. DEM BVH/LBVH search, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.11 `search_direction`

Scope. MPM, DEM

Accepted value. string, callable, path, or enumerated choice

Meaning. controls spatial search or broad-phase neighbor construction

Numerical or workflow effect. trades neighbor build cost against candidate count and contact-list stability

Related theory. DEM BVH/LBVH search, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py`

29.7.12 `select_save_data(...)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.13 `select_save_data(particle=True,grid=False,object=True)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. sets geometry, interpolation, grid spacing, or boundary classification

Numerical or workflow effect. changes support size, active node count, pressure-cell classification, and transfer accuracy

Related theory. MPM weak form and transfers.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.14 `select_save_data(ppcontact=False,pwcontact=False)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.15 `SemiImplicit`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.16 `servo_switch(status)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.17 `ServoFile`

Scope. MPM, DEM

Accepted value. string, callable, path, or enumerated choice

Meaning. selects input or output files

Numerical or workflow effect. controls restart, geometry loading, recorder writes, and postprocessing data flow

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/WallGenerator.py`

29.7.18 `set_configuration`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.19 `set_configuration(**kwargs)`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.20 `set_configuration(dimension,solver_type,configuration,ipc,static_twophase,**kwargs)`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. chooses a solver, assembly format, or preconditioner parameter

Numerical or workflow effect. affects compilation path, memory layout, convergence, and boundary-condition application

Related theory. Implicit solid MPM, MGPCG multilevel solver.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.21 `set_configuration(dimension=3,solver_type="Implicit",**kwargs)`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. chooses a solver, assembly format, or preconditioner parameter

Numerical or workflow effect. affects compilation path, memory layout, convergence, and boundary-condition application

Related theory. Implicit solid MPM, MGPCG multilevel solver.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.22 `set_fbar_parameters`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.23 `set_fbar_parameters(**kwargs)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.24 `set_implicit_solver_parameters`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. chooses a solver, assembly format, or preconditioner parameter

Numerical or workflow effect. affects compilation path, memory layout, convergence, and boundary-condition application

Related theory. Implicit solid MPM, MGPCG multilevel solver.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.25 `set_implicit_solver_parameters(**kwargs)`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. chooses a solver, assembly format, or preconditioner parameter

Numerical or workflow effect. affects compilation path, memory layout, convergence, and boundary-condition application

Related theory. Implicit solid MPM, MGPCG multilevel solver.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.26 `set_semi_implicit_solver_parameters`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. chooses a solver, assembly format, or preconditioner parameter

Numerical or workflow effect. affects compilation path, memory layout, convergence, and boundary-condition application

Related theory. Implicit solid MPM, MGPCG multilevel solver.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.27 `set_semi_implicit_solver_parameters(dict)`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. chooses a solver, assembly format, or preconditioner parameter

Numerical or workflow effect. affects compilation path, memory layout, convergence, and boundary-condition application

Related theory. Implicit solid MPM, MGPCG multilevel solver.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.28 `set_solver`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. chooses a solver, assembly format, or preconditioner parameter

Numerical or workflow effect. affects compilation path, memory layout, convergence, and boundary-condition application

Related theory. Implicit solid MPM, MGPCG multilevel solver.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.29 `set_solver(**kwargs)`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. chooses a solver, assembly format, or preconditioner parameter

Numerical or workflow effect. affects compilation path, memory layout, convergence, and boundary-condition application

Related theory. Implicit solid MPM, MGPCG multilevel solver.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.30 `set_solver(dict)`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. chooses a solver, assembly format, or preconditioner parameter

Numerical or workflow effect. affects compilation path, memory layout, convergence, and boundary-condition application

Related theory. Implicit solid MPM, MGPCG multilevel solver.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.31 `set_solver(solver=None,**kwargs)`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. chooses a solver, assembly format, or preconditioner parameter

Numerical or workflow effect. affects compilation path, memory layout, convergence, and boundary-condition application

Related theory. Implicit solid MPM, MGPCG multilevel solver.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.32 `set_static_wall(static_wall=True)`

Scope. DEM

Accepted value. boolean, default true when called without arguments

Meaning. marks walls as static after insertion.

Numerical or workflow effect. static facet/patch walls do not need to be reinserted into linked-cell bins or rebuilt in the wall BVH batch after the initial search structure is built. Particle batches are still rebuilt or refit as needed.

Related theory. DEM BVH/LBVH search, Contact Search.

Source hint. `src/dem/mainDEM.py;src/dem/neighbor/BoundingVolumeHierarchy.py`

29.7.33 `set_THB`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.7.34 `set_window(window)`

Scope. shared

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.35 `Shape`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. sets geometry, interpolation, grid spacing, or boundary classification

Numerical or workflow effect. changes support size, active node count, pressure-cell classification, and transfer accuracy

Related theory. MPM weak form and transfers.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.36 `shape_function`

Scope. MPM, DEM LSMPM

Accepted value. For MPM, values accepted by the MPM element/shape-function branch. For `DEM.set_configuration(scheme="LSMPM")`: `Linear`, `QuadBSpline`, or `CubicBSpline`; aliases such as `quadraticbspline` are normalized.

Meaning. selects the interpolation basis used by particle-grid transfer. In LSMPM soft bodies it controls the precomputed TLMPM shape support for soft material points and soft surface points.

Numerical or workflow effect. `Linear`, `QuadBSpline`, and `CubicBSpline` use 2^3, 3^3, and 4^3 support nodes respectively in the LSMPM soft grid. Wider support smooths stress/contact transfer but increases memory and per-step work.

Related theory. MPM weak form and transfers, Level-set MPM soft bodies.

Source hint. `src/mpm/mainMPM.py;src/dem/Simulation.py;src/dem/generator/InsertionKernel.py`

29.7.37 `shape_smooth`

Scope. MPM

Accepted value. string, callable, path, or enumerated choice

Meaning. sets geometry, interpolation, grid spacing, or boundary classification

Numerical or workflow effect. changes support size, active node count, pressure-cell classification, and transfer accuracy

Related theory. MPM weak form and transfers.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.7.38 `ShapeFactor`

Scope. MPM, DEM

Accepted value. string, callable, path, or enumerated choice

Meaning. sets geometry, interpolation, grid spacing, or boundary classification

Numerical or workflow effect. changes support size, active node count, pressure-cell classification, and transfer accuracy

Related theory. MPM weak form and transfers.

Source hint. `manual/reference;src/dem/contact/JiangRolling.py`

29.7.39 `ShapeFunction`

Scope. MPM

Accepted value. string, callable, path, or enumerated choice

Meaning. sets geometry, interpolation, grid spacing, or boundary classification

Numerical or workflow effect. changes support size, active node count, pressure-cell classification, and transfer accuracy

Related theory. MPM weak form and transfers.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference`

29.7.40 `ShearModulus`

Scope. DEM, MPDEM/CFDEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets elastic constants used by stress increments and tangent stiffness

Numerical or workflow effect. controls wave speed, timestep restrictions, incompressibility, and implicit matrix conditioning

Related theory. Constitutive and contact models, Implicit solid MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/dem/contact/HertzMindlin.py;src/mpdem/contact/HertzMindlin.py`

29.7.41 `shell_offset`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/generator/WallGenerator.py`

29.7.42 `ShellOffset`

Scope. MPM, DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/WallGenerator.py`

29.7.43 `Sign`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/boundaries/BoundaryConstraint.py`

29.7.44 `Simulation`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.45 `SimulationTime`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py;src/mpdem/mainDEMPM.py;src/mpm/mainMPM.py`

29.7.46 `smooth_setting`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/mpm/PostPlot.py`

29.7.47 `SmoothLinear`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.48 `SoftType`

Scope. Shared incremental materials, including `MPM.add_material.material.SoftType`.

Accepted values. `Linear`, `Exponential`, or absent for no softening.

Linear inputs. Peak/residual strengths and `material.PlasticDevStrain`, `material.ResidualPlasticDevStrain` fully determine the bounded line. There is no additional scale parameter; the former `SoftenParameter` key is no longer read.

Migration. For a previously clipped multiplier $\eta \geq 1$, use $\varepsilon_{res,new} = \varepsilon_{peak} + (\varepsilon_{res,old} - \varepsilon_{peak})/\eta$. For example, endpoints 0 and 0.02 preserve a previous clipped factor-5 curve over 0 and 0.1. The paper-based sensitive-clay control uses zero-angle `DruckerPrager`, `MPM.add_material.material.dpType=Inscribed`, and cohesion slope -50 kPa: its accumulated equivalent plastic history reaches residual at `material.ResidualPlasticDevStrain=0.32`. It uses a uniform grid and `MPM.set_configuration.stress_integration=SubStepping`, which refreshes strength at the current local plastic history. The classical closed-form DP path still freezes strength during its step. The separate native soft-MC invariant would require endpoint $0.32/\sqrt{2}$; that conversion does not apply to DP. The derivative is zero on the plateaus.

The sensitive-clay script selects this corrected MC control with `--materialmc`; it keeps the DP control's 0.2 m uniform grid and boundaries.

Exponential inputs. `material.alpha` and `material.beta` remain unchanged.

Local integration. Small-strain DP, MC and softening `ElasticPerfectlyPlastic` refresh strength from the current local plastic history; generic returns also refresh before evaluating yield residuals. MC/Mises tensor-history normalization and recorded state schemas are unchanged.

Random-field Mises. Its initializer reads `material.PlasticDevStrain` and `material.ResidualPlasticDevStrain` from the same dictionary, with the particle yield-stress peaks loaded from file.

Related theory. Bounded softening and plastic modulus.

Source. `src/physics_model/consititutive_model/SoftenModel.py` and `infinitesimal_strain/ElasPlasticity.py`.

29.7.49 `Solid`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.50 `SoftBody`

Scope. DEM LSMPM

Accepted value. `BodyType="SoftBody"` in `DEM.create_body` or `DEM.add_body` when `scheme="LSMPM"`.

Meaning. inserts or generates deformable level-set MPM particles rather than rigid LS-DEM bodies.

Numerical or workflow effect. creates soft material points, compact soft MPM grids, precomputed TLMPM shape functions, a deforming local SDF, soft surface point ids for contact, and `LSMPMSoftPoint*.npz` output. The same level-set template can be used for rigid and soft bodies.

Related theory. Level-set MPM soft bodies.

Source hint. `src/dem/generator/BodyGenerator.py;src/dem/SceneManager.py;src/dem/Recorder.py`

29.7.51 `solid_cut_cell_min_fraction`

Scope. MPM

Accepted value. positive real fraction, typically small

Meaning. lower bound on the open face/cell fraction used by solid SDF cut-cell pressure coefficients

Numerical or workflow effect. prevents very small cut fractions from producing ill-conditioned pressure stencils; increasing it makes the boundary more conservative but less geometrically sharp

Related theory. SDF and level set geometry, Incompressible fluid MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.7.52 `solid_sdf_cut_cell`

Scope. MPM

Accepted value. boolean

Meaning. enables signed-distance solid cut-cell handling in implicit fluid pressure solves

Numerical or workflow effect. modifies pressure face coefficients and solid boundary classification from the supplied SDF/level-set geometry

Related theory. SDF and level set geometry, Incompressible fluid MPM, MGPCG pressure solver.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.7.53 `solid_sdf_cut_cell=True`

Scope. shared

Accepted value. boolean literal used in examples

Meaning. example shorthand for enabling `solid_sdf_cut_cell`

Numerical or workflow effect. same as `solid_sdf_cut_cell`; it activates SDF-based solid cut-cell pressure coefficients

Related theory. SDF and level set geometry, Incompressible fluid MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.54 `SolidCell`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. controls spatial search or broad-phase neighbor construction

Numerical or workflow effect. trades neighbor build cost against candidate count and contact-list stability

Related theory. DEM BVH/LBVH search, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.55 `SolidCellThickness`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. controls spatial search or broad-phase neighbor construction

Numerical or workflow effect. trades neighbor build cost against candidate count and contact-list stability

Related theory. DEM BVH/LBVH search, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/boundaries/BoundaryConstraint.py`

29.7.56 `SolidDensity`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets mass density or phase density

Numerical or workflow effect. scales particle mass, body force, wave speed, pressure response, drag, and virtual-mass forces

Related theory. Constitutive and contact models.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/physics_model/consititutive_model/MaterialModel.py`

29.7.57 `solver_type`

Scope. MPM

Accepted value. string, callable, path, or enumerated choice

Meaning. chooses a solver, assembly format, or preconditioner parameter

Numerical or workflow effect. affects compilation path, memory layout, convergence, and boundary-condition application

Related theory. Implicit solid MPM, MGPCG multilevel solver.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpm/mainMPM.py`

29.7.58 `solver_type="Explicit"`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. chooses a solver, assembly format, or preconditioner parameter

Numerical or workflow effect. affects compilation path, memory layout, convergence, and boundary-condition application

Related theory. Implicit solid MPM, MGPCG multilevel solver.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.59 `solver_type="Implicit"`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. chooses a solver, assembly format, or preconditioner parameter

Numerical or workflow effect. affects compilation path, memory layout, convergence, and boundary-condition application

Related theory. Implicit solid MPM, MGPCG multilevel solver.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.60 `solver_type="SemiImplicit"`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. chooses a solver, assembly format, or preconditioner parameter

Numerical or workflow effect. affects compilation path, memory layout, convergence, and boundary-condition application

Related theory. Implicit solid MPM, MGPCG multilevel solver.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.61 `space`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/sdf/LevelSetGrid.py`

29.7.62 `sparse_grid`

Scope. MPM, MPDEM/DEMPM through the MPM subsolver

Accepted value. boolean or dict, for example `\{"Enabled":True,"Backend":"BlockScan","BlockSize":4,"CapacityFactor":1.25,"MaxActiveBlocks":0,"VisualizeActiveBlocks":False\}`

Meaning. enables the BlockScan compact-node background grid for supported MPM branches.

Numerical or workflow effect. active particle support nodes mark active blocks, the block prefix scan builds a compact node map, and particle `LnID` entries are remapped from dense physical node ids to compact sparse node ids before P2G/G2P kernels run. Supported public branches are `solver_type="Explicit"` and implicit solid MPM on Taichi CPU or CUDA. Explicit sparse-grid coverage includes ordinary 2D/3D solid ULMPM/TLMPM, ordinary 3D fluid ULMPM, and the separate explicit `TwoPhaseSingleLayer` Cartesian path. Implicit solid sparse-grid coverage uses the same compact node stride in residual, mass, tangent, Dirichlet, and reaction-force kernels and supports `assemble_type="MatrixFree"`, `"COO"`, and `"HashTriplet"`. Contact-based MPDEM/DEMPM may use the sparse MPM background grid with Lagrangian coupling because DEM and coupled contact search still use DEM/coupled neighbor buffers. The sparse-grid path rejects implicit FDM fluid MPM, double-layer semi-implicit two-phase MPM, MPM internal `contact_detection`, CFDEM, PBC, and absorbing boundaries; those branches use dense, FDM/MAC, contact, or fluid-

coupling layouts rather than BlockScan compact node numbering.

Related theory. MPM weak form and transfers, compatibility matrix.

Source hint. `src/mpm/Simulation.py;src/mpdem/Simulation.py;src/mpm/SceneManager.py;src/mpm/sparse_grid/BlockSparseGrid.py;src/mpm/Recorder.py`

29.7.63 `sparse_grid.VisualizeActiveBlocks`

Scope. MPM `sparse_grid` dictionary

Accepted value. boolean; default `False`

Meaning. when grid output and VTK visualization are enabled, writes one clipped quad or hexahedron per active BlockScan storage block instead of serializing the full dense background grid.

Numerical or workflow effect. this is an output-only switch. It adds `active_block_id`, `compact_block_id`, and `block_index` cell fields to `GraphicMPMGrid*.vtu`; it does not change the block scan, compact node map, or solver state.

Related theory. MPM weak form and transfers.

Source hint. `src/mpm/Simulation.py;src/mpm/Recorder.py`

29.7.64 `SphereFile`

Scope. DEM

Accepted value. string, callable, path, or enumerated choice

Meaning. selects input or output files

Numerical or workflow effect. controls restart, geometry loading, recorder writes, and postprocessing data flow

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/dem/generator/LoadFromFile.py`

29.7.65 `sphereIndex`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/PostPlot.py;src/dem/generator/LoadFromFile.py`

29.7.66 `Spheroid`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.67 `src`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.68 `src.igampm.mainIGAMPM.IGAMPM(...)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. current IGA-MPM coupling facade implementation.

Numerical or workflow effect. builds or reuses implicit IGA/MPM engines and assembles IPC contact coupling.

Related theory. IGA-MPM.

Source hint. `src/igampm/mainIGAMPM.py`

29.7.69 `src/dem/mainDEM.py`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.70 `src/iga/mainIGA.py`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.71 `src/igampm/mainIGAMPM.py`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. source file for the public `IGAMPM` facade.

Numerical or workflow effect. coordinates sub-solver build, contact-model synchronization, and run dispatch.

Related theory. IGA-MPM.

Source hint. `src/igampm/mainIGAMPM.py`

29.7.72 `src/mpdem/mainDEMPM.py`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.73 `src/igampm/engines`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. IGA-MPM coupling engine with closest-point contact, barrier/friction assembly, and monolithic system helpers.

Numerical or workflow effect. evaluates contact geometry and sparse contact blocks for IGA-MPM coupling.

Related theory. IGA-MPM IPC contact model.

Source hint. `src/igampm/engines`

29.7.74 `src/igampm/ContactManager.py`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. IGA-MPM IPC contact configuration manager.

Numerical or workflow effect. creates IPC barrier/friction objects and freezes their settings at build time.

Related theory. Contact Model.

Source hint. `src/igampm/ContactManager.py`

29.7.75 `src/mpm/mainMPM.py`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.76 `STABBICG`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `manual/reference`

29.7.77 `stabilize`

Scope. MPM

Accepted value. boolean switch or small enumerated mode

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpm/mainMPM.py`

29.7.78 `Stable`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.79 `Staggered`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.80 `StaggeredRigidTwoPhaseDriver`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.81 `startGrid`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. sets geometry, interpolation, grid spacing, or boundary classification

Numerical or workflow effect. changes support size, active node count, pressure-cell classification, and transfer accuracy

Related theory. MPM weak form and transfers.

Source hint. `src/dem/generator/LoadFromFile.py`

29.7.82 `startIndex`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/PostPlot.py;src/dem/generator/LoadFromFile.py;src/dem/generator/WallGenerator.py`

29.7.83 `startNode`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/PostPlot.py;src/dem/generator/LoadFromFile.py`

29.7.84 `StartPoint`

Scope. MPM

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/boundaries/BoundaryConstraint.py`

29.7.85 `StartTime`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/utils/functions/SinFunction.py`

29.7.86 `state_vars`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/mpm/PostPlot.py;src/mpm/generator/LoadFromFile.py`

29.7.87 `StateDependentMohrCoulomb`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.88 `static_twophase`

Scope. MPM

Accepted value. boolean switch or small enumerated mode

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `manual/reference`

29.7.89 `StaticFriction`

Scope. DEM, MPDEM/CFDEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets friction coefficient or tangential resistance parameter

Numerical or workflow effect. limits tangential contact forces, changes plastic strength, or activates IPC friction

Related theory. DEM dynamics and contact, IGA-MPM IPC coupling, Constitutive and contact models.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/dem/contact/HertzMindlin.py;src/dem/contact/Linear.py;src/mpdem/contact/HertzMindlin.py;src/mpdem/contact/Linear.py;src/physics_model/consititutive_model/infinitesimal_strain/RateDependent.py`

29.7.90 `StaticTwoPhaseRigidAdapter`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.91 `StaticTwoPhaseULMPM`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.92 `Status`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/engines/ElementTest.py`

29.7.93 `stiffness`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/mpm/Contact.py`

29.7.94 `Stiffness`

Scope. DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/contact/EnergyConservation.py`

29.7.95 `StiffnessRatio`

Scope. DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/contact/EnergyConservation.py;src/dem/contact/JiangRolling.py`

29.7.96 `strain`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/mpm/PostPlot.py`

29.7.97 `strain_smoothing`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. chooses a solver, assembly format, or preconditioner parameter

Numerical or workflow effect. affects compilation path, memory layout, convergence, and boundary-condition application

Related theory. Implicit solid MPM, MGPCG multilevel solver.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpm/mainMPM.py`

29.7.98 `stress`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `src/mpm/PostPlot.py;src/mpm/generator/LoadFromFile.py`

29.7.99 `stress_integration`

Scope. MPM

Accepted value. `"ReturnMapping"`, `"ImplicitIntegration"`, `"SubStepping"`, or a model-specific value such as `"ImplicitIntegrationAL"` when implemented by that constitutive class

Meaning. selects the local stress-update algorithm for small-strain plastic material models

Numerical or workflow effect. `ReturnMapping` and `ImplicitIntegration` call `ElasPlasticity.PlasticMaterial.ImplicitIntegration`; `SubStepping` calls `ElasPlasticity.PlasticMaterial.ExplicitIntegration`, whose active path uses adaptive `NBURKDP2` and whose source also contains `RKDP2`/`RKDP6` Runge-Kutta helpers; `ImplicitIntegrationAL` is model-specific and currently implemented for `ModifiedCamClay`

Related theory. Return mapping and explicit/substepping stress integration, MCC return mapping.

Source hint. `src/physics_model/consititutive_model/infinitesimal_strain/ElasPlasticity.py;src/physics_model/consititutive_model/infinitesimal_strain/ModifiedCamClay.py;src/mpm/mainMPM.py`

29.7.100 `StressRatio`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `manual/reference;src/physics_model/consititutive_model/infinitesimal_strain/ModifiedCamClay.py`

29.7.101 `SubStepping`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.102 `surface_node_number`

Scope. DEM

Accepted value. integer count, index, level, or interval

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/dem/mainDEM.py`

29.7.103 `surface_num`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/generator/LoadFromFile.py`

29.7.104 `surface_tension`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/physics_model/consititutive_model/strain_rate/Newtonian.py`

29.7.105 `SurfaceFile`

Scope. MPM, DEM

Accepted value. string, callable, path, or enumerated choice

Meaning. selects input or output files

Numerical or workflow effect. controls restart, geometry loading, recorder writes, and postprocessing data flow

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/LoadFromFile.py`

29.7.106 `SurfaceNodeNumber`

Scope. MPM, DEM

Accepted value. integer count, index, level, or interval

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/GeneralShapeTemplate.py`

29.7.107 `SurfaceResolution`

Scope. MPM, DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/GeneralShapeTemplate.py`

29.7.108 `SurfaceTension`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `manual/reference;src/physics_model/consititutive_model/strain_rate/Newtonian.py`

29.7.109 `SymplecticEuler`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.110 `sync_settings()`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.111 `t_current`

Scope. MPM, DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/generator/LoadFromFile.py;src/dem/generator/WallGenerator.py;src/mpm/generator/LoadFromFile.py`

29.7.112 `tangential_force`

Scope. DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `src/dem/contact/ContactModelBase.py;src/dem/contact/RollingModelBase.py`

29.7.113 `TangentialStiffness`

Scope. DEM, MPDEM/CFDEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/dem/contact/EnergyConservation.py;src/dem/contact/Linear.py;src/dem/contact/LinearRolling.py;src/mpdem/contact/EnergyConservation.py;src/mpdem/contact/Linear.py`

29.7.114 `TangentialViscousDamping`

Scope. DEM, MPDEM/CFDEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/dem/contact/EnergyConservation.py;src/dem/contact/JiangRolling.py;src/dem/contact/Linear.py;src/dem/contact/LinearRolling.py;src/mpdem/contact/EnergyConservation.py;...`

29.7.115 `target_stress`

Scope. DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `src/dem/generator/WallGenerator.py`

29.7.116 `TargetStress`

Scope. MPM, DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/WallGenerator.py`

29.7.117 `Taylor`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.118 `Template`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/BodyGenerator.py;src/dem/generator/LoadFromFile.py;src/mpdem/generator/LSBodyGenerate.py;src/mpm/generator/BodyGenerator.py;...`

29.7.119 `templateID`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/generator/LoadFromFile.py`

29.7.120 `Tensile`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `manual/reference;src/physics_model/consititutive_model/finite_strain/Hydrogel.py;src/physics_model/consititutive_model/infinitesimal_strain/DruckerPrager.py;src/physics_model/consititutive_model/infinitesimal_strain/MohrCoulomb.py;src/physics_model/consititutive_model/infinitesimal_strain/StateDependentMohrCoulomb.py`

29.7.121 `Tensile1`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `manual/reference;src/physics_model/consititutive_model/finite_strain/Gent.py`

29.7.122 `Tensile2`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `manual/reference;src/physics_model/consititutive_model/finite_strain/Gent.py`

29.7.123 `three_invariants`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.124 `ThreeInvariants`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/physics_model/consititutive_model/infinitesimal_strain/ModifiedCamClay.py`

29.7.125 `TimeStamp`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/utils/functions/LinearFunction.py`

29.7.126 `Timestep`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py;src/mpdem/mainDEMPM.py;src/mpm/mainMPM.py`

29.7.127 `TLMPM`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference`

29.7.128 `TorqueLocalDamping`

Scope. DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/structs/BaseStruct.py`

29.7.129 `total_displacement`

Scope. MPM, DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/PostPlot.py;src/mpm/PostPlot.py`

29.7.130 `total_grid_num`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. sets geometry, interpolation, grid spacing, or boundary classification

Numerical or workflow effect. changes support size, active node count, pressure-cell classification, and transfer accuracy

Related theory. MPM weak form and transfers.

Source hint. `src/dem/generator/LoadFromFile.py`

29.7.131 `total_surface_num`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/PostPlot.py;src/dem/generator/LoadFromFile.py`

29.7.132 `TotalParticles`

Scope. MPM, DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/BodyGenerator.py`

29.7.133 `track_energy`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. boolean switch or small enumerated mode

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py;src/mpdem/mainDEMPM.py;src/mpm/mainMPM.py`

29.7.134 `Traction`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/generator/BodyGenerator.py;src/mpm/generator/LoadFromFile.py`

29.7.135 `TractionConstraint`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.136 `Translation`

Scope. MPM, DEM

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/LoadFromFile.py;src/dem/generator/WallGenerator.py`

29.7.137 `TrianglarPrism`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.138 `Triangle2D`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.139 `TryNumber`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. integer count, index, level, or interval

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/BodyGenerator.py;src/dem/generator/ClumpTemplate.py;src/mpdem/generator/LSBodyGenerate.py`

29.7.140 `TwistingFriction`

Scope. DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets friction coefficient or tangential resistance parameter

Numerical or workflow effect. limits tangential contact forces, changes plastic strength, or activates IPC friction

Related theory. DEM dynamics and contact, IGA-MPM IPC coupling, Constitutive and

contact models.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/dem/contact/LinearRolling.py`

29.7.141 `TwistingStiffness`

Scope. DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/dem/contact/LinearRolling.py`

29.7.142 `TwistingViscousDamping`

Scope. DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/dem/contact/LinearRolling.py`

29.7.143 `TwoPhaseBondedNearestParticleSupport`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.144 `TwoPhaseDoubleLayer`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.145 `TwoPhaseParticleCloud`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.146 `TwoPhaseSingleLayer`

Scope. MPM material type value

Accepted value. literal value paired with `material_type`

Meaning. one material point carries both solid skeleton and pore-fluid variables.

Numerical or workflow effect. in `Explicit`, this selects the Cartesian ULMPM single-layer mixture path in two or three dimensions with PIC/FLIP transfer.

Related theory. MPM weak form and transfers.

Source hint. `src/mpm/mainMPM.py;src/mpm/Simulation.py;src/mpm/SceneManager.py`

29.7.147 `Type`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/utils/RegionFunction.py`

29.7.148 `ULMPM`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference`

29.7.149 `Up`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.150 `update_contact_properties`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.151 `update_contact_properties(...)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.152 `update_material_properties`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.153 `update_particle_properties`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.154 `update_particle_properties(...)`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.155 `update_wall_status`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.156 `UserDefined`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.157 `UserDefined`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.7.158 `USF`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference`

29.7.159 `USL`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference`

29.8 V–Z Keywords and Calls

29.8.1 `Value`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/utils/functions/LinearFunction.py`

29.8.2 `velocity`

Scope. MPM, DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `src/dem/PostPlot.py;src/dem/generator/LoadFromFile.py;src/dem/generator/WallGenerator.py;src/mpm/PostPlot.py;src/mpm/generator/LoadFromFile.py`

29.8.3 `Velocity`

Scope. MPM, DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/dem/structs/Geometry.py;src/mpm/boundaries/BoundaryConstraint.py`

29.8.4 `velocity_gradient`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `src/mpm/generator/LoadFromFile.py`

29.8.5 `velocity_projection`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/mainMPM.py`

29.8.6 `velocity_projection="Affine"`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.8.7 `VelocityConstraint`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.8.8 `VelocityProjection`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `manual/reference`

29.8.9 `VelocityVerlet`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. controls spatial search or broad-phase neighbor construction

Numerical or workflow effect. trades neighbor build cost against candidate count and

contact-list stability

Related theory. DEM BVH/LBVH search, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.8.10 `VelocityX`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/boundaries/BoundaryConstraint.py`

29.8.11 `VelocityY`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/boundaries/BoundaryConstraint.py`

29.8.12 `VelocityZ`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/boundaries/BoundaryConstraint.py`

29.8.13 `verlet_distance_multiplier`

Scope. MPM, DEM

Accepted value. value type defined by the owning call site

Meaning. controls spatial search or broad-phase neighbor construction

Numerical or workflow effect. trades neighbor build cost against candidate count and contact-list stability

Related theory. DEM BVH/LBVH search, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py;src/mpm/mainMPM.py`

29.8.14 `vertice_index`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/generator/LoadFromFile.py`

29.8.15 `vertices`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/PostPlot.py;src/dem/generator/LoadFromFile.py`

29.8.16 `Vertices`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/mpm/generator/LoadFromFile.py;src/utils/RegionFunction.py`

29.8.17 `ViewAngle`

Scope. MPM, DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/Simulation.py;src/mpm/Simulation.py`

29.8.18 `Virtual`

Scope. shared

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.8.19 `VirtualForce`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. defines the virtual body-force term used with virtual stress loading

Numerical or workflow effect. enters the equivalent volume load int N b_v dV

Related theory. Virtual stress boundary.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpm/boundaries/BoundaryConstraint.py`

29.8.20 `VirtualStress`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. defines the virtual stress tensor used by the virtual stress boundary method

Numerical or workflow effect. enters the equivalent volume load -int B^T sigma_v dV

Related theory. Virtual stress boundary.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpm/boundaries/BoundaryConstraint.py`

29.8.21 `Viscosity`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets fluid, pore-pressure, porous-media, or viscous parameter

Numerical or workflow effect. enters pressure projection, Darcy or Ergun drag, pressure wave speed, and mixture mass balance

Related theory. Incompressible fluid MPM, Semi-implicit two-phase MPM.

Source hint. `manual/reference;src/physics_model/consititutive_model/MaterialModel.py;src/physics_model/consititutive_model/strain_rate/Bingham.py;src/physics_model/consititutive_model/strain_rate/Newtonian.py`

29.8.22 `visualize`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. boolean switch or small enumerated mode

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py;src/mpdem/mainDEMPM.py;src/mpm/mainMPM.py`

29.8.23 `Visualize`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. boolean switch or small enumerated mode

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/BodyGenerator.py;src/dem/generator/WallGenerator.py;src/mpdem/generator/LSBodyGenerate.py;src/mpm/generator/BodyGenerator.py`

29.8.24 `visualize_interval`

Scope. MPM

Accepted value. integer count, index, level, or interval

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/mpm/mainMPM.py`

29.8.25 `VisualizeInterval`

Scope. MPM, DEM

Accepted value. integer count, index, level, or interval

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/Simulation.py;src/mpm/Simulation.py`

29.8.26 `VisualizeMode`

Scope. MPM, DEM

Accepted value. string, callable, path, or enumerated choice

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/GeneralShapeTemplate.py`

29.8.27 `void_ratio_ref`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/physics_model/consititutive_model/infinitesimal_strain/ModifiedCamClay.py`

29.8.28 `volume`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/mpm/PostPlot.py;src/mpm/generator/LoadFromFile.py`

29.8.29 `Volume`

Scope. MPM, DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/ClumpTemplate.py`

29.8.30 `wall_coordination_number`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. integer count, index, level, or interval

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py;src/mpdem/mainDEMPM.py`

29.8.31 `wall_interaction`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/mpdem/mainDEMPM.py`

29.8.32 `wall_per_cell`

Scope. MPM, DEM

Accepted value. value type defined by the owning call site

Meaning. controls spatial search or broad-phase neighbor construction

Numerical or workflow effect. trades neighbor build cost against candidate count and contact-list stability

Related theory. DEM BVH/LBVH search, DEM dynamics and contact.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/mainDEM.py`

29.8.33 `WallBase`

Scope. MPM, DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/WallGenerator.py`

29.8.34 `WallCenter`

Scope. MPM, DEM

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/WallGenerator.py`

29.8.35 `WallFile`

Scope. MPM, DEM

Accepted value. string, callable, path, or enumerated choice

Meaning. selects input or output files

Numerical or workflow effect. controls restart, geometry loading, recorder writes, and postprocessing data flow

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/WallGenerator.py`

29.8.36 `WallHeight`

Scope. MPM, DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/WallGenerator.py`

29.8.37 `WallID`

Scope. MPM, DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/WallGenerator.py`

29.8.38 `wallID`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/generator/WallGenerator.py`

29.8.39 `WallRadius`

Scope. MPM, DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/WallGenerator.py`

29.8.40 `WallShape`

Scope. MPM, DEM

Accepted value. string, callable, path, or enumerated choice

Meaning. sets geometry, interpolation, grid spacing, or boundary classification

Numerical or workflow effect. changes support size, active node count, pressure-cell classification, and transfer accuracy

Related theory. MPM weak form and transfers.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/WallGenerator.py`

29.8.41 `WallType`

Scope. MPM, DEM

Accepted value. string, callable, path, or enumerated choice

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/WallGenerator.py`

29.8.42 `WallVertice`

Scope. MPM, DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/WallGenerator.py`

29.8.43 `WindowSize`

Scope. MPM, DEM

Accepted value. coordinate vector, bounds, or geometric tuple

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/Simulation.py;src/mpm/Simulation.py;src/mpm/mainMPM.py`

29.8.44 `write_angular_velocity`

Scope. DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `src/dem/PostPlot.py`

29.8.45 `write_background_grid`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. sets geometry, interpolation, grid spacing, or boundary classification

Numerical or workflow effect. changes support size, active node count, pressure-cell classification, and transfer accuracy

Related theory. MPM weak form and transfers.

Source hint. `src/mpm/PostPlot.py`

29.8.46 `write_bodyID`

Scope. MPM, DEM

Accepted value. value type defined by the owning call site

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `src/dem/PostPlot.py;src/mpm/PostPlot.py`

29.8.47 `write_boundary_condition(output_path)`

Scope. shared

Accepted value. string, callable, path, or enumerated choice

Meaning. selects input or output files

Numerical or workflow effect. controls restart, geometry loading, recorder writes, and postprocessing data flow

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.8.48 `write_bounding_box`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/PostPlot.py`

29.8.49 `write_bounding_sphere`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/PostPlot.py`

29.8.50 `write_displacement`

Scope. MPM, DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/PostPlot.py;src/mpm/PostPlot.py`

29.8.51 `write_external_force`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `src/mpm/PostPlot.py`

29.8.52 `write_force_chain`

Scope. DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `src/dem/PostPlot.py`

29.8.53 `write_free_surface`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/mpm/PostPlot.py`

29.8.54 `write_groupID`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `src/dem/PostPlot.py`

29.8.55 `write_materialID`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. identifies body, region, material, or group affected by an operation

Numerical or workflow effect. wrong identifiers give empty selections, wrong material assignment, or missing contact properties

Related theory. Theory overview.

Source hint. `src/mpm/PostPlot.py`

29.8.56 `write_mean_stress`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `src/mpm/PostPlot.py`

29.8.57 `write_outer_norm`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/mpm/PostPlot.py`

29.8.58 `write_pressure`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets fluid, pore-pressure, porous-media, or viscous parameter

Numerical or workflow effect. enters pressure projection, Darcy or Ergun drag, pressure wave speed, and mixture mass balance

Related theory. Incompressible fluid MPM, Semi-implicit two-phase MPM.

Source hint. `src/mpm/PostPlot.py`

29.8.59 `write_radial_stress`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `src/mpm/PostPlot.py`

29.8.60 `write_radii`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/PostPlot.py`

29.8.61 `write_state_variables`

Scope. MPM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/mpm/PostPlot.py`

29.8.62 `write_strain_component`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/mpm/PostPlot.py`

29.8.63 `write_stress_component`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `src/mpm/PostPlot.py`

29.8.64 `write_strong_force_chain`

Scope. DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `src/dem/PostPlot.py`

29.8.65 `write_velocity`

Scope. MPM, DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `src/dem/PostPlot.py;src/mpm/PostPlot.py`

29.8.66 `write_volume`

Scope. MPM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/mpm/PostPlot.py`

29.8.67 `write_wall`

Scope. DEM

Accepted value. value type defined by the owning call site

Meaning. provides a user option consumed by the listed call site or example workflow

Numerical or workflow effect. effect occurs during configuration, allocation, insertion, solve, output, or postprocessing

Related theory. Theory overview.

Source hint. `src/dem/PostPlot.py`

29.8.68 `WriteFile`

Scope. MPM, DEM, MPDEM/CFDEM

Accepted value. boolean switch or small enumerated mode

Meaning. selects input or output files

Numerical or workflow effect. controls restart, geometry loading, recorder writes, and postprocessing data flow

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/BodyGenerator.py;src/dem/generator/GeneralShapeTemplate.py;src/mpdem/generator/LSBodyGenerate.py;src/mpm/generator/BodyGenerator.py`

29.8.69 `WritePath`

Scope. MPM, DEM

Accepted value. string, callable, path, or enumerated choice

Meaning. selects input or output files

Numerical or workflow effect. controls restart, geometry loading, recorder writes, and postprocessing data flow

Related theory. Theory overview.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/generator/BodyGenerator.py;src/mpm/generator/BodyGenerator.py`

29.8.70 `YieldStress`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets a prescribed mechanical vector, tensor, or initial condition

Numerical or workflow effect. changes momentum, boundary work, stress update, or virtual-stress volume loading

Related theory. MPM weak form and transfers, Virtual stress boundary.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;src/physics_model/consititutive_model/infinitesimal_strain/ElasticPerfectlyPlastic.py;src/physics_model/consititutive_model/strain_rate/Bingham.py`

29.8.71 `young_modulus`

Scope. shared

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets elastic constants used by stress increments and tangent stiffness

Numerical or workflow effect. controls wave speed, timestep restrictions, incompressibility, and implicit matrix conditioning

Related theory. Constitutive and contact models, Implicit solid MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex`

29.8.72 `YoungModulus`

Scope. MPM, DEM

Accepted value. real scalar, vector, or material parameter with physical units from the problem

Meaning. sets elastic constants used by stress increments and tangent stiffness

Numerical or workflow effect. controls wave speed, timestep restrictions, incompressibility, and implicit matrix conditioning

Related theory. Constitutive and contact models, Implicit solid MPM.

Source hint. `docs/helper/geotaichi_user_theory_manual.tex;manual/reference;src/dem/contact/JiangRolling.py;src/physics_model/consititutive_model/finite_strain/Gent.py;src/physics_model/consititutive_model/finite_strain/HenckyElastic.py;src/physics_model/consititutive_model/finite_strain/Hydrogel.py;...`

29.9 Soft Particle and IPC Keywords

29.9.1 `soft_particle`

Scope. MPM configuration and MPDEM/DEM bridge workflows.

Accepted value. Boolean or dict with `Enabled` and optional `Backend`.

Meaning. Enables the MPM soft-particle mode and stores the backend name, currently `LevelSetTLMPM` for level-set soft bodies.

Numerical or workflow effect. Selects compact TLMPM soft-body fields, soft level-set advection/redistance, and soft contact kernels. Pure `MPM.run()` still rejects this mode until MPM owns the full level-set scene; current runnable examples use the DEM/MPDEM bridge.

Related theory. MPM soft-particle backend.

Source hint. `src/mpm/Simulation.py;src/mpm/soft_particle`

29.9.2 `soft_particle_backend`

Scope. MPM configuration.

Accepted value. String, default `LevelSetTLMPM`.

Meaning. Names the backend used after `soft_particle=True`.

Numerical or workflow effect. The current backend stores level-set soft bodies as compact TLMPM grids and material points; unsupported backend names are only stored and will fail when a corresponding engine is requested.

Related theory. MPM soft-particle backend.

Source hint. `src/mpm/Simulation.py`

29.9.3 `ipc_contact`

Scope. MPM configuration and direct/soft-particle engines.

Accepted value. Boolean; parameters are supplied through `MPM.add_contact("IPC",...)`.

Meaning. Enables the IPC contact branch and stores barrier/friction parameters for direct MPM or soft-particle contact engines.

Numerical or workflow effect. Selects IPC barrier/friction energy assembly instead of ordinary MPM contact detection.

Related theory. MPM self-collision and soft-particle IPC.

Source hint. `src/mpm/mainMPM.py;src/mpm/Simulation.py`

29.9.4 `soft_rigid_contact`

Scope. DEM/MPDEM LSMPM configuration.

Accepted value. `DEM` or `IPC`; alias `lsmpm_soft_rigid_contact`.

Meaning. Chooses the contact route for level-set MPM soft bodies against rigid or affine rigid bodies.

Numerical or workflow effect. `DEM` uses explicit LSDEM/DEM contact forces. `IPC` routes soft-affine workflows to the mixed implicit operator that assembles energy, gradient, Hessian, friction, damping, and CCD in one Newton system.

Related theory. Mixed LSMPM soft-affine IPC.

Source hint. `src/dem/Simulation.py;src/mpdem/engines/SoftAffineIPCOperator.py`

29.9.5 `soft_background_damping`

Scope. `DEM.set_affine_body_parameters` for mixed LSMPM soft-affine IPC.

Accepted value. Nonnegative real scalar.

Meaning. Soft MPM grid background damping coefficient used by the mixed IPC operator.

Numerical or workflow effect. Adds damping residual and diagonal Hessian contribution on soft grid degrees of freedom. It is separate from contact damping and separate from DEM local damping on affine bodies.

Related theory. Mixed LSMPM soft-affine IPC.

Source hint. `src/dem/Simulation.py;src/mpdem/engines/SoftAffineIPCOperator.py`

29.9.6 `contact_damping_stiffness`

Scope. AffineBody and mixed soft-affine IPC parameters and material-pair contact properties.

Accepted value. Nonnegative real scalar. Aliases include `contact_damping`, `k_CD`, and `k_cd`; pair property spelling may use `ContactDampingStiffness`.

Meaning. Contact-pair damping stiffness multiplying the active normal contact Hessian.

Numerical or workflow effect. Adds a damping residual $\Delta t k_D K_B(y - \hat{y})$ and associated Hessian contribution for active contacts. It should be grouped with contact/barrier properties, not with global affine-body material parameters.

Related theory. Mixed LSMPM soft-affine IPC.

Source hint. `src/dem/Simulation.py;src/dem/engines/AffineBodyEngine.py;src/mpdem/engines/SoftAffineIPCOperator.py`

29.9.7 `estress`

Scope. soft-particle state variables and output.

Accepted value. Computed scalar state variable; users do not prescribe it directly.

Meaning. Von-Mises equivalent stress derived from the current stress tensor for soft-particle output.

Numerical or workflow effect. Provides a compact scalar stress field for `LSMPMSoftPoint*.npz` and visualization. It is diagnostic and does not replace the finite-strain material class PK1 stress or tangent used by TLMPM force/Hessian assembly.

Related theory. Finite-strain material class contract.

Source hint. `src/mpm/MaterialManager.py;src/mpm/PostPlot.py`

30 Detailed Model Reference

30.1 Solid and Fluid Constitutive Models

30.1.1 `LinearElastic`

Scope. small-strain solid MPM

User parameters. `Density,YoungModulus,PoissonRatio`

Theory. Hooke elasticity with Lame constants and a symmetric tangent.

Implementation effect. The model determines state variables, stress or contact update, tangent symmetry, timestep estimates, recorder fields, and meaningful keyword checks.

Related theory. Constitutive and contact models, MPM weak form and transfers, Implicit solid MPM.

$$r_i^{int} = \sum_p V_p B_i(x_p)^T \sigma_p, \qquad K_{ij} = \frac{\partial r_i^{int}}{\partial u_j}.$$

30.1.2 `ElasticPerfectlyPlastic`

Scope. small-strain plastic MPM

User parameters. `Density,YoungModulus,PoissonRatio,YieldStress,ResidualYieldStress,PlasticDevStrain`

Theory. Trial stress returned to a von-Mises-like surface with peak-to-residual softening.

Implementation effect. The model determines state variables, stress or contact update, tangent symmetry, timestep estimates, recorder fields, and meaningful keyword checks.

Related theory. Constitutive and contact models, Implicit solid MPM.

$$r_i^{int} = \sum_p V_p B_i(x_p)^T \sigma_p, \qquad K_{ij} = \frac{\partial r_i^{int}}{\partial u_j}.$$

30.1.3 `MohrCoulomb`

Scope. geomechanics MPM

User parameters. `Density,YoungModulus,PoissonRatio,Tensile,Cohesion,Friction,Dilation`

Theory. Pressure-dependent frictional surface with tensile cutoff; nonassociated flow can be nonsymmetric.

Implementation effect. The model determines state variables, stress or contact update, tangent symmetry, timestep estimates, recorder fields, and meaningful keyword checks.

Related theory. Constitutive and contact models, Implicit solid MPM.

$$r_i^{int} = \sum_p V_p B_i(x_p)^T \sigma_p, \qquad K_{ij} = \frac{\partial r_i^{int}}{\partial u_j}.$$

30.1.4 `DruckerPrager`

Scope. Native small-strain geomechanics MPM and Direct implicit ULMPM finite-strain plasticity; these are separate implementations selected by the backend.

User parameters. Native: `Density,YoungModulus,PoissonRatio,Tensile,Cohesion,Friction,Dilation,dpType`. Direct: `Density,YoungModulus,PoissonRatio,Cohesion,FrictionAngle,DilationAngle,dpType`.

Theory. Both use a smooth conical approximation to Mohr–Coulomb with selectable mapping. The Direct model is multiplicative finite strain, supports associated or nonassociated constant-dilation flow, is perfect plastic, and uses no separate tensile cutoff. Its material map is always 3D; Cartesian 2D uses plane strain and axisymmetric 2D uses no-swirl kinematics.

Implementation effect. Direct ULMPM supplies material-owned plastic history and an analytic physical PK1 tangent. In standalone lagged MPM–IPC, monolithic FEM–MPM and IGA–MPM, and Direct MPM–ABD, nonassociated DP recomputes the physical return and plastic volume at each trial, includes the volume derivative in the nonsymmetric Jacobian, and automatically uses device BiCGSTAB with node-block Jacobi preconditioning (scalar Jacobi for COO). Residual Armijo retains conservative contact and material CCD; no material-consistency outer loop runs. Associated DP retains PCG and its existing frozen-plastic-volume consistency iteration. Other Direct-MPM consumers retain their frozen-potential contract with particle-local Aitken relaxation; this numerical relaxation does not evolve dilation. Zero dilation uses a tensile apex cap without a separate cutoff parameter. In monolithic FEM–MPM and IGA–MPM IPC the FEM or IGA child remains elastic.

Related theory. Constitutive and contact models, Finite-strain associated plasticity, Implicit solid MPM, FEM–MPM IPC, and IGA–MPM.

$$r_i^{int} = \sum_p V_p B_i(x_p)^T \sigma_p, \qquad K_{ij} = \frac{\partial r_i^{int}}{\partial u_j}.$$

30.1.5 `StateDependentDruckerPrager` (Direct finite strain)

Scope. Direct implicit ULMPM within monolithic FEM–MPM, IGA–MPM, or three-dimensional MPM–AffineBody IPC. Ordinary `DruckerPrager` is unchanged.

User parameters. `MPM.add_material(model="StateDependentDruckerPrager",...)` accepts the finite-strain DP elastic/density/cohesion parameters and `fai_c` (critical friction angle in

degrees; `FrictionAngle` is an alias). `e0,e_Tao,lambda_c,ksi,nd,nf` each default to 0.3 and are dimensionless. `dpType` defaults to `MiddleCircumscribed`; its $3 + \sin\phi$ mapping matches the small-strain SDMC implementation. `DilationAngle` does not control the evolving dilation law.

Theory. The state law of `StateDependentMohrCoulomb` uses previous compression-positive Cauchy pressure, clamped below at 1000 Pa, and a 101000 Pa reference pressure. Void ratio follows the total finite volume increment, including hoop stretch in axisymmetry, and is clamped to [0.1, 1.5]. Friction and dilation vary at every trial; elastic moduli and cohesion remain constant. The finite-strain DP tensile apex is retained, with no independent SDMC tensile cutoff.

Implementation effect. The analytic physical PK1 Jacobian includes angle evolution and plastic-volume derivatives, and is generally nonsymmetric. The coupled solver uses residual Newton/BiCGSTAB with its existing conservative CCD and independent lagged-friction loop. There is no constitutive outer loop or incremental energy potential. The 14-entry history adds `void_ratio,committed_jacobian,state_pressure`; accepted steps commit it and failed steps restore it. VTU output adds `void_ratio`. The IGA–MPM CPT script enables this model with `--contactipc--state-dependent` and persists all added history in its checkpoint. Standalone Direct stepping, Direct TLMPM, Native MPM, LSMPM soft bodies, and parameter/history adjoints are unsupported.

Related theory. State-dependent finite-strain DP.

Source hint. `src/physics_model/consititutive_model/finite_strain/StateDependentDruckerPrager.py`.

30.1.6 `VonMises` (Direct finite strain)

Scope. Direct implicit ULMPM and monolithic FEM–MPM or IGA–MPM IPC.

User parameters. `Density,YoungModulus,PoissonRatio,YieldStress`; optional `HardeningModulus=0`.

Theory. Multiplicative Hencky associated J_2 return using the standard $\sqrt{3J_2} = \sigma_y$ convention and optional linear isotropic hardening. The local incremental potential has an analytic symmetric PK1 tangent; plastic MPM defaults to PSD-projected Newton. The material map is always 3D, with plane-strain embedding for Cartesian 2D and no-swirl embedding for axisymmetric 2D.

Implementation effect. The accepted-step Taichi kernel stores the projected elastic deformation gradient and equivalent plastic strain. Direct TLMPM is rejected until explicit plastic-gradient/rest-basis storage is implemented.

Related theory. Finite-strain associated plasticity, Implicit solid MPM, and FEM–MPM IPC, and IGA–MPM.

30.1.7 `ModifiedCamClay`

Scope. critical-state soil MPM

User parameters. `Density,PoissonRatio,StressRatio,lambda,kappa,OverConsolidationRatio,pc,ConsolidationPressure,void_ratio_ref,pressure_ref,void_ratio_initial,porosity,three_invariants,subloading,subloading_u,bonding,bounding_surface,mc_a,mc_b,mc_c,mc_d,m_degradation,m_shear,s_h`

Theory. Critical-state clay with pressure-dependent elasticity, preconsolidation hardening, optional Lode-angle stress ratio, subloading, and bonding shifts.

History consistency. The generic plastic modulus differentiates the current flow-driven history, including bond shifts and the subloading ratio. A zero plastic increment does not reinitialize that ratio.

Implementation effect. The model determines state variables, stress or contact update, tangent symmetry, timestep estimates, recorder fields, and meaningful keyword checks.

Related theory. Constitutive Equations, MCC return mapping and augmented Lagrangian view, Return mapping and substepping, Semi-implicit two-phase MPM.

$$r_i^{int} = \sum_p V_p B_i(x_p)^T \sigma_p, \qquad K_{ij} = \frac{\partial r_i^{int}}{\partial u_j}.$$

Stress integration. `ReturnMapping` and `ImplicitIntegration` use the MCC nested scalar return mapping. `ImplicitIntegrationAL` uses the implemented four-unknown local solve $[p, q, p_c, \Delta\lambda]$ with finite-difference Jacobian and safeguards. `SubStepping` uses the generic explicit integration path.

30.1.8 `StateDependentMohrCoulomb`

Scope. state-dependent granular MPM

User parameters. `Density,YoungModulus,PoissonRatio,e0,e_Tao,lambda_c,ksi,nd,nf,fai_c`

Theory. Friction and dilation depend on void ratio and state parameter.

Implementation effect. The model determines state variables, stress or contact update, tangent symmetry, timestep estimates, recorder fields, and meaningful keyword checks.

Related theory. Constitutive and contact models.

$$r_i^{int} = \sum_p V_p B_i(x_p)^T \sigma_p, \qquad K_{ij} = \frac{\partial r_i^{int}}{\partial u_j}.$$

30.1.9 `GranularMaterial`

Scope. small-strain granular solid MPM

User parameters. `Density,YoungModulus,PoissonRatio`

Theory. Pressure-aware granular continuum update for simple granular material behavior without the full MC/DP critical-state return mapping.

Implementation effect. The material factory treats this as a solid model with its own state and stress update; use it when a lightweight granular continuum model is intended.

Related theory. State-dependent and critical-state sand models, Constitutive Equations.

$$r_i^{int} = \sum_p V_p B_i(x_p)^T \sigma_p, \qquad K_{ij} = \frac{\partial r_i^{int}}{\partial u_j}.$$

30.1.10 `NorSand`

Scope. critical-state sand MPM

User parameters. `Density,G0,kappa,lambda,M,N,beta,vc0,v0,h,p_ref`

Theory. NorSand-style critical-state sand model with pressure-dependent elasticity, state parameter, image stress/pressure, dilatancy, and hardening controls.

Implementation effect. Adds NorSand state variables and a model-specific plastic update; random-field parameters are not accepted by this source path.

Related theory. State-dependent and critical-state sand models, Return mapping and substepping.

$$r_i^{int} = \sum_p V_p B_i(x_p)^T \sigma_p, \qquad K_{ij} = \frac{\partial r_i^{int}}{\partial u_j}.$$

30.1.11 `SanisandMS`

Scope. advanced sand plasticity

User parameters. `G0,nu,Mc,c,lambda_c,e0,ksi,m,h0,ch,nb,A0,nd,zeta,mu0,beta`

Theory. Bounding-surface sand model with pressure-dependent elasticity and fabric.

Implementation effect. The model determines state variables, stress or contact update, tangent symmetry, timestep estimates, recorder fields, and meaningful keyword checks.

Related theory. Constitutive and contact models.

$$r_i^{int} = \sum_p V_p B_i(x_p)^T \sigma_p, \qquad K_{ij} = \frac{\partial r_i^{int}}{\partial u_j}.$$

30.1.11.1 Finite-strain class output. The finite-strain models below are class-owned hyperelastic models. Their primary production outputs are $\boldsymbol{P} = \partial\psi/\partial\boldsymbol{F}$ and the material tangent $\partial\boldsymbol{P}/\partial\boldsymbol{F}$; LSMPM soft particles and implicit soft-affine IPC consume those class outputs directly. The scalar `estress` is a derived von-Mises output state variable, not an input material parameter and not a replacement for PK1. See Finite-strain material class contract.

30.1.12 `HenckyElastic`

Scope. finite-strain hyperelastic MPM

User parameters. `Density,YoungModulus,PoissonRatio`

Theory. Uses logarithmic strain and is objective under large rotations.

Implementation effect. The model determines state variables, stress or contact update, tangent symmetry, timestep estimates, recorder fields, and meaningful keyword checks.

Related theory. Constitutive and contact models, MPM weak form and transfers.

$$r_i^{int} = \sum_p V_p B_i(x_p)^T \sigma_p, \qquad K_{ij} = \frac{\partial r_i^{int}}{\partial u_j}.$$

30.1.13 `NeoHookean`

Scope. finite-strain hyperelastic MPM

User parameters. `Density,YoungModulus,PoissonRatio`

Theory. Compressible hyperelastic baseline with deviatoric and volumetric response.

Implementation effect. The model determines state variables, stress or contact update, tangent symmetry, timestep estimates, recorder fields, and meaningful keyword checks.

Related theory. Constitutive and contact models, MPM weak form and transfers.

$$r_i^{int} = \sum_p V_p B_i(x_p)^T \sigma_p, \qquad K_{ij} = \frac{\partial r_i^{int}}{\partial u_j}.$$

30.1.14 `Gent`

Scope. finite-strain rubber MPM

User parameters. `Density,YoungModulus,PoissonRatio,Tensile1,Tensile2`

Theory. Limits chain extensibility and stiffens near locking.

Implementation effect. The model determines state variables, stress or contact update, tangent symmetry, timestep estimates, recorder fields, and meaningful keyword checks.

Related theory. Constitutive and contact models.

$$r_i^{int} = \sum_p V_p B_i(x_p)^T \sigma_p, \qquad K_{ij} = \frac{\partial r_i^{int}}{\partial u_j}.$$

30.1.15 `MooneyRivlin`

Scope. finite-strain rubber MPM

User parameters. `Density,YoungModulus,PoissonRatio,Coefficient`

Theory. Uses first and second isochoric invariants.

Implementation effect. The model determines state variables, stress or contact update, tangent symmetry, timestep estimates, recorder fields, and meaningful keyword checks.

Related theory. Constitutive and contact models.

$$r_i^{int} = \sum_p V_p B_i(x_p)^T \sigma_p, \qquad K_{ij} = \frac{\partial r_i^{int}}{\partial u_j}.$$

30.1.16 `Hydrogel`

Scope. soft material MPM

User parameters. `Density,YoungModulus,PoissonRatio,Tensile`

Theory. Soft finite-strain model with tensile limits.

Implementation effect. The model determines state variables, stress or contact update, tangent symmetry, timestep estimates, recorder fields, and meaningful keyword checks.

Related theory. Constitutive and contact models.

$$r_i^{int} = \sum_p V_p B_i(x_p)^T \sigma_p, \qquad K_{ij} = \frac{\partial r_i^{int}}{\partial u_j}.$$

30.1.17 `Newtonian`

Scope. fluid MPM

User parameters. `Density,Modulus,Viscosity,cL,cQ,ElementLength,atmospheric_pressure,SurfaceTension`

Theory. Stress is pressure plus viscous deviatoric stress.

Implementation effect. The model determines state variables, stress or contact update, tangent symmetry, timestep estimates, recorder fields, and meaningful keyword checks.

Related theory. Incompressible fluid MPM, Constitutive and contact models.

$$r_i^{int} = \sum_p V_p B_i(x_p)^T \sigma_p, \qquad K_{ij} = \frac{\partial r_i^{int}}{\partial u_j}.$$

30.1.18 `Bingham`

Scope. viscoplastic fluid MPM

User parameters. `Density,Modulus,Viscosity,YieldStress,CriticalStrainRate,gamma`

Theory. Regularized yield-stress fluid.

Implementation effect. The model determines state variables, stress or contact update, tangent symmetry, timestep estimates, recorder fields, and meaningful keyword checks.

Related theory. Incompressible fluid MPM, Constitutive and contact models.

$$r_i^{int} = \sum_p V_p B_i(x_p)^T \sigma_p, \qquad K_{ij} = \frac{\partial r_i^{int}}{\partial u_j}.$$

30.1.19 `Rigid`

Scope. rigid or prescribed MPM body

User parameters. `Density`

Theory. Provides mass and body identity without deformable stress evolution.

Implementation effect. The model determines state variables, stress or contact update, tangent symmetry, timestep estimates, recorder fields, and meaningful keyword checks.

Related theory. MPM weak form and transfers, Constitutive and contact models.

$$r_i^{int} = \sum_p V_p B_i(x_p)^T \sigma_p, \qquad K_{ij} = \frac{\partial r_i^{int}}{\partial u_j}.$$

30.1.20 `RigidBody`

Scope. rigid or prescribed MPM body

User parameters. `Density`

Theory. Alias accepted by the material factory for the rigid model path.

Implementation effect. `model="RigidBody"` and a missing or `None` model select `RigidModel` in the current MPM material factory.

Related theory. MPM weak form and transfers, Constitutive models.

30.1.21 `UserDefined`

Scope. solid or fluid MPM material extension

User parameters. user-provided model data required by the registered hook

Theory. Delegates stress update and state handling to a user-defined constitutive implementation while preserving the MPM material interface.

Implementation effect. Accepted by the solid, two-phase, and fluid material factory paths; the user implementation must provide the state variables, stress update, tangent or timestep behavior expected by the selected solver.

Related theory. Constitutive models, Stress integration.

30.2 DEM and MPDEM Contact Models

30.2.1 `Linearcontact`

Scope. DEM and MPDEM

User parameters. `NormalStiffness,TangentialStiffness,Friction,dampingkeys`

Theory. Spring-dashpot contact with Coulomb truncation.

Implementation effect. The model determines state variables, stress or contact update, tangent symmetry, timestep estimates, recorder fields, and meaningful keyword checks.

Related theory. DEM dynamics and contact, Constitutive and contact models.

$$r_i^{int} = \sum_p V_p B_i(x_p)^T \sigma_p, \qquad K_{ij} = \frac{\partial r_i^{int}}{\partial u_j}.$$

30.2.2 `LinearRollingcontact`

Scope. DEM

User parameters. `RollingStiffness,TwistingStiffness,RollingFriction,TwistingFriction`

Theory. Adds rolling and twisting resistance.

Implementation effect. The model determines state variables, stress or contact update, tangent symmetry, timestep estimates, recorder fields, and meaningful keyword checks.

Related theory. DEM dynamics and contact, Constitutive and contact models.

$$r_i^{int} = \sum_p V_p B_i(x_p)^T \sigma_p, \qquad K_{ij} = \frac{\partial r_i^{int}}{\partial u_j}.$$

30.2.3 `HertzMindlincontact`

Scope. DEM and MPDEM

User parameters. `ShearModulus,Poisson,Friction,RollingFriction,Restitution`

Theory. Nonlinear contact stiffness from effective radius and modulus.

Implementation effect. The model determines state variables, stress or contact update, tangent symmetry, timestep estimates, recorder fields, and meaningful keyword checks.

Related theory. DEM dynamics and contact, Constitutive and contact models.

$$r_i^{int} = \sum_p V_p B_i(x_p)^T \sigma_p, \qquad K_{ij} = \frac{\partial r_i^{int}}{\partial u_j}.$$

30.2.4 `JiangRollingResistance`

Scope. DEM

User parameters. `YoungModulus,StiffnessRatio,Friction,ShapeFactor,CrushFactor`

Theory. Rolling resistance law for spherical approximations of angular grains.

Implementation effect. The model determines state variables, stress or contact update, tangent symmetry, timestep estimates, recorder fields, and meaningful keyword checks.

Related theory. DEM dynamics and contact, Constitutive and contact models.

$$r_i^{int} = \sum_p V_p B_i(x_p)^T \sigma_p, \qquad K_{ij} = \frac{\partial r_i^{int}}{\partial u_j}.$$

30.2.5 `EnergyConservationcontact`

Scope. DEM

User parameters. `NormalStiffness,TangentialStiffness,Stiffness,NormalCutOff,StiffnessRatio,Friction`

Theory. Energy-consistent or barrier-style contact family.

Implementation effect. The model determines state variables, stress or contact update, tangent symmetry, timestep estimates, recorder fields, and meaningful keyword checks.

Related theory. DEM dynamics and contact, Constitutive and contact models.

$$r_i^{int} = \sum_p V_p B_i(x_p)^T \sigma_p, \qquad K_{ij} = \frac{\partial r_i^{int}}{\partial u_j}.$$

30.3 Fluid Coupling and Force Models

30.3.1 `ParticleFluid`

Scope. MPDEM/CFDEM

User parameters. `NormalStiffness,NormalViscousDamping,NonSlipFraction`

Theory. Semi-resolved particle-fluid interface property set.

Implementation effect. The model determines state variables, stress or contact update, tangent symmetry, timestep estimates, recorder fields, and meaningful keyword checks.

Related theory. DEM-MPM and CFDEM coupling, Incompressible DEM-sphere coupling.

$$r_i^{int} = \sum_p V_p B_i(x_p)^T \sigma_p, \qquad K_{ij} = \frac{\partial r_i^{int}}{\partial u_j}.$$

30.3.2 `Abrahamdrag`

Scope. MPDEM/CFDEM

User parameters. `DragForceModel,DragLaw,ParticleReynold`

Theory. Reynolds-dependent drag coefficient.

Implementation effect. The model determines state variables, stress or contact update, tangent symmetry, timestep estimates, recorder fields, and meaningful keyword checks.

Related theory. DEM-MPM and CFDEM coupling, Incompressible DEM-sphere coupling.

$$r_i^{int} = \sum_p V_p B_i(x_p)^T \sigma_p, \qquad K_{ij} = \frac{\partial r_i^{int}}{\partial u_j}.$$

30.3.3 `VirtualMassforce`

Scope. MPDEM/CFDEM

User parameters. `virtualmasscoefficient`

Theory. Adds displaced-fluid inertia proportional to relative acceleration.

Implementation effect. The model determines state variables, stress or contact update, tangent symmetry, timestep estimates, recorder fields, and meaningful keyword checks.

Related theory. DEM-MPM and CFDEM coupling, Incompressible DEM-sphere coupling.

$$r_i^{int} = \sum_p V_p B_i(x_p)^T \sigma_p, \qquad K_{ij} = \frac{\partial r_i^{int}}{\partial u_j}.$$

31 Geometry API Addendum

This addendum collects the public geometry utilities that are spread across `src/sdf`, `src/levelset`, and `src/nurbs`. These classes are used by MPM free surfaces and cut cells,

DEM/LSDEM geometry, IGA primitives, IGA-MPM closest-point contact, and standalone geometry generation.

31.1 SDF Constructors and Operations

Entry point	Main input	User-visible role
`circle`	center/radius	2D disk SDF; used for round regions, particles, and cross sections.
`line`	point/normal	2D signed half-plane boundary.
`slab`	bounds/axis	2D or 3D finite strip/slab between two planes.
`rectangle`	size/corners	2D axis-aligned box SDF.
`rounded_rectangle`	size/radius	2D rectangle with rounded corners.
`equilateral_triangle`	size	2D regular triangle SDF.
`hexagon`	radius	2D regular hexagon SDF.
`rounded_x`	width/radius	2D rounded X-shaped primitive.
`cross`	size	2D cross-shaped primitive.
`polygon`	vertices	2D polygon SDF from ordered vertices.
`sphere`	center/radius	3D ball SDF.
`plane`	point/normal	3D signed half-space boundary.
`box`	size/corners	3D rectangular box SDF.
`rounded_box`	size/radius	3D box with rounded edges.
`box_frame`	size/thickness	Hollow rectangular frame primitive.
`torus`	major/minor radius	Ring torus SDF around the local axis.
`capped_torus`	radii/angle	Torus segment with end caps.
`heart`	scale	Analytic heart-shaped 3D primitive.
`link`	radii/length	Chain-link style swept torus primitive.
`hexagonal_prism`	radius/height	Hexagonal column SDF.
`capsule`	endpoints/ radius	Line segment swept by a sphere.
`cylinder`	radius/axis	Infinite or oriented cylinder SDF.
`capped_cylinder`	endpoints/ radius	Finite cylinder with planar caps.
`rounded_cylinder`	radius/height/ rounding	Cylinder with rounded cap edges.
`cone`	angle/radius	Infinite or oriented cone SDF.
`capped_cone`	endpoints/radii	Truncated cone between two cap planes.

Entry point	Main input	User-visible role
`rounded_cone`	radii/height/ rounding	Cone/frustum with rounded transition.
`revolved_vesica`	radius/offset	Revolved vesica profile for lens-like solids.
`octahedron`	size	Regular octahedron SDF.
`pyramid`	height/base	Pyramid SDF.
`rhombus`	diagonals/ height	Rhombic primitive used for faceted solids.
`tetrahedron`	size	Regular tetrahedron SDF.
`dodecahedron`	size	Regular dodecahedron approximation.
`icosahedron`	size	Regular icosahedron approximation.
`arbitrarily`	mesh/function	Mesh-backed arbitrary SDF wrapper.
`polyhedron`	vertices/faces	Polyhedral solid wrapper.
`surfacefunction`	callable	Uses a user function as an implicit surface.
`polysuperellipsoid`	superquadric parameters	Superellipsoid-like particle template.
`polysuperquadrics`	superquadric parameters	General superquadric/poly-superquadric template.
`pointcloud`	sample points	Point-cloud based distance wrapper.
`image`	image path/-data	Converts a binary or grayscale image into a 2D SDF.
`text`	string/font	Converts rendered text into a 2D SDF.

Entry point	Main input	User-visible role
`union(a,*bs,k)`	SDF objects	Hard or smoothed minimum; combines bodies into one solid.
`difference(a,*bs,k)`	SDF objects	Subtracts one or more SDF solids from the first.
`intersection(a,*bs,k)`	SDF objects	Keeps the common volume of all operands.
`blend(a,*bs,k)`	SDF objects	Smooth blend useful for rounded unions.
`negate(other)`	SDF object	Reverses inside/outside sign.
`dilate(other,r)`	offset radius	Expands the zero level surface outward by r.
`erode(other,r)`	offset radius	Shrinks the solid inward by r.

Entry point	Main input	User-visible role
`shell(other,thickness)`	thickness	Builds a narrow shell around the zero isosurface.
`repeat(other,spacing,count,padding)`	spacing/count	Periodically repeats an SDF in a grid.
`translate(other,offset)`	vector	Rigid translation by inverse mapping.
`scale(other,factor)`	scalar/vector	Scales geometry and distance consistently.
`rotate(other,angle,...)`	angle/axis	Rotates a 2D or 3D SDF.
`rotate_to(other,a,b)`	two directions	Rotates a 3D SDF so direction a aligns with b.
`orient(other,axis)`	axis	Orients a 3D shape to a target coordinate axis.
`circular_array(other,count,offset)`	count/offset	Replicates a 3D object around an axis.
`elongate(other,size)`	size vector	Stretches a primitive by inserting a flat interior region.
`twist(other,k)`	twist rate	Applies a coordinate twist before SDF evaluation.
`bend(other,k)`	curvature	Bends a 3D shape by a constant curvature map.
`bend_linear(other,p0,p1,v,e)`	endpoints/ vector	Blends a linear bend between two stations.
`bend_radial(other,r0,r1,dz,e)`	radii/height	Radial bend/transition for annular shapes.
`transition_linear(f0,f1,p0,p1,e)`	two SDFs	Linearly transitions between two shapes across a band.
`transition_radial(f0,f1,r0,r1,e)`	two SDFs	Radially transitions between two shapes.
`wrap_around(other,x0,x1,r,e)`	interval/radius	Wraps a straight geometry around a circular path.
`slice(other,slab)`	slab SDF	Clips a 3D object by a 2D/3D slab.
`extrude(other,h)`	height	Turns a 2D SDF into a straight 3D extrusion.
`extrude_to(a,b,h,e)`	two 2D SDFs	Morphs between two cross sections while extruding.
`revolve(other,offset)`	offset	Revolves a 2D profile around an axis.
`linear(t)`	$t \in [0,1]$	Linear easing for transition weights.
`in_quad`, `out_quad`, `in_out_quad`	$t \in [0,1]$	Quadratic ease-in, ease-out, and symmetric ease.
`in_cubic`, `out_cubic`, `in_out_cubic`	$t \in [0,1]$	Cubic easing for smoother starts/stops.

Entry point	Main input	User-visible role
`in_quart`, `out_quart`, `in_out_quart`	$t \in [0,1]$	Quartic easing with stronger acceleration.
`in_quint`, `out_quint`, `in_out_quint`	$t \in [0,1]$	Quintic easing for high-order smooth transitions.
`in_sine`, `out_sine`, `in_out_sine`	$t \in [0,1]$	Trigonometric easing.
`in_expo`, `out_expo`, `in_out_expo`	$t \in [0,1]$	Exponential easing.
`in_circ`, `out_circ`, `in_out_circ`	$t \in [0,1]$	Circular-arc easing.
`in_elastic`, `out_elastic`, `in_out_elastic`	t, k	Overshooting elastic easing.
`in_back`, `out_back`, `in_out_back`	t	Backtracking ease curves.
`in_bounce`, `out_bounce`, `in_out_bounce`	t	Bounce-style transition weights.
`in_square`, `out_square`, `in_out_square`	t	Square-root based easing.

Common object methods. SDF geometry objects are callable distance functions. Their shared methods are:

Method or property	Meaning
`__call__(point)`	Evaluates signed distance after applying the object's transform.
`__or__`, `__add__`	Operator aliases for union.
`__and__`	Operator alias for intersection.
`__sub__`	Operator alias for difference.
`resolutions(resolution)`	Sets surface/grid sampling resolution.
`rays(ray)`	Sets ray count for ray-tracing based surface sampling.
`smears(smear)`	Sets smoothing width used by grid sampling.
`grids(space,extent)`	Allocates a `LocalGrid` over the SDF bounding box.
`generate_sdf()`	Samples the SDF on the allocated local grid.
`ray_tracing()`	Builds surface nodes by casting rays against the implicit surface.
`marching_cube()`	Extracts a triangle mesh from the gridded distance field.
`generate()`	Convenience pipeline for grid/surface generation.
`save(path)`	Saves mesh/grid output generated from the SDF.
`show()` and `show_slice()`	Displays the generated 3D surface or a 2D SDF slice.

Method or property	Meaning
`dump_files()`	Writes generated intermediate files for downstream workflows.
`visualize()`	Visualization helper for generated geometry.
`tetrahedization()`	Produces tetrahedralization data where the workflow supports it.
`center`, `volume`, `eqradius`	Stored mass-property estimates for the generated shape.
`inertia`	Principal inertia estimate from generated geometry.
`lower_bound`, `upper_bound`	Bounding-box limits used by grid allocation and sampling.

31.2 LocalGrid and Level Set API

Entry point	Main input	User-visible role
`LocalGrid.read_grid(space,extent)`	spacing/buffer	Creates a regular signed-distance grid sized from the owning SDF bounds.
`LocalGrid.set_distance_field(...)`	arrays/ metadata	Imports an externally computed grid, node coordinates, origin, size, and center of mass.
`LocalGrid.set_grid(start_point,region_size)`	origin/size	Defines local grid origin, physical extent, and resolution-derived node layout.
`LocalGrid.build_node_coords()`	none	Fills the node-coordinate array used by interpolation and export.
`LocalGrid.generate_sdf()`	SDF owner	Samples or finalizes a distance field for solver use.
`LocalGrid.distance(p)`	point	Returns interpolated signed distance at point `p`.
`LocalGrid.move()`	transform state	Updates the local-grid position for moving geometry paths.
`LevelSet`	dimension/grid	Base Taichi level-set storage and interpolation object.
`FastSweepingLevelSet`	iteration count/band	Adds fast-sweeping reinitialization to signed-distance form.
`FastSweepingLevelSet.redistance()`	current ϕ	Reinitializes ϕ while preserving the interface sign.
`FastMarchingLevelSet`	seed interface	Alternative heap/front based signed-distance reinitialization.

Entry point	Main input	User-visible role
`FiniteDifference`	grid/velocity	Finite-difference Hamilton-Jacobi update utilities.
`GeometryMethod`	geometry field	Geometry-based level-set update helper.
`SemiLagrangian`	grid/order/RK	Level-set advection class using backtraced interpolation.
`SemiLagrangian.run(grid_v,dt)`	velocity/time step	Advects the level set by one step; `runge_kutta`=1–4 selects RK1–RK4 backtracing.
`SemiLagrangianMultiGrid`	multigrid fields	Semi-Lagrangian advection variant for multigrid/structured velocity layouts.
`Fluid_Surface`	fluid SDF	Builds fluid surface samples and curvature data from a signed-distance field.
`Fluid_Surface.build_surface()`	current ϕ	Extracts interface data used by surface-tension and visualization workflows.
`linear1d`, `bilinear`, `trilinear`	grid samples	First-order interpolation used when `order=2`.
`weno31d`, `biweno3`, `triweno3`	grid samples	WENO3 interpolation used when `order=3`.
`weno41d`, `biweno4`, `triweno4`	grid samples	WENO4 interpolation used when `order=4`.
`weighted_essentially_non_oscillatory`	values/spacing	NumPy WENO derivative helper for offline or Python-level level-set work.
`essentially_non_oscillatory`	values/spacing	NumPy ENO derivative helper.

Key constructor options for `SemiLagrangian` are `dimension`, `grid_size`, `resolution`, `ghost_cell`, `order`, `runge_kutta`, `iteration_num`, `local_band`, `grid_type`, and `visualize`. In the current interface, `runge_kutta` selects RK1–RK4 backtracing and `order` selects linear, WENO3, or WENO4 interpolation.

31.3 NURBS and B-Spline API

Entry point	User-visible role
`find_span_linear(num_ctrlpts, knot,degree,knot_vector)`	Linear search for the knot span containing a parameter value.
`find_span(...)`	Binary-search span lookup used by evaluation and element extraction.

Entry point	User-visible role
`find_element(knot_vector)`	Extracts nonzero knot intervals/elements from a knot vector.
`find_multiplicity(knot_vector)`	Counts repeated knots for continuity and refinement logic.
`basis_functions(span,knot,degree,knot_vector)`	Evaluates the nonzero B-spline basis functions on one span.
`basis_function_ders(...,order)`	Evaluates B-spline basis derivatives up to the requested order.
`one_basis_function(...)`	Evaluates a single basis function without forming the full local vector.
`one_basis_function_ders_1st(...)`	First-derivative variant for a single basis function.
`BsplineBasis1d`, `BsplineBasis2d`, `BsplineBasis3d`	Tensor-product non-rational B-spline basis vectors in 1D, 2D, and 3D.
`BsplineBasisDers1d`, `BsplineBasisDers2d`, `BsplineBasisDers3d`	First-derivative tensor-product B-spline bases.
`BsplineBasis2ndDers1d`, `BsplineBasis2ndDers2d`, `BsplineBasis2ndDers3d`	Second-derivative B-spline basis blocks for curvature/Hessian use.
`BsplineBasisInterpolations1d/2d/3d`	Interpolates control-point or field values with B-spline basis functions.
`BsplineBasisInterpolationsDers1d/2d/3d`	Interpolates first derivatives of a B-spline field.
`BsplineBasisInterpolations2ndDers1d/2d/3d`	Interpolates second derivatives of a B-spline field.
`NurbsBasis1d`, `NurbsBasis2d`, `NurbsBasis3d`	Rational basis evaluation using control weights.
`NurbsBasisDers1d/2d/3d`	First derivatives of rational NURBS bases.
`NurbsBasis2ndDers1d/2d/3d`	Second derivatives of rational NURBS bases.
`NurbsBasisInterpolations1d/2d/3d`	Rational interpolation of control-point fields.
`NurbsBasisInterpolationsDers1d/2d/3d`	First derivatives of rational interpolated fields.
`NurbsBasisInterpolations2ndDers1d/2d/3d`	Second derivatives of rational interpolated fields.
`NurbsDxDweight1d/2d/3d`	Sensitivity of a NURBS point with respect to control weights.
`Spline`, `SplineCurve`, `SplineSurface`, `SplineVolume`	Base geometry classes that store degree, knot vectors, control points, and common evaluation methods.
`BSplineCurve`, `BSplineSurface`, `BSplineVolume`	Non-rational spline primitives; all weights are effectively one.

Entry point	User-visible role
`bspline_curve`, `bspline_surface`, `bspline_volume`	Factory functions for B-spline primitives.
`NurbsCurve`, `NurbsSurface`, `NurbsVolume`	Rational spline primitives with explicit weights.
`nurbs_curve`, `nurbs_surface`, `nurbs_volume`	Factory functions for NURBS primitives.
`Rectangle`, `Ring`	Surface primitive constructors for common 2D NURBS patches.
`Cube`, `Tube`, `Cylinder`	Volume primitive constructors for common NURBS solids.
`single_point`	Evaluates one physical point at a parameter coordinate.
`derivative`	Evaluates first derivatives/tangent vectors.
`second_derivative`	Evaluates second derivatives for curvature and Newton projections.
`normal`	Computes surface normals from derivative cross products.
`tangent`	Computes curve or boundary tangent directions.
`curvature`	Computes curvature quantities from first and second derivatives.
`interpolation`	Interpolates arbitrary control-point fields through the spline basis.
`dxdknot`, `d2xd2knot`	Sensitivities with respect to knot values.
`dxdctrlpts`	Sensitivity of evaluated coordinates with respect to control points.
`dxdweight`	Sensitivity of rational geometry to NURBS weights.
`ctrlpt_id`	Maps tensor-product indices to flat control-point IDs.
`connectivity`	Builds element or control-point connectivity arrays.
`gather_boundary_ctrlpts`	Collects control points on selected boundaries.
`activate_boundary`	Marks spline boundaries for boundary conditions or contact.
`update_boundary`	Updates cached boundary geometry after control-point changes.
`boundary_normal`	Computes normals on active spline boundaries.
`boundary_tangent`	Computes tangents on active spline boundaries.
`boundary_inversion`	Finds boundary parameter coordinates closest to a physical point.

Entry point	User-visible role
`insert_knot`, `knot_insertion`	Inserts knots while preserving geometry.
`refine_knot`, `knot_refinement`	Adds multiple knots/elements for h-refinement.
`projection`	Closest-point projection wrapper on the primitive.
`inversion`	Solves for parameter coordinates corresponding to a physical point.
`distance`	Returns closest distance and associated parameter data.
`curve_inversion`	Newton/initial-guess closest-point inversion for curves.
`surface_inversion`	Newton/initial-guess closest-point inversion for surfaces.
`curve_curve_distance`	Computes closest distance between two spline curves.
`surface_surface_distance`	Computes closest distance between two spline surfaces.
`surface_ground_distance`	Computes closest distance between a spline surface and a ground plane.
`global_interpolate_bspline_curve`	Builds a B-spline curve passing through input points.
`global_interpolate_bspline_surface`	Builds an interpolating B-spline surface from gridded data.
`global_approximate_bspline_curve`	Least-squares approximation curve with optional fixed points.
`global_approximate_bspline_surface`	Least-squares approximation surface.
`bspline_curve_fitting_least_squares`	Fits an existing B-spline curve to point data.
`nurbs_curve_fitting_gauss_newton`	Gauss-Newton rational curve fitting with weights/control points.
`bspline_surface_fitting_least_squares`	Fits a B-spline surface by linear least squares.
`bspline_multisurface_fitting_least_squares`	Fits multiple connected B-spline surfaces.
`bspline_surface_fitting_gauss_newton2D`	Nonlinear 2D surface fitting with Gauss-Newton updates.
`bspline_surface_fitting_newton_raphson2D`	Newton-Raphson surface fitting with Hessian information.
`read`, `write`	Geometry I/O methods where implemented by the primitive or plotting helper.
`visualize`	Plots or exports curves, surfaces, control nets, and fitted geometry.

Entry point	User-visible role
`estimate_length`, `estimate_area`	Numerical quadrature estimates for curve length and surface area.
`convex_hull`	Diagnostic hull computation used by fitting and boundary classification.

NURBS primitives are the geometry objects passed to IGA and IGA-MPM workflows. Use B-spline primitives when all weights are one; use NURBS primitives when exact conics, rational weights, or weight sensitivities are needed.

32 Compatibility Matrix

Feature combination	Required entry points and options	Why the rule matters
`TLMPM`	`MPM.set_configuration(solver_type="Explicit",material_type="Solid",configuration="TLMPM",discretization="FEM")`	Total-Lagrangian kernels assume solid FEM shape functions and explicit update storage.
`Lightweight` MPM	`MPM.set_configuration(mode="Lightweight",solver_type="Explicit",configuration="ULMPM")`	Lightweight mode skips fields required by implicit, TL, and coupled branches.
`fluid_level_set`	`MPM.set_configuration(solver_type="Implicit",material_type="Fluid",discretization="FDM",fluid_level_set=True)` and `MPM.add_element(ElementType="Staggered")`	The free-surface θ is assembled into a staggered-grid pressure projection, not the FEM solid matrix.
`density_projection`	Same implicit fluid FDM setup as `fluid_level_set`; add `density_projection=True` and its tolerance/shift limits.	The projection reuses fluid/air/solid cell classification and pressure-face data.
`solid_sdf_cut_cell`	Same implicit fluid FDM setup; add `solid_sdf_cut_cell=True` and optionally `solid_cut_cell_min_fraction`.	Cut-cell coefficients require solid SDF face fractions on MAC pressure faces.
Explicit MPM	`MPM.set_configuration(solver_type="Explicit",discretization="FEM")` and `MPM.set_solver`. Do not use `assemble_type="HashTriplet"`.	Explicit update does not build the implicit sparse tangent path.

Feature combination	Required entry points and options	Why the rule matters
Explicit `TwoPhaseSingleLayer`	`solver_type="Explicit"`, `material_type="TwoPhaseSingleLayer"`, Cartesian 2D or 3D, ULMPM/FEM, PIC/FLIP transfer.	Axisymmetric runs, 3D B-Bar, 3D Affine/Taylor projection, MLS, Gauss-cell force assembly, contact, and double-layer explicit runs are rejected in the current source.
Implicit solid MPM	`MPM.set_configuration(solver_type="Implicit",material_type="Solid",discretization="FEM")`, `set_implicit_solver_parameters(integration_scheme="Newmark",assemble_type=...)`	The global solve is a Newmark displacement increment solve with material/geometric/Jaumann tangent blocks.
Implicit fluid MPM	`MPM.set_configuration(solver_type="Implicit",material_type="Fluid",discretization="FDM")`, `add_element(ElementType="Staggered")`, `set_implicit_solver_parameters(linear_solver="PCG"or"MGPCG")`	The pressure unknowns live at cell centers and velocities on MAC faces.
Double-layer `SemiImplicit` MPM	`material_type="TwoPhaseDoubleLayer"`, top-level `discretization="FEM"`, `pressure_solver="MGPCG"`, and `set_semi_implicit_solver_parameters`.	Solid particles use the collocated MPM grid, fluid velocities use the MAC grid, and pressure uses the cell-centered MGPCG hierarchy; axisymmetry is rejected.

Feature combination	Required entry points and options	Why the rule matters
`BlockScansparse_grid`	`MPM.set_configuration(sparse_grid=\{...\})` with `solver_type="Explicit"` or implicit solid MPM on Taichi CPU/CUDA, normal mode, non-adaptive FEM elements, no `G2P2G`, no MLS, no Gauss-cell force assembly, no MPM internal `contact_detection`, no CFDEM, no PBC/absorbing boundary. Contact-based MPDEM/DEMPM Lagrangian coupling is allowed because the sparse grid only changes the MPM background grid. Covered explicit paths include ordinary 2D/3D solid ULMPM/TLMPM, ordinary 3D fluid ULMPM, and explicit `TwoPhaseSingleLayer`. Covered implicit solid assembly paths are `MatrixFree`, `COO`, and `HashTriplet`.	BlockScan remaps particle support nodes to compact sparse node ids. Implicit solid uses compact-node strides for residuals, tangents, constraints, and reactions. Double-layer semi-implicit pressure, implicit FDM fluid, and CFDEM use different unknown indexing and are rejected.
Adaptive implicit solid	`MPM.add_element(...,AdaptiveGrid={...})` plus implicit FEM; use shape-function hanging constraints.	Penalty and bridging-domain adaptive constraints are rejected in the current implicit solid path.
DEM `scheme`	`DEM.set_configuration(scheme=...)` with `DEM`, `LSDEM`, `LSMPM`, `PolySuperEllipsoid`, `PolySuperQuadrics`, or `AffineBody`.	The scheme selects body storage, template type, contact geometry kernels, and for `AffineBody` the affine engine/-solver/recorder path.
`LSMPM` soft bodies	`DEM.set_configuration(scheme="LSMPM",shape_function=...)`, allocate `max_soft_body_number`, `max_material_point_number`, `levelset_grid_number`, `surface_node_number`, and use `create_body/add_body` with `BodyType="SoftBody"`.	Soft bodies use TLMPM material points plus a deforming local level set. Contact search reuses DEM neighbor classes, but the LSMPM contact-node table includes rigid surface nodes and soft surface material points; soft output is stored in `LSMPMSoftPoint*.npz`.

Feature combination	Required entry points and options	Why the rule matters
`MPMsoft_particle` mode	`MPM.set_configuration(mode="SoftParticle")` or `soft_particle=True`, dimension 3, `configuration="TLMPM"`, `solver_type="Explicit"`, `material_type="Solid"`, `discretization="FEM"`, `stabilize=None`, `gauss_number=0`.	The soft-particle kernels live in `src/mpm/soft_particle`; current runnable level-set examples use the DEM/MPDEM bridge until MPM owns the full level-set scene.
`LSMPMsoft-affineIPC`	`DEM.set_configuration(scheme="LSMPM",soft_rigid_contact="IPC")`, add `SoftBody` and `AffineBody` through generate/create, set `DEM.set_affine_body_parameters`, and run through `MPDEM/DEMPM`.	The mixed operator solves soft MPM grid dofs and affine control points in one Newton/PCG system and uses point-level-set, point-triangle, and edge-edge IPC geometry.
DEM neighbor search	`DEM.choose_neighbor()` or configuration search values `Brust`, `LinkedCell`, `HierarchicalLinkedCell`, `BVH`.	Search data structures must match size distribution and wall/level-set geometry.
CFDEM implicit incompressible coupling	Configure the MPM subsolver as implicit fluid FDM in 3D, configure DEM bodies, then create `DEMPM` with `coupling_scheme="CFDEM"`.	The coupling samples a 3D fluid grid and exchanges drag/pressure forces with DEM spheres.
IGAMPM explicit DEM law	Construct `gt.IGAMPM(iga,mpm,contact_model="Linear")` or `"HertzMindlin"` before MPM particle allocation, configure two explicit children in 3D Cartesian coordinates, and call `add_property` for every active MPM-material/IGA-patch pair.	The native Lagrangian MPM branch exchanges finite-radius point–NURBS contact forces with IGA through the shared DEM law and rational basis. Rolling friction is unavailable because IGA control points have no rotational DOFs.
IGAMPM IPC	Use `gt.IGAMPM(iga,mpm)`, configure both sub-solvers as implicit, choose `contact_model="IPC"`, and select `assemble_type="COO"` or `"HashTriplet"`. The Direct ULMPM child accepts `NeoHookean`, finite-strain `DruckerPrager`, or associated finite-strain `VonMises`.	IPC coupling assembles closest-point barrier/friction blocks and rejects explicit or mixed solver pairs. Plastic MPM requires ULMPM, keeps IGA elastic, and commits or restores its history together with `F0`.

Feature combination	Required entry points and options	Why the rule matters
Random-field MPM	`MPM.set_configuration(random_field=True)` before `MPM.add_material`; each supported material needs `MaterialFile`.	Per-particle material state is allocated and filled during material/body insertion.

33 Practical Workflows

33.1 Explicit Solid MPM

Entry point	What it contributes
`gt.init(dim=2or3)`	Sets global dimension before Taichi fields and kernels are specialized.
`MPM(...)`	Creates the facade, simulation state, scene manager, and empty material/body registries.
`set_configuration`	Use `solver_type="Explicit"`, `material_type="Solid"`, `discretization="FEM"`, and choose mapping/stabilization.
`set_solver`	Defines `Timestep`, `SimulationTime`, `CFL`, output interval, and save path.
`memory_allocate`	Reserves material slots, particle capacity, constraint capacity, and optional neighbor/coupling buffers.
`add_material`	Registers the solid constitutive model; random-field files must already be enabled by configuration.
`add_element`	Builds `Q4N2D`/`R8N3D` background elements and shape functions.
`add_region` and `add_body`	Define sampling regions and insert material points.
`add_boundary_condition`	Adds grid velocity, reflection, friction, traction, displacement, or solid-cell constraints.
`select_save_data`	Chooses particle, grid, and object output channels before the recorder is finalized.
`run`	Calls missing essentials, checks timestep if requested, advances explicit particle-grid updates, and writes output.

Prefer `MUSL` mapping for general large-deformation examples. Enable `B-BarMethod` or `F-Bar Method` only for material and deformation regimes where the corresponding strain/Jacobian smoothing is valid.

33.2 Implicit Solid MPM

Entry point	What it contributes
`set_configuration`	Use `solver_type="Implicit"`, `material_type="Solid"`, `discretization="FEM"`, and the intended small- or finite-strain material family.
`set_implicit_solver_parameters`	Selects Newmark parameters, residual/displacement tolerances, linear solver, line search, and `assemble_type`.
`add_material`	Must provide a constitutive tangent when assembled stiffness is required; see implicit tangent theory.
`add_element`	Defines the FEM background grid and, for adaptive runs, the hierarchy and hanging-node constraints.
`add_boundary_condition`	Strong/penalty Dirichlet choices determine how matrix rows, RHS, and reaction forces are handled.
`add_contact`	Optional IPC/contact branch; contact capacities and CCD options must be sized before assembly.
`run`	Repeatedly assembles internal force, mass/damping/stiffness blocks, solves the Newton increment, and updates particles.

`MatrixFree` lowers matrix memory but exposes less assembled diagnostic state. `COO` is easier for strong Dirichlet conditions and reaction-force extraction. `HashTriplet` reduces duplicate triplets when capacity and symmetry assumptions are correct.

33.3 Incompressible Fluid MPM

Entry point	What it contributes
`set_configuration`	Use `solver_type="Implicit"`, `material_type="Fluid"`, `discretization="FDM"`; add `fluid_level_set`, `density_projection`, or `solid_sdf_cut_cell` only on this path.
`set_implicit_solver_parameters`	Selects the pressure linear solver, usually `PCG` or `MGPCG`, and pressure iteration tolerances.
`add_material`	Registers `Newtonian` or supported fluid material parameters, including density, bulk/viscosity, atmospheric pressure, and surface tension.
`add_element`	Use `ElementType="Staggered"` so pressure cells and MAC velocity faces exist.
`add_body`	Inserts fluid particles; these are also used to rebuild `fluid_sdf` for free-surface projection.

Entry point	What it contributes
`body.Template.nParticlesPerCell`	Sets samples per cell axis, so the full cell count is its square in 2D or cube in 3D. Increase it until cell occupancy, volume, pressure, and free-surface observables converge, and size `max_particle_number` accordingly.
`add_boundary_condition`	`SolidCell` boundaries and velocity constraints define wall and no-through faces.
`GEOTAICHI_INCOMPRESSIBLE_P2G`	Optional implementation diagnostic: `auto` (default), `atomic`, `shared`, `shared_memory`, or `cell_reduced`. Auto uses CUDA shared-memory cell reduction on CUDA and atomic scatter elsewhere; forcing a shared alias on a non-CUDA backend selects the cell-reduced fallback.
`run`	Builds or updates `fluid_sdf`, assembles the pressure projection, corrects face velocities, optionally density-projects particles, and advects.

`fluid_level_set=True` activates the ghost-fluid θ treatment documented in fluid level-set pressure projection. `density_projection=True` should be treated as a particle regularization step that depends on the same classification quality.

Current incompressible SDF/IBM examples under `examples/mpm/IncompressibleFluid` are:

Script	What it validates or demonstrates
`fixed_sdf_coupling_2d.py`	2D fixed solid SDF coupling on the staggered incompressible MPM grid.
`validate_fixed_sdf_coupling_2d.py`	Re-reads saved NPZ output from the 2D fixed-SDF case and checks final velocity/-pressure fields and solid-cell classification.
`lid_driven_cavity_2d.py`	2D lid-driven cavity benchmark for no-slip wall handling and incompressible pressure projection.
`cylinder_flow_ibm_2d.py`	2D cylinder flow with immersed-boundary/SDF solid treatment and periodic-flow style setup for wake testing.
`fixed_sdf_coupling_3d.py`	3D fixed SDF coupling path on the incompressible fluid grid.
`lsdem_coupling_3d.py`	3D LSDEM volume-fraction IBM with fictitious interior fluid, mixed-density pressure projection, and fixed-wall cut cells.
`surface_tension_laplace_validation.py`	Surface-tension Laplace pressure validation using the fluid level-set surface workflow.

33.4 Semi-Resolved Spheres and CFDEM

Entry point	What it contributes
`DEM(...)` and `MPM(...)`	Construct and configure the sub-solvers before coupling.
`DEMPM(dem,mpm,...)`	Wraps the DEM and MPM facades and selects coupling state.
`set_configuration(coupling_scheme="CFDEM")`	Enables the CFDEM branch and synchronizes shared domain/coupling settings.
`add_property`	Registers DEM-MPM interaction parameters and drag/contact property pairs.
`add_essentials` or `run(drag_model=...)`	Builds coupling data and selects the drag, buoyancy, pressure-gradient, and virtual-mass closures.
`update_contact_properties`	Allows post-setup property changes before the relevant contact/coupling kernels use them.

The MPM side must be an implicit 3D fluid FDM solve for incompressible CFDEM. The DEM side must already have material attributes, bodies, walls, search, and contact properties configured.

For ordinary DEM spheres, the incompressible branch is dense semi-resolved: the Gaussian solid fraction enters $\partial_t \epsilon_f + \nabla \cdot (\epsilon_f \boldsymbol{u}_f) = 0$, and drag reaction uses porosity-weighted MAC mass. For LSDEM, the same public coupled workflow selects the fully resolved volume-fraction IBM described in Fully Resolved LSDEM Volume-Fraction IBM. Moving LSDEM bodies do not populate the cut-cell SDF; `solid_sdf_cut_cell=True` therefore applies only to separately declared fixed `SolidCell` walls in that route.

33.5 DEM Packing and Contact

Entry point	What it contributes
`DEM(...)`	Creates DEM scene, material, body, wall, contact, and recorder managers.
`set_configuration`	Sets domain, gravity, boundary behavior, scheme, search mode, and energy/output flags.
`set_solver`	Sets timestep, CFL/adaptive controls, simulation time, save interval, and output path.
`memory_allocate`	Reserves particle, sphere, clump, wall, material, and contact-history capacities.
`add_region`, `add_attribute`, `add_template`	Define insertion regions, material/body attributes, and clump/level-set/general templates.
`create_body`, `add_body`, `add_body_from_file`	Generate or import particles/clumps/rigid bodies.

Entry point	What it contributes
`add_wall`, `add_wall_from_file`	Generate walls or read wall geometry.
`choose_neighbor`	Builds the selected broad-phase search structure.
`choose_contact_model` and `add_property`	Select contact laws and assign material-pair surface parameters.
`run`	Finalizes essentials, updates contact lists, integrates rigid-body motion, and records output.

Use `LinkedCell` for relatively uniform spheres, `HierarchicalLinkedCell` for strong polydispersity, and `BVH` for nonuniform or level-set/clump geometry. Contact surface properties should be assigned before the first run.

33.6 DEM AffineBody Workflow

Entry point	What it contributes
`DEM.set_configuration`	Use `scheme="AffineBody"` or `affine_body=True`. Optional `affine_assemble_type` forwards to the AffineBody linear-system backend selector. The DEM `search` value may be `LinkedCell`, `HierarchicalLinkedCell`, or `BVH`; the affine neighbor helper reuses those DEM search classes on affine surface primitives.
`DEM.set_affine_body_parameters` or `affine_body_parameters`	Sets IPC and Newton controls such as `assemble_type`, `YoungModulus`, `dhat`, `barrier_stiffness`, `contact_damping_stiffness`, `friction_epsv`, Newton/PCG tolerances, `hessian_shift`, `max_step`, and CCD/ACCD options. Level-set templates may additionally enable `levelset_auto_initialize` and set the initial anchor stiffness, continuation ratio, positive-gap tolerance, stiffness growth, stage cap, and iteration cap.
`DEM.memory_allocate`	Use `max_affine_body_number`; if omitted, the AffineBody branch falls back to `max_rigid_body_number`. Wall and material capacities are still allocated by the ordinary DEM memory keys.

Entry point	What it contributes
`DEM.add_template`	Register a template whose `Object` provides a non-empty triangular surface mesh. The template vertices, faces, volume, and surface measures define the affine basis, mass matrix, quadrature, and VTK output mesh. The default `TriangleMesh` representation also uses mesh contact primitives. With `ContactRepresentation="LevelSet"`, the object must provide its signed-distance grid; that field defines body–body narrow phase.
`DEM.create_body` and `DEM.add_body`	Insert affine bodies from the template. Body keys follow DEM style: `Name`, `BodyPoint`, `GroupID`, `MaterialID`, `ScaleFactor`/`Radius`/ `BoundingRadius`, `BodyOrientation`, initial translational/angular velocity, `FixMotion`, and optional per-body `YoungModulus` and `Friction`.
`DEM.add_body` with `GenerateType="Distribute"`	For `BodyType="AffineBody"`, samples intentionally overlapping particles to the requested `Porosity`, then runs a frictionless IPC feasibility projection before the generated state is exposed to dynamics. Its optional packing-projection controls include `Dhat`, `BarrierStiffness`, `AnchorStiffness`, `ContinuationRatio`, `StiffnessGrowth`, `MaximumStages`, `MaximumIterations`, `GradientTolerance`, `GapTolerance`, and Krylov/line-search limits. Native level-set templates use their SDF directly. Watertight, consistently wound TriangleMesh templates remain meshes: `GaussianEpsilon`, `GaussianWeight`, `GaussianCutoff`, `MinkowskiWeight`, `InclusionWeight`, `SeparationWeight`, `SeparationStiffnessGrowth`, and `Untangle*` controls govern direct surface untangling (`UntangleStiffnessGrowth` remains a compatibility alias); strict point–triangle and edge–edge IPC then uses `CCDType="ACCD"`, `CCDEta`, `ACCDTolerance`, and `CCDMaximumIterations`. Untangling, strict IPC, ACCD, matrix-free Hessian-vector, Krylov, and line-search iterations are Taichi device kernels; TriangleMesh generation does not build an SDF proxy.
`DEM.add_wall` and `DEM.add_wall_from_file`	Defines analytic plane walls or triangular facet walls for body-wall IPC contact. Plane walls use a gap barrier; facet walls use point-triangle distances.

Entry point	What it contributes
`DEM.add_property`	For `scheme="AffineBody"`, properties with `dType="particle-particle"`, `"particle-wall"`, or `"all"` provide the friction coefficient consumed by the affine IPC friction model. The ordinary DEM spring-slider model selected by `choose_contact_model` is not used by AffineBody.
`DEM.run`	Each step predicts affine control points, rebuilds or refits neighbor candidates, assembles inertia, body force, rigidity, and IPC normal barriers, solves the Newton PCG system, applies CCD/ACCD and line search, accepts the affine state, and records output. The mesh representation may also assemble friction and contact damping. The level-set representation assembles lagged IPC friction for particle–particle, particle–wall, and mixed soft–affine contact; fully implicit level-set friction and level-set contact damping remain unsupported. Level-set motion may use ACCD conservative advancement.

AffineBody output is separate from ordinary DEM sphere/clump output. `AffineBody*.npz` stores the affine surface vertices, faces, body/group metadata, four control points $\boldsymbol{y}$, control velocities, and material ids. `GraphicAffineBody*.vtu` stores the reconstructed surface mesh for visualization. The engine also exposes centers, bounding boxes, bounding radii, mass matrices, Young moduli, friction values, and solver diagnostics through the recorder snapshot.

33.7 DEM Soft-Rigid LSMPM Workflow

Entry point	What it contributes
`DEM.set_configuration`	Use `scheme="LSMPM"` and optionally `soft_shape_function="Linear"`, `"QuadBSpline"`, or `"CubicBSpline"`. `shape_function` and `lsmpm_shape_function` remain aliases. Soft-level-set reinitialization is disabled by default for ReferenceMap and can be controlled by `soft_levelset_reinitialization` or the flat `soft_levelset_reinit_*` keys. Select level-set transport with `soft_levelset_advection_scheme="ReferenceMap"` (default), `"MacCormack"`, or `"WENO5"`; `"SemiLagrangian"` aliases ReferenceMap rather than MacCormack. WENO5 uses `soft_levelset_advection_cfl` for subcycling. Compact mechanical-grid storage, grid topology, PIC fraction, and domain checks are controlled by the separate `soft_grid_*`, `soft_pic_fraction`, and `soft_levelset_domain_*` keys. The DEM `search` value may be `LinkedCell`, `HierarchicalLinkedCell`, or `BVH`; these existing DEM neighbor classes are reused for soft-rigid and soft-soft broad phase.
`DEM.memory_allocate`	Allocate ordinary level-set storage `max_rigid_body_number`, `levelset_grid_number`, and `surface_node_number`, plus LSMPM storage `max_soft_body_number` and `max_material_point_number`. Increase `point_coordination_number` when a surface point may see many neighboring bodies or walls. A separately refined/compact mechanical grid may also need `soft_grid_number`; allocate `max_soft_velocity_constraint` when soft grid velocity constraints are used. Cached support capacities can be provided explicitly or inferred from templates preprocessed before allocation.
`DEM.preprocess_soft_grid_template`	Precomputes LSMPM material points, mechanical-grid topology, interpolation support, reference volumes, and exact capacity counts for a registered template. Use the returned `material_point_number`, `soft_grid_number`, and template-support counts to size `memory_allocate`.
`DEM.add_attribute`	Provide density, local damping, and soft finite-strain parameters. `ConstitutiveModel` accepts `NeoHookean`, `HenckyElastic`, `MooneyRivlin`, `Gent`, and `Hydrogel`.
`DEM.add_template`	Register the level-set template mesh/SDF exactly as for rigid LSDEM. Soft bodies do not require a separate template type; the same template supplies the initial local level set, surface mesh, inertia, and sampling grid.

Entry point	What it contributes
`DEM.create_body`	Insert rigid bodies with `BodyType="RigidBody"` and soft bodies with `BodyType="SoftBody"`. Soft templates use `MaterialPointsPerCell` or `ParticlePerCell` to choose the MPM particle density.
`DEM.add_body`	Generate or lattice-pack multiple soft bodies with `BodyType="SoftBody"` and `GenerateType="Generate"` or `"Lattice"`. This route uses the same region, radius distribution, orientation, overlap, and history-check logic as rigid level-set generation.
`DEM.choose_contact_model`, `DEM.add_property`	Linear, Hertz-Mindlin, energy-conserving penalty, and barrier LS contact models can be assigned to soft-rigid and soft-soft contacts. For zero-cutoff models every candidate surface point with $g < 0$ is active.
`DEM.select_save_data`	Use `particle=True` or `soft=True` to save `LSMPMSoftPoint*.npz`. Use `surface=True` and `grid=True` when PostPlot should reconstruct the deforming soft surface from the saved level set.
`DEM.run`	Each step resets contact forces, updates DEM/LSMPM contact lists, assembles contact forces, advances soft bodies by TLMPM, reconstructs each soft level set by ReferenceMap or uses the selected incremental transport, and, when redistance is enabled, monitors $\max \lvert \lVert \nabla \phi \rVert - 1 \rvert$ and reinitializes in a narrow band when the threshold is exceeded. It updates soft mass centers and bounds from material points, and writes the selected outputs.

The soft TLMPM `shape_function` controls material-point/grid interpolation and stress update. Supported choices are `Linear`, `QuadBSpline`, and `CubicBSpline`; GIMP is intentionally not part of this LSMPM branch. This setting is independent of the soft contact force distribution. For contact, stored soft surface material points are used as contact nodes, each with contact radius $\frac{1}{2}(V^0)^{1/3}$ and surface weight $(V^0)^{2/3}$. The deforming local SDF is reconstructed after the TLMPM step by default ReferenceMap, with MacCormack and WENO5–SSPRK3 as incremental alternatives. ReferenceMap samples the initial SDF stored once at body insertion, and redistance or volume shifts do not modify that source. Its affine material-point inverse patches and local metric correction do not imply an exact curved-surface distance or exact finite-band normal. Redistance is disabled by default for ReferenceMap; when explicitly enabled, it runs only when the monitored signed-distance error exceeds the configured threshold. The WENO5 default maximum Courant number is 0.20 and large mechanical steps are subcycled. Defaults are `grad_threshold=0.2`, `check_interval=5`, `monitor_band_cells=4`, `reinit_band_cells=8`, `pseudo_time_cfl=0.3`, and `iterations=0`; zero iterations selects $\lceil W_r/(c_\tau \Delta x_\phi) \rceil$ pseudo-time steps. When a contact force must be applied to a soft body, the kernel finds the nearest soft surface point, uses support radius $2\max(\Delta x_\phi, (V^0)^{1/3})$, and distributes the force to neighboring soft surface points with

compact weights $(1 - r^2/h^2)^2$. If no positive weight is found, the nearest point receives the full force. This is a local kernel spreading method, not a polynomial MLS reconstruction and not the MPM `moving_least_square` transfer path.

Recorder and PostPlot output are intentionally split. Rigid level-set surfaces are written as `GraphicLSDEMSurface*.vtu`; in LSMPM this VTK surface excludes soft bodies. Soft material points are written as `LSMPMSoftPoint*.npz` and, when visualization is enabled, `GraphicLSMPMPoint*.vtu`. PostPlot can also read `LSDEMGrid*.npz` and `BoundingBox*.npz`, run marching cubes on each soft body's saved SDF, and write `GraphicLSMPMSoftSurface*.vtu`. If `write_von_mises_stress=True`, PostPlot computes von-Mises stress from the stored soft material-point stress tensor and writes it on `GraphicLSMPMPoint*.vtu`; the recorder itself keeps the raw stress tensor in the NPZ file.

Soft-body contact forces should be inspected on `LSMPMSoftPoint*.npz`. A soft body row in `LSDEMRigid*.npz` may have zero `contact_force` because soft contact forces are accumulated on material points and then mapped to the soft TLMPM grid. Rigid bodies in the same simulation still receive ordinary body resultant force and torque in `LSDEMRigid*.npz`.

33.8 IGA and IGAMPM

Entry point	What it contributes
`IGA(...)`	Creates the NURBS/IGA facade.
`IGA.set_configuration`	Selects dimension and solver family.
`IGA.add_primitives`	Registers NURBS/B-spline curves, surfaces, or volumes.
`IGA.add_boundary_condition`	Adds Dirichlet/Neumann conditions on geometric entities.
`IGA.add_material`	Registers material energy or constitutive parameters.
`IGA.add_element`	Defines polynomial degree/element extraction data.
`IGA.set_solver`	Sets nonlinear/linear solver and time controls.
`IGA.build` and `IGA.run`	Construct the solver objects and advance the analysis.
`IGAMPM`	Use `geotaichi.IGAMPM` to couple ordinary `IGA` and `MPM` setup flows. IPC requires two implicit children; shared DEM Linear/Hertz–Mindlin contact requires two explicit children.
`IGAMPM.add_property`	Registers one explicit MPM-material/IGA-patch DEM law before build.
`IGA-MPMIPCCoupling`	Low-level helper for closest-point contact, barrier/friction assembly, and monolithic block extraction.

Use pure `IGA` for NURBS-based solid analysis. Use `IGAMPM` when a script needs NURBS/IGA geometry coupled to MPM particles through explicit DEM-law contact or implicit IPC. The MPM sub-solver should still be configured through the normal `MPM` facade.

For standalone `IGA.add_boundary_condition(dirichlet=boundary)`, `DirichletBoundary.append` values mean total displacement from the initial control points in explicit IGA, and displacement increments in implicit IGA. Implicit Newton initializes constrained increments before line search and minimizes over the remaining free degrees of freedom; prescribed motion can therefore supply positive mechanical work. The square-column example under `examples/iga/twisting_column_3d/` supports both solvers and provides a separate saved-state evaluator. See the constrained energy discussion in IGA and IGA–MPM.

34 Working-Tree Schema Additions

This section records keys found by the generated working-tree capability audit that require explicit path-level coverage in addition to the preceding model tables. Canonical lower-case material spellings remain preferred when both forms are accepted.

34.1 Constitutive and random-field keys

Full path	Accepted value/default	Meaning and effect
`MPM.add_material.material.InitialVoidRatio`	real; default -1	Modified Cam-Clay alias for `void_ratio_initial`; positive `porosity` overrides it through $e_0 = n/(1-n)$.
`MPM.add_material.material.Subloading`	boolean; default `False`	Modified Cam-Clay alias that activates the subloading-surface branch.
`MPM.add_material.material.SubloadingU`	real; default (0)	Modified Cam-Clay alias for the subloading evolution parameter `subloading_u`.
`MPM.add_material.material.Bonding`	boolean; default `False`	Modified Cam-Clay alias enabling the bonded/bounding-surface extension.
`MPM.add_material.material.BoundingSurface`	boolean; defaults to `Bonding`	Explicit bounding-surface switch; when supplied it takes precedence over the preceding bonding activation value.
`MPM.add_material.material.MCA`	real; default (0)	Upper-case alias for Modified Cam-Clay bounding/bonding coefficient `mc_a`.
`MPM.add_material.material.MCB`	real; default (0)	Upper-case alias for Modified Cam-Clay bounding/bonding coefficient `mc_b`.
`MPM.add_material.material.MCC`	real; default (0)	Upper-case alias for Modified Cam-Clay bounding/bonding coefficient `mc_c`.

Full path	Accepted value/default	Meaning and effect
`MPM.add_material.material.MCD`	real; default (0)	Upper-case alias for Modified Cam-Clay bounding/bonding coefficient `mc_d`.
`MPM.add_material.material.Degradation`	real; default (0)	Modified Cam-Clay alias for the bonding degradation coefficient `m_degradation`.
`MPM.add_material.material.BondedShearModulus`	real; default (0)	Modified Cam-Clay alias for the bonded shear parameter `m_shear`.
`MPM.add_material.material.HydrateSaturation`	real; default (0)	Modified Cam-Clay alias for hydrate saturation `s_h`.
`MPM.add_material.material.PressureRef`	positive pressure; default (100)	NorSand alias for `p_ref`; it sets the reference pressure used to initialize pressure-dependent bulk stiffness.
`MPM.add_material.material.e_init`	real; required	SanisandMS initial void ratio, also accepted as `InitialVoidRatio`.
`MPM.add_material.material.emax`	real; required	SanisandMS maximum void-ratio bound used by its state relation.
`MPM.add_material.material.emin`	real; required	SanisandMS minimum void-ratio bound used by its state relation.
`MPM.add_material.random_field.ResidualFraction`	real; default (1)	Residual fraction stored by the material random-field reader together with `MaterialFile`; confirm the selected model implements random-field input.

34.2 LSMPM template and solid-plane boundary keys

These template keys are consumed by the DEM-owned public LSMPM workflow even though their preprocessing implementation resides under `src/mpm`.

Full path	Accepted value/default	Meaning and effect
`DEM.create_body.Template.CenterMaterialPointQuadrature`	boolean; default `False`	Recenters automatically sampled hexahedral LSMPM material points so their mean coincides with the template center.
`DEM.create_body.Template.MaterialPointCoordinates`	finite array of shape $(N, 3)$; optional	Replaces automatic LSMPM material-point sampling. It is rejected for the tetrahedral grid route.

Full path	Accepted value/default	Meaning and effect
`DEM.create_body.Template.MaterialPointVolume`	positive scalar or positive length-N array; optional	Supplies reference volume for each custom material point; a scalar is broadcast to all points.
`DEM.create_body.Template.MechanicalGridSpacing`	positive finite length; template/configuration fallback	Sets LSMPM mechanical-grid spacing, independent of the level-set grid spacing.
`DEM.create_body.Template.MechanicalGridRefinement`	refinement dict; optional	Selects local tetrahedral mechanical-grid refinement. The hexahedral LSMPM route rejects this key.
`DEM.create_body.Template.ReferenceVolume`	positive finite volume; template-volume fallback	Renormalizes all material-point reference volumes so that their sum equals the requested body reference volume.
`DEM.create_body.Template.MechanicalGridBoundary`	region dict or list of region dicts; optional	Selects compact mechanical-grid nodes for fixed velocity constraints; allocate `max_soft_velocity_constraint` accordingly.
`MPM.add_boundary_condition.boundary.PlanePoint`	coordinate vector; `StartPoint` fallback	Alias for `Point` in a two-dimensional `SolidPlaneCell` boundary.
`MPM.add_boundary_condition.boundary.PlaneNormal`	nonzero coordinate vector; required unless `Norm` is supplied	Alias for the normalized solid-plane normal used to classify the double-layer cell boundary.

34.3 DEM batch insertion and local-damping aliases

Full path	Accepted value/default	Meaning and effect
`DEM.create_body_batch.body.Template.BodyPoints`	finite array $(N, 3)$; required	Batch center/mass-center coordinates for rigid or soft level-set bodies.
`DEM.create_body_batch.body.Template.BoundingRadii`	positive finite length-N array; required	Per-body bounding radii used to scale the registered level-set template.
`DEM.create_body_batch.body.Template.BodyOrientationsRadians`	finite array $(N, 3)$; zeros by default	Per-body Euler-angle orientation in radians for batch insertion.

Full path	Accepted value/default	Meaning and effect
DEM.create_body.Template.force_local_damping	real scalar; LocalDamping fallback	Lower-case compatibility alias for ForceLocalDamping on AffineBody templates; it damps translational force updates.
DEM.create_body.Template.torque_local_damping	real scalar; LocalDamping fallback	Lower-case compatibility alias for TorqueLocalDamping on AffineBody templates; it damps rotational torque updates.

A Condensed Keyword Index

This index lists keyword names extracted from user-facing readers. Some keys are internal restart or output fields and should not be used unless a loader or postprocessor asks for them.

A.1 MPM Keyword Names

AOSOA, AdaptiveGrid, AdaptiveStep, BackgroundColor, BodyID, BoundaryType, BridgingCells, BridgingDomain, BufferCells, CFL, CameraUp, CellThickness, CheckHistory, ConfiningPressure, Density, Displacement, DisplacementX, DisplacementY, DisplacementZ, ElementSize, ElementType, EndPoint, ExternalForce, File, FileType, FixVelocity, FluidPressure, Friction, GhostCell, HangingConstraintMode, HangingShapeCapacityFactor, InitialVelocity, InternalStress, LookAt, LookFrom, MaterialFile, MaterialID, MoveVelocity, NLevel, Norm, Offset, Orientation, OuterNormal, ParticleFile, ParticleNumber, ParticleSplitBatch, ParticleSplitInterval, ParticleStress, Penalty, PenaltyBeta, PenaltyIterations, Period, Phase, PointLight, PorePressure, Pressure, RefineCriterion, RefineInterval, RefineParticles, RefineRatio, RefineThreshold, RegionFunction, RegionName, Restart, RigidBody, SaveInterval, SavePath, ScaleFactor, Sign, SimulationTime, SolidCellThickness, SplitInteriorOnly, SplitInteriorVolumeFraction, StartPoint, Template, Timestep, Traction, Velocity, VelocityX, VelocityY, VelocityZ, Vertices, VirtualForce, VirtualStress, Visualize, VisualizeInterval, WindowSize, WriteFile, WritePath, alphaPIC, assemble_type, background_damping, boundary, boundary_direction_detection, calculate_reaction_force, configuration, coupling, delayed_fluid_advection, density_projection, density_projection_error_clamp, density_projection_interior_only, density_projection_max_shift_ratio, density_projection_tolerance, discretization, displacement_tolerance, dof_multiplier, domain, drift_correct, fbar_fraction, fluid_domain_volume_fraction, fluid_level_set, fluid_level_set_function, free_surface_detection, function, gauss_number, gravity, gravity_field, ibm_function, ibm_source_function, integration_scheme, is_2DAxisy, jacobian_clamp, linear_solver, linear_solver_relative_tolerance, mapping, material_type, matrix_symmetric, max_constraint_number, max_material_number, max_particle_number, mode, moving_least_square, multilevel, newmark_parameter, particle_shifting, particle_traction_method, pre_and_post_smoothing, pressure_beta, pressure_solver, pressure_smoothing, quasi_static, random_field, residual_tolerance, set_THB, shape_function,

shape_smooth, solid_cut_cell_min_fraction, solid_sdf_cut_cell, solid_sdf_function, solver_type, sparse_grid, stabilize, strain_smoothing, stress_integration, track_energy, symmetrize_matrix_free_tangent, use_elastic_matrix_free_tangent, velocity_projection, verlet_distance_multiplier, visualize.

A.2 DEM Keyword Names

AdaptiveStep, Alpha, Axis, BackgroundColor, BodyNumber, BodyOrientation, BodyPoint, BodyType, BoundingBoxFile, BoundingRadius, BoundingSphereFile, CFL, CameraUp, CellSize, CheckHistory, CheckOverlap, ClumpFile, ClumpProperty, ControlType, Counterclockwise, DigitalElevation, Direction, DiscretizeDomain, File, FileType, FixAngularVelocity, FixMotion, FixVelocity, Fraction, Gain, GenerateType, GeometryFile, GridFile, GridNumber, GroupID, Index, InertiaMoment, InitialAngularVelocity, InitialVelocity, LengthSize, LimitVelocity, LookAt, LookFrom, MainAxis, MaterialID, MaxBoundingRadius, MaxRadius, MaterialPointsPerCell, MinBoundingRadius, MinRadius, MoveVelocity, NSphere, Name, NoData, Object, Orientation, OuterNormal, ParticleColor, ParticleFile, ParticleNumber, ParticlePe rCell, Pebble, PebbleFile, Period, PointLight, PoissonSampling, Porosity, Position, Radius, RadiusDistribution, RayPath, RegionName, Resolution, RigidFile, SaveInterval, SavePath, ScaleFactor, ServoFile, ShellOffset, SimulationTime, SoftBody, SphereFile, SurfaceFile, SurfaceNodeNumber, SurfaceResolution, TargetStress, Template, Timestep, TotalParticles, Translation, TryNumber, ViewAngle, Visualize, VisualizeInterval, VisualizeMode, Volume, WallBase, WallCenter, WallFile, WallHeight, WallID, WallRadius, WallShape, WallType, WallVertice, WindowSize, WriteFile, WritePath, body_coordination_number, boundary, bvh_rebuild_interval, calm, compaction_ratio, domain, enable_shell, engine, function, gravity, hierarchical_level, hierarchical_size, iterative_model, levelset_grid_number, max_clump_number, max_digital_elevation_facet_number, max_facet_number, max_material_num ber, max_material_point_number, max_particle_number, max_patch_number, max_plane_number, max_rigid_body_number, max_rigid_template_number, max_servo_wall_number, max_soft_bod y_number, max_soft_velocity_constraint, max_sphere_number, point_coordination_number, scheme, search, search_direction, shape_function, soft_grid_number, soft_grid_storage, soft_grid_type, soft_levelset_advection_cfl, soft_levelset_advection_interval, soft_level set_advection_scheme, soft_levelset_domain_check, soft_levelset_domain_tolerance_cells, soft_levelset_reinitialization, soft_mechanical_grid_spacing_ratio, soft_pic_fra ction, soft_template_levelset_node_number, soft_template_material_point_number, soft_template_surface_node_number, surface_node_number, template_type, track_energy, verlet_distance_multiplier, visualize, wall_coordination_number, wall_per_cell.

A.3 MPDEM/CFDEM Keyword Names

AdaptiveStep, BodyOrientation, BodyPoint, BoundingRadius, CFD_coupling_domain, CFL, CheckHistory, FixMotion, GenerateType, GroupID, InitialAngularVelocity, InitialVelocity, MaterialID, Name, Period, PoissonSampling, Radius, RegionName, SaveInterval, SavePath, ScaleFactor, SimulationTime, Template, Timestep, TryNumber, Visualize, WriteFile, alphaPIC, background_damping, body_coordination_number, calm, compaction_ratio, coupling_scheme, domain, enable_shell, enhanced_coupling, function, gravity, nParticlesPerCell, particle_int eraction, ppcontact, pwcontact, search, track_energy, visualize, wall_coordination_number, wall_interaction, DragForceModel, DragLaw, ParticleReynold.

A.4 Implicit FEM and Coupling Engine Keyword Names

The capability index also follows controlled `dict.get` reads at the engine boundary. These names are valid nested solver/contact dictionary keys; they are listed here separately from the facade tables so agent-generated models can distinguish engine controls from top-level configuration.

MPM finite-strain material construction may pass a nested `material_parameters` dictionary. Standalone FEM accepts `absolute_tolerance`, `cloth_energies`, `enforce_stable_time_step`, `initial_velocity`, `line_search_minimum_step`, `line_search_reduction`, `line_search_sufficient_decrease`, `linear_solver_max_iters`, `linear_solver_tolerance`, `material_parameters`, `max_iters`, `minimum_jacobian`, `project_hessian_pd`, `raise_on_nonconvergence`, `save_path`, and `time_step`.

FEM–DEM reuses `absolute_tolerance`, `line_search_minimum_step`, `line_search_reduction`, `line_search_sufficient_decrease`, `linear_solver_max_iters`, `linear_solver_tolerance`, and `raise_on_nonconvergence`. FEM–MPM accepts the same set plus `max_iters` and `inexact_newton`. The latter defaults to false and requires nonassociated DP with a Taichi linear solver. It adapts linear accuracy while retaining free-force, Dirichlet, and lagged-friction convergence requirements. The true linear error is bounded by $\max(\varepsilon_{\text{lin,a}}, \eta_k \|\boldsymbol{R}_{free,k}\|)$ before Dirichlet elimination, also for Direct MPM–AffineBody below. Direct MPM–AffineBody accepts `inexact_newton` under the same material restriction, plus `absolute_tolerance` and `residual_tolerance` for its force stopping criterion (defaults 10^{-10} and 10^{-8}). Its public Direct route enables Inexact Newton by default for nonassociated DP; `inexact_newton=False` opts into strict linear solves. For all lagged-friction routes, the outer probe is unapplied and unclamped, and uses the same physical correction norm as inner Newton: FEM nodes, ABD surface vertices, and MPM/soft-point motion plus mechanical-grid-spacing scaled displacement gradients. Axisymmetric MPM also includes the hoop increment; see physical convergence. Force references persist through friction refreshes and reset per attempted step. Nonassociated DP requires force balance and correction convergence simultaneously, including after refresh. Terminal inexact corrections are checked at configured strict linear accuracy. Standalone lagged MPM–IPC uses BiCGSTAB and residual Armijo, with the free-force bound $10^{-10} + 10^{-8}\max(R_0, 1)$ and its existing `residual` velocity tolerance. No new public setting is required.

The complete new solver key paths are:

Key path	Values/default	Numerical effect
`FEMPM.set_solver.solver.inexact_newton`	Boolean; false	Adaptive Taichi Krylov tolerance for nonassociated DP; force convergence remains mandatory.
`MPDEM.set_solver.solver.inexact_newton`	Boolean; true for nonassociated DP, false otherwise	Direct 3D MPM–ABD adaptive Krylov forcing; requires nonassociated DP when enabled.

Key path	Values/default	Numerical effect
`MPDEM.set_solver.solver.absolute_tolerance`	Finite non-negative real; 10^{-10}, assembled-residual units	Absolute free-force stopping threshold for the nonassociated Direct route.
`MPDEM.set_solver.solver.residual_tolerance`	Finite non-negative real; 10^{-8}, dimensionless	Relative free-force threshold, using the initial norm clamped below by one.

Related FEM–MPM controls are:

Key path	Values/default	Numerical effect
`FEMPM.set_solver.solver.correction_velocity_tolerance`	Positive finite real; 10^{-7} m/s	Strict-mode correction stopping tolerance; also bounds prescribed-displacement error by Δt times this tolerance.
`FEMPM.set_solver.solver.newton_velocity_tolerance`	Same type/default	Accepted alias, used only when the canonical correction tolerance is absent.
`FEMPM.memory_allocate.memory.max_point_triangle_pairs`	Positive integer; `max_contact_pairs`	Allocated 3D point–triangle capacity; overflow must fail.
`FEMPM.memory_allocate.memory.max_point_edge_pairs`	Positive integer; `max_contact_pairs`	Allocated 2D point–edge capacity; overflow must fail.

Related theory. FEM–MPM and MPM–AffineBody coupling.

These controls apply only to the implicit transactional engines; explicit branches reject enabled step retry rather than pretending to have a Newton rollback boundary.

A.5 IGA–MPM Engine and Contact Keyword Names

The monolithic driver accepts `monolithic_max_iterations`, `monolithic_tolerance`, `monolithic_linear_solver_max_iters`, `monolithic_linear_solver_tolerance`, `armijo_c1`, `armijo_max_backtracks`, `strict_feasibility_tolerance`, `contact_ccd_safety`, `contact_ccd_min_step`, and `contact_ccd_max_iterations`. Fully implicit friction additionally accepts `fully_implicit_armijo_reduction`, `fully_implicit_dirichlet_atol`, `fully_implicit_force_atol`, `fully_implicit_force_rtol`, and `fully_implicit_velocity_predictor`.

Surface selection accepts `contact_surface_include`, `contact_surface_exclude`, `include_surfaces`, `exclude_surfaces`, `contact_surface_duplicate_policy`, and `contact_surface_duplicate_tolerance`. Friction dictionaries accept `friction_profile`, `static_friction`, `dynamic_friction`, `viscous_friction`, `mu_static`, `mu_dynamic`, `mu_viscous`, `stribeck_velocity`, `friction_fixed_point_iterations`, `friction_fixed_point_tolerance`, `friction_max_iterations`, and `friction_tolerance`.

B Maintenance Commands

The keyword tables can be refreshed with commands like:

```
python agent/geotaichi-model-builder/scripts/build_capability_index.py \
  --output agent/geotaichi-model-builder/references/capability-index.json
python agent/geotaichi-model-builder/scripts/audit_api_docs.py --kind all
python agent/geotaichi-model-builder/scripts/self_test.py
```

After editing this file, compile with:

```
latexmk -xelatex -interaction=nonstopmode -halt-on-error \
  -outdir=docs/helper docs/helper/geotaichi_user_theory_manual.tex
```